

Swiss German Intonation Patterns

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Abbreviations used

AC	accent command	$T1dist_rise$	temporal distance
AM	autosegmental-metrical		between accent command
ASCII	American Standard Code for		onset and segment onset
	Information Interchange	T_2	accent command offset
BE	Bern Swiss German	$T2dist_rise$	temporal distance
f_0	fundamental frequency		between accent command
Fb	baseline value of fundamental		offset and segment offset
	frequency	$T2dist_fall$	temporal distance
GR	Grisons Swiss German		between accent command
IP	intonation phrase		offset and segment
IPO	Instituut voor Perceptie		offset
	Onderzoek	ToBI	Tone and Break Indices
LFC	Low Frequency Contributions	TTS	text-to-speech
LPC	Linear Predictive Coding	VIF	Variance Inflation Factor
MLR	Multiple Linear Regression	VS	Valais Swiss German
PC	phrase command	ZH	Zurich Swiss German
RFC	rise/fall/connection	α :	natural angular frequency
SDS	Sprachatlas der Deutschen		of the phrase control
	Schweiz (Linguistic Atlas of	β :	mechanism
	German-speaking Switzerland)		natural angular frequency
SNF	Swiss National Science		of the accent control
	Foundation		mechanism
T_0	phrase command onset	γ :	relative ceiling level of the
T_1	accent command onset		accent component

SAMPA reference

Throughout this study, *Speech Assessment Methods Phonetic Alphabet* (SAMPA) transcription is used. The applied 7-bit ASCII Swiss German SAMPA was developed at the *Laboratoire d'Analyse Informatique de la Parole* (LAIP), Lausanne (Siebenhaar, Zellner, Keller 2002). It largely resembles the SAMPA originally developed by Wells (1997). The key differences are: Swiss German SAMPA /&/ for Wells' SAMPA /{/ , /*/ for /@/, /G/ for /N/, and /'/ for /"/. Lexical *stress* is conceived of as a feature of vowels. _(ptk) stands for the occlusion phase preceding fortis plosives, V(bdg) stands for the occlusion phase preceding lenis plosives, #& represents a filled pause, #h a respiration pause, and # a silent pause.

CHAPTER 1

Introduction

Intonation, or *voice fundamental frequency*, is a key feature of speech found in all languages around the globe (Hirst & Di Christo 1998: 1). Mainly, intonation is a tool for the structuring of information in spoken language, much like punctuation and formatting in a written text. Its scope, however, is much wider: intonation is the ubiquitous, underlying score of speech. After all, it is no coincidence that intonation is commonly referred to as “the melody of speech”. It is a channel of information that exceeds the realm of phonology, grammar or semantics: it is a means to express emotion, humor, and attitude, and hint at characteristics of the speaker. This is exactly why Stalder (1819: 7–8), in his early observation on the intonation differences of Swiss German dialects, alluded to “the stiffness and seriousness of the Bernese, – the hasty and quick of the Entlibucher, – the sluggishness in the articulation of the upper Freiämter, – the singing of the shepherds in the high mountains of Uri, Bern, Appenzell, and the Valais” (translation by AL). Speech melody seems to be an indicator of much more than information structuring and literally sets the “tone” in a conversation. Because it seems so deeply rooted in human behavior and is thus such a natural and integral part of speech, its existence in daily life is very rarely perceived and truly acknowledged as such. How many foreign language classes actually take the time to teach you the speech melody of the language you are learning? How often in our daily conversations do we actually take note of the intonational features of language?

The present study aims to do exactly that: to take note of the *fundamental frequency* [*f*0] *contours of spontaneous speech*. To be more precise, this study examines the *f*0 contours of the Bern, Grisons, Valais, and Zurich Swiss German dialects. As of today, no systematic account of Swiss German dialectal intonation exists, not to mention a systematic study of *f*0 behavior that is based on spontaneous speech. Hotzenköcherle (1962: 240) therefore rightly notes in the introductory volume of the *Linguistic Atlas of German-speaking Switzerland* (*Sprachatlas der Deutschen Schweiz*, 1962–2003) that the study of suprasegmental features of Swiss German dialects is considered “an important and alluring task for future monographic research”.

The study of intonation is, however, a difficult endeavor because intonation is truly multifunctional. As Vaissière (2004:256) puts it: “[o]f all dimensions of speech, intonation is clearly the most difficult to study”. Intonation can vary according to a number of dimensions, the most central of which are social factors, such as *sex* and *age* (for example, see Bolinger 1989:9ff.), language-dependent factors, such as *tones* or *pitch accents* (see Cruttenden 1986:1ff.), linguistic and paralinguistic functions like *phrase* and *lexical accents*, *prominence*, *information structure*, *focus*, *contrast*, or *conversational setting* (Baumann 2006a; Selting 1995), as well as *emotion* and *attitude* (see Murray & Arnott 1993; Kehrein 2002). In short, intonation is conditioned by a large number of variables on the linguistic, the paralinguistic, and the non-linguistic level of speech which is why, according to Bolinger (1989), intonation constitutes the “greasy part of speech”. This, of course, poses major difficulties for both a definition as well as an empirical investigation of intonation. This is why, until today, no universally accepted definition of intonation exists. There is neither consensus as to the object of research and the aim of intonational studies, nor agreement as to how intonation should be represented (Vaissière 2004:256).

This study sets out to provide an adequate description of the *f0* contours of spontaneous speech of the Bern, Grisons, Valais, and Zurich dialect and aims to fill the gap on Swiss German dialectal intonation. The interdisciplinary character of this study, incorporating acoustics, dialectology, as well as sociophonetics, hopefully contributes to a better understanding of the internal mechanisms of *f0* in spontaneous speech and sheds light on the weight of the different variables that interact in order to shape the *f0* contours of speech.

Within the framework of a research project at the University of Bern, speech data of 40 subjects from four different regions of German-speaking Switzerland were retrieved.¹ These regions include the Canton of Bern, Grisons, Valais, and Zurich. A vast majority of studies on intonation resort to the study of laboratory speech. In this study, however, the study of spontaneous material was preferred for two reasons: spontaneous speech data allows for the idiosyncratic dialectal features to permeate uninhibitedly (see Gilles 2005). Secondly, and equally important, it permits an investigation of *f0* variability in the context of linguistic, as well as paralinguistic and non-linguistic functions of intonation, since some of these components of intonation only surface in natural speech and informal settings (unless they are artificially induced or portrayed, of course).

1. SNF-Project 100011-116271/1: “Quantitative Ansätze zu einer Sprachgeographie der schweizerdeutschen Prosodie”. Department of Linguistics, University of Bern, 2005–2008.

The theoretical framework chosen to analyze and represent f_0 contours is an analysis-by-synthesis (Bell et al. 1961) procedure using the Command-Response model (Fujisaki & Hirose 1982), developed in Tokyo (Japan) and originally created for the Japanese language. This bridges yet another gap in the research literature on intonation: this is the first large-scale study that applies the Command-Response approach on a corpus of spontaneous, dialectal speech. The variables in this study are of linguistic, paralinguistic, as well as non-linguistic nature. f_0 variation in the variables was tested via an analysis of two global intonation parameters as well as three local parameters. Since this study aims to provide a fertile basis for future intonation research on Bern, Grisons, Valais, and Zurich Swiss German, as many linguistic, paralinguistic, and non-linguistic variables were incorporated as the scope of the study would allow. Hence, in the light of the numerous findings as well as the mostly inexistent material for comparison, the interpretations given in this study are by all means hypothetical and require further support. After all, the main goal of this study is to provide a first, thorough description of the f_0 behavior of four Swiss German dialects, not to formulate fully-fledged, linguistic interpretations. Rather, they are meant to spur new, exciting research geared solely towards the understanding of specific variables.

This study is structured into a total of 14 chapters. The second chapter will provide an introduction of the key concepts of intonation. After that, we will turn to a discussion of different ways of representing intonation contours in the section on intonation models. Chapter 4 will then present the model chosen for this study: the Command-Response model. Subsequently, we will review the rather scarce account of existing literature on Swiss German intonation before we turn to the statistical analysis. The statistics section is headed by Chapter 6, which specifies the methods used in this study. Chapter 7 illustrates the application of the Command-Response model and the modeling constraints applied. Note that the modeling approach applied in this study has a strong explorative character, since the analysis method is not formally defined in the papers presented by Fujisaki and co-workers, particularly not with regard to spontaneous speech. Chapters 8 to 11 give the results of the statistical analyses, starting with overall results and then progressing according to type of variable: linguistic variables, paralinguistic variables, and non-linguistic variables. The global and local f_0 behavior in each of the variables is analyzed in bivariate tests using parametric and non-parametric statistical tests against the background of detecting dialect-specific patterns as well as cross-dialectal differences. Each of these chapters features an immediate discussion of the findings. In Chapter 12, I will present dialect-specific, multiple linear regression models and logistic regressions for the investigated model parameters. The regressions allow for a distillation of the relative contribution

of independent variables towards explaining f_0 variability in a given parameter value. At the heart of this study are the *dialect profiles*, presented in Chapter 13, in which the major findings are revisited, discussed, and placed into a broader, dialectological and sociolinguistic context. Finally, we proceed to concluding remarks and a brief outlook.

CHAPTER 2

Intonation

The *fundamental frequency* [hereafter f_0] of spoken language, is commonly referred to as the *intonation* of language. The term *intonation*, which is derived from Greek *tonos* (tension), denotes the rise and fall of voice pitch over entire phrases and sentences. It is generally considered to belong to the category of *suprasegmentals* which, as the term implies, include those properties of an utterance that lay beyond a single segment. Rather, they affect a string of segments, ranging from one syllable to entire sentences. The term *suprasegmentals* is often used synonymously with the term *prosody*, derived from Greek *prosodia*, which Allen (1973: 3) describes as “a ‘tune’ to which speech is intoned, [...] the melodic accent which characterized each full word in Greek”. Apart from intonation, prosody further includes *loudness*, *quantity*, *speech rate*, *rhythm*, *voice quality*, *phrasing*, and *pausing* (see Möbius 1993a: 9). These prosodic parameters are inter-dependent: duration and intensity, for example, can affect the perception of pitch (see Lieberman 1980; Niebuhr 2007).

Every language exhibits intonational features. In fact, it was shown that prosodic features are among the first linguistic features children acquire, and – as observed in aphasic patients – they also seem to be the last feature lost (Hirst & Di Cristo 1998: 2). From an evolutionary perspective, it is theorized that the underlying principle of pitch is to convey a physical impression of the sound source, i.e. the size of the vocalizer (see Vaissière 2004: 252). Because of the sexual dimorphism in the vocal anatomy of humans, the f_0 of males is generally lower than the f_0 of females. If threatened, males tend to make use of these low-pitched sounds in order to protect their family and to intimidate their enemies. Cross-culturally and cross-linguistically, low pitch therefore is associated with dominance, physical largeness, and potential threat, while high pitch conveys submissiveness, small physique, and harmlessness. This phenomenon is known as the frequency code (see O’Hala 1983).

The extraordinary characteristic about intonation is that it carries *meaning*. A falling intonation, for example, more often than not denotes finality, whereas a rising intonation signals incompleteness (Fox 2000: 269ff.). This is not the case, however, for other suprasegmental features. For example, *stress* and *tone* or, on the

segmental level, *phonemes* and *syllables*, are not inherently meaningful. Furthermore, intonation is *gradient* and not discrete, meaning that the shape of a fall or rise can be described accurately and numerically. This gradience also translates into gradience in meaning (Fox 2000).

Generally, one can distinguish between three types of languages with respect to how intonation modulations are exploited: *tone languages*, *pitch accent languages*, and *intonation languages*. In tone languages, tone is a characteristic of the lexicon: the change of a tone in an otherwise unaltered segmental environment can trigger a change in meaning. For tone languages, *f0* is thus primarily used to signal lexical contrasts (Vaissière 2004: 242). In pitch accent languages, on the other hand, tonal patterns of the word represent the most basic constituent for the shape of *f0* movements (see Vaissière 2004: 242). In intonation languages, the speech melody is a feature of phrases and sentences, which, in languages without lexicalized tones, is manifested in terms of a rather intricate intonational system. Intonation and tone are not mutually exclusive, yet tone languages make less rigorous use of intonation patterns than intonation languages (Cruttenden 1986: 9ff.). The majority of intonation languages are characterized by a default, low, falling pitch and a contrasting, raised pitch. Most frequently, the latter is used in questions, whereas the former contour indicates statements (Hirst & Di Cristo 1998: 1). There are, however, many varieties that show reverse tendencies, such as the rising intonation of Belfast English declaratives or the falling interrogatives of Bengali (Gussenhoven 2004: 54). So evidently, there are language-inherent differences in intonation in the world's languages, which means that intonational features may not always translate into the same linguistic functions.

Intonation research is a highly inter-disciplinary field of research, involving disciplines such as linguistics (phonetics in particular), speech pathology, foreign language education, and speech technology. Over the past decades, the study of *intonation* has continually gained impetus and the number of conferences and workshops in this area of research is growing steadily. In foreign language education, for example, intonation and prosody research finds direct application in classrooms: students can be given acoustic as well as visual feedback on their prosodic performance (Siepmann 2001: 13ff.). Speech technology represents another multifaceted branch of intonation research, which includes speech analysis, synthesis, and speech recognition (Mixdorff 1998: 9). Speech synthesis research, for example, is geared towards improving the naturalness of synthetic speech. Due to this interdisciplinary character of intonation research, the terminology used in this field has become exceedingly hybrid. Möbius (1993: 7) aptly points out that intonation research is marked by a Babylonian confusion of tongues. Hence, in order to minimize possible confusion, the

following sections are geared at introducing and defining the terminology and concepts used throughout this study.

The organization of the subsequent sections mirrors the structural hierarchy of intonation. Since this study is based on a corpus of spontaneous speech, special attention is paid to the intonation of spontaneous speech. After a definition of the broad concept of intonation and its underlying physiology, the focus will shift to the topmost overall structure, the intonation phrase. The subsequent concepts include the declination of the intonation phrase, pitch reset, and – on the level of global intonation – the phenomenon of pitch range. On the local intonational level, the often-debated terms prominence, accent, and stress will be introduced and discussed. Accent is then further subcategorized into word accent and sentence accent. Having arrived at the lowest intonational level, we will take a closer look at microprosodic intonational variation. After this introduction to the key concepts of intonation, we will touch upon the relevant informational, paralinguistic, and non-linguistic functions of intonation.

Before delving into the matter of defining intonation, it is crucial to draw attention to a couple of important issues. Because the present study is mainly concerned with the *phonetic* modeling of Swiss German dialects, the following section is weighed accordingly. In other words, the terminology introduced corresponds to the terminology used in intonational phonetics as opposed to intonational phonology. Furthermore, the next section neither sets out to cover the vast array of definitions presented in previous literature on intonation nor to contribute to the unification of terminologies.¹ Finally, it should be borne in mind that the definitions used and discussed in the present study largely stem from the German and English research tradition and are thus mainly based on the German and English language.

2.1 Defining intonation

Baumann (2006a:4) points out that, in most literature, intonation is either understood in a broad or a narrow sense.² In a narrow sense, intonation is conceptualized as the continuous contour of speech melody, with f_0 as the acoustic correlate, which designates the quasi-periodic number of cycles per second of the

1. For a detailed discussion of intonation-related terminology, see Inozouka 2003.

2. Beckman (1986) provides a definition of intonation that is situated halfway between these two extreme points of view. She advocates a pluriparametric view of intonation by underlining the critical role of intensity and durational features in intonational representation.

speech signal, measured in Hertz (Botinis et al. 2001:264; Gilles 2005:3). In contrast, intonation in a broad sense is equated with prosody, including pitch range, phrasing, stress, accentuation, rhythm, and tempo. The view adopted in the present study corresponds to the narrow view, with f_0 being the primary acoustic parameter of intonation. The author believes that a systematic link between intonation and the grammatical system can only be detected by means of such a restricted definition (Günther 1999:62). It is this notion of intonation, realized with all its manifold linguistic functions on both the word and sentence level, which constitutes the core of this study.

The physiological mechanisms indispensable to the production of speech sounds are the supralaryngeal vocal tract, the larynx, and the subglottal system. Intonation is generated for the most part by the laryngeal structure (i.e. length of the vocal cords and muscular tension) and by subglottal pressure. The larynx, a feature found in all terrestrial animals, was originally devised to protect the lungs of the primal lungfish from water intrusion but, as evolution progressed, has adapted for phonation (Lieberman & Blumstein 1988: 11). In the words of Lieberman and Blumstein (1988), speech sound is generated as follows:

The primary role of the larynx in the production of speech is to convert a relatively steady flow of air out from the lungs into a series of almost periodic, i.e. “quasi-period,” puffs of air. The larynx does this by rapidly closing and opening the airway by moving the vocal cords together or pulling them apart.

(Lieberman & Blumstein 1988:4)

Vocal cord frequency thus refers to the number of complete cycles in a given period of time, usually within a second. The required time to complete one full cycle is called a period and is indicated as 1 Hertz (Hz). The faster the vocal fold vibration, the higher the periods per second, i.e. the higher the produced sound (Gussenhoven 2004:2). The male f_0 usually varies between 60 Hz and 240 Hz, while the female f_0 generally falls somewhere between 180 Hz and 400 Hz (Cruttenden 1986:4). If the frequency is lower than 40 Hz, the vocal fold vibration is perceived as a series of separate events. If it exceeds 40 Hz, however, it is considered continuous.

Fundamental frequency is only found in voiced sounds. Interestingly enough, the human ear perceives continuous speech as a nonstop f_0 pattern, albeit about one quarter of the sounds in English, for example, is voiceless (Cruttenden 1986:4). In the same vein, human perception allows us to perceive a male f_0 on the telephone, despite the fact that male voices are normally below 300 Hz and telephones usually only transmit frequencies between 300 Hz and 3400 Hz. The correct f_0 is calculated via periodicity analysis (see Kohler 1977).

The perceptive equivalent of f_0 is referred to as *pitch* (see Inozouka 2003: 53; Baumann 2006a: 4). Lehiste (1970) points out that the lowest pitch still perceivable to humans is at approximately 16 Hz, the highest at 20,000 Hz. In the present study, f_0 is understood as the actual vibration of the vocal folds, whereas pitch is used in the sense of the f_0 as perceived by the human ear.

f_0 variation results from the modification of the length and tension of the vocal cords brought about by cartilage movement and intrinsic muscles activity (Möbius 1993: 72). Figure 2.1 shows the cartilage and the intrinsic muscles of the human larynx (adopted from Lieberman & Blumstein 1988: 98).

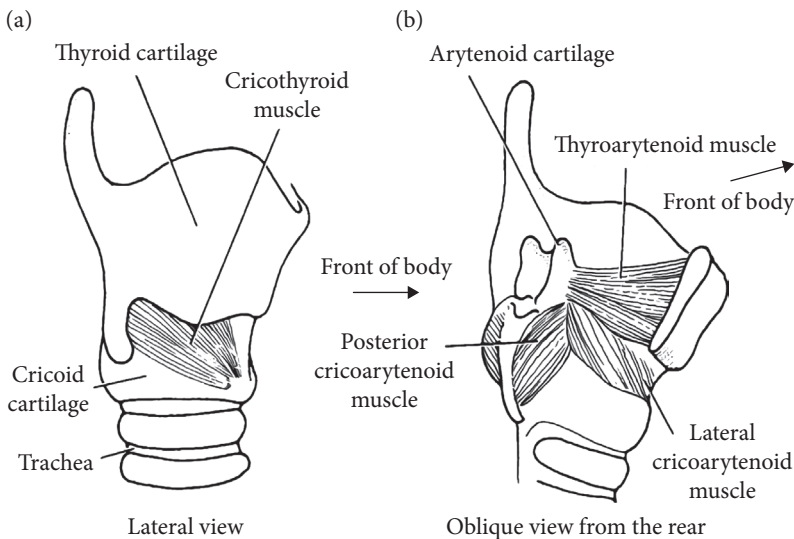


Figure 2.1. The human larynx (adopted from Lieberman & Blumstein 1988: 98)

The major constituent of the larynx is the ring-shaped cricoid cartilage, attached to the top part of the trachea. The thyroid cartilage, identifiable from the outside as the male Adam's apple sits on top of the cricoid cartilage. The male thyroid cartilage grows larger during puberty and causes an elongation of the vocal folds. This does not only result in the development of the Adam's apple but also generates an overall lower f_0 . The vocal cords – a combination of ligaments, muscles, and tissue – are located inside the thyroid cartilage, attached to the two arytenoid cartilages, the thyroid and cricoid cartilage (Lieberman & Blumstein 1988: 100). Via electromyography (EMG), a technology for measuring electrical muscular activity, Collier (1975) discovered that the cricothyroid muscle is mostly responsible for the direction, the range, and the speed of f_0 changes. Muscular contraction brings about a higher f_0 , while relaxation causes a lowering of f_0 .

Further, he shows that a decrease of subglottal air pressure results in a gradually falling baseline. Thus, whenever the cricothyroid muscle is inactive, f_0 is governed by subglottal air pressure (Collier 1975:254). Atkinson (1978) further provides evidence that laryngeal tension has a strong impact on higher leveled f_0 movements, whereas pulmonary pressure dominates f_0 at lower values. More specifically, the cricothyroid muscle, supported by the lateral cricoarytenoid muscles (see Figure 2.1), controls the highest f_0 values, whereas mid-range f_0 values are governed by the sternohyoid muscle (an extrinsic muscle to the larynx). What both Collier (1975) and Atkinson (1978) were able to demonstrate is that intonation production is the product of a joint effort of several different components of an intricate, underlying physiological mechanism (Möbius 1993:74).

2.2 Intonation phrase

The most significant production unit of intonation is the *intonation phrase* [hereafter referred to as IP]. This term was introduced originally by Pierrehumbert in 1980 and is often used synonymously with the term *phrase*. Each IP contains at least one accent that serves as the tonal anchor point for the intonation contour (Gilles 2005:6). An IP may consist of one syllable only or may span syntactic phrases, clauses, or entire sentences. A larger utterance may therefore contain more than one IP (Botinis et al. 2001). In general, IPs in spontaneous conversation are shorter than IPs in prepared speech, yet the duration of the IPs depends on the idiosyncratic style of the speaker. Some speakers realize comparatively shorter IPs, others produce longer IPs (Féry 1988:43). Shorter IPs are usually accompanied by shorter pre- and post-boundary pauses, while longer IPs are typically associated with longer pre- and post-boundary pauses (Krivokapic 2007). The boundaries of IPs usually feature clearly perceivable, either high or low boundary tones in phrase-final position (see Pierrehumbert 1980). Phrase-initial tones are possible as well but are not necessarily required (Féry 1988:45). Unfortunately, however, IP boundary marking is not always quite as straightforward since it involves a number of factors apart from intonation as well.

IPs are delimited by a bundle of different prosodic features which can either occur together or independently.³ This cluster of prosodic features entails *f_0 movements* at the beginning and end of an IP, *pausing*, *phrase-final syllable*

3. The recognition of prosodic boundaries is one of the major concerns in automatic speech processing research (see Ostendorf 2000).

lengthening, decrease of articulation rate, and pitch resets (see Cruttenden 1986). Further, Peters et al. (2005) note that laryngealization, glottalization, and change of intensity also contribute to IP boundary marking. What is still unclear is the weight of these individual features, however. For example, Vaissière (1983:57) points out that “[t]he physiological basis of the relations among pause, breathing, declination and resetting is difficult to establish, since speakers may pause without breathing, or reset the baseline without pausing.” As a result, the prosodic parameters responsible for IP demarcation are discussed and weighed differently in the literature.

Further cues for IP boundary marking can also be derived from syntactic structure. This is especially true for prepared speech and, to a lesser extent, also for spontaneous speech. Taylor (1994:16) notes that it is unclear whether syntax determines prosody via complex mapping or whether syntax constitutes one pillar of IP placement. It seems quite likely, however, that prosodic structure is affected by metrical factors. Accordingly, prosodic boundaries can be placed in the middle of syntactic constituents (Taylor 1994). The correlation of IP boundaries and syntactic boundaries therefore is not causal but casual (Botinis et al. 2001:269). Because syntax and IP often do not overlie, the IP is considered by some scholars to be associated with sense groups, i.e. words grouped by meaning (see for example O’Connor and Arnott 1961:3).

Lieberman (1980:192) emphasizes yet another factor of IP demarcation, namely the importance of breathing. Breath groups are used by the speakers to divide a train of words into sentences, where speakers usually draw breaths at the end of conceptual units, such as sentences and clauses (ibid.; Vaissière 1983:54; for Swiss German see Hove 2004). Typically, breathing and pauses trigger a reset in pitch (Vaissière 1983:57). Hence, Cruttenden (1986) and Peters et al. (2005) consider pauses (including respiratory pauses) as a reliable cue for IP boundary marking. Additionally, pauses frequently seem to accompany new conversation topics and often take place after the first word of an IP. Cruttenden (1986:38) attributes such pauses, which provide the speaker with additional time to plan the rest of the sentence, to typical “performance errors”. He adds, however, that pausing is highly idiosyncratic, and that pauses cannot always be taken as cues for IP boundaries. Duez (1982) demonstrates that pauses in political speech are more aligned with the underlying grammatical structures, whereas in spontaneous speech, grammatical structure and pause placement must not necessarily coincide. Since repetitions, incomplete sentences, and false starts are a feature of unmonitored speech, IP boundaries in spontaneous speech are often vague (Cruttenden 1986:36). A case in point is the incongruence between syntactic structure and IP placement. In German, for example, IP boundaries can but do not have to coincide with larger syntactic boundaries (see Bierwisch 1966).

2.3 Declination and pitch reset

Declination refers to the gradual lowering in f_0 during the continuation of an utterance. In the words of Lieberman (1980:195) “[t]he falling fundamental frequency contour, which is structured by the vegetative aspects of respiration, is the universal language signal signifying the end of an ‘ordinary’ sentence”. The easiest way for a person to make a sound during expiration will involuntarily result in this kind of f_0 declination (ibid.). Hence, physiologically, declination is grounded in the decreasing transglottal air pressure in the course of an utterance. Some scholars therefore argue that declination is not controlled by the speaker and should be interpreted as a result of physiological speech production mechanisms (see Lieberman 1967; Vaissière 1983). Conversely, Pierrehumbert (1980) assumes that the speaker has intentional control over declination, and that declination should be considered a means of vocal expression. Not denying the principle of phonetic declination, she argues that downdrift also bears a phonological function, which she calls *downstep*. This so-called downstep constitutes a grammaticalized version of declination.

As is illustrated in Figure 2.2, the declination tendency is typically framed by a topline, the realization of f_0 peaks, and a baseline, which serves as a reference point for the valleys of an intonation contour (see ‘t Hart et al. 1990).

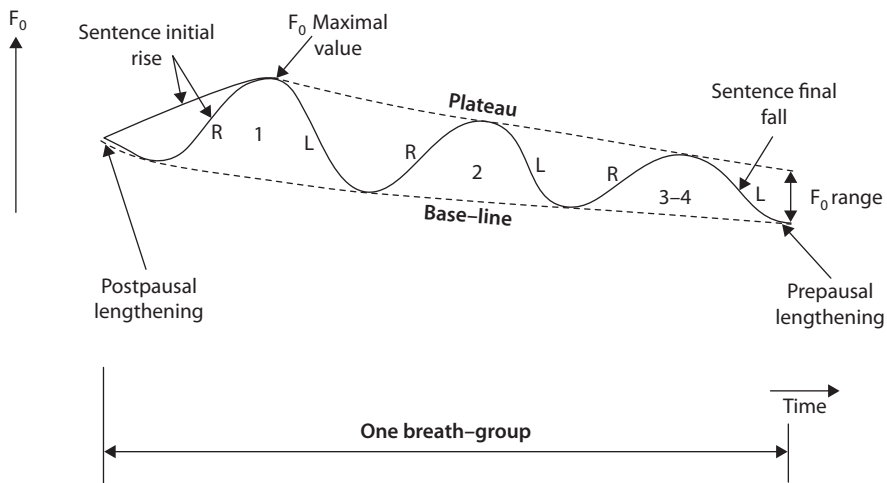


Figure 2.2. Properties of an intonational phrase as observed in several languages (adopted from Vaissière 1983:55). The dotted lines represent top- and baseline of the IP

Declination applies to both the top- as well as the baseline of the intonation contour. Baseline declination is apparent in that the fundamental frequency of unaccented syllables is higher for IP initial position than for phrase-final position. The declination of the topline, on the other hand, denotes the decline of absolute f_0 for pitch prominent syllables in the course of an IP. On a higher level, declination also operates across intonation-groups or whole paragraphs (Cruttenden 1986:126ff.). The degree of declination normally depends on the length of the utterance, with declinations being steeper for shorter utterances and less steep for longer utterances (see Vaissière 1983; see Adriaens 1991:60). It is crucial to note, however, that declination does not occur in all intonational phrases. Declination effects are high for laboratory speech this is not the case for spontaneous speech in which the declination tendency seems to be less obvious or sometimes not present at all (Vaissière 1983:57).

Declination is closely linked with a *reset in pitch*, i.e. the readjustment of the declination line. Since declination causes the f_0 to reach a fairly low point, the f_0 needs to be reset. Generally, this happens after a pause but the resetting of the baseline can also occur in the absence of pauses (see Fujisaki et al. 1979). Therefore, the relationship between pitch reset and pausing is still unclear (Vaissière 1983:57). According to Pierrehumbert (1979), speakers of English frequently reset their pitch in such a way that it coincides with a syntactic boundary. In this sense, resetting is used as a boundary marker and the degree of the reset is indicative of the strength and importance of the syntactic boundary (Vaissière 1983:57). Pitch resets thus suggest special prominence, which is why they carry a direct communicative function (Fox 2000:309).

As has already been mentioned, pitch resets in spontaneous speech need not coincide with larger syntactic boundaries. This loose link between syntax and pitch reset is exemplified in the following Figure 2.3.⁴

4. The figure illustrates a contour that was generated with the Command-Response model. This type of figure occurs commonly in the present study. The top panel illustrates the speech waveform, the +++ show the extracted f_0 . The bold line displays the model f_0 , while the dotted line stands for the phrase component. At the bottom of the figure, the accent commands are depicted in rectangular shapes. Syllable boundaries are marked with the vertical dotted lines. In all of these figures, the contour is displayed in the log F domain (see Mixdorff 1998:8). For more information about the Fujisaki model see Chapter 4.

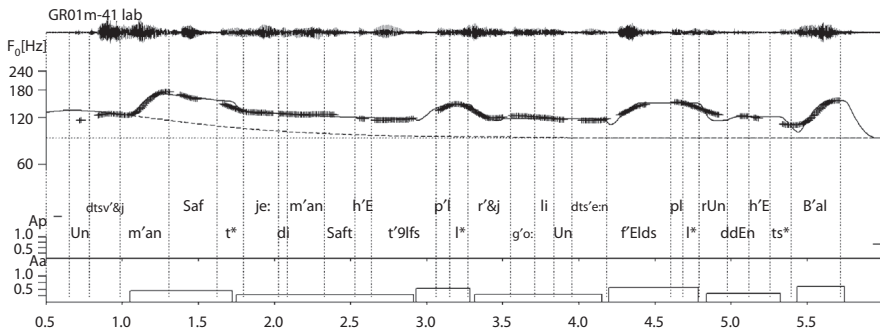


Figure 2.3. Example of a Grison speaker's IP – containing several syntactic phrases

In German, the speech sequence in the above Figure denotes the following (syntactic phrases are marked off with square brackets): [*und zwei Mannschaften*] [*jede Mannschaft hat elf Spieler*] [*einen Goalie und zehn Feldspieler*] [*und dann hat's einen Ball*] (Engl.: [*and two teams*] [*each team has eleven players*] [*one goal-keeper and ten field players*] [*and then there's a ball*]). One would expect that at the introduction of the new topic [*and then there is a ball*], the speaker would start a new IP since the larger syntactic unit, the main clause [*and two teams*] followed by two subordinate clauses [*each team has eleven players*] [*one goal-keeper and ten field players*], is completed. Instead, the f_0 continues to decline even into the following phrase and the new topic [*and then there is a ball*] and thus stretches over the entire utterance. A further problem arises regarding the distinction between pitch resets and word accents. For the syllable *f'EldS*, for instance, one could assume that there is a minor pitch reset. On the other hand, the rise in f_0 could also be interpreted as the pitch movement of the word accent for this particular syllable.⁵

2.4 Stress and accent

Werner (2000: 3) does not exaggerate by saying that the definitions for *accent* and *stress* (or the German counterparts *Akzent* and *Betonung*) are hotly fought over in intonation research. For the most part, theoretical debates on accent refer to the acoustic correlates of accent, yet they also refer to its phonological significance (2000: 114). In the words of Ladd (1996: 286), “the terminology in the general area

5. As illustrated by the modeling of this phrase in Figure 2.3, the pitch movement of the syllable *f'EldS* was interpreted as the pitch movement of the word accent.

of ‘accent’ is really a mess”. In order to shed some light on these terminological issues, the following section will introduce some of the expressions frequently used in the intonation literature or related to accent in a broader sense. These terms include prominence, stress and accent, accent group, and phrase accent. Naturally, it is difficult to tackle these issues as separate concepts since the terms and definitions overlap. Prominence and stress, for example, to some degree overlap conceptually: a stressed syllable is a prominent syllable, and a syllable can be made prominent by marking it with stress. Nevertheless, it makes sense to try to uphold narrow definitions, as this is the only way in which subsequent findings can be discussed in a precise manner.

2.4.1 Prominence

Prominence is a means of marking syllables and placing emphasis. A prominent syllable stands out of a given context, implying increased importance (Cruttenden 1986: 7). Mayer (1997: 15) distinguishes between two types of prominence: *lexical prominence* and *phrasal prominence*. If prominence occurs on the word level, it is normally referred to as word accent or lexical accent. In reference to the discussion of declination, it should be noted that the later a prominent syllable occurs in an IP, the lower its peak f_0 value (Taylor 1994: 17). This of course is due to the declination tendency of the IP.

The acoustic correlates of prominence are complex and language-dependent. More importantly, it is imperative to distinguish between *prominence production* and *prominence perception*. For prominence production, the most critical indicator for varieties of English is duration, followed by intensity and, least importantly, f_0 . For prominence perception, on the other hand, f_0 was shown to occupy a much more critical role (see Kochanski et al. 2005; Hirst 1983).⁶ Since the aim of this study is to model the produced intonation contours of speakers of Swiss German dialects, the following discussion will mostly focus on prominence production.

Prominent syllables are commonly assumed to be louder, longer, and higher in f_0 .⁷ Since both f_0 as well as intensity are produced by increasing and decreasing pulmonary effort, subglottal pressure, and vocal fold tension, those two parameters

6. However, Bannert (1983) discovered that f_0 may also be perceived as higher in constituents with higher intensity – thus, underlining the role of intensity in intonation perception.

7. It should be borne in mind, however, that prominence can also be achieved by a reversal of these parameters. Isačenko and Schädlich (1970: 24ff.), for example, have shown that a lowering of the f_0 on specific segments – if the surrounding segments show high pitch – has the same effect as increasing the pitch in a low f_0 segmental environment.

prone to correlate positively in most instances (Möbius 1993a:15). However, the role of intensity for prominence marking is not yet clear and its linguistic function in continuous speech is difficult to determine. Intensity is also the least studied of the three parameters (Vaissière 1983:63). Not all languages mark prominence concurrently with all three above-mentioned parameters. Rather, a number of languages have ascertained their own suprasegmental code (*ibid.*). Dutch and English have been reported to feature a concurrence of all three parameters in lexically stressed syllables (Vaissière 1983:62–63). In these languages, the prosodic parameters for marking prominence are timed according to the lexically strong syllables. Hence, they are referred to as stress-timed languages. The temporal interval between stressed syllables is uniform, i.e. these languages show a tendency towards isochrony based on stress (Cruttenden 1986:25). An example language with reduced correlation of the three prosodic parameters for prominence marking is French. In the case of French, a syllable-timed language, the parameters for prominence marking are bound to the first and last syllable of the word, or, the rhythmic group: on the first syllable, we typically find a minor rise in f_0 , while the word-final syllable may exhibit a variety of prominence contrasts (Vaissière 1983:63).⁸ These examples show that the relative weight of each of the three prosodic factors for prominence marking is language-specific and that it is the interrelations of these features which, according to Vaissière (1983), are the most salient characteristics of a language variety:

It is possible that the specific interrelations between the three suprasegmental features (f_0 , duration and intensity) [...] are the most salient characteristics differentiating between languages, dialects and individual ways of speaking. If this is true, most of the existing descriptions of prosodic systems [...] are incomplete, since they describe only one parameter at the time. (Vaissière 1983:66)

This finding provides fertile grounds for future research on Swiss German dialects. Earlier literature on the Valais dialect, for example, suggests that intensity and pitch do not coincide frequently in prominence marking (see Wipf 1910). Instead, other compensatory means for marking prominence seem to be applied. The representation of f_0 is the only prosodic parameter that will be under scrutiny in this study. In future research, however, the study of intensity and its (possible) correlation with f_0 might prove to be of interest.

8. The question of whether or not isochrony in fact holds in reality is another issue, since it is claimed that for syllable-timed languages, for example, acoustic correlates of stress-to-stress intervals cannot be verified decisively. This issue is, however, not pursued further in the present paper (see Auer & Uhmann 1988; Ramus 1999; Low et al. 2000).

2.4.2 Stress

Stress is one of the most elusive prosodic features (Lehiste 1970: 106). Adding to the terminological confusion, stress and accent are often used interchangeably in the literature. Stress, often used synonymously with lexical accent or lexical stress, is determined by the lexicon of a language and is marked by prosodic prominence. In a given linguistic context, the syllables that carry stress are perceived as more salient. Commonly, one distinguishes between word stress as an abstract feature, as proposed by Trager and Smith (1951), Chomsky and Halle (1968), and re-interpreted by Liberman and Prince (1977), and the actual phonetic realization of stress (see Günther 1999: 48ff.).

Trager and Smith (1951) as well as Chomsky and Halle (1968) suggest that stress is primarily a property of single vowels and that prominence placement is governed by stress rules. Based on this concept, Liberman and Prince (1977) elaborate this approach and show that stress is not an absolute feature of vowels but is, in fact, relative. Stress is assigned according to strong and weak syllables, which are present on both the lexical and the phrasal level. This approach falls under the general taxonomy of metrical phonology. In this theoretical framework, prominence is understood as an abstract feature, deriving from the metrical strength of syllables in any given utterance (Niebuhr 2007: 6). Spoken in isolation, each word carries a lexical accent which can be predicted by means of metrical rules.

Generally, disyllabic words (e.g. *carrot*) consist of a relatively stronger syllable (*s*) and a weaker syllable (*w*). In words with more than two syllables (e.g. *Pamela*), the relative strength of the syllables is assigned by metrical theory. In words with primary and secondary stress (e.g. *sensibility*), the main stress of the word is placed on the syllable that is governed by *s* nodes only (Hayes 1985: 1ff.).⁹

The correspondence between abstract, prosodic characteristics and acoustic features is a very intricate matter (see Hirst & Di Cristo 1998: 5ff.). With respect to stress, it is the role of intensity, in particular, which is a source of ambiguity. In the case of German, the acoustic correlates of stress normally entail an increased *f*₀, as well as higher intensity and longer duration (see Mixdorff 1998). Isačenko and Schädlich (1970), too, advocate that intensity changes are only a secondary means for placing stress, the most central factor being changes in *f*₀. Usually, stress falls on the first syllable of the stem of a native word, which in German generally constitutes the penultimate syllable of the word. In the English language, however, Trager and Smith (1951) point out the significance of intensity as a means

9. Psycholinguistic studies have shown that such prominence patterns are represented in the mental lexicon of the speaker (see Levelt 1989).

of marking stress. This claim is also supported by Beckman (1986) and Silipo and Greenberg (1999) for American English and, recently, by Kochanski et al. (2005) for British English. In the latter study, intensity and duration appear to carry more information on stress than *f₀*. Ivic and Lehiste (1963), on the other hand, conclude that the reliable cue for stress marking in Serbo-Croatian is not intensity but, in fact, duration (ibid. 1963, quoted in Lehiste 1970: 134). In other words, acoustic correlates of stress seem to be highly language-dependent.

2.4.3 Accent

Cruttenden (1986: 16) equates *stress* with any kind of prominence, whereas the term *accent* is only used if prominence is achieved through *f₀* movements. Word-stress syllables, so Cruttenden (1986: 17), are marked as *stressed* in the dictionary and are potential carriers of *accents*, i.e. *f₀* movements – given certain exceptions (see Féry 1988; Günther 1999). In theory, any syllable can be accented (e.g. in cases of contrastive accents where lexically weak syllables can be accented) but, normally, it is the metrically strong syllables which carry accent. Mixdorff (1998) argues along similar lines and reports that, in German, accented syllables feature a distinct *f₀* movement, higher intensity, and longer duration. If *f₀* movements are not present, he refers to that syllable as stressed but “de-accented” (ibid. 1998: 19).

Given the above discussions, there seems to be little evidence to suggest that stress indeed constitutes a prerequisite of accent. For this reason, accent, in the present study, is defined independently of stress. *Accent*, in the current study, refers to a realized *f₀* movement, regardless of the presence of lexically strong (i.e. stressed) or weak (i.e. unstressed) elements. This *f₀* movement can concern only one, or several segments, which by being “accented” are understood as being made *f₀* prominent. *Stress*, on the other hand, refers to the abstract word stress, which is understood as being governed by stress rules (see Trager & Smith 1951; Chomsky & Halle 1968). *Stress* can be signaled by any, or a combination, of the three prosodic parameters intensity, duration, and *f₀*. More often than not, *stress* and *accent* do, but need not, correlate, since both *f₀* as well as intensity are produced by an increase and decrease of pulmonary effort, subglottal pressure, and vocal fold tension. Moreover, I agree with Vaissière (1983), who proposes that each of the three prosodic parameters for marking prominence (intensity, duration, and *f₀*) should be regarded independently as their relative weight seems to be language-specific.

2.5 Pitch range

Pitch range, often used synonymously with pitch span, refers to the variation in pitch height between *f₀* maxima and *f₀* minima in speech and has to be defined

separately from pitch level (Ladd 1996). Pitch level denotes the baseline, i.e. local f_0 minima, which can be raised or lowered from utterance to utterance. Pitch range carries paralinguistic functions: a higher pitch range is often associated with emotions such as joy, anger, fear, or surprise (see Murray & Arnott 1993). Furthermore, it also exhibits a communicative function, in that it can mark the beginnings and endings of new topics. A high pitch range, for example, introduces a new topic, whilst a low pitch span signals the end of a topic (Cruttenden 1986: 129). Pitch range variation may apply only to parts of an IP (Gussenhoven 2004: 76). Thus, the boundaries between pitch range and local accents often are not clear-cut, which makes it difficult to judge whether a high accent, for example, is due to local f_0 prominence or a momentary shift in pitch range (Taylor 1994: 20). Mennen Schaeffler, and Docherty (2008) explain that pitch range is particularly difficult to quantify. Most commonly, long-term distributional measures are used for pitch span measurement (2008: 527).

2.6 Functions of intonation

The information carried by speech can be categorized into three major groups. Following Fujisaki's (2004) distinction, these groups are:

Linguistic information: "the symbolic information that is represented by a set of discrete symbols and rules for their combination" (Fujisaki 2004: 1).

Paralinguistic information: "the information that is not inferable from the written counterpart but is deliberately added by the speaker to modify or supplement linguistic information" (Fujisaki 2004: 2).

Non-linguistic information: "these factors are not directly related to the linguistic and paralinguistic contents of the utterances and cannot generally be controlled by the speaker" (ibid.).

Intonation operates on all of these three levels of information. On the *linguistic* level, intonation – or, to be more precise, stress and its possible acoustic correlate f_0 – can serve as a means of marking metrically strong syllables. Further, f_0 is indicative of word class demarcation: grammatical words frequently feature lower f_0 than lexical words (see Mixdorff 1998; Möbius 1993a, for example). On a *paralinguistic* level, intonation is a tool for conversational structuring in that it can highlight certain constituents that carry important information. The term *non-linguistic* refers to features over which the speaker has no direct control. These include physical factors such as sex, age, anatomical idiosyncrasies, or speech habits, such as the rate of articulation. The non-linguistic function of intonation further refers to the ability of intonation to express, for instance, emotion.

It should be pointed out that the above categorization is idealistic, since there is no consensus regarding the association of intonational features with specific

functions. This is due to the nature of the phenomenon under scrutiny. Intonation is nestled in a complex network of overlapping and mutually influencing linguistic and paralinguistic structural levels (see Gilles 2005: 16). The question of whether and how one can demarcate the linguistic from the paralinguistic dimension of intonation is a major concern of current intonation research (Schmidt 2001: 10). Thus far, a universally applicable method that allows the separation of paralinguistic and non-linguistic aspects from linguistic functions of intonation is yet to be found (Lehiste 1970: 96). For this reason, the overarching-function of intonation, namely information structuring, is foregrounded in this section. Subsequently, the focus will rest on paralinguistic and non-linguistic functions, since these are imperative in the context of the spontaneous speech data on which the present study is based.

2.6.1 Information structuring

Intonation is a means of creating perceptively intelligible units. The stream of speech is divided into IPs by means of global intonation contours, which facilitates parsing on the part of the listener. The clearer the breaks, the better the listener will be able to follow (see Bannert & Schwitalla 1999). In addition, IPs are strongly related to coherence structuring (Botinis et al. 2001: 276). In ambiguous phrases such as “the queen, said the knight, is a monster” versus “the queen said the knight is a monster”, it is the prosodic structure which can reveal the intended meaning, which is reflected indirectly by the different syntactic trees of the two phrases (Nooteboom 1997: 671). These different types of information structure allow the speaker to group together the information to his liking.

Baumann (2006b: 154) proposes three fundamental dimensions of information structure. He sustains his argument by postulating three dichotomies: *theme* – *rheme*, *given information* – *new information* (also referred to as anaphoric and non-anaphoric), and *background* – *focus*.¹⁰ The theme stands for the opening element of a clause: it refers to that which is being talked about or to “the point of departure for the clause as a message” (Halliday 1967: 212, quoted in Baumann 2006b: 154). The rheme, on the other hand, is the complement of the theme, i.e. the actual message itself (ibid.). Information that had already been mentioned or implied is referred to as given information. Given information need not necessarily be explicit but can also refer to implicit knowledge shared

10. The use of the terminology is not unequivocal, however. *Theme* – *rheme* is also sometimes referred to as *topic* – *focus*, while *background* – *focus* is termed *given* – *new* or *theme* – *rheme* (Baumann 2006b).

by the speaker and listener or may simply refer to common sensical information (Mixdorff 1998:21). The dichotomy of background and focus information describes the juxtaposition of uninformative and informative parts of an utterance (Baumann 2006b:155).

Information structure can be created by means of syntax (word order phenomena, passives), by morphology or morphosyntax (pronominalization, the use of specific particles), or by means of prosodic strategies (Baumann 2006b:159). The following section will of course be concerned with the third category, the prosodic strategies of information structuring. In particular, we will dwell on the function of *focus*, which is understood as a cover term for the above-mentioned phenomena rheme, new information, and background information (see Féry 1993:13). Note that in the present study, *focus* is understood as a paralinguistic variable, since *focus* is intentionally added by the speaker as a means to enhance the linguistic information.

2.6.1.1 *Phrase accent and focus*

Phrasal stress, or phrase accent, which is governed by metrical rules, is a phenomenon similar to word stress and is closely linked with the concept of focus (see Günther 1999:48). Thorsen (1988b) distinguishes between syntactically determined default accents, which normally occur in utterance-final position, and semantically/pragmatically determined focal accents. In the present study, the term focal accent is adopted (or will simply be referred to as focus), and default accents are termed phrase accents. Neither of these two accents is universal as languages or regional varieties do not necessarily have to signal phrase or focal accents (*ibid.*). Some authors, such as Botinis et al. (2001) or Nöth (1991), treat focus as being nearly synonymous with nucleus, sentence stress, and focal accent. In what follows, the two concepts will be discussed and described as two conceptually linked yet not identical linguistic means of highlighting information in an utterance.

It is common to distinguish between normal stress, or broad focus (see Cruttenden 1987:81), and narrow focus. Broad focal accents can be derived for every sentence by a set of rules – this pattern contains one primary stress, i.e. a sentence stress (Ladd 1993:160) – while narrow focus is determined pragmatically or semantically. In broad focus, no particular word or phrase is in focus, hence the term normal stress, “whereas in narrow focus a grammatical constituent which forms only a part of the intonation-group is brought into focus” (Cruttenden 1987:81). It is crucial to note that broad focus does not have a particular meaning or function (Ladd 1993:160). Cruttenden further notes that broad focus refers to “out-of-the-blue” phrases. Hence, the terms broad and narrow focuses are

connected with Thorsen's (1988b) syntactically determined default accents and semantically determined focal accents. In the present study, only semantically determined focal accents are taken into consideration.

2.6.1.2 *Semantically determined focal accents*

Prominence marking on the phrasal level does not only correlate with syntactic structure but also with sentential semantic structure. In other words, semantic foci can be emphasized by means of intonational marking. In this sense, a semantically foregrounded component can receive focus, whereas other constituents of the same sentence are inevitably put in the background since they are less prominent acoustically. This abstract focus domain is realized with semantically determined focal accents. Thus, local pitch movements take on the function of information structuring, in which focus occupies a highlighting function that is associated with the most important information unit of an utterance (Botinis et al. 2001: 268; Gilles 2005: 17ff.).

According to Nootboom (1997: 670), the reasons why a specific constituent of an utterance is marked with a focal accent is not entirely clear. However, he hypothesizes that one of the causes for focal accent placement seems to be the "newness" of information (ibid.). Hence, he argues that new information is often associated with +focus (Nootboom 1997: 671). Accordingly, old information normally falls outside the scope of focus. In cases of contrastive focus, however, even old or given information can of course be placed in focus. Féry (1993: 17) gives the following definition of newness of information:

- a. information that the speaker takes to be either anaphorically or situationally recoverable from the preceding discourse;
- b. presupposed information;
- c. information that the speaker assumes to be currently in the foreground of his interlocutor's consciousness;
- d. information previously evoked or retrievable from the context of the utterance, as well as information inferentially related to some evoked entity (shared knowledge).

Consider the following example:

Here is the *book* you asked me to bring. (Roach 1983: 144; emphasis added).

The information that the speaker was asked bring a certain book is not new, which is why *you asked me to bring* is de-accented while the noun *book* is put into focus (see Roach 1983: 144). Generally, given information is articulated with lower pitch and weaker stress (see Chafe 1974). Yet, the demarcation of given and new information by means of intonation is not binary, as there are different levels

of givenness (see Baumann 2006a). Hence, different accent types reflect different levels of givenness in a continuum from new to given information, ranging from H^* (new information) to L^* (given information) (see Pierrehumbert & Hirschberg 1990; Kohler 1991a). This acoustic realization of givenness in speech bridges the gap between the theory underlying phrase accent and focal accents, as presented up to now, and their actual acoustic correlates, which shall be introduced in the following subchapter.¹¹

2.6.1.3 *Focus effects*

The acoustic correlates of narrow focus appear to be language-specific. In German, for example, focus is associated with a number of parameters; most importantly duration and intensity (see Batliner & Nöth 1989). Heldner (1996) demonstrates that focus accents in Swedish, too, are realized with f_0 rises and lengthening of the segments in both focus-medial and final sentence position. For Hong Kong Cantonese, however, the most important acoustic correlate is duration: f_0 is marginal and intensity irrelevant (see Bauer et al. 2004). Xu et al. (2004) show that apart from affecting the focused constituent, focus also results in a compressed pitch in the post-focal region, while pre-focal accents largely remain neutral. Botinis et al. (2001: 274), too, confirm that focus does not only affect the narrow focus domain, i.e. a word or, in case of broad focus, a sequence of words which receives focus. Instead, post-focal syllables are flattened and lowered in f_0 and the pre-focal domain is compressed noticeably. With regard to contrastive focus, Cooper et al. (1985) discovered a number of relevant results. The duration and the relative prominence of f_0 in contrastive focus turn out to be a function of the location of the focus constituent. Cooper et al. (1985: 2153) point out that “the influence of focus on the content word following the focused word is more distinctive than the influence on the focused word itself”, a result that is particularly relevant to f_0 changes. Most importantly, Cooper et al. (1985) observe a post-focal f_0 drop for content words if the focus lies on the first or the middle content word. They discovered that the f_0 of the focused word did not differ much from the f_0 of the non-focal condition, while the f_0 of post-focal content words, on the other hand, was significantly lower than the f_0 of non-focal control versions (ibid.). As far as spontaneous speech focus placement is concerned, Cooper et al. (1985) as well as Xu (1999) have made an interesting observation. According to their findings, focus in spontaneous speech is placed in unpredictable positions – without any instruction given by the researchers. This is an obstacle which cannot easily be

11. In what follows, the presented acoustic correlates are in reference to semantically determined focal accents, if not stated otherwise.

avoided and which needs to be addressed when dealing with spontaneous speech (see Xu 1999:72).

Finally, let us briefly consider phrase accents in the context of German. Both Mixdorff (1998:21) and Möbius (1993a) concede that the concept of phrase accent is somewhat problematic for German because the last accent in an utterance usually determines the communicative function of the utterance, e.g. terminating or continuing, and utterance-final accents are generally less prominent acoustically. In reference to Kohler (1977), Möbius (1993:22) notes that even though the existence of the sentence accent in German is undeniable, it is yet unclear whether a sentence accent is obligatory or not. Grønnum-Thorsen (1989:9) reports similar concerns and notes that “[t]he difficulty is in ascertaining the presence or not of final (be they default or focal) sentence accents” because German phrase accents are not highly perceptible acoustically. Thus, in this study, attention will be paid to semantically and pragmatically determined focal accents only.

2.6.2 Paralinguistic

2.6.2.1 *Prosodic paragraphing*

In written language, a text is structured by typographic means such as indentation, punctuation, and capitals. This allows for the distinction of paragraphs, for example, and thus gives the text an overall information structure. In a similar way, the structure of spoken language is dependent on suprasegmentals. Longer stretches of speech are characterized by an overall suprasegmental structure also referred to as a paragraph or, in the present study, a prosodic paragraph (Van Donzel 1999:3). The respective beginnings and endings of a prosodic paragraph are clearly marked prosodically. Interestingly, sentences that occur paragraph initially, medially, or finally, exhibit a different f_0 pattern when they are spoken in isolation (see Lehiste 1975). Tseng and Su (2008) demonstrate that the overall f_0 is highest in paragraph initial position and gradually decreases throughout the remainder of the paragraph. This tendency reflects the speakers’ anticipation of a gradual declination of f_0 throughout the paragraph. With regard to pause length, Lehiste & Wang (1977) report that pauses tend to be longer if they occur in paragraph final position as opposed to in between IPs (Lehiste & Wang 1977). The average f_0 for paragraph-initial IPs is generally higher than for paragraph-final IPs (see Hirschberg & Nakatani 1996; Leemann & Siebenhaar 2006). Spontaneous speech seems to be subject to similar paragraphing patterns, but the structure is somewhat less strict due to the high number of disfluencies and hesitation (Van Donzel 1999:69).

From an auto-segmental metrical point of view, Pierrehumbert and Hirschberg (1990:308) propose that speakers structure paragraphs intonationally by the

composition and selection of pitch accents, phrase accents, and boundary tones. Thereby, the speakers indicate the relationship between the current utterances and previous and subsequent utterances respectively (ibid.). Along similar conceptual lines, Tseng et al. (2005) analyze data derived from read discourses and provide a framework for fluent speech prosody that seems very promising for application to the present study. The model reflects the planning of fluent speech production and the cognitive constraints that apply, with a major emphasis on cross-phrase prosodic relations while also accounting for individual intonation patterns (Tseng et al. (2005:286). Following the concept of prosodic hierarchy, which proposes that utterances are structured hierarchically by a number of constituents on different prosodic levels, they devise a hierarchical cluster of units based on perceptual observation. On the highest prosodic level, they identify speech paragraphs, which they term *phrase groups* (PG). A PG can entail the following units (from top to bottom): *breath groups* (BG), *prosodic phrases* (PPh), *prosodic words* (PW), and *syllables* (SYL) (see Tseng and Chou 1999). These units are located within boundary breaks – loosely based on ToBI breaks – with a distinction between smaller boundary breaks on the lowest prosodic unit, the syllable (break 1), and a large boundary break (break 5), which stands for the break between prosodic paragraphs.

Two adjacent PGs can be set apart by that break (break 5), which is perceptually most relevant and marks the sharpest contrast. Consequently, where one PG ends another PG begins. Tseng et al. (2005:289) note that in their corpora, a PG consists of at least three PPhs but can consist of up to twelve PPhs. They add that for short paragraphs, which only require one breath of air, the BG layer collapses into the PG layer, in which case the PGs necessarily end and begin with a new breathing cycle (Tseng et al. 2005:288). New breath groups usually occur when the BG exceeds five PPhs.

The authors highlight the importance of the first and last PPh in a PG: “the[ir] pitch contours [...] could be described to possess distinct and identifiable intonation patterns, and in fact, are the ONLY positions where such intonations occur across phrases” (Tseng et al. 2005:289, emphasis as in the original). They explain that the major function of the first and last PPh is to signal the beginning and ending of a PG. Based on this assumption, Tseng et al. (2005) perform a categorization of PG-initial, PG-medial, and PG-final prosodic phrases. Both PG-initial and PG-final PPhs are characterized by distinct pitch resets, with either a rapid decline (PG-initial PPh) or a slow, elongated, final fall (PG-final PPh). Because of these distinct patterns, the listener can clearly distinguish paragraph-initial from paragraph-final PPhs (Tseng et al. 2005:289). Most importantly, however, there are no distinct *f0* patterns for the trajectories of the PG-medial PPhs and they are generally flatter than PG-initial and PG-final PPhs. This explains why many IPs

in fluent speech do not exhibit distinct intonation patterns. The authors note that for PGs with more than three PPhs, the additional PPhs should be categorized as PG medially.

The framework proposed by Tseng et al. (2005:286) accounts for the perception of speech paragraphs in fluent speech. Conceptual levels beyond the paragraph are not further accounted for, a gap which is filled by Fujisaki (2008). While Fujisaki's (1992, 1997) understanding of prosodic hierarchy differs to some degree from Tseng et al. (2005) – introducing prosodic clauses and prosodic sentences into the hierarchy, which correspond syntactically to clauses and sentences – the two approaches conceptually argue along similar lines (1997:38). Most importantly, Fujisaki (2008) describes levels beyond the paragraph, which he refers to as major discourse segments. Major discourse segments consist of minor discourse segments, units that are located syntactically between sentence and paragraph. A sequence of major discourse segments constitutes an entire discourse, which corresponds syntactically to a text (Fujisaki 2008:5).

2.6.2.2 *Conversational*

Apart from acting as a paragraph-structuring device, intonation also contributes to conversational turn taking. A turn refers to “[t]he talk of one party bounded by the talk of others [...], with turn-taking being the process through which the party doing the talk of the moment is changed” (Goodwin 1981:2). Sacks et al. (1974) designed a set of rules that describe at which point in a conversation a turn might be anticipated. These instances of possible turn-takings, called transition-relevance places, are characterized by particular signals, such as discourse markers, syntactic and semantic features, or prosodic features.

As early as 1964, Von Essen (1964:15) points out the significance of phrase endings. In German for instance, phrase endings indicate whether an utterance is a statement, a form of address, a request, or a question. Apart from determining the sentence mode, phrase endings are also important indicators of either the termination or continuation of the speaker's turn. Gilles (2005) shows that prototypical turn-endings usually occur at the end of a contribution to a conversation and the phrase endings indicating a termination are generally marked with a falling intonation (ibid; Lieberman 1967). Continuations, on the other hand, signal that a speaker intends to hold the floor. They serve the purpose of creating cohesion between a number of IPs and conversation contributions. Cross-linguistically, this intonational function is predominantly realized with a rise or fall rise in phrase-final position (ibid.; Cruttenden 1971). More specifically, it is the f_0 of the boundary tones that signals continuity or finality (see for example Pierrehumbert 1980; Brown, Currie & Kenworthy 1980). Low boundary tones more often than not signal finality, while high-pitched boundary tones indicate continuity. In the

context of finality marking, Hirschberg and Pierrehumbert (1986) also demonstrated that the more extensive the final lowering, the higher the degree of finality, i.e. the clearer the message that a turn is completed.

2.6.3 Non-linguistic functions

Intonation also operates on a non-linguistic level, a characteristic that many linguists and phoneticians would rather discard (Ladd et al. 1986: 125¹²). The most important non-linguistic components include fundamental features of interpersonal communication, such as happiness and solidarity, or hints about the speaker's emotional state (Ladd 1996: 33). We perceive non-linguistic effects even if we cannot understand the linguistic information, i.e. the language itself. For example, we can roughly tell someone's emotional state even if that person speaks Zulu and we do not understand a single word of it. Since the cues for non-linguistic information are loudness, voice quality, and pitch range, this means that, being a non-linguistic parameter, fundamental frequency is directly involved. Therefore, it is imperative to distinguish between the linguistic and the non-linguistic component of intonation, a distinction that is of even more importance in spontaneous, conversational speech (Vaissière 1983: 53).

The essential distinction between linguistic and non-linguistic messages, so Ladd, is that non-linguistic messages are gradient and linguistic messages are categorical (1996: 36ff.). In other words, a non-linguistic message can be articulated in a very angry, semi-angry, or mildly angry way. For linguistic messages, however, the alteration of the segments /f/ to /s/ in *fin* and *sin*, for example, triggers a change in meaning. Ladd (1996: 41) adds that, in the same vein, intonation also “has a categorical linguistic structure, consisting of a linear sequence of phonological events that occur at well-defined points in the utterance”. Yet, some scholars argue that intonation, in a linguistic sense, is gradient and not discrete and that this gradience also translates into gradience in meaning (see Fox 2000: 269ff.). Hereafter, non-linguistic gradience parallels linguistic gradience. What is crucial to note at this point is the evident connection between linguistic intonation and non-linguistic intonation in that they are both communicated by means of *f0* changes and they are both distinct. In the words of Ladd et al. (1986):

[A]n approach that carefully distinguishes intonation from paralinguistic cues and designs its studies with that distinction in mind will be the most productive way to investigate the role of intonation in expressing attitude. (Ladd et al. 1986: 137)

12. Note that Ladd et al. (1986) as well as Ladd (1996) use the term “paralinguistic,” which in the present study is referred to as “non-linguistic”.

As a listener, it is not difficult to distinguish linguistic from non-linguistic intonation. Listeners are usually able to make correct inferences about the emotional state of a speaker based on prosodic features (Bänziger Scherer 2005:256). This is not the case for a phonetic analysis of f_0 , however. Consequently, the vocal correlates of different emotional states in the context of modeling spontaneous dialectal intonation are highly pertinent. This section is concerned with intonation as a means of expressing emotion.

Research on the vocal expression of emotion nearly always calls attention to the difficulty of defining the object under scrutiny, namely emotion. This is the case not least because many disciplines are involved in the study of emotion, such as psychology, linguistics, neuroscience, speech science, and phonetics, which of course renders a straightforward definition of the term difficult (Redecker 2006:39). Problematic issues in researching emotion recur around the nature of emotional experience, the function of emotional experience, the origin of emotional experience, and the domain of inquiry, i.e. whether one should study a handful of primary emotions or assume an infinite continuum of emotions, for example (Metts & Bowers 1994:509). It is essential to distinguish between the concept of emotion and related notions, including mood or attitude. Moods are affective states of little intensity yet long duration and need not be triggered by a specific event. Attitudes are even more persistent and reflect belief systems and preferences. Emotions, in contrast, are normally set off by a specific event and are relatively short in duration (see Kranich 2003:42). Hence, *emotion* can be described as a series of psycho-physiological changes of one's state, caused by outside stimuli (e.g. sensory perception), inner stimuli (e.g. bodily apperception), or cognitive processes (e.g. appraisals or expectations). Consequently, emotional responses are generated which are characterized by emotional expressions (Fröhlich 2000:39). Should a person's emotion be identified via external apperception, the literature suggests three types of emotion indicators:

1. Verbal expression of emotion (e.g. "I am scared")
2. Neuro-physiological changes (e.g. blushing, increased heart rate etc.)
3. Motor-expressive indicators (rising of the eyebrows when surprised, for example)

The third group also comprises non-verbal communication, made manifest by prosodic clues. Therefore, prosody – and in the context of the present study also *intonation* – can act as one possible indicator for emotion.

Intonational correlates of emotion are predominantly documented with global descriptors of f_0 , such as f_0 range, average f_0 , f_0 level, as well as intensity contour and segmental duration measurements (Bänziger & Scherer 2005:254;

Schröder 2004:52ff.). In spite of many efforts over the past 40 years to detect straightforward relationships between emotion and prosody, the fragmented literature has achieved no such thing (Bänziger & Scherer 2005). This result does not come as a surprise, considering that the nature of emotions and attitudes has also not yet been defined adequately (Ladd et al. 1986:127). There appear to be certain tendencies of how emotional states translate into different phonetic features, yet it remains unanswered how specific acoustic features distinguish the quality of emotions (see Stibbard 2001). The following Table 2.1 provides a summary of the phonetic features of five primary emotions, published in a survey article on vocal correlates of emotion by Murray and Arnott (1993). The authors note that despite the different techniques applied in the surveyed studies, the vocal effects of emotion are comparatively consistent between authors.

Table 2.1. Human vocal effects of primary emotions (Murray & Arnott 1993:1106)

	Anger	Happiness	Sadness	Fear	Disgust
Speech rate	slightly faster	faster or slower	slightly slower	much faster	very much slower
Pitch average	very much higher	much higher	slightly lower	very much higher	very much lower
Pitch range	much wider	much wider	slightly narrower	much wider	slightly wider
Intensity	higher	higher	lower	normal	lower
Voice quality	breathy, chest tone	breathy, blaring	resonant	irregular voicing	grumbled, chest tone
Pitch changes	abrupt, on stressed syllables	smooth, upward inflections	downward inflections	normal	wide, downward terminal inflections
Articulation	tense	normal	slurring	precise	normal

Neutral speech is not displayed in this chart. However, Murray and Arnott (1993:1103) point out that neutral speech is frequently characterized by a narrow pitch range where f_0 is normally distributed around the average pitch level. In sum, Murray and Arnott (1993:1106) conclude that the most crucial parameter for a distinction between the primary emotions appears to be the pitch envelope, i.e. level, range, shape, and timing of the f_0 contour.

As for the modeling of non-linguistic intonation, it was shown that the Fujisaki model provides a means to accurately model this kind of intonation (see Fujisaki & Hirose 1993; Higuchi et al. 1994; Hirose et al. 2005; O'Reilly & Chasaide 2007). Fujisaki and Hirose (1993) study the intonational correlates of several attitudes/intentions by means of analyzing Japanese data. These states of emotion include default (the speaker reports a fact), assertion (the speaker is

committed to a fact), interrogation (the speaker addresses a question to another person), exhortation (the speaker signals an invitation), and hesitation (the speaker expresses reluctance in accepting a positive response from his/her question) (Fujisaki & Hirose 1993: 254–255). Interrogation, exhortation, and hesitation are characterized by utterance-final rises, modeled with a high f_0 local accent. The assertive contour shows a longer local accent for the verb of the utterance. Hesitative intonation indicates a significantly longer final mora, which translates into a delayed onset of the final local accent. Higuchi et al. (1994), too, work with Japanese data and investigate how different speaking styles – normal, kind, hurried, and angry – affect the model parameter. The investigated parameters – base fundamental frequency and local and global accent amplitudes – varied significantly in different speaking styles. Angry utterances exhibit a high base fundamental frequency, as well as low local and global accent amplitudes, resulting in flat f_0 contours. Kind speech is characterized by high pitched local accents, while the f_0 for hurriedly spoken sentences are somewhat lower than for normal speech. The other parameters that were investigated remained similar. Most relevant for the present study, however, is O'Reilly and Chasaide's (2007) study on the intonational correlates of the portrayed emotions surprised, bored, neutral, angry, happy, and sad in Hiberno-English. In the context of the global contour, they discover that high activation emotions (happy, surprised, and angry) and low activation emotions (bored, neutral, and sad) have different overall declination contours: the declinations are steep for the former, but flat for the latter category. Low activation emotions further exhibit low f_0 local accents, in addition to generally demonstrating longer accent groups.

CHAPTER 3

Intonation models

To this day, there is no single, exclusively used model of intonation. There is, however, an array of diverse approaches to intonation. Due to the heated debate on the functions and, particularly, the representation of prosody, the field of intonation research is characterized by a variety of schools and modeling approaches. Despite a thorough discussion of the more popular models and approaches towards intonation in the literature, a direct comparison remains difficult because of the vastly different meta-theoretical concepts, theoretical and methodological approaches, and types of categories and parameters used for the codification of intonational parameters (Werner 2000; Vaissière 2004:238).

Models of intonation can generally be categorized into *phonetic* or *phonological* models. The former make claims about the concrete, close-to-the-signal, phonetic form of intonation, whereas the latter are geared towards explaining language-systematic descriptions about the distinctive features of phonetic forms. By means of a formulation of rules, the phonological, symbolic approach transposes the abstract phonological description of intonation contours into its concrete phonetic form (Siepmann 2001: 8ff.). Intonation is mainly viewed as the addition of atomistic local events: phrase accents and pitch on the one hand, and boundary tones on the other (Pierrehumbert 1980). Most studies working in this symbolic framework have a tendency to be descriptive and theoretical. Their basic endeavor is to analyze the prosodic structure and its relation to phonology and other aspects of grammar. In contrast, advocates of the phonetic representation of intonation view the final *f₀* contour as the addition of multiple components, including baselines, globally declining phrase components, and local word accents (at the uppermost level) (cf. Öhman 1967). Close-to-the-signal, phonetic studies on intonation tend to experiment with instrumental measures and essentially aim at the quantification of acoustic correlates. In a final step, perceptual responses on behalf of listeners are explored regularly.

Cutler and Ladd have introduced the label “concrete” and “abstract” for the phonetic and phonological approach, respectively (1983: 2ff.). Both of these

approaches differ vastly with regard to both the degree of implied abstractness of prosodic representation as well as the representation and function of intonation. In the abstract approach, prosody is understood in the framework of linguistic structure. Prosodic features are classified as “any phenomena that involve phonological organization at levels above the segment” (ibid.). The aim here is to establish an inventory of abstract categories, ultimately creating a formalization of intonational function and form. The concrete approach, on the other hand, refers to prosody in its physical sense, i.e. “those phenomena that involve the acoustic parameters of pitch, duration, and intensity” (ibid.). The link between function and form is understood as a straightforward mapping between tangible meaning and precise acoustic events. Here, the realization of intonation rather than the representation of intonation constitutes the primary scientific goal.

3.1 Autosegmental – metrical phonology: ToBI

The autosegmental-metrical [hereafter AM] framework constitutes the most dominant approach in the description of intonation. The AM approach originated as a break from Chomsky and Halle’s (1968) linear phonology, shows heavy influences of Goldsmith’s (1976) insights gained from investigating tone languages, and contains fundamental principles from metrical phonology (see Liberman 1975; Liberman & Prince 1977).

3.1.1 Fundamental principles

Chomsky and Halle’s (1968) *Sound Pattern of English* proposes that words are characterized as a linear sequence of phonemes, which are in turn made up of bundles of binary, distinctive, segmental and suprasegmental features. These features do not, however, allow for an optimal description of gradient suprasegmental features such as intonation. For example, a tonal movement on one segment itself, such as a rise from low pitch to high pitch in the vowel /a:/ cannot be represented by a linear arrangement of binary features because this would entail both ([–high] and [+high]), two opposing features. Furthermore, suprasegmental features are typically determined on the level of syllables not on the phoneme level (Baumann 2006a:13ff.). In his investigation of tone languages, Goldsmith (1976) observed that a system which only allows a one-to-one correspondence between segments and tones, as suggested in the linear system proposed by Chomsky and Halle, could not capture the dissociation between the segmental and tonal level found in tone languages, where single segments

syllable in a phrase, while the compound stress rule attributes greater prominence to the left constituent in compounds. On the basis of these principles, Janet Pierrehumbert (1980) introduced a powerful and efficient system for the analysis and description of English intonation and thereby laid the foundation for one of the most influential description frameworks in intonation research. This system views intonation as a sequence of independent, categorical tones (or autosegments) that are represented on their own tone tier.¹ Apart from the tone tier, Pierrehumbert (1980) proposes a text tier and a metrical tier. The local events on the tone tier are composed of High (H) and Low (L) tones, which stand in phonological opposition to each other. These tones can have two major functions: they either lend prominence through pitch, in which case they are called *pitch accents*, or they delimit phrases, in which case they are referred to as *boundary tones*. H and L tones on the tone tier are associated with prominent syllables or the end of phrases in the text tier. This means that the position of pitch accents can be predicted by the metrical structure of an utterance, a principle adopted from Liberman and Prince (1977). Boundary tones are associated with the edges of the intonation units but need not denote metrically strong syllables. They are assigned for structural demarcation reasons, as opposed to prominence reasons, and signal the edge of phrases. After a pause, it is also possible that a phrase-initial boundary tone is placed (Féry 1993: 51). A third type of tone, the *phrase accent*, determines the pitch value between nuclear accents – i.e. the last fully-fledged pitch accent of a phrase – and the boundary tone. With regard to phrasing, Pierrehumbert (1980) only integrates one level, namely the IP. In later work, however, intermediate phrases and accentual phrases are introduced as well. While boundary tones demarcate the edges of IPs, phrase accents delimit the edges of intermediate phrases, which prosodically represent a smaller constituent (Kügler 2007: 6).

As for the formal notation, a minimalistic labeling inventory is employed. Pitch accents associated with a stressed syllable are marked (*), boundary tones of IPs are symbolized with a (%), and phrase accents are marked with a (–). Pitch accents consisting of more than one tone are notated with a (+) between the two tones. Pierrehumbert (1980) distinguishes between seven types of accents:

L* and H* consist of one tone only, which aligns with the metrically strong syllable.

1. Intonation is viewed “as the sum of atomistic local events” (Vaissière 2004: 238).

L^*+H , $L+H^*$, H^*+L , $H+L^*$, and H^*+H are bitonal accents, consisting of an accent tone (marked with $*$) and a floating tone that is not directly linked to a specific syllable.²

The principles for the association between pitch accents and metrically strong syllables are represented in Figure 3.2, accompanied by idealized f_0 contours (adopted from Mayer 1997:29). The subsequent Figure 3.3, on the other hand, gives an exemplary tonal representation of the utterance *I really believe Ebenezer was a dealer in magnesium* (Pierrehumbert 1980:56).

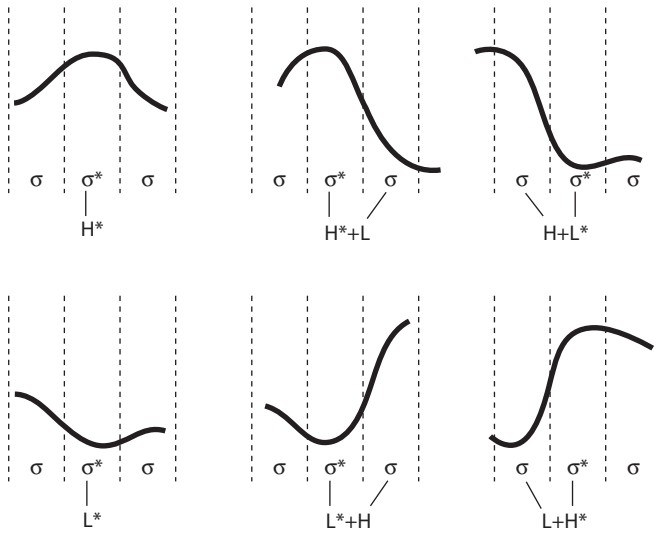


Figure 3.2. Tone-syllable association and the corresponding idealized f_0 contours (adopted from Mayer 1997:29)

I really believe Ebenezer was a dealer in magnesium.
 $\begin{array}{ccccccc} | & | & | & | & | & | & | \\ H^* & H^-+L^* & H^-+L^* & H^-+L^* & H^-+L^* & L^- & L\% \end{array}$

Figure 3.3. Tonal representation of the utterance *I really believe Ebenezer was a dealer in magnesium* (adopted from Pierrehumbert 1980:56)

Pierrehumbert (1980) does not suggest a strict division of the contour into nuclear and pre-nuclear accents, which is why the nucleus does not occupy a special status. Instead, she notes that the nuclear accent is simply the last fully-fledged

2. The H^*+H accent is abolished in the revised standard proposed by Beckman and Pierrehumbert in 1986.

pitch accent in a phrase. Both types of accents, nuclear and pre-nuclear, can be any of the above mentioned pitch accents (Taylor 1994: 29). A finite state grammar mechanism thereby generates well-formed tonal sequences of an IP, as shown in Figure 3.4.

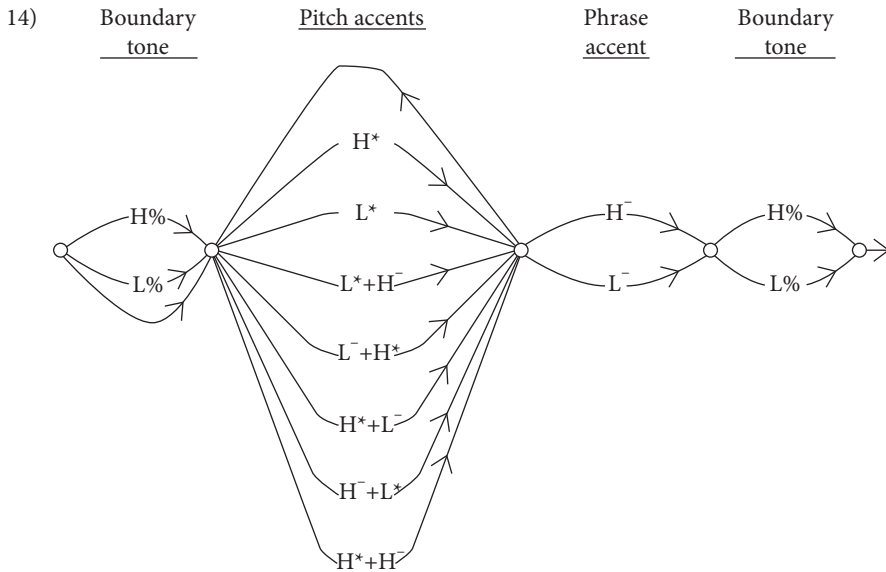


Figure 3.4. Finite state grammar of an IP (adopted from Pierrehumbert 1980: 29)

Altogether, this intonational taxonomy illustrates 22 different, well-formed tunes for American English. The grammar postulates that an IP consists of one or more pitch accents, followed by a compulsory phrase accent and a compulsory boundary tone (Mayer 1997: 27). Furthermore, contours are made up of strings of tones that occur at well-defined points in an utterance (Ladd 1996: 80). As Féry (1993: 54) explains, the meaning of the tunes can be determined “compositionally from the pitch accents, the phrase accent, and the boundary tone”.

By means of context-sensitive rules, this underlying phonological representation of tones translates into concrete phonetic events, i.e. *f0* representations (Werner 2000: 4ff.). This process is referred to as interpolation. On the one hand, the phonetic rules that determine the *f0* values of H and L tones are based on the metrical strength of the syllables with which the tones are associated. On the other hand, the *f0* values are determined by the *f0* values of the preceding tone, meaning that the calculation of the tones depends on the preceding and not the following tones in the utterance (Botinis et al. 2001: 281). The central, context-sensitive rule is the downstep rule that models the declination tendency of IPs (see 2.3).

Essentially, the downstep rule lowers the accents (*downstep*) by a constant value throughout the course of a declarative utterance which, in the AM-framework, is annotated by the diacritic (!). Pierrehumbert (1980) argues that the downdrift observed in *f0* contours is predominantly a phonological effect and is controllable directly by the speaker (Taylor 1994: 29).

Pierrehumbert's (1980) model is essentially sequential. Even tonal correlates of the phrase structure of an utterance, i.e. phrase and boundary tones, are conceptualized as elements of a tonal sequence (Möbius 1993a: 57). Therefore, the above-mentioned downstep results from repetitively applied rules on the local tonal level, realized as a sequence of H L H accents for example, as opposed to conceiving of declination as a phrase-level, intonational phenomenon that affects all pitch accents (see Lieberman 1967). Liberman and Pierrehumbert (1984) even go so far as to claim that there is no evidence for declination in English. Consequently, Pierrehumbert (1980) objects to the assumption that speakers plan the structure of intonation and thus rejects the idea of look-ahead or pre-planning mechanisms in language production. Hence, one of the core arguments of Pierrehumbert's Thesis is that intonation is essentially locally determined. Ladd (1983) highlights this fundamental principle as follows:

[...] the pitch movements associated with accented syllables are themselves what make up sentence intonation [...] there is no layer or component of intonation separate from accent: intonation consists of a sequence of accents, or, to put it more generally, a sequence of tonal elements. (Ladd 1983: 40)

In other words, the assumption that intonation consists of layers, such as superposed global and local components, as it is conceived of in the present study, is rejected by the tone sequence approach.

3.1.2 Tone and Break Indices (ToBI)

ToBI (Price et al. 1991; Silverman et al. 1992) represents one of the most prominent descriptive systems for intonation that has evolved from the AM-framework. ToBI largely relies on Pierrehumbert's (1980) minimalistic labeling inventory and was established by a group of American researchers whose goal was to create a system that could be commonly used for the prosodic transcription of labeled speech corpora (Ladd 1996: 94). The ToBI system contains Pierrehumbert's (1980) *tone tier*, complemented with a *break index tier*, a component that indicates breaks, a *miscellaneous tier*, a component for miscellaneous information, and an *orthographic tier*. The *miscellaneous tier* contains transcriptions of linguistic phenomena such as breathing, laughter, disfluencies and so on. As for the *break index tier*, Price et al. (1991) distinguish between five different break indices which

represent the strength of the prosodic dissociation between adjacent words in the *text tier*. In other words, they represent the perceived strength of phrase boundaries, ranging from clitic group boundaries (break 0) to word boundaries that coincide with an IP boundary (break 4).

To this day, ToBI is considered the informal standard for prosodic transcriptions and stands for a promising framework of cross-linguistic intonational comparisons. The most impressive feature of this labeling system is its sparse use of parameters: a given contour can be described by means of merely two tonal elements/primaries. Another advantage of the ToBI labeling system is its compatibility with ASCII (Mayer 1997:73). These are likely the reasons for its ongoing success in intonation research (Werner 2000:6). Nonetheless, a direct comparison of studies that apply the ToBI system is made difficult in that the basic assumptions of this descriptive framework are implemented differently. The Dutch version of ToBI (ToDI), for example, does not provide a transcription/annotation of break indices (see Gussenhoven et al. 1999).

The tone sequence approach has found extensive use in research on German intonation, most notably in the studies by Uhmman (1988) and Féry (1988). Much work has also been put into the adaptation of the ToBI labeling system to German. Currently, there are three variants to choose from: GToBI(S), a system developed in Stuttgart (see Mayer 1995), GToBI(VM), the Verbmobil GToBI system (see Reyelt & Batliner 1994), and the Saarbrücken GToBI(SB) system (Grice and Benz Müller 1995). Of all these, the work of Grice and Benz Müller (1995) currently seems to provide the most comprehensive description of German intonation.

A derivative yet modulated form of ToBI has been proposed by Grabe et al. (1998b), who introduced IViE (Intonational Variation in English). IViE heavily draws on ToBI in offering tools for the transcription of standard varieties; at the same time, however, they present tools for a prosodic modeling of dialectal variants. Intonational structure is described on a number of levels (phonological, phonetic, and rhythmic tiers), as opposed to ToBI's original single tier transcription. The IViE project is relevant for the present study because it set out to test for intonational variation across dialects as well as variation across different speaking styles. Approaches up to 1998 were limited to the comparison of one specific English dialect and the standard variety of Southern England. On the phonetic level, the IViE approach deduces that dialectal speakers of Cambridge and Newcastle English compress, while speakers from Belfast and Leeds truncate (cf. Grabe et al. 2000). The phonetic accounts of intonational variation in the British Isles retrieved from controlled speech furthermore specify on particular phrase types such as for instance declaratives or yes/no questions (cf. Fletcher et al. 2004). Phonologically, they demonstrate significant cross-dialectal differences in nuclear accent realization in declaratives, WH-questions, yes/no questions, as well

as declarative questions. While the IViE project is relevant to the current study in terms of its research object, cross-comparisons are extremely difficult because of the vastly different nature of the theoretical approach (meta-theoretical concepts, theoretical approaches, and types of categories and parameters for the codification of intonational parameters).

3.1.3 Shortcomings

The tone sequence approach and its labeling system ToBI have received criticism from a number of angles, much like any other approach towards describing intonation. For example, Ladd (1996) points out the lack of phonetic specification. He highlights that it is rather difficult to establish the criteria by which to determine whether a given pitch accent consists of one or two tones. Furthermore, he puts forth the question of how “we determine that there is not a tone at any given point in a string” (Ladd 1996: 103). Similarly, in Grabe’s (1998: 31–32) opinion, it is unclear “whether ToBI is intended to provide phonetic transcriptions of intonation, phonological transcriptions, or possibly neither”. ToBI was originally created to provide a standard for prosodic transcription analogous to the IPA. However, Grabe (1998) questions this analogy because the IPA is used to transcribe non-sensical sounds phonetically without making linguistic decisions, whereas a ToBI transcription requires linguistic decisions. For instance, such decisions include the identification of the stressed syllables associated with pitch accents, an issue which lies solely in the hand of the researcher.

With regard to the semantic interpretation of the model, Taylor (2000: 1709) emphasizes that “there has been no evidence to show that there are strict boundaries between intonational units which signal abrupt changes in meaning”. More specifically, “if intonational sound S_A gives rise to meaning M_A and sound S_B gives rise to meaning M_B , then a sound half-way between S_A and S_B can certainly give rise to a meaning somewhere between M_A and M_B ” (ibid.). In the same vein, Fox (2000) adds:

[T]he continuous phonetic scale is reflected in a parallel continuous scale of meaning. It is therefore difficult to identify on the basis of the criterion of distinctiveness of meaning a restricted number of phonologically distinct entities which underlie the very large number of occurring manifestations (275).

Other criticism concerns the framework’s strong emphasis on the local determination of intonation. Regardless of the framework’s preference to view intonation as locally determined, there is evidence for non-local influences on intonation. In utterances with parentheses, for example, the *f0* contour is interrupted by the parenthesis, yet the speaker subsequently continues with an *f0* nearly equivalent to the *f0* contour present in the same utterance without parentheses (see

Kutik et al. 1983). Furthermore, some scholars have uttered doubt as to whether the AM-framework could accurately capture certain tones in tone languages. Kehrein (2002: 26), for example, observes that with only two tones (H and L), an accurate modeling of tones in tone languages, particularly the lexically-distinctive tones, is not possible. As an example, he presents the Miao-Yao languages, in which tones are made distinct only by altering the pitch range, while the direction of the tonal movement is essentially the same. Ladd (1996: 252ff.), too, notes that pitch range poses a fundamental problem not only for the autosegmental-metrical approach, but for prosody research in general.

3.2 Other intonation models

Naturally, there are further approaches towards prosodic description that do not simply fall into one of the previously mentioned categories. There are proponents for a morphological or pragmatic account of intonation. Here, cues on different levels, i.e. pragmatic, syntactic, facial, and gestural levels, are taken into account (e.g. Rossi 1999). Further approaches include perception-based frameworks like the *IPO approach* (see 't Hart et al. 1990), Kohler's (1991a) *Kiel Intonation Model* (KIM), which evolves around the functional aspect of intonation (Botinis et al. 2001: 280), as well as the TILT model established in Edinburgh by Taylor (1992, 1998, 2000). TILT is based on Taylor's earlier works involving a simpler intonation model, referred to as the *rise/fall/connection model* [hereafter RFC] (Taylor 1995). The TILT model provides a rule system for the codification of *f0* contours that was created for the purpose of synthesis and analysis. The model established at Aix-en-Provence, referred to as *International Transcription System for Intonation* [hereafter INTSINT] (Hirst & Di Cristo 1998), provides a system for the labeling of *f0* contours by means of absolute and relative tones. INTSINT uses the tonal symbols *T(op)*, *M(id)*, *B(ottom)*, *H(igher)*, *S(ame)*, *L(ower)*, *U(pstepped)*, and *D(ownstepped)*. *T*, *M*, and *B* represent absolute symbols that refer to the pitch range of a speaker, and *H*, *S*, *L*, *U*, and *D* are relative tones that refer to their tonal environments (Siepmann 2001: 48). Other comprehensive models include the *Lund model* (see Bruce 1977; Garding & Bruce 1981; Bruce 1997) as well as Yi Xu's *PENTA (target approximation) model* (see Yi Xu 1999). Approaches which are fundamentally different from the models discussed so far are those emerging from application-oriented research, e.g. research conducted by telephone companies or speech syntheses research on high-quality intonational representation. In these approaches, large data sets are searched for systematic intonational similarities by the implementation of neural networks.

CHAPTER 4

Command-Response model: Fujisaki

The Command-Response model structures f_0 contours of an utterance into intonational units that can be subsequently referred to linguistic categories. Firstly, the origins of the model are introduced, followed by a presentation of the mathematical formulation and the underlying physiological principles of the model. Then, the model parameters and their linguistic interpretation, as conceived of by Fujisaki and co-workers, are discussed in detail. Subsequently, the discussion covers the general procedure steps in the application of the model as well as earlier applications to German. The chapter ends with a subsection on the model's advantages and shortcomings.

4.1 Origins

The *Command-Response model*, also known as the *Fujisaki model*, was first introduced by Prof. Fujisaki at the University of Tokyo in 1969 and has been continuously developed over the past two decades (Fujisaki & Nagashima 1969). The publication of 1982 by Fujisaki and Hirose is most frequently used as a reference for the current version of the model. In contrast to previously introduced models, the Fujisaki model centers on intonation production. Based on insights provided by Öhman (1965, 1967), Fujisaki developed an algorithm which simulates the intonation-relevant articulatory mechanisms and thus allows for a precise generation of continuous, natural f_0 contours (Werner 2000: 10).

The fundamental principle of the model is that intonation is hierarchically structured. Although the conceptualization of superposed components in intonation – a global intonation contour onto which local word accents are placed – had been around for quite a while, Öhman (1965, 1967) was the first to provide a mathematical model that allowed for the calculation of the corresponding functions. Öhman's (1967) *Functional model of larynx control*, as displayed in Figure 4.1 is capable of generating f_0 contours of separate, stepwise sentence and word intonation inputs, based on plausible assumptions about the physiology of the larynx (Werner 2000: 11).

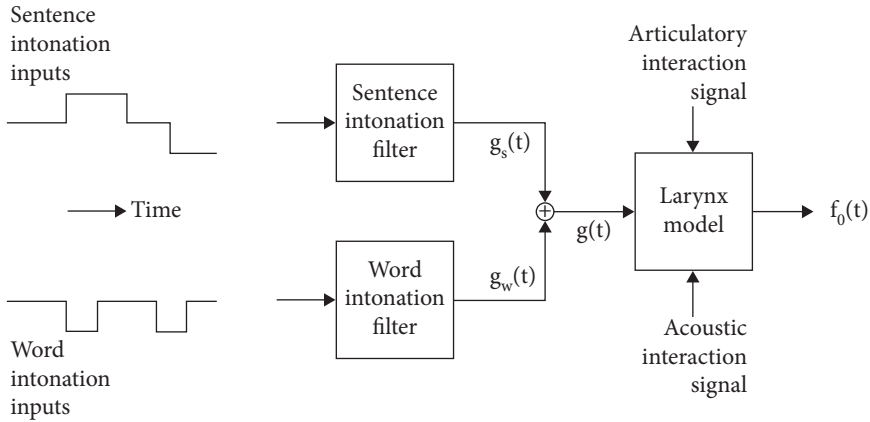


Figure 4.1. Functional model of larynx control, as suggested by Öhman (1967: 20)

Fujisaki and Nagashima (1969) extended Öhman's (1967) model while maintaining the core principle of the larynx control mechanism. As one of most important modifications, the Fujisaki model posits that all commands for the generation of voicing and accent are binary input functions, while Öhman (1967) postulated step functions that can occur in an arbitrary number of levels and amplitudes (see Figure 4.1). Furthermore, Fujisaki and Nagashima added the criterion of hysteresis to the glottal control mechanism to account for the declination tendency of f_0 contours, which in Öhman's (1967) approach may have been modeled by step functions on arbitrary levels and amplitudes (Fujisaki & Nagashima 1969: 58–59).

4.2 Mathematical formulation

As mentioned earlier, the model is hierarchically structured and formulated as a linear model that consists of two second-order, critically damped filters. For synthesis purposes, the system calculates the continuous concatenation of f_0 contours based on two input signals which are additively linked. As input signals, the model receives *phrase commands* [hereafter PCs] in the form of an impulse function and *accent commands* [hereafter ACs] in the form of a rectangular function. The input signals are then processed by the phrase and accent control mechanisms. The output signal of the two mechanisms is added onto the smallest asymptotic value of the f_0 contour that is to be generated, which is either referred to as base f_0 , F_{min} , or Fb [hereafter Fb]. For analysis purposes, the processing follows the reversed order of the described processes. Figure 4.2 shows a block diagram of the model.

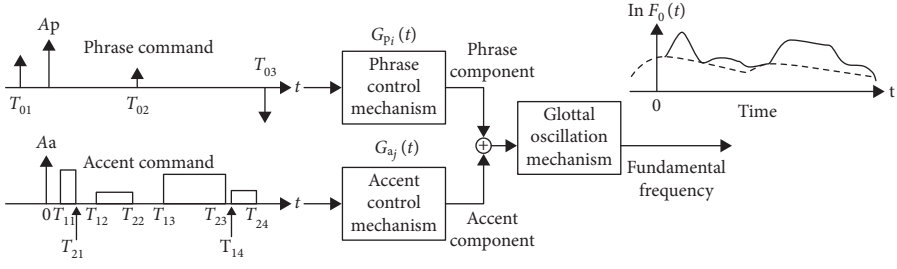


Figure 4.2. Block diagram of the Fujisaki intonation model (adopted from Fujisaki & Hirose 1984: 235)

The mathematical formulation of the model describes the resulting contour of $\ln f_0(t)$ as the sum of the base fundamental frequency F_b and the phrase and accent components governed by their respective $G_p(t)$ and $G_a(t)$ *control mechanisms*:

$$\ln F_0(t) = \ln F_b + \sum_{i=1}^I A_{p_i} G_p(t - T_{0i}) + \sum_{j=1}^J A_{a_j} [G_a(t - T_{1j}) - G_a(t - T_{2j})]$$

$$G_p(t) = \begin{cases} a^{-2} t \exp(-a t), & \text{for } t \geq 0, \\ 0, & \text{for } t < 0. \end{cases}$$

$$G_a(t) = \begin{cases} \min[1 - (1 + \beta t) \exp(-\beta t), \gamma], & \text{for } t \geq 0, \\ 0, & \text{for } t < 0. \end{cases}$$

where

F_b : baseline value of fundamental frequency

I : number of phrase commands

J : number of accent commands

A_{p_i} : magnitude of the i^{th} phrase command

A_{a_j} : amplitude of the j^{th} accent command

T_{0i} : timing of the i^{th} phrase command

T_{1j} : timing of the j^{th} accent command

T_{2j} : end of the j^{th} accent command

α : natural angular frequency of the phrase control mechanism

β : natural angular frequency of the accent control mechanism

γ : relative ceiling level of accent components¹

1. The relative ceiling value γ was not incorporated into the Fujisaki model until 1992 (Fujisaki 1992b). In this publication, the value is generally set to a constant of 0.9. The purpose of this value is to guarantee that the accent component reaches its maximum in finite time.

4.3 Underlying physical and physiological principles

f_0 contours are the result of different physiological mechanisms involved in the production of intonation (Collier 1975; Atkinson 1978). Global intonation contours seem to be a product of the decreasing subglottal pressure during phonation, while quick f_0 movements are actively controlled by the cricothyroid muscle (see 2.1). Fujisaki's production model of f_0 , however, is based on somewhat different physiological and physical causes.

In one of Fujisaki's earliest works (Fujisaki & Nagashima 1969), he explains why he opted for the logarithmic scale in the mathematical formulation of his model. The observed pitch contour of a sample word contained considerable f_0 variation between the subjects. Thus, the f_0 contour was plotted on a logarithmic scale that made the contours appear very similar, which "justifies the use of logarithmic pitch contours in the formulation of prosodic rules" (Fujisaki & Nagashima 1969: 55). In later publications (see Fujisaki, Tatsumi, & Higuchi 1980), Fujisaki modifies this rationale and states that the use of the natural logarithmic domain in his model stems from the stress-strain relationship of the human vocalis muscle. It approximates the linear relationship between muscle tension and muscle stiffness: the tenser the skeletal muscle, the stiffer it becomes (see Fink & Demarest 1978). This phenomenon of course also applies to the human vocalis muscle. The approximation of this linear relationship between tension (T) and elongation (x) is as follows (based on Fujisaki 2004: 4ff.):

1. $dT/dl = a + bT$,
 T : tension
 l : length of the muscle
 a : the stiffness at $T = 0$

This leads to the following definition of tension (T):

2. $T = (T_0 + a/b)\exp\{b(l-l_0)\} - a/b$
 T_0 : static tension applied to the vocal cords
 l_0 : length of the tension at $T = T_0$
 a : the stiffness at $T = 0$

If $T_0 \gg a/b$, equation (2) can be approximated by

3. $T = T_0 \exp(bx)$
 x : change in vocal cord length if T is changed from T_0

Fundamental frequencies of elastic membranes vary proportionately to the square root of their tensions:

$$4. \quad f_0 = c_0 \sqrt{T/\sigma}$$

σ : density per unit area of the membrane

c_0 : constant that is inversely proportionate to the size of the membrane

If (3) and (4) are combined, we obtain the following formula:

$$5. \quad \log_e f_0(t) = \log_e \{ c_0 \sqrt{T/\sigma} \} + (b/2)x$$

Hence, $\log_e f_0(t)$ is the sum of a constant term and a time-varying parameter (elongation). The constant $c_0 \sqrt{T_0/\sigma}$ can be rewritten as Fb , which represents the baseline of f_0 onto which the second term in equation (5), the time-varying parameter, is added. This results in equation (6).

$$6. \quad \log_e f_0(t) = \log_e + Fb + (b/2)x(t)$$

In the Fujisaki model, this constant corresponds to the baseline value Fb . It is further argued that the time-varying component of $\log_e f_0(t)$ is represented as the sum of two time-varying components.

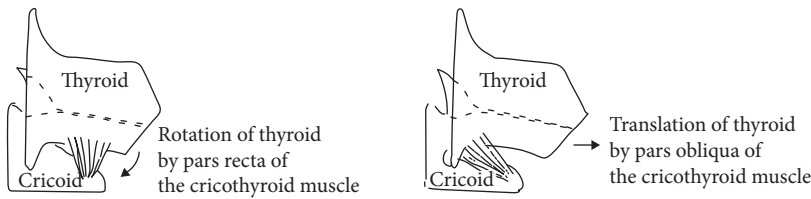


Figure 4.3. Rotation and translation movement of the thyroid cartilage (adopted from Fujisaki 1987: 171). In the context of his model, Fujisaki associates rotation with local f_0 movements and translation with global intonation movements

Based on the insight gained from Electromyography studies, Fujisaki emphasizes the significance of the function of the cricothyroid muscle for f_0 control. Because of the activity of the cricothyroid, the thyroid cartilage translates forward and rotates downwards, which causes an elongation of the vocal cords. In his model, the accent component is believed to result from the rotation of the thyroid cartilage, an assumption that is also supported by Collier (1975) and Atkinson (1978), who found that the cricothyroid muscle controls the highest f_0 movements. Fujisaki further hypothesizes, that global f_0 contours – or phrase components, in his model – are a result of the translation of the thyroid cartilage, a translation observed in radiographic studies (see Fink & Demarest 1978). Both movements of the thyroid cartilage, rotation and translation, additively affect vocal tension, as depicted in Figure 4.3.²

2. For negative ACs, a laryngeal mechanism for the active lowering of f_0 is postulated. This involves muscles other than the cricothyroid, such as the sternohyoid and the thyrohyoid, which are normally not active in the production of standard spoken Japanese (Fujisaki et al. 1998).

Thus, the resulting change in f_0 , given by the time-varying second term in $\log_e f_0(t)$, constitutes the sum of these two movements, resulting in the following equation:

$$7. \quad \log_e f_0(t) = \log_e Fb + (b/2)\{x_1(t) + x_2(t)\}$$

The physiological justification for the natural angular frequencies α and β , postulated in the equation in Section 4.2, are given in the different muscular reaction times for rotation and translation of the thyroid. Studies have shown that the activation time during the process of voicing varies for different larynx muscles. Kakita and Hiki (1976), for example, studied the physiological correlates of Japanese word accents by closely observing intrinsic and extrinsic laryngeal muscle activity. They provide evidence that the larynx muscles become active at different points in time. Most importantly, they were able to show that the cricothyroid muscle is activated approximately 50ms before any of the other muscles involved in the voicing process. Their findings are as follows:

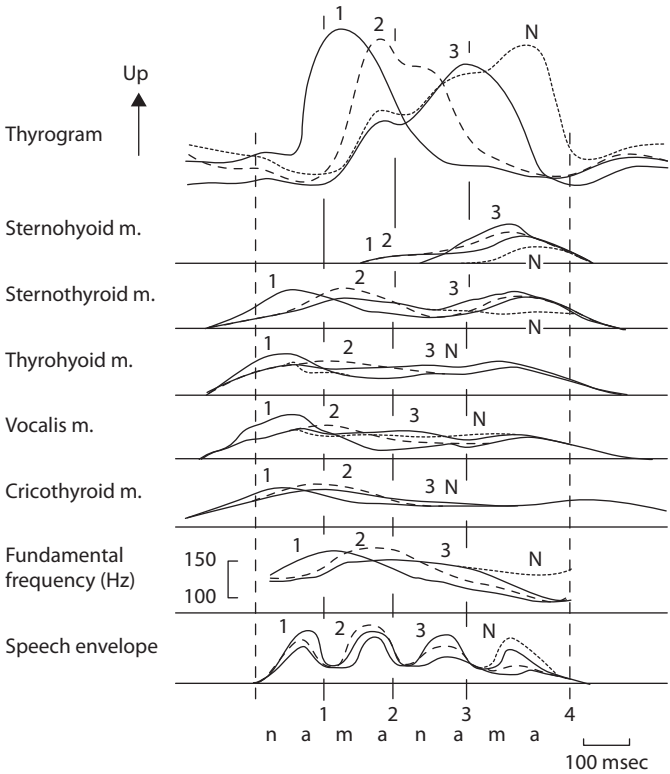


Figure 4.4. Activity patterns of laryngeal muscles. The cricothyroid muscle becomes active approximately 50ms before any of the other muscles involved in voicing (adopted from Kakita & Hiki 1976:44)

Based on these findings, Fujisaki hypothesizes that the muscular reaction time for rotation and translation of the thyroid cartilage – one component of the cricothyroid muscle – may be different as well. A calculation of the natural angular frequencies for rotation (ACs) and translation (PCs) results in a ratio of 3 to 1. However, in his analyses and publications, the ratio of the damping factors β and α often is at 7 to 1. With regard to this issue, he admits that “whether or not the 7 to 1 ratio commonly observed for β and α in the analysis of actual f_0 contours can be fully accounted for by the suggested mechanism, calls for further study” (Fujisaki 1987:174).

4.4 Model parameters: Characteristics and linguistic interpretation

This section gives a detailed account of the model components and their linguistic interpretation as presented by Fujisaki and co-workers. In the majority of their publications, Fujisaki and his co-workers do not dwell on specifications as to the linguistic interpretation of the model parameters. However, some quite thorough linguistic explanations of the model are provided in Fujisaki et al. (1990) and elaborated on in Fujisaki (1992a and 1997). Hence, much of what follows is based on these articles.

4.4.1 *Fb*

As pointed out in the previous section, on the underlying physiological principles of the model, $\log_e f_0(t)$ is the sum of a constant term and a time-varying parameter (elongation). This was shown in the formula $\log_e f_0(t) = \log_e F_b + (b/2)x(t)$ where the constant was rewritten as F_b , representing the baseline of f_0 to which the second term in the equation, the time-varying parameter, is added. In the Fujisaki model, this constant corresponds to the baseline value F_b .

The base f_0 has seen different definitions over the years. In 1979 and 1981, F_b is described as “the lowest frequency of the vocal cord vibration” (1979:166) and as “the lower limit of fundamental frequency below which vocal fold vibration cannot be sustained in the glottis of a speaker” (1981:3). In 1987, however, Fujisaki (1987:167) defines F_b as the “asymptotic value of fundamental frequency in the absence of accent components”, whereas in 2008, it is referred to as the “baseline value of fundamental frequency” (2008b:2). In the first two definitions, F_b is understood in a physiological sense: it represents the lowest possible f_0 produced by a speaker. In this sense, it is speaker-specific. The definition of F_b provided in 1987, on the other hand, can be understood as utterance or IP-specific.

Interestingly enough, in the 2008 definition both a speaker-specific and an utterance-specific interpretation of *Fb* are entailed. However, the assumptions that *Fb* is speaker-specific or utterance-specific are incompatible (Möbius 1993a: 80) since the parameter configurations are different for both conditions as shown in Figure 4.5.

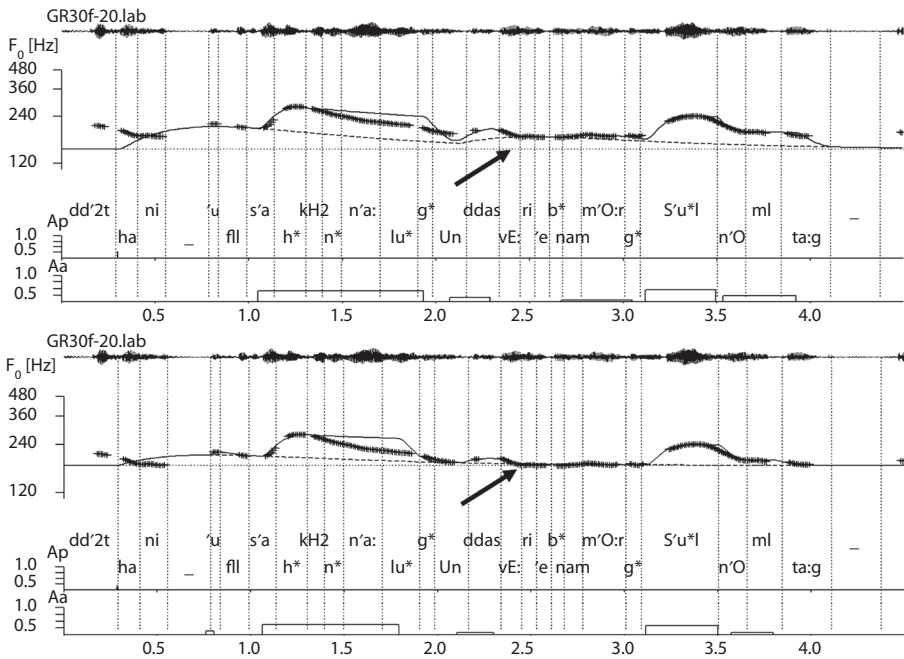


Figure 4.5. *Fb* speaker-specific (top) and *Fb* utterance-specific (bottom). The arrows point to where the different parameter setting is most obvious: in the speaker-specific condition, an additional PC is needed for the modeling

The modeled utterances are *u fil s'ah* kH9n*n 'a:lu*g* Un, das vEri 'eb* am m'Orge* S'u*! n'Omitag* (German: *sehr viele Sachen anschauen können und, das wäre eben am Morgen Schule und am Nachmittag*; Engl.: *could look at very many things and, this would be in the morning school and in the afternoon*). The modeling for both conditions is different in so far as, for the utterance-specific *Fb* condition, fewer PCs and ACs with reduced magnitudes are needed, since the base frequency is higher than in the speaker-specific condition. As shown in Figure 4.5, the utterance-specific *Fb* condition does with only one PC and five rather low-amplitude ACs, while the speaker-specific condition requires two PCs and five rather distinct ACs to model the *f0* in a sensible way.

4.4.2 Phrase component

The phrase component is one addend in the time-varying second term of $\log f_0(t)$. The phrase component $Gp(t)$ describes the slow, global changes in the f_0 of an utterance and includes the parameters Ap , the impulse's magnitude (in this study referred to *PC magnitude*), its onset time T_0 , and α , which stands for the time constant of the input signal and determines the degree to which the phrase contour is damped. Figure 4.6 gives an example of the impulse response of the phrase control mechanism for varying PC magnitudes with an alpha of 2.0/sec (adopted from Mixdorff 1998: 49), while Figure 4.7 displays the impulse response of alpha 1/sec, 2/sec, and 3/sec for constant PC magnitudes.

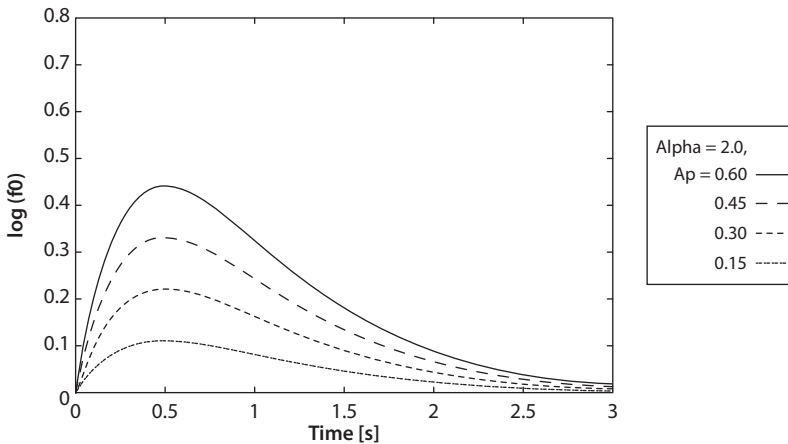


Figure 4.6. Impulse response of the phrase control mechanism with constant alpha and varying magnitudes (adopted from Mixdorff 1998: 49)

With respect to the location of T_0 in the context of the phrase it models, Fujisaki (1987) observes that PC onsets, T_0 , are located approximately 200ms before the onset of an utterance, which reflects the activation time for the translation of the thyroid cartilage. With regard to α , Fujisaki, Hirose and Ohta (1979: 169) explain that parameters that characterize the speaker's physiological mechanism – such as α in the case of the phrase component – remain constant in one sentence or over several sentences. Fujisaki and Hirose (1982) demonstrate that the time constant is approximately the same in speakers of English, Japanese, and Estonian. They add, however, that the mechanisms underlying the phrase component are not universal “[...] since there may be languages which may involve laryngeal controls not formulated in our present model” (Fujisaki & Hirose 1982: 67). Parameters that

convey linguistic information, such as T_0 and Ap , on the other hand, “are expected to vary over a wide range depending on the information content of each element” (Fujisaki, Hirose, & Ohta 1979: 169).

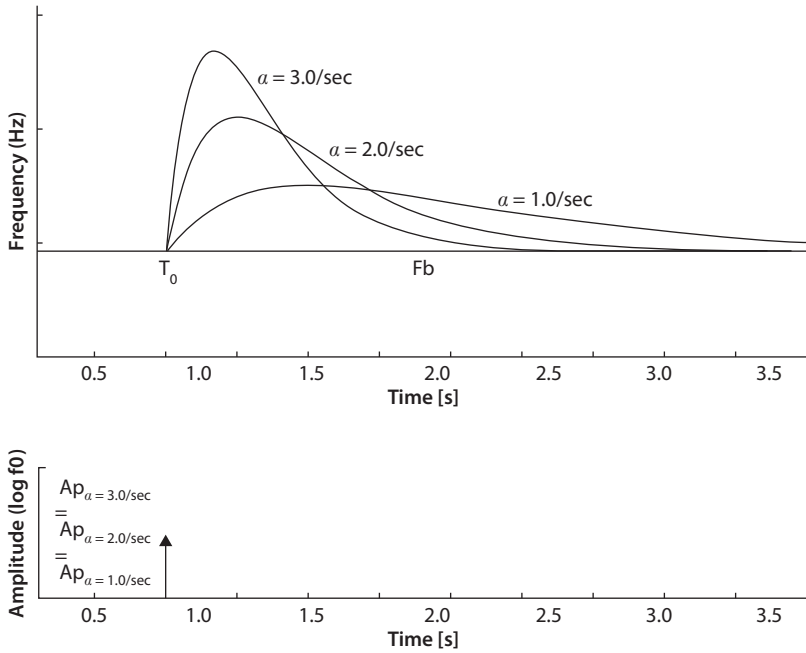


Figure 4.7. Impulse response of the phrase control mechanism with constant magnitudes and varying alpha

4.4.2.1 Linguistic interpretation

The phrase component is suitable to describe the general declination tendency in intonation contours since the contour of a phrase component rises quickly and decreases gradually towards the asymptotic value Fb . In the words of Fujisaki and Hirose (1982: 63):

[D]eclination is an inherent element in our model, viz. the shape of the impulse response $Gp(t)$ itself declines with time. Except for the initial plateau, which may or may not be observed in the f_0 contour depending on the amount of lead of the phrase command. (Fujisaki & Hirose 1982: 63)

Thus, the impulse phrase input is a reset of the declination line, where “an increase in the magnitude of the phrase command causes an increase in the slope of the declination” (Fujisaki & Hirose 1982: 65; emphasis as in the

original). Perceptually, this means that an increase in the magnitude of the phrase component implies an increase in the pitch range of an utterance or a speaker (Taylor 1994: 40).

PCs are located before major syntactic boundaries, referred to as *prosodic phrases* by Fujisaki and co-workers (termed *IPs* in the present study), such as between subject and predicate phrases. Each major syntactic phrase is believed to contain one PC, which conveys largely syntactic information (see Fujisaki 1990, 1992a, and 1997). Fujisaki, Hirose, and Takahashi (1990: 488) point out, however, that there are also cases “where prosody fails to meet all the requirements presented by word accent, syntax, and discourse”. In other words, PC placement is also subject to speaking style and respiration, for instance.

It was shown that PC magnitude is subject to phrase length, speaker rate, pragmatic context of the prosodic phrase, and the position of the command in an utterance. Initial PCs show higher amplitudes than subsequent PCs of the same utterance (Fujisaki 1987: 167). Fujisaki (1987: 167) notes that “[t]here are cases, however, where pragmatic factors call for the occurrence of a large phrase command at a sentence-medial position”. Similarly, they investigate the relationship between speech rate and declination as given in PC magnitude values. Fujisaki and Hirose (1982) report that magnitude is lower for speakers with a fast articulation rate than for speakers with a normal and slow rate. That is, the faster the speaker articulates, the more the declination decreases.

4.4.3 Accent component

The accent component is the third and last term in the sum $\log_e f_0(t)$. The accent control mechanism $Ga(t)$ includes the parameters T_{1j} , T_{2j} , A_{aj} , and β and describes the local, relatively quick f_0 movements superposed onto the phrase component. The time constant β governs the steepness of the rise and fall of the accent. Generally, β is much higher than α , which gives the filter a quicker response to the model’s rapidly rising and falling word accents (Taylor 1994: 39).

The timing parameters for the ACs, T_{1j} , and T_{2j} , as well as the Aa (referred to as *AC amplitude* in this work) vary from sentence to sentence since they reflect the lexical information of the words (Fujisaki 1981: 11). The parameter that characterizes the speaker’s physiological mechanism – β in the case of ACs – is assumed to be constant between utterances as well as between speakers on the grounds that “the mechanical properties of the human larynx should be approximately the same regardless of language” (Fujisaki 1992a: 172). Figures 4.8 and 4.9 show the step response of the accent control mechanism for varying amplitudes as well as

varying durations (adopted from Mixdorff 1998:50), while Figure 4.10 shows the step response of beta 16/sec, 20/sec, and 23/sec with a constant AC amplitude and duration.

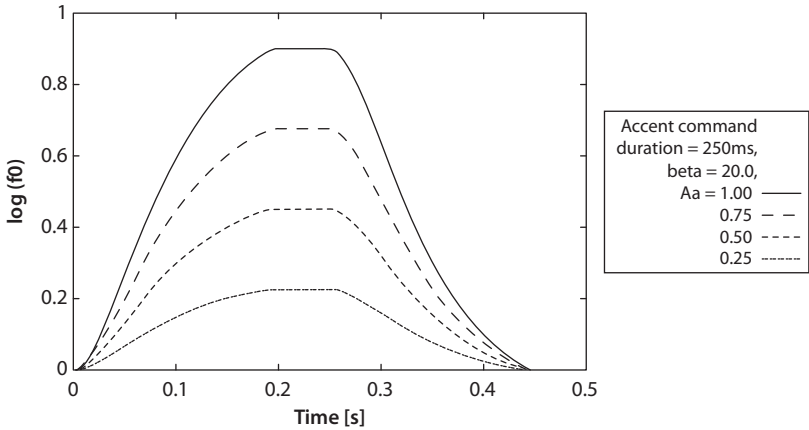


Figure 4.8. Response of the accent control mechanism with varying amplitude and constant beta (adopted from Mixdorff 1998:50)

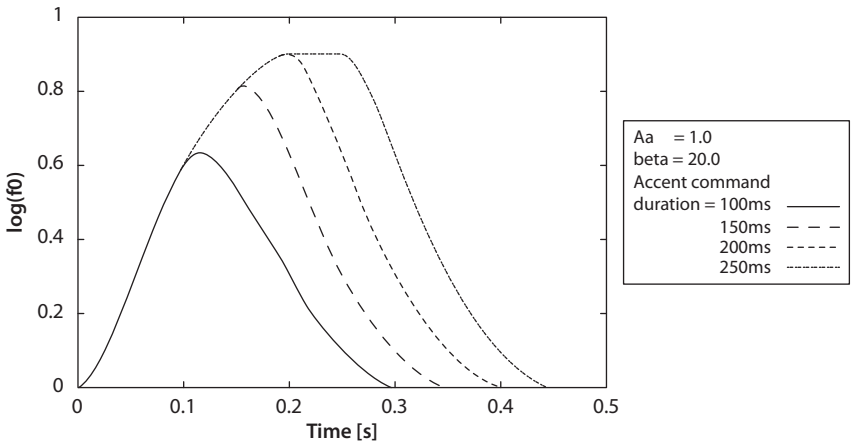


Figure 4.9. Response of the accent control mechanism with varying durations, constant beta and AC amplitude (adopted from Mixdorff 1998:50)

T_1 and T_2 are the timing parameters, T_1 being the onset time of the AC at which the f_0 contour begins to rise, T_2 being the offset time of the AC at which the f_0 contour starts to fall. T_1 is commonly associated with the beginning of the accented syllable. Fujisaki (1987:167) states that the AC onsets in Japanese are

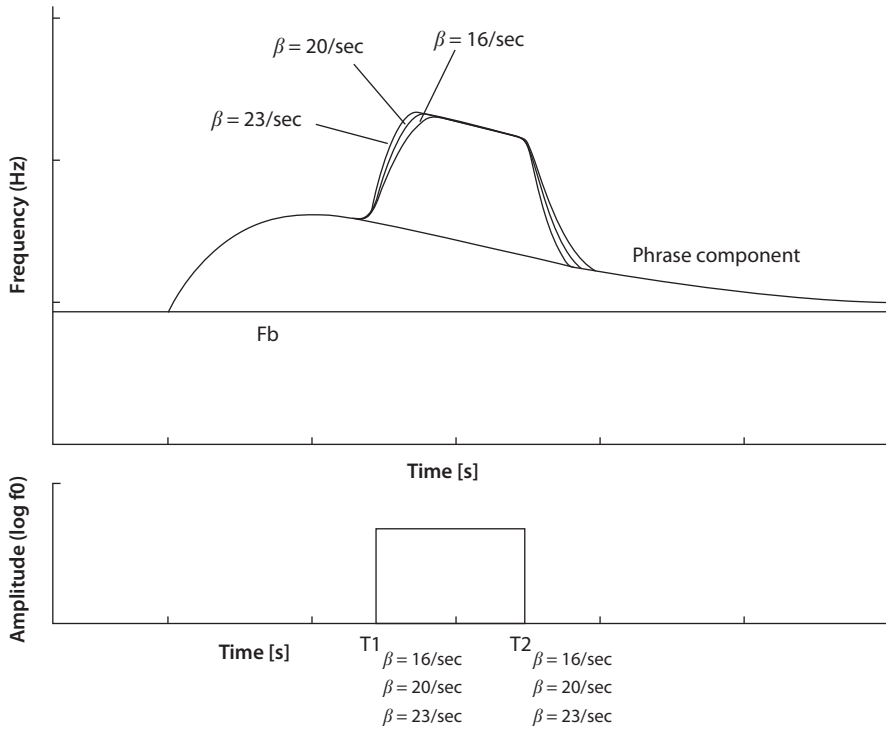


Figure 4.10. Step response of the accent control mechanism with constant AC amplitudes and varying beta

located approximately 40–50ms before the segmental onset of a high-pitched mora, and they also end approximately 40–50ms before the segment end of a high-pitched mora, mirroring the short activation time for the rotation of the thyroid cartilage. If the accent starts with a relatively high f_0 value, such as is the case for falls for instance, T_1 occurs early relative to segment onset of the accented syllable, so as to render the larynx muscle tense in order to begin at a high f_0 . If the accent group starts with a low f_0 value, such as for rises for instance, the time to set the muscles at the right position is not required and the voicing begins thus after segment onset.

As for the polarity of ACs, Fujisaki (2003) notes that it is of course possible for a language to have a laryngeal control distinct from that of other languages. Positive local commands are common in English, German, Japanese, Greek, Korean, Polish, and Spanish, for instance. However, a number of other languages, including Swedish, Chinese, Thai, and also a number of Japanese dialects, exhibit local commands of both positive and negative polarity (Fujisaki et al. 1998;

Fujisaki 2004). For Standard Chinese, on the other hand, a good approximation of the four tones (high, rising, low, and falling) can be achieved by application of a combination of negative and positive commands, as shown in Figure 4.11 (adopted from Fujisaki 2008b: 5).

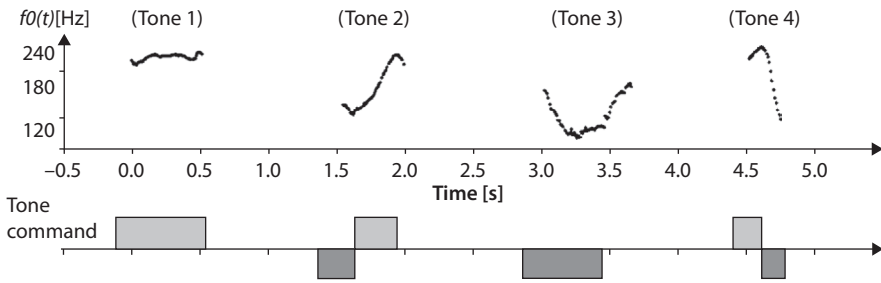


Figure 4.11. Generating local f_0 contours with ACs of negative and positive polarity, as proposed for the four tones of Standard Chinese (adopted from Fujisaki 2008b: 5)

A single positive tone models the high tone, a negative tone command followed by a positive tone command generates the rising tone, a single negative tone command can be used to model the low tone, and a positive tone command followed by a negative tone command is most appropriate to model falling tones (Fujisaki 2008b:5).

4.4.3.1 *Linguistic interpretation*

The accent control mechanism is responsible for prominence marking with f_0 on the local level. “Prominence is expressed generally by increasing the amplitude of the AC [...] and/or by suppressing the amplitude of the immediately following accent component” (Fujisaki 1992a: 167). In other words, a change of amplitude of the local accent component results in changes of f_0 of the accent peak, thus rendering the accent more or less prominent (Taylor 1994: 40). From a linguistic point of view, Fujisaki (1987: 166) observes that the step response mechanism with its short rise time can adequately generate the f_0 movements that accompany prosodic words, i.e. the local rises and falls caused by word accents. Fujisaki and Hirose (1982) have shown that AC amplitudes tend to decrease in fast-articulated speech, in opposition to normal and slow speech.

4.5 Earlier applications to German

After the Fujisaki model had been applied to Japanese, it was also used to analyze the intonation of a number of other languages, including Chinese (Fujisaki et al.

1990), Korean (Fujisaki 1996), and Swedish (Fujisaki et al. 1993). The first application of the model to German was conducted by Antoniadis (1984), followed by more extensive applications by Möbius (1993a, 1993b), Möbius et al. (1990, 1993), Mixdorff (1997, 1998, 2002a, 2008), Mixdorff and Fujisaki (1994), and Mixdorff and Pfitzinger (2005). In what follows, the most relevant results of Möbius' (1993a, 1993b) as well as Mixdorff's (1998) works are presented along with notes on the authors' parameter configurations, linguistic interpretations of the model, and the most important steps employed in the f_0 modeling procedure.³

4.5.1 Möbius

Möbius (1993a, 1993b) and his co-workers (Möbius et al. 1990, 1993) applied the Fujisaki model to German data in the context of a research project at the University of Bonn, which aimed at the production of a text-to-speech system for German. As speech material, Möbius uses declarative and interrogative sentences (WH-questions, yes/no questions, and echo questions with one or two IPs) that were produced in High German by three male and two female speakers with little regional coloring. The test phrases are phonologically balanced, i.e. they reflect the frequency distribution of German phonemes.

As for the interpretation of the Fujisaki model, Möbius strongly foregrounds the linguistic interpretation and develops a rule system for the synthetic generation of f_0 contours by means of Fujisaki PCs and ACs. He further considers the addition of the phrase component to Fb as a reliable indicator for both declination and sentence mode. As for the linguistic interpretation of the AC, Möbius relies on Grønnum's (1988a) concept of *stress groups*, which he refers to as *accent groups*, i.e. the grouping of a stressed syllable and its following unstressed syllables.

With regard to the parameter configuration and modeling, Möbius uses constant α and β values without compromising the quality of the generated f_0 contours. By determining the damping factor β of the Fujisaki model software program, he arrives at optimal β of 16.1. For α , Möbius (1993a: 106) uses the set value of 3.1, the average value of all alphas applied by Fujisaki et al.. Linguistically, he interprets Fb as an utterance-specific parameter and points out that a calculation of the physically lowest Fb of each speaker would involve considerable measurement effort (Möbius 1993a: 80). He applies PCs only to larger syntactic phrases, such as between a main clause and a subordinate clause. T_0 is placed 323ms before utterance, phrase boundary, and utterance ending. A constant, negative (-0.1 magnitude) PC is added in the utterance-final position of declaratives

3. For the most part, the results presented in this subsection are based on Möbius' (1993) and Mixdorff's (1998) Ph.D. Theses.

and WH-questions so as to simulate the typical final lowering in these sentences types. The temporal distance between T_0 and utterance or phrase onset is kept constant because of the application of an alpha of 3.1 in all instances. Further, Möbius postulates that the magnitude peak of the PC must be reached in the utterance-initial position in order to warrant a gradual decline from beginning to end of the phrase contour. Pätzold (1991, quoted in Möbius 1993a: 105) shows that T_0 is thus fixed and equals $1/\alpha$. On the local level, he applies exactly one AC for one accent group, where the accent group contains one word accent and all subsequent unstressed syllables.

The most relevant results of Möbius' project include the following findings: in terms of the *phrase component*, Möbius shows that the magnitudes differ from speaker to speaker and that magnitude values are a function of the accent distribution in the utterance as well as a function of sentence mode. In single-phrase utterances with a phrase-initial accent syllable and non-accented syllable in phrase-final position, the PC magnitudes are approximately $\frac{1}{4}$ higher than in phrases with different accent distributions. As for interrogative sentence modes, Möbius shows that WH-questions demonstrate a higher magnitude (.35) than yes/no (.27) or echo questions (.14). Sentence mode in interrogatives is triggered by the magnitude of the phrase component, as well as by the phrase-final f_0 movement. Both yes/no questions and echo-questions end with a sharp rise, while WH-questions fall sharply. Neither was a link established between PC magnitude and phrase length nor does speaking rate have a significant effect on PC magnitude. Most notably, Möbius concludes that PC magnitude in utterance-initial position is not systematically linked to utterance duration in his data. Instead, his data suggests an association of PC magnitude with the distribution of accents in the utterance (1993a: 183). As for the *AC parameters*, Möbius shows that shorter accent groups generally demonstrate higher amplitudes than longer accent groups. Also, he observes effects between word class and AC amplitude: the AC amplitudes of nouns are 7.5% higher, those of verbs, adverbs, and prepositions 5% lower than the mean of response, while adjectives are right at the mean of response. Furthermore, Möbius proves that the duration of an AC is strongly linked to the duration of the accent group. Also, phrase-final accent groups are only about $\frac{3}{4}$ of the length of ACs in other positions and their f_0 movements are predominantly falling (76% of all accent groups). In non-phrase-final position, on the other hand, the vast majority of the accent groups are rising or rising-falling. Consequently, the position of the AC determines the distance between T_1 and accent group onset: the further back in the phrase, the shorter the distance between T_1 and accent group onset. This is because falling f_0 movements are modeled with the falling part of the accent component, conditioning the AC to occur early. Lastly, Möbius postulates a number of rules for the generation of German intonation contours, which he then

tests by means of synthesis experiments to ensure the perceptual equivalence of the original f_0 contours to the corresponding rule-generated f_0 contours.

4.5.2 Mixdorff

Mixdorff (1998) developed a quantitative model of German intonation based on the results of a series of production and perception experiments conducted for his Ph.D. He investigated the f_0 contours for sentence mode, broad and narrow focus, and conducted perception experiments of the retrieved material. Subsequently, he analyzed the f_0 contours of complex utterances by means of the Command-Response model and tested for phrase duration, speech rate, pause duration, and a number of other linguistic and paralinguistic variables, in addition to testing the response of the model parameters.

On a conceptual level, he links the accent component of the Fujisaki model with the tone switch concept introduced by Isačenko and Schädlich (1970) as well as Stock and Zacharias' (1982) concept of phonologically distinctive intonemes. Intonemes are quasi-discrete, phonologically distinctive elements of intonation "characterized by the occurrence of a tone switch at an accented syllable" (Mixdorff 1998: 33). In other words, tone switches are viewed as discrete intonational events. By the application of the Fujisaki model, Mixdorff can describe intonemes not only qualitatively but also quantitatively by means of the AC parameters they are connected to, namely amplitude, duration, and timing (Mixdorff 1998: 119). Three types of intonemes are distinguished – all of which differ in their communicative function:

Information intoneme (I): an intoneme associated with a falling tone switch in utterance-final position. The speaker intends to convey a message.

Contact intoneme (C): an intoneme associated with a rising tone switch in utterance-final position. It serves as a question marker if morphosyntactic markers are not present. The speaker intends to establish contact.

Non-terminal intoneme (N): an intoneme with a rising f_0 movement. Frequently occurs in non-final accents and serves to signal incompleteness of the phrase/utterance.

Mixdorff (1998: 69) models local accents in such a way that every tone switch is allocated at an onset or an offset of an AC, where the AC amplitude corresponds to the interval the tone switch takes at the accented syllable. He further explains that the timing of the tone switch is closely linked to the nuclear vowel of the accented syllable and "can be appropriately described relative to the duration of the word where it occurs" (Mixdorff 1998: 122). The PC, on the other hand, is understood as representing the global declination trend (Mixdorff 1998: 73).

With regard to the configuration of the model parameters and modeling, Mixdorff, too, uses constant α and β values, where α is set at 2.0/s and β at 20.0/s. For short, single-word utterances, however, an α of 3.0/s is used. Contrary to the settings used by Möbius, Fb is kept constant and is set to the mean of f_0 minima of the declarative sentences produced by the speaker. This means that Fb is understood as denoting the speaker-dependent asymptotic value of f_0 , and not as utterance-dependent. The initial values of the phrase component are modeled according to the unstressed syllables of the f_0 contour, which, in complex utterances, leads to a PC onset time of about 300ms before the onset of the utterance (Mixdorff 1998: 143). The cues for additional PCs are primarily resets of the declination line, syntactic criteria, or other criteria to obtain decent modeling accuracy, admitting that “the location of additional commands within the utterance is sometimes difficult to estimate” (Mixdorff 1998: 70).

In the first experiment, Mixdorff (1997) analyzed the f_0 contours of different sentences modes, as well as narrow and broad focal condition based on lab speech produced by 14 native German speakers (see Mixdorff 1997). In statements, the narrow focus element shows a rise-fall f_0 movement, while unaccented syllables remain flat. In the question condition, the narrow focus element shows a rising tone switch with an f_0 height that is sustained until the end of the phrase, at which point an additional AC results in a boost in amplitude so as to indicate the question-final rise. Narrow-focused constituents generally demonstrate higher AC amplitudes and reduced amplitudes for secondary accents. Amplitudes in elements with broad focus are lower and tend to be characterized by AC merging (hat pattern), as is displayed in Figure 4.12.

Figure 4.12 shows the f_0 contours of the phrase *Sie haben den Wagen geliehen* (Engl.: *They rented the car*). The top panel represents the narrow focus condition, the bottom panel the broad focal condition. The focused element *geliehen* (*rented*) clearly stands out being marked with high AC amplitude, whereas the broad focal condition exhibits an accent merging on the words *Wagen* and *geliehen*. The durations of narrow and broad focused constituents follow a similar pattern: while both conditions are accompanied by lengthening, narrow focused elements are subject to lengthening to a higher degree than broad focused constituents. Terminating and non-terminating broad focus phrases differ primarily in that non-terminal utterances show a rising tone switch on the last accent syllable, while the statement condition demonstrates a falling tone switch in the utterance-final syllable accent. This means that in the non-terminating condition, the AC timing parameters T_1 and T_2 occur later in relation to the accented syllable. With regard to PC magnitudes, Mixdorff notes a decrease for narrow focus conditions, which is strongest if the narrow focus is placed on the first accent syllable in the phrase. This is true especially for statements.

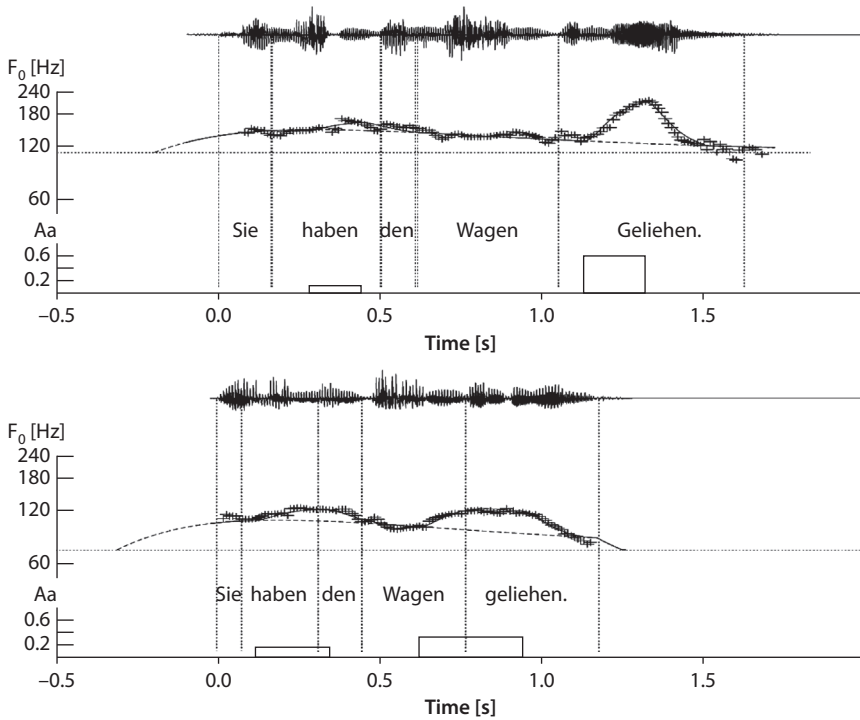


Figure 4.12. Narrow (top) and broad (bottom) focus effects on the f_0 contour (adopted from Mixdorff 1998: 84, 89)

In his third series of experiments, Mixdorff analyzed the f_0 contours of complex German utterances and tested for a number of linguistic and paralinguistic effects. 20 German speakers were asked to read a short text made up of sentences with varying syntactic features. The aim of this experiment was to develop a set of rules for the synthesis of German sentence intonation, as well as to create intonational reference points for a comparison of German utterances produced by Germans and Japanese.⁴

With regard to the statistical evaluation of the phrase component locations, Mixdorff observed that most prosodic phrases occur together with syntactic phrases. He did find, however, that some long syntactic phrases are split into two or more prosodic phrases, each of which was modeled with a separate PC. He discovered that 80% of the phrases were approximately 1.7 seconds in length, which corresponds approximately to 8 or less syllables (Mixdorff 1998: 140). With regard

4. The present summary will not elaborate on the results of these two research aims which are covered in the latter parts of Mixdorff's Ph.D. (1998).

to PC magnitude, Mixdorff showed positive correlations for duration of the preceding phrase, the number of syllables in the preceding phrase, and pause duration at PC onset. He found a negative correlation for the number count of the current phrase in the utterance when counting from the head of the utterance: the later the phrase in the utterance, the lower the PC magnitude (see Mixdorff & Fujisaki 1995). On the local level, Mixdorff found different effects for parts of speech and AC amplitude: the amplitudes are highest for proper nouns, lower for numerals, adjectives and nouns, and lowest for verbs. As for f_0 movements, Mixdorff observes that approximately 99% of all N-intonemes exhibit a rising tone switch.⁵

Unlike Möbius, Mixdorff only found a weak positive correlation between AC amplitude and AC duration.⁶ He further investigated the timing of I-intonemes and showed that 58% of these tone switches occur early on in the accented syllables, a position which corresponds to an early peak in Kohler's (1991a) conception of peak alignment, i.e. a falling f_0 through the nucleus. The remaining 42% of I-intonemes occur late with regard to the accented syllable, corresponding to Kohler's medial peak, which is a rise/fall through the nucleus. Finally, Mixdorff investigated the effect of different speech rates on the model parameters and found that for fast rates (8 syllables/sec), PC as well as AC amplitude of N- and I-intonemes is lower than for medium (6 syllables/sec) and slow rates (5 syllables/sec). Furthermore, increased speaking rate is characterized by a reduction of the number of prosodic phrases, a decrease in pause duration, as well as merging of ACs. Based on these results, Mixdorff formulated a set of rules for the generation of f_0 contours in the context of TTS. In his subsequent work, particularly in his post-doctoral Thesis (2002), Mixdorff developed an additional prosodic module which takes into account the connection between f_0 and the rhythmic/durational dimension of speech. In more recent work (Mixdorff & Pfitzinger 2005; Mixdorff 2008), he analyzes the modeling of fundamental frequency of spontaneous German utterances.

4.5.3 Shortcomings of the model

The Command-Response model is frequently criticized for its incapability of modeling certain low or low-rising accents, which are found for instance in American and British English (Liberman & Pierrehumbert 1994; Taylor 1994). Taylor (1994) and Ladd (1996) point out in their discussion of the shortcomings of

5. Mixdorff (1998) also observed a small number of falling tone switches in utterance-medial position.

6. Möbius (1993a) discovered that the shorter the AC in his data, the higher its amplitude.

the model that while the model's accent component can capture H* accents fairly accurately, the modeling of rise-fall contours poses certain difficulties. Both the rise and the fall of the accent component are governed by the same time constant but the rise time is in fact usually longer than the fall time (Taylor 1994: 41; Ladd 1996: 285). Hence, only the rise of a local accent can be modeled accurately, while the following fall is determined by the global parameters, and thus is not under the control of the AC. Furthermore, according to Taylor (1994) and Ladd (1996), it is even more difficult to model L* accents as well as slow-rises, such as those found in interrogatives, for example. Taylor (1994) illustrates this difficulty on the example of the contour presented in Figure 4.13.

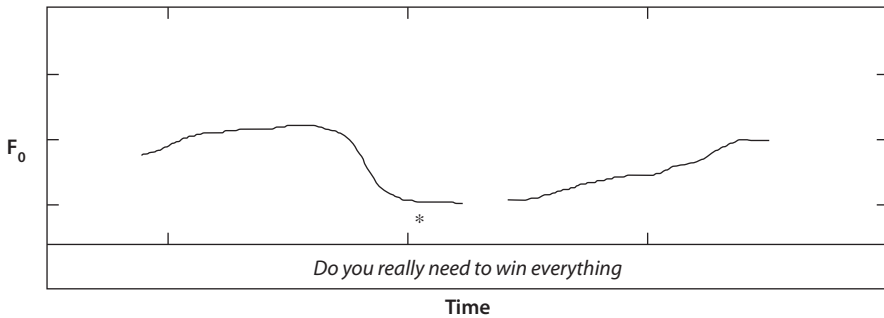


Figure 4.13. The L* accent on *need* as well as the following slow-rise are difficult to capture with the Fujisaki model (adopted from Taylor 1994: 42)

When considering the modeling procedure in such cases, Taylor (1994) suggests the implementation of negative step functions. The problem is that negative step functions do not adequately generate the shape of low accents, since low accents are frequently preceded by a rapid fall but followed by a slow-rise. The requirement for the negative accent with the same time constant for the fall and the rise would result in a rise which is just as rapid as the preceding fall. Moreover, he discusses the implementation of backward PCs or of a combination of an AC and a PC, in which the fall of the AC models the fall into the low accent and the PC models the subsequent slow-rise. Both of these techniques are less than optimal, however. The use of negative PCs would require a “rethink as to how to make the production mechanism behave in a plausible manner” (Taylor 1994: 41), and the combination of AC and PC would impede the alignment of accent shape with the stressed syllable. Also, the phrase component would not adequately capture the slow-rise since “the phrase shape is often too rapid for the gradual rises which follow low accent syllables” (Taylor 1994: 41).

Fujisaki et al. (1998) are aware of this problem and have suggested strategies for the modeling of slow-rises and low accents. They proposed to model slow-rises

by means of a short-interval sequence of PCs, as shown in Figure 4.14 on the example of the sentence *You've probably done better than you think* in British English, in which the phrase-final slow-rise is generated by a succession of PCs.

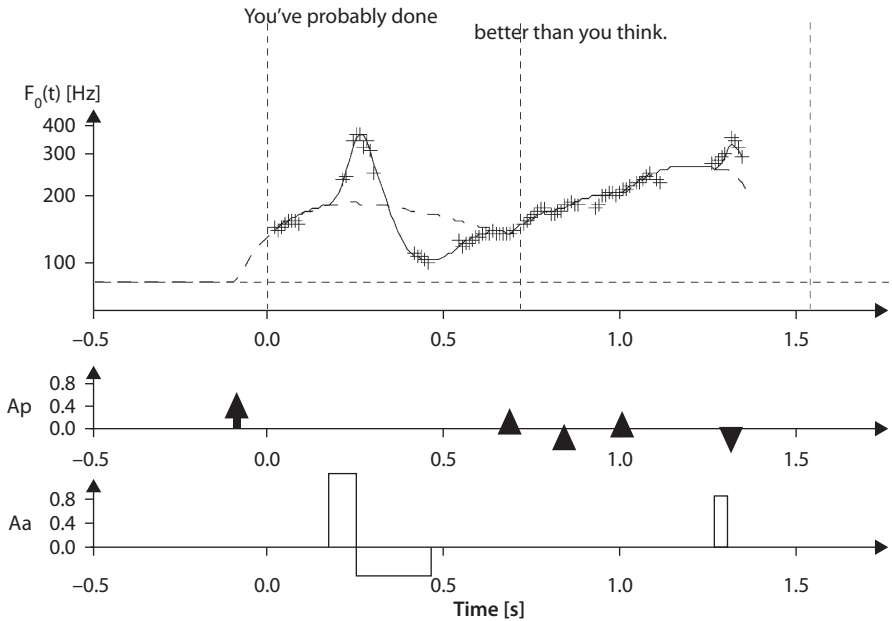


Figure 4.14. The modeling of slow-rises according to best fit (adopted from Fujisaki et al. 1998)

Taylor (2000:1711) justifiably notes, however, that albeit this procedure creates a good fit of the f_0 contour, “it has a severe price, in that all reference between the position of the phrase component and its linguistic meaningfulness has been lost”. Apparently, the modeling of the f_0 contours is done according to the criterion of an *optimal fit*, while only marginally taking into account the linguistic constraints (see Fujisaki et al. 1998). This, in turn, poses difficulties for a linguistic interpretation since the number of the model commands in an optimal approximation often exceeds the number of prosodic words and prosodic phrases in an utterance. As for local dips, Fujisaki et al. (1998) propose modeling with negative step functions in phrase-medial position. As mentioned earlier, Taylor (1994) does not agree with negative step functions for L^* modeling based on the argument that they do not mimic the f_0 contour appropriately. Taylor (1994: 42) thus concludes that “[i]t doesn’t seem possible to model low accents or gradually rising intonation effects in the Fujisaki system without radically changing the model”. In short, both Ladd (1996) and Taylor (1994) question the claimed

generalizability of the model beyond Japanese, a language which does not feature L^* and L^*+H accents.

Another problematic area, as pointed out by Ladd (1996:30), is the relationship between the phrase component and the prosodic structure of an utterance. Albeit, linguistically, phrase components should be used to model IPs, this condition cannot always be met. In order to capture the right shape of an f_0 contour, PCs sometimes need to be placed in positions where they do not make sense linguistically, as shown in Figure 4.14. This observation ties in nicely with one of the criticisms or limitations of the model encountered in the present data: nearly flat sections at relatively low f_0 values (close to Fb) and nearly flat sections at relatively high f_0 values (distant from Fb). This is particularly common in spontaneous speech where declination is only present to a minimal degree, if at all. Fujisaki et al. (1998) have suggested a way of modeling these two common f_0 contours: low flat f_0 contours should be modeled with a low-magnitude PC and a sequence of ACs that increase in amplitude towards the end of the phrase (let us call this *method I*), while high flat f_0 contours are to be modeled with a high-magnitude, phrase-initial impulse command, followed by a sequence of small PCs in short intervals (*method II*). This way of modeling is illustrated in Figure 4.15.

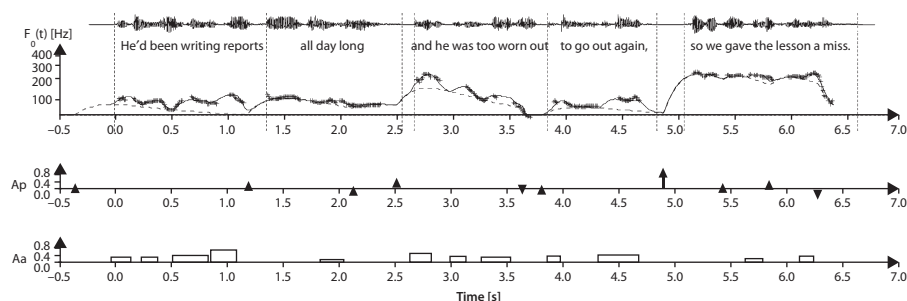


Figure 4.15. Suggested modeling of nearly flat sections at low f_0 values (0.0–1.3s) and modeling of nearly flat sections at high f_0 values (5.0–6.5s), according to Fujisaki et al. (1998)

In the author's view, both approaches are linguistically dubious: in the contour modeling of the time interval 0.0–1.3s with *method I*, the phrase-initial, small-magnitude PC is necessary for the modeling of the verb phrase *He'd been writing reports*, and the four ACs increasing in amplitude towards the end of the phrase are necessary to model the low flat f_0 contour. In this case, however, the increasing amplitude of the ACs does not imply an increase in prominence in the way inherently entailed in AC amplitudes. On the contrary, the ACs only increase in amplitude so as to compensate for the slope of the phrase component. *Method II*, applied to model the high flat f_0 values between 5.0 and 6.5s, is just

as questionable linguistically. While the first, high-magnitude PC is adequately placed to model the phrase *so we gave the lesson a miss*, the subsequent small PCs are used to model the high, flat *f0* contour. Yet, the small PCs do neither model IPs nor declination trends in the way inherently entailed in PCs. Instead, they are also placed to compensate for the steep slope of the first high-magnitude PC. Unfortunately, in order to model long, flat high or low *f0* contours, one has to opt either for *method I* or *method II* because the model does not offer any alternative way of modeling.⁷

4.6 Strengths – why the Fujisaki model was chosen for this study

The decision for or against a specific intonation model depends largely on the aim of the study. The reasoning underlying the choice of a phonetic, quantitative approach as opposed to a phonologically driven method of analysis in the current study is threefold.

The first reason concerns the intimate link between phonetics and phonology, phonology being the function of phonetic events. Since Trubetzkoy (1939), we distinguish between phonetics and phonology in that phonology categorically interprets language-specific, continuous, acoustic signals and thereby adds a component of meaning to the stream of speech. Today, however, this allegedly obvious separation is being questioned on a number of different levels. Neurological and psycholinguistic research, for instance, questions whether and in what way gradient processes can be represented mentally. This softening of formerly assumed rigid boundaries between phonetics and phonology is particularly prevalent in a description of *f0* contours. The continuous variation of fundamental frequency forms the basis for phonological interpretation. In the light of the linguistic importance of intonation and the numerous phonological functions of intonation discussed in earlier sections, a cross-dialectal, phonological analysis of *f0* contours is long overdue. By virtue of the close relationship between phonetics and phonology – phonology being the function of intonation – a thorough phonetic description of *f0* contours is a necessary prerequisite for the derivation of the functions of fundamental frequency contours.

Secondly, the use of a quantitative model constitutes the optimal method to study variation – one of the central problems in linguistics. An essential feature of phonetics is that no two utterances are identical. Utterances differ in infinite

7. For the analysis of the data in the current study, modeling *method I* was chosen for reasons which will be explained in Chapter 7.

ways, including prosody, segmental phonology, syntax, or pragmatic content. Consequently, it is impossible to produce an exact repetition of an utterance (cf. Labov 2004). In this sense, prosody provides the optimal grounds for the study of language variation: variation is accounted for by analysis, pattern recognition, and interpretation of a representative corpus of data. By choosing a quantitative method, we corroborate Fujisaki's (1997:27–28) claim that “the study of human communicative behavior through language belongs to the empirical sciences, where one needs to obtain, first and foremost, clear and objective knowledge of the phenomena through *making measurements*” (emphasis as in the original). Since, until today, no systematic, close-to-the-signal study on the f_0 contours of Swiss German dialects has been conducted – even despite an abundance of often impressionistic descriptions of dialectal intonation – the establishment of precise measurements of the f_0 contours by means of a quantitative intonation model was considered useful. These objective measures may then serve as a basis for subsequent phonological interpretations.

Thirdly, a quantitative, phonetic account of prosodic features of Swiss German constitutes an important counterpart to the vast majority of intonation studies working in abstract and symbolic frameworks. In the study at hand, the first methodological step consists of analyzing and parametrizing the f_0 contour, thus providing a mathematical description of the contour. Only in a second step is the linguistic analysis of these mathematical parameters and their relation to the individual segments established. This provides innovative insight into dialectal f_0 contours that is not conceivable with abstract symbolic, syntactic, or functional conversational analytical analyses. Hence, the findings in the current study can complement, specify, and support existing findings on f_0 patterns of Swiss German. Most crucially, however, even minor differences in f_0 realizations, albeit on a subphonemic level, may in the end be perceptually relevant for a cross-dialectal comparison (cf. Haas 1979).

The reasons for choosing the Fujisaki and Hirose Command-Response model are as follows:

4.6.1 High degree of accuracy of generated f_0 contours

While tone-sequence approaches use binary, high-low descriptions of intonation, the Fujisaki model is able to generate any contour with a close match and precision with only a few parameters because of its underlying mathematical formulation. Such a degree of accuracy is imperative to obtain numerical data on pitch excursions, which constitute a critical parameter for cross-dialectal comparisons. As mentioned earlier, pitch range poses a fundamental problem in the autosegmental-metrical approach.

4.6.2 Superposition

As was mentioned earlier, there is evidence for non-local influences on intonation caused by speech planning. The rate of declination, for instance, varies with sentence length, which means that the speaker must possess a *look-ahead* mechanism to control the rate according to the length of a given utterance (see Kutik et al. 1984). In contrast to other models, which also propose mathematical formulations of the f_0 contour, e.g. Taylor's (1994) *TILT model*, the Command-Response model mimics this global declination trend by means of the phrase component, which generates a declination trajectory onto which local accents are superposed. The model proposes a sensible, analytic separation of two individual components: the phrase component and the accent component.

4.6.3 Selective concatenation with segments

The analytic separation between the phrase and accent components is extremely useful as it permits a selective concatenation of the individual components with the segments. After the segments have been annotated with linguistic, paralinguistic, and non-linguistic variables, this concatenation between model parameters and segments is very practical because it allows an analysis of the effects of these variables on the global and/or local components of the f_0 contour. All effects can unequivocally be explained by their underlying causes (Möbius 1993a: 66).

4.6.4 Resynthesis

A vital strength of the Fujisaki model is the opportunity of resynthesis of the generated f_0 contour. Resynthesis is highly beneficial to f_0 analysis because one can systematically change one parameter at a time, so that the contribution of each parameter to f_0 can be evaluated individually (Vaissière 2004: 241). As will be illustrated at a later point, resynthesis constitutes a central component of the modeling procedure: resynthesis on both the syllable level as well as the phrase level is applied repeatedly. If the synthesized version of the f_0 contour is not satisfactory, the commands can, and ought to be, adjusted accordingly.

4.6.5 Replication

The analysis procedure is verifiable and can be replicated, with the exception of the last step in which resynthesis is applied. This property is not possible in other approaches. The IPO approach, for instance, does not allow for replication and verification and is thus criticized for its perception criterion: the researcher is the only subject of the perception experiments (Inozouka 2003: 96).

4.6.6 Physiological justification

Fujisaki and Hirose (1982:58) remark that the majority of intonation models “are based either on a crude approximation of the observed characteristics of f_0 contours or on their perceptual impressions, and are constructed without an awareness of the control mechanism of f_0 ”. Apart from early conceptualizations of the model for larynx control (see Öhman 1965, 1967), the Command-Response model is currently the only model to propose a plausible physiological interpretation by connecting f_0 movements with the activity of intrinsic larynx muscles. The physiological explanation of and basis for the process of f_0 production is an attractive facet of the Command-Response model.

CHAPTER 5

Swiss German

Switzerland has four official languages: German, French, Italian, and Romansh (*Federal Constitution of 1999*, Article 4). All of these languages were officially recognized back in 1848, apart from Romansh, which was added in 1938 (Ris 1979: 41). However, the term German does not provide an adequate representation of the spoken dialect of the Swiss people. Rather, the variety of German spoken by the Swiss people is referred to as Swiss German, of which there are approximately 4.5 million speakers (Lüdi & Werlen 2005).¹ Swiss German is prevalent in the Midlands, French is the dominant language in the West, and Italian is the language of the Southern Canton of Ticino and three valleys in the Grisons Canton. Romansh is spoken in several places in the Canton of Grisons. In the Swiss National Census in 2000, 63.7% of the Swiss population indicated German as their primary language, 20.4% indicated French, 6.5% indicated Italian, and Romansh was indicated by only 0.5% of the population (Lüdi & Werlen 2005).² According to Löffler (1997: 1858), the Swiss people believe Swiss German to occupy the most dominant role with respect to the other three national languages.

The variety of Swiss German is comprised of a great number of dialects, spoken in German-speaking Switzerland. Oftentimes, the dialect regions are not clearly delineated (Lötscher 1983: 141), but the rule of thumb holds that the dialects are named according to the Canton in which they are spoken. There are 26 Cantons in Switzerland, 19 of which declare German an official language, hence, the often-encountered question, whether the Swiss are able to understand one another. Normally, mutual intelligibility is assured due to extensive dialect contact between migrating Swiss people. Additionally the mass media, too, contribute their share to a mutual dialectal understanding, as the majority (an estimated 60%) of the Swiss National Television programs are currently broadcast in dialect (Siebenhaar & Wyler 1997). It is possible, however, that speakers from the Midlands, for example,

1. Swiss German dialects belong to the Alemannic languages, a group of varieties of the Upper German branch of the Germanic language family. Swiss German is not easily intelligible to most speakers of Standard German, although it is intelligible to other Alemannic dialects.

2. A total of 9% indicate as their first language a language other than the official languages of Switzerland (Lüdi & Werlen 2005).

have difficulty understanding the speech of a Valais speaker. A survey, conducted by the Swiss National Public Radio in 1975 shows that Valais Swiss German is the least understood dialect, while North Eastern and central dialects of Switzerland are understood best (see Werlen 1985:203–204). The Valais speakers are aware of their seemingly exotic dialect and are used to accommodating when in dialect contact with speakers of what they refer to as the “Üsserschwyz” (“outer Switzerland”), i.e. Swiss German speakers who live outside the Rhone valley (see Schnidrig 1986; Werlen et al. 2002; Werlen 2005b; Clyne 1984).

Typologically, the term Swiss German encompasses any of the Alemannic dialects spoken in Switzerland (Ris 1979:50). The dialects are categorized into three dialect groups: *Low Alemannic* (to which only Basel Swiss German belongs), *High Alemannic* (e.g. Bern Swiss German, Zurich Swiss German and Grisons Swiss German)³ and *Highest Alemannic* (e.g. Valais Swiss German as well as in a number of villages in the Grisons).⁴ The only exception to this rule is the Swiss German spoken in the municipality of Samnaun in the Grisons. Due to its topographical location and the community’s ties with neighboring Tyrol, the former Rhaeto-Romance dialect was replaced with the adjacent Tyrolean, Southern Bavarian dialect.

The geolinguistic structuring of Swiss German can be illustrated in a simplified way along five tenets (adopted from Siebenhaar & Wyler 1997), see Figure 5.1.

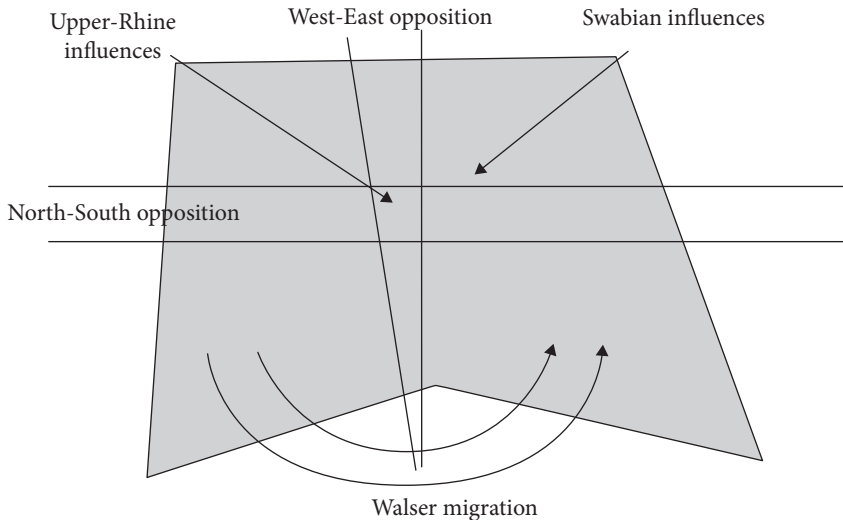


Figure 5.1. Geolinguistic structuring of Swiss German (adopted from Siebenhaar & Wyler 1997, translation by AL).

3. Only the four dialects under scrutiny are categorized here.

4. Although it is generally agreed to refer to the varieties of German spoken in Switzerland as *dialects*, Grimes (1984) strongly advocated that Swiss German be considered a language proper.

At the most fundamental level, there exists a North-South and an East-West division. In addition, we find linguistic influences from the Northeast and the Northwest (see Hotzenköcherle 1961, 1984; Lötscher 1983). A special case poses the Walser migration that interrupts the given West-East divide (see Russ 2002; Bohnenberger 1913). From the viewpoint of geolinguistic structuring of the dialects, the North-South and East-West oppositions are most essential. The North-South divide largely reflects a difference between archaic and more modern forms used in the dialects (Wiesinger 1983:829). As early as 1616, the Grisons historian J. Guler pointed out these differences between Alpine and Midland varieties, which he referred to as “natural effects of the geographical environment” (quoted in Zinsli 1946:11). In other words, he believed that, due to the severe weather and life conditions in the Alps, the variety spoken in the mountains developed into stronger, rawer, and more masculine dialects. Whether these varieties can in fact be referred to as stronger or more masculine remains unanswered here. However, the fact that topographic isolation plays a crucial role in dialect formation or preservation cannot be denied. The Upper-Valais dialect still retains forms from Old High German, most likely because these mountainous valleys were long secluded (Lötscher 1983:148). The division between East-West dialect regions is equally significant, as the contrasting dialectal features are as numerous and diverse.⁵ The causes for different dialects in these areas are largely due to the political landscapes of the past 200-500 years. In the West, the Bernese subject region was characterized as a politically unified area that favors linguistic homogeneity. On the other hand, Zurich in the East, too, exerted political and linguistic influence (Lötscher 1983:158). The discrepancies between the dialects of the East and the West lie primarily in the morphological realm. The plural of verbs in the East, for example, are patterned homogeneously, while the Western varieties use a differentiated plural system, similar to that of Standard German (Siebenhaar & Wyler 1997: 26). The third and fourth components of the geolinguistic structuring are the German influences from the North. Firstly, we find Upper-Rhine influences, which primarily affect North-western varieties. These influences are essentially present in lengthening phenomena, in the same way they have asserted themselves in Standard German (Siebenhaar & Wyler 1997). It is argued that, among other factors, these influences result from the “natural openness” of the area around Basel which made (and still makes) possible language contact with areas of the North (Hotzenköcherle 1984:76, translation AL). The second Northern influence is the impact of Swabian German on the Northeastern varieties of Swiss German. Finally, the East-West divide is

5. The East-West contrast is often referred to as the Brünig-Napf-Reuss line, which is said to stand for linguistic and cultural differences between Eastern and Western Switzerland (see Weiss 1947).

disrupted by the Walser migration, in which Valais residents left their valleys and settled in the Grisons in around about the 13th and 14th centuries. Consequently, the dialects spoken by these migrants still bear Western, Highest Alemannic, Swiss German characteristics, which clearly stand in contrast with the dialects spoken in the Grisons (see Löttscher 1983; Russ 2002; Bohnenberger 1913).

The Swiss are well aware of the abundant dialect differences and they all have their own, particular opinions about them. In general, it can be said that Zurich German is often perceived as fast, neutral, modern, and adaptive. Valais German, on the other hand, is considered lovely, indigenous, yet unintelligible (Ris 1992), and Grisons German is often described as original, warm, and melodic (Schwarzenbach 1969). Bern German, which enjoys the status of being Switzerland's most popular variety (Schwarzenbach 1969), is described as snug, homely, and slow (see Berthele 2006; Hengartner 1995; Werlen 1985).

5.1 Language use

The situation in German-speaking Switzerland is diglossic: both Standard German as well as Swiss German idioms are used (see Ferguson 1959). The relationship between these two varieties has been the topic of many debates over the past years (see Werlen 1988; Werlen 1998; Ris 1979; Schläpfer 1987). Two forms of the same language, high variety and vernacular, co-exist and they each have their specific scopes and domains of application. In the case of Switzerland, the varieties are usually clearly separated, meaning that hybrids rarely exist (see Rash 1998; Ammon 1995).

Swiss German is understood as the unconfined means of communication amongst Swiss German speakers and meets high approval in society. It is even viewed as more prestigious than Standard German (Sieber & Sitta 1986). In Germany or Austria, where we find a continuum between Standard German and dialect, the use of dialect or standard is functionally and stylistically determined and happens naturally most of the time. In Switzerland, however, we do not find such a continuum (Löffler 2000: 2044). The Swiss are typically aware of their language choice and the switch from dialect to standard is usually thematized and justified. Siebenhaar (1997) describes this issue as follows:

[The fact that], in German-speaking Switzerland, the professor and the untrained worker, the farmer and the priest can converse in the same language, is very important for the self-image of the Swiss German people. The use of the same variety is an expression of a democratic tradition, which distinguishes Switzerland from countries such as Germany or England, for example [...]

(Siebenhaar 1997: 11, translation AL)

Dialect use in Switzerland is important for expressing one's identity, as the Swiss anchor themselves linguistically in using their dialects (see Werlen 2005a; Christen 1998). The use of the standard is largely restricted to writing and reading, as Swiss German does not have a formal writing system (see Dieth 1938). However, recent technological inventions, such as the use of email or mobile text messages, have led to increased use of written Swiss German, which is inevitably categorized by a number of highly individualistic orthographic styles and is often an immediate pointer to the writer's dialect (see Christen 2004). In terms of its oral use, Standard German is restricted to the school context, mass media (see Ramseier 1988) and public speeches (Löffler 1997: 1858). Statistically, Standard German is used most extensively in schools (see Rash 1998; Löffler 1997: 1858; Sieber 1988), with the caveat, however, that Standard German is normally only spoken during school lessons. After-school meetings or announcements before the start of a lesson are held in dialect (Siebenhaar & Wyler 1997). As for the kind of Standard German spoken on the radio, the official guidelines for the presenters of Swiss National Public Radio (DRS) state that the presenters are by all means allowed to apply regional dialect coloring, since this does not inhibit communication on a cross regional level (Buri et al. 1993). In both chambers of parliament, Standard German is applied as members from all four linguistic areas of Switzerland are present. Over the past decades, there has been an increase in dialect use (Ris 1979: 44-45; Schläpfer 1987: 170). This development of course poses difficulties for the French, Italian, and Romansh speakers of Switzerland, who feel left out because the language they are taught in school is Standard German (see Clyne 2000: 2011; Kolde 1986).

5.2 Existing literature on Swiss German dialects

The dialects of Swiss German have been studied very thoroughly and extensively for the past 200 years (Lötscher 1983).⁶ The prime examples of the work on Swiss German dialects are the *Idiotikon* (1881ff.) and the *Sprachatlas der Deutschen Schweiz* (SDS) (*Linguistic Atlas of German-speaking Switzerland*) (1962–2003).

The *Idiotikon*, also referred to as *Das Wörterbuch der schweizerdeutschen Sprache*, is a collection of Swiss German vocabulary retrieved from written documents starting from the 13th century. The roughly 8000 sources of the entries in the *Idiotikon* are comprised of either historic lexemes (“ältere Sprache”) or pieces of living dialect (“lebende Mundart”) (Haas 1981: 34). The historic entries, dated

6. For an overview of the existing literature between 1800 and 1959, see Sonderegger (1962).

between 1300 and 1799, stem from literary excerpts, chronicles, archives, lexica etc., whereas the entries for living dialect consist of either direct contributions to the editors or excerpts from dialectal or scientific literature (*ibid.*).

The *Sprachatlas der Deutschen Schweiz* (SDS) attends to the geographic distribution of vowel quantity and quality, consonants, nouns, verbs, pronouns, adjectives, and counters. Most importantly, however, the main part of the SDS is dedicated to the word geography of Swiss German dialects. Both works were initiated out of the (mis)perception that Swiss German dialects are endangered and doomed (see Tappolet 1901). The SDS is considered to be one of the most significant regional atlases of the German language (Trüb 1982: 151). The initiators of the SDS, Rudolf Hotzenköcherle, Heinrich Baumgartner, and a group of fellow researchers, visited nearly 600 places in German-speaking Switzerland and some places in the Ticino and upper-Italy, where the Walser had emigrated in the 13th and 14th centuries. They were equipped with a questionnaire that included around 2500 questions, aiming at a comprehensive representation of the current state of the Swiss German dialects (see Hotzenköcherle 1962). The atlases include 1500 dialect maps as well as 600 figures. The atlas is distinguished among dialectologists for its recording location density as well as its highly differentiated results (Sonderegger 1977: 135–136).

Apart from these major works on Swiss German dialects, there are myriad publications that assess Swiss German dialects from various perspectives. Most pertinent for the present study are dialectal grammars (*Mundartgrammatiken*), which began to emerge towards the first half of the 20th century. These grammars offer more or less comprehensive accounts of the dialects in question. For the present study, the most pertinent descriptions are: Baumgartner (1922) for Bern Swiss German; Wipf (1910) for Valais Swiss German; and Brun (1910) and Meinherz (1920) for the dialect spoken in the Grisons. Weber's (1987) normative grammar, which is not part of the dialectal grammars series, is relevant for Zurich Swiss German.

Despite the abundance of studies on dialectal differences, most of these works only marginally hint at syntactic or prosodic features.⁷ As far as prosodic and particularly intonational features are concerned, there are only a few studies on Swiss German dialects, which suggests that the description of Swiss German dialects has thus far neglected to take into account intonational differences.

7. Even though Baumgartner and Hotzenköcherle were aware of the need to consider syntactic differences in their SDS, they only collected a small number of items because of the intricacy of eliciting dialectal, syntactic data. This gap was filled by a project at the University of Zurich (2000–2008) on the geolinguistic structuring of Swiss German dialectal syntax (see Glaser 2006).

5.3 Previous work on Swiss German intonation

The study of suprasegmental features of German dialects represents one of the areas of research that has long been neglected (see König 2007). To this day, no systematic account of Swiss German dialectal intonation exists. This does not mean that the differences between Swiss German dialects were not recognized. As early as 1819, Stalder made note of dialectal differences in tone and tune and adds that it is tremendously difficult to pinpoint what precisely these differences are:

[t]his diversity of the dialects mostly affects the sounding, i.e. the odd sounds of vowels and diphthongs, which are boldly sung, soon screeched, boldly lengthened into an ear-adverse fashion, boldly shortly aspirated [...] who wants to and can draw these sounds and tones and speech melodies with their rises and falls on paper? Who can visually display with dead letters and other inanimate symbols [...] for example the stiffness and seriousness of the Bernese, – the hasty and quick of the Entlibucher, – the sluggishness in the articulation of the upper Freiamter, – the singing of the shepherds in the high mountains of Uri, Bern, Appenzell, and Valais, particularly the Lötscher? (Stalder 1819:7-8, translated by AL)

Stalder (1819) concludes by saying that even if the melodic aspects were drawn or annotated, the various tunes would only remain shadows. 20 years later, Mörikofer (1838) states that it is the Alpine varieties, in particular, which are characterized by a distinct rising and falling in intonation (2nd ed. 1864: 161). Even though melodic features of dialectal speech have long been perceived as prominent, intonation research was not established as a discipline in its own right until the 1970s. This is largely due to the fact that the technological tools for a systematic analysis and description of fundamental frequency have only been introduced in the middle of the 20th century.

The work on dialectal intonation can be subdivided into a handful of categories, which will be discussed in more detail in the following sections. Chronologically, these include the collection of studies called *Contributions to Swiss German Grammar* (1910–1941), various MA theses (1971–2000), Fitzpatrick's (1999) recent study on Bern Swiss German, works on the intonation of Swiss Standard German (Panizzolo 1982; Stock 2000; Ulbrich 2005), and descriptions originating from speech synthesis research (Siebenhaar et al. 2004a; Siebenhaar 2004; Häsler, Hove, Siebenhaar 2005). These descriptions from speech synthesis research are most pertinent to the present study because of the application of the same intonation model, the Fujisaki model (Fujisaki & Hirose 1982).

5.3.1 Contributions to Swiss German grammar

In the late 19th century, a tradition of writing dialect grammars and dialect descriptions was initiated. Under the supervision of Albert Bachmann (1910–1941),

the series entitled *Contributions to a Swiss German Grammar* was geared at the synchronic description of a variety of dialects. Pieced together in a mosaic fashion and based on Winteler's (1876) structuralist descriptions of the Kerenzer dialect spoken in central Switzerland, these studies were aimed to give a comprehensive description of Swiss dialects. The studies relevant for the present study concern the descriptions of dialectal, intonational features of Bern Swiss German (Baumgartner 1922), Zurich Swiss German (Weber 1987, not part of the dialect grammar series), Valais Swiss German (Wipf 1910), as well as the dialect spoken in the Grisons (Brun 1910; Meinherz 1920). The authors of these grammars mainly distinguish between dynamic accent ("exspiratorischer Akzent") and musical accents ("musikalischer Akzent"). I understand "dynamic accents" in the sense of *stress* and "musical accents" in the sense of *accent* as defined in the present study in Section 2.4.3. In what follows, the focus will be placed primarily on pitch accents. If available and appropriate, additional references about the dialect's intonation are included.

5.3.1.1 Bern Swiss German

The variety of Bern Swiss German is not clearly delimited and basically includes all of the dialects spoken in the Canton of Bern (Keller 1961:87). The dialect described by Baumgartner (1922) is spoken in the Northwest of the Canton. He discusses extensively the dynamic accent and its form of occurrence in simple words, tri-syllabic words, noun-compounds, multi-syllabic words, and words of foreign origin. Baumgartner (1922: 19ff.) distinguishes between strongest (1) and weakest (5) expiratory accents. His findings can be summarized as follows:⁸

1. The first syllable in simple words carries the stress. Schwa syllables are only weakly stressed, as in *f'at*r* (German: *Vater*; Engl.: *father*), with a stressed/unstressed ratio of 2:5 between *f'a* and *t*r*.
2. Unstressed syllables with a full vowel, such as *v'irtSaft* (German: *Wirtschaft*; Engl.: *restaurant*), correspond to stress level 4.
3. In tri-syllabic simple words, the medium-syllabic schwa carries the lowest dynamic accent, while final syllable vowels are articulated proportionately with more intensity, resulting in a ratio of 2:5:4 (first syllable:schwa:final syllable), as in *m'Umpf*IE* (German: *einen Mund voll*; Engl.: *a mouthful*).
4. The relationship between prefix/pre-nuclear syllable and stressed syllable is 4:2, for example in *Erl'Oub** (German: *erlauben*; Engl.: *allow*).
5. Noun-compounds show a ratio of 2:3 if both constituents are mono-syllabic, for instance in *h'u:shUnd* (German: *Haushund*; Engl.: *house dog*).

8. For a more thorough discussion of the Bernese dynamic accent system, see Baumgartner (1920: 19–22).

6. In polysyllabic words, the dynamic accents are distributed in a number of ways. For the most part, however, the patterns are 2:5:3:4, 2:4:3:5, and 2:5:3:5, as in *kx'9d*rhuf:** (German: *Kehrrichthaufen*; Engl.: *garbage pile*), exhibiting a dynamic accent pattern 2:5:3:5.

Baumgartner (1922: 19) adds that the difference between a strong dynamic accent and a weak dynamic accent in Bern Swiss German is less distinct than in Eastern Swiss dialects.

With regard to pitch accents, Baumgartner (1922: 22) writes that the dialects he investigated do not belong to the “singing” varieties.⁹ Instead, he discovers a gradual fall and rise from syllable to syllable. We normally find a high dynamic accent coinciding with a musical high tone, as is the case in nearly all other Swiss German dialects, except for the Valais dialect (Baumgartner 1922: 23). In the early 20th century, Haldimann (1903: 296) had come to similar results after investigating the vocalization of the Bernese spoken in Goldbach, nearly 20 kilometers east of Bern. She concluded that this dialect does not belong to the singing varieties either, since dynamic and pitch accents usually coincide. Word accents are either flat or falling, while lengthened affirmation answers or lengthened markers to indicate doubt can have a twin peak *f0* contour, high-low-high in the former and low-high-low in the latter case (Haldimann 1903: 298). Declarative phrases usually show fall-ing intonation (*ibid.*). Later, in 1985, Marti (1985: 11) observed that the only region where singing is quite common in Bern Swiss German is the Bernese Oberland.

5.3.1.2 Grisons Swiss German

The term “Grisons Swiss German” is mostly associated with the variety of Swiss German spoken in the Chur Rhine valley. This taxonomy primarily encompasses the Chur dialect and the dialectal variants thereof spoken in adjacent valleys and villages.¹⁰ Additionally, however, also the Bavarian and the Walser dialects spoken in Samnaun are included in this category.¹¹ In 1918, Brun investigated exactly this latter variety of the Grisons dialect, the Walser dialect. This Highest Alemannic variety is spoken in Obersaxen (with a population of 636 in 1910, according to the 1910 census), in the central part of the Grisons Canton, Southwest to Chur. The dialect was brought into the community of Obersaxen by the Walser who emigrated from

9. For a description of “singing” varieties in German, see Zimmermann (1998).

10. The city of Chur plays a significant role in this geolinguistic mapping of the Grisons dialect because it is the capital of the Grisons.

11. For a more detailed account on the linguistic situation in the Grisons Canton, see Section 6.1.3.

the Valais in the 12th and 13th century. Meinherz (1920), on the other hand, studied the dialect of the Bündner Herrschaft, an area located in the Northern part of the Rhine valley, closer to the Canton of St. Gallen. As a general observation, Brun (1918:27) notes that the differences between dynamic and non-dynamic accents are not as distinct as in the Midland dialects. In contrast, Meinherz (1920:34) provides a different account and, like other authors of dialect grammars before him, introduces several levels of strength in dynamic accents, 1 being the strongest, 6 the weakest. The descriptions of the dynamic accents in the dialectal variants studied by Meinherz (1920:34ff.) and Brun (1918:27) can be subsumed as follows:

1. Meinherz (1920:34) discovered that there are twin peak dynamic accents, similar to those of the Valais dialect, where the main stress falls on the first peak. This is the case for monosyllabic words that are elongated extensively, such as *ja*: (German: *ja*; Engl.: *yes*).
2. In bisyllabic simple words with an *i* or *** in the second syllable, Meinherz (1920:35) found a ratio of 2:5 or 2:6, as in *v'ag** (German: *Wagen*; Engl.: *car*). If there is a full vowel in the second syllable, the ratio is 2:4 or 2:5, such as in *v'YrtSaft* (German: *Wirtschaft*; Engl.: *economy*).
3. Most prefix/pre-nuclear accents carry a low dynamic accent of 6, like *f*rbr'En** (German: *verbrennen*; Engl.: *to burn*) (Meinherz 1920:35).
4. Brun (1918:27) established that in compounds, the stress falls on the first syllable, e.g. *h'ys:ha:ltig* (German: *Haushaltung*; Engl.: *housekeeping*). However, accent shifts to the second element are common, as in *h'e'rts'as*: (German: *Herzass*; Engl.: *heart of ace*). This finding was also corroborated by Meinherz (1920:36).
5. Brun (1918:27) notes that there are a number of words which exhibit stress assignments retained from older forms of the dialect, e.g. *mit'ag* (German: *Mittag* [first syllable stress]; Engl.: *noon*).
6. Brun (1918:27) further shows that old words of foreign origin generally have a high dynamic accent on the first syllable, e.g. *kx'ap:ə'lə* (German: *Kapelle* [second syllable stress]; Engl.: *chapel*).

As for the musical accent, Meinherz (1920) attempted a description of the dialects' intonation, although he admits that an investigation of the musical accents is much more difficult than assessing the dialects' dynamic accents.¹² He opts for musical annotation and thus describes impressionistically that "[t]he musical progression of speech is slightly cradling, comparable to a wavy line" (Meinherz 1920:37, translation AL). In Meinherz' (1920:38ff.) opinion, contrary to Brun's

12. This investigation is particularly difficult, so Meinherz, because musical accents are dependent on the mental disposition of the speaker (1920:37).

(1918) beliefs, the Grisons dialect should be classified as a borderline case of a “singing” variety due to the following characteristics:

1. In case of a twin peak dynamic accent, there is tonal gradation, such as for *jO:* (German: *ja*; Engl.: *yes*), where we find a rising pitch accent from C to F.
2. As is the case in Valais German, musical and dynamic accent do not always coincide. In a bisyllabic word with no sentence stress, for example, we can find a rising musical accent along with a falling dynamic accent, such as in *er hEt kHɛ:r* m'y*s** (German: *er musste umkehren*; Engl.: *he had to turn back*), with an *f0* rise from E to F and a simultaneous decrease in expiratory accent on *m'y*s**.
3. Another phenomenon that also occurs in the Valais variety is the falling musical accents in questions, such as *vi tu: d'O: hE:r*?* (German: *willst du hier hin?*; Engl.: *do you want to come here?*), where *hE:r** carries a falling intonation from C to G.
4. Oddly enough, the Grisons speakers seem to prefer to place phrase accents at the beginning of a phrase, even though the sense of this word does not require the emphasis.

As for Brun (1918), he unfortunately did not tackle the issue of the musical accent. He did stress, however, that a high dynamic accent and a high tone usually do not coincide in the Grisons dialect of Obersaxen. Weak syllables often bear a higher pitch accent than highly dynamic ones. The dialect belongs to the singing varieties, so Brun (1918: 28).

Apart from the above accounts on Grisons Swiss German, Cavigelli (1969) also made some observations on the intonation of a dialect spoken in Bonaduz, roughly twelve kilometers Southeast of Chur. Bonaduz poses a special case of Swiss German as Rhaeto-Roman was spoken in the village until the late 19th century before Swiss German gradually prevailed. Cavigelli's (1969: 326–328) most relevant contribution to the present study is the accent shift triggered by the language change: the accent shifted from the ultimate or the penultimate syllable in Romansh to an initial syllable stress in Swiss German.¹³ It is particularly the younger generations of speakers of the Bonaduz dialect who adhere to this accent shift and apply it regularly. There are, however, many cases where the Rhaeto-Roman original, penultimate and ultimate stress patterns are imposed onto German compounds. Such stress-misplacements are typical for particularly the older generation of Bonaduz dialect speakers (Cavigelli 1969).

13. It remains unclear whether the author was referring to pitch or dynamic accents.

5.3.1.3 Valais Swiss German

Valais Swiss German is commonly thought of as melodic and exotic (see for example Werlen and Matter 2004: 264; Schlegel 2006: 10; Bösch 1964: 31). This salient feature of the dialect is likely to be one of the reasons for Elisa Wipf's (1910) interest in Valais Swiss German.¹⁴ She set out to provide a fairly far-reaching description of the dynamic accents and the pitch accents of Valais Swiss German, based on the dialect of Visperterminen (a village in the Southeast of the Canton of Valais). At the time of research, this dialect was spoken by merely 600 speakers, whereas today the number of speakers is estimated at 1450.¹⁵ She chose this village because, at that time, it had not yet become a tourist attraction and was linguistically isolated at a high mountainside. After noting in the preface that Valais Swiss German has long been attracting the attention of linguists, particularly with regard to its unique use of *dynamic accents*, Wipf (1910: 17ff.) then goes on to describe those accents as follows:

1. Twin peak dynamic accents are rather common on monosyllabic words with a low vowel if they occupy the function of a full sentence, such as in *ja*: (Engl.: *yes*), which functions as an approval-particle. They may further occur on sentence-final, long vowels, e.g. *ix xum:**u** bald*¹⁶ (German: *ich komme bald*; Engl.: *I will be coming soon*). The latter of the twin peak accents, although weaker, is still strong enough to nearly be realized as a new syllable.
2. The difference in intensity between strong and weak dynamic accents is significantly less when compared to Standard German as well as other Swiss German dialects. In the word *h'im:El* (German: *Himmel*; Engl.: *heaven*) for example, the ratio between the strong dynamic accent (*hi*) and the weak dynamic accent (*m:El*) parallels the one found in the German word *Schönheit* (Engl.: *beauty*).
3. The intensity difference between schwa-syllable and stressed syllable is slightly more present, e.g. *bits'i** (German: *beziehen*; Engl.: *to refer to something*).
4. Like their Old High German etymological predecessors, Valais vowels in final syllables carry a strong dynamic accent.

14. Bohnenberger (1913) also gives an account of Valais Swiss German, yet for the main part steers clear of an intonational description of the dialects.

15. Number retrieved from http://www.heidadorf.ch/content/de_dorf_portrait_leute.html, 16.04.2012.

16. In this example, the bold print highlights the vowel with twin peak dynamic accents.

5. In non-compounds it is always the root syllable that carries the strongest dynamic accent, while the secondary stress in multi-syllabic words often falls on the last syllable. For instance, in the word *t'ErbinEr* (name of the people living in Visperterminen), the primary dynamic accent falls on *t'Er* and the secondary on *nEr*.
6. Words of foreign origin show primary stress on the first syllable, or a consecutive syllable thereafter. In the case of *m'ilitEr* (German: *Militär*; Engl.: *military*), it is the first syllable *mi* that is stressed and not the ultimate syllable *t'Er/t'Er* as is the case in Standard German (and most other Swiss German dialects).

Wipf (1910) notes that the switch between strong and weak dynamic accents found in Standard German is not present in Valais German. Instead, the dialect spoken in the Valais is marked by a calm and gradual progression. This claim is also corroborated by Steiner (1921: 211) who states that the strong and weak dynamic accents of Valais Swiss German are the least distinguished, compared to other Swiss German dialects. Wipf (1910) points out the following, however:

When first listening [to Valais Swiss German speakers], one does not, however, obtain this pleasant, harmonious impression. Instead, after realizing that they are in fact speaking German and not Romansh, one is overcome with an almost annoying sensation, as if the people place accents as strongly as possible on the most irrelevant of syllables. (Wipf 1910: 19, translation AL)

Because the dynamic accents are also placed on unstressed syllables, Wipf (1910) claims that there are only three or four levels of dynamic accents as opposed to the five suggested by Baumgartner (1922) for Bern Swiss German.

Moreover, Wipf (1910: 21) claims that Visperterminen Valais Swiss German belongs to the “singing” varieties.¹⁷ She argues that, while dynamic gradation is low, speech melody plays a much more crucial role in the structuring of the parts of speech. Despite the dialect’s nearly level dynamic accents patterns, this variety sounds very melodic – a characteristic which is due to its high pitch movements. Because of the fact that nearly evenly distributed dynamic accents are combined with high pitch accents and that full syllable-final vowels carry a dynamic accent, this variety has an exceptionally romance-like sound to it.

17. For or a description of “singing” varieties in German, see Zimmermann (1998).

Wipf (1910:22ff.) further lists a number of *pitch accentuation* characteristics of the Visperterminen dialect:

1. Most relevant for the present study is Wipf's (1910:22, translation AL) observation that "[t]he general rule that dynamic accent and musical accent coincide and that unstressed syllables carry a lower tone applies to all of the Swiss German dialects except for [the variety spoken in] Visperterminen". She continues to say that the distribution of high and low pitch accents on high dynamic and low dynamic accent syllables is entirely free.
2. In final position of declarative sentences, a rising tone is predominant. For instance in the case of *Er g'ëjd g'ë:ru* (German: *er geht gerne*; Engl.: *he likes to go*), we encounter a tone shift of a fourth (C to F) or possibly only a third.
3. In questions, the last word is characterized primarily by falling pitch movements. The example *vas m'axt d=r 'ot:o?* demonstrates a musical fall from A to F (German: *was macht der Otto?*; Engl.: *what is Otto doing?*).
4. With regard to sentence or phrase accent, she generally finds phrase-final rises, regardless of whether the sentence ends in a stressed or unstressed syllable, which she however does not investigate any further due to a lack of technical means. Her final remark concerns the fact that phrase accents are not governed by the semantic value of the syllable in such a way that the semantically most significant syllable carries the highest or lowest pitch accent.

5.3.1.4 Zurich Swiss German

Weber (1987:21) observes in 1948 that Zurich Swiss German represents a heterogeneous dialect region, heavily influenced by Northeastern Swiss German dialectal varieties.¹⁸ Weber (1987) noted that there is a more Northern and a more Southern Zurich Swiss German group, the latter of which he decided to focus on. In particular, he decided to write a grammar about one particular, namely his native dialect Rüti-Hinwil in the Southeastern part in the Canton of Zurich. As for the dynamic accents, he only made note of two cases of deviations from the standard:¹⁹ compound-adjectives, such as *vorzüglich* (Engl.: *excellent, exquisite*), bear first syllable stress in Zurich German, while in Standard German, the dynamic accent is placed on the second syllable. Secondly, he mentions words of foreign origin such as *Musik* (Engl.: *music*), or *Schokolade* (Engl.: *chocolate*), in which dynamic stress

18. This section refers to the 3rd edition (1987) of Weber's study, which was originally published in 1948.

19. The fact that only minor deviations were found is on a par with Ris' (1992) observation that Zurich German is often perceived as neutral and remains fairly close to the Standard German variety.

also lies on the first syllable, as opposed to the Standard German version of *Musik* and *Schokolade*. Weber (1987:52) pointed out that Zurich Swiss German follows the general Swiss German pattern, exceptions granted, that the parts of speech with high dynamic accents are articulated with a higher tone, whereas the less dynamic parts are marked by a lower pitch. In addition, he found that the gradation of dynamically strong and weak accents is not distinct.

As for the melodic, intonational side of speech, Weber (1987:52) described the fundamental frequency movements as generally “calm and gradual”, without any quick intonation movements, except for emotionally charged language. Thus, in Weber’s (1987:53) opinion, Zurich Swiss German stands in contrast to the “singing” Swiss German varieties (e.g. Appenzell, Toggenburg, Glarus, Uri). Figure 5.2 exemplifies his way of illustration of typical Zurich Swiss German intonation contours by means of musical annotation.

Gewöhnlicher Aussagesatz:



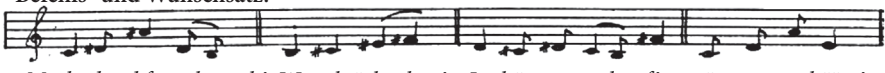
Lüüt hät s ghaa wie Vö - ge - li im Hauf-saa-me. Säl-ber äs-se macht fäiss.
Leute hat es gehabt wie Vögelein im Hanfsamen. Selber essen macht fett.

Ausrufsatz (Beteuerungen):



Si - cher uf Eer und häi - lig! Will s Gott isch waar! Ganz si-cher!
Sicher, auf Ehre und Heilig(keit)! Will es Gott, ist es wahr! Ganz sicher!
(Bei Gott..)

Befehls- und Wunschsatz:



Mach das d furt chunsch! Wotsch ächt choo! Jez hör e-maal uuf! wänn er nu chëem!
Mache, dass du fortkommst! Willst du wohl kommen! Jetzt höre einmal auf!
Wenn er nur käme!

Fragesatz:



Bisch es duu? Fry - li jaa! Chasch nüd Grüe - zi sää - ge?
Bist du es? Freilich ja! Kannst du nicht, Grüss Gott' sagen?
Freundlich fragend: Antwort:
Chasch au sin - ge? lich, ja fry - li chan i sin - ge!
Kannst du auch singen? Ich, ja freilich kann ich singen!

Figure 5.2. Weber’s (1987:53) exemplary Zurich Swiss German intonation contours

Weber (1987:53) admits that the type of melodic annotation he used only gives a crude account of the Zurich German intonation contours. Nevertheless, it is remarkable that a description of intonational features of a dialect, even if only two pages long, would surface in a popular-oriented dialect grammar of the mid-20th century at all.²⁰

Fleischer & Schmid (2006) by and large corroborate Weber’s (1987) findings. In their article “Zurich German”, which is particularly concerned with segmental features of the dialect, they note that Zurich German lexical stress is similar to that of Standard German in that word stress falls on the lexical root, i.e. the first syllable of the word, in most cases. Thus, word stress in the lexicon is largely determined morphologically (Fleischer & Schmid 2006:250). As with Bernese German, the authors assume as a default accent a low rising pitch accent (see Fitzpatrick 1999). This is illustrated in the following Figure 5.3.

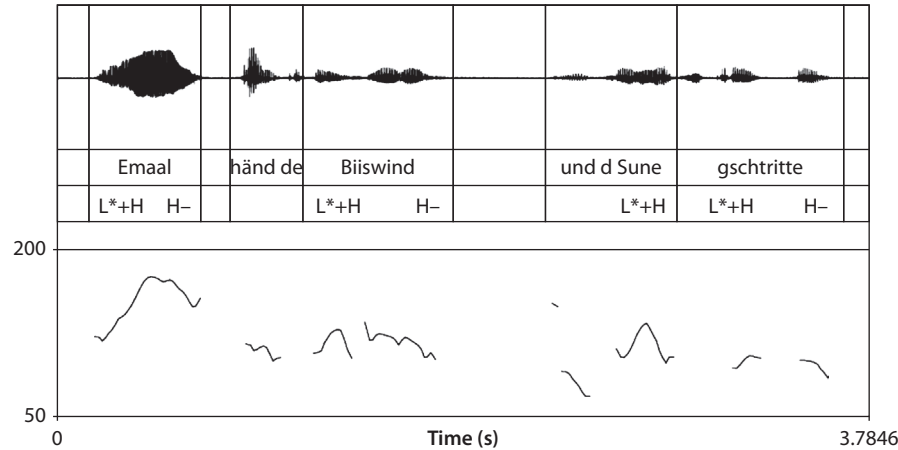


Figure 5.3. Low-rising accents, as described by Fleischer and Schmid (2006:250)

In addition to these observed slow-rising pitch accents, Fleischer and Schmid (2006:250) note that Zurich German reaches as a larger overall *f*₀ range than Northern Standard German (see Fitzpatrick 1999).

20. As a comparison, the *Duden* pronunciation lexicon (2005, 6th ed.) dedicates nearly 100 pages to prosodic features alone.

5.3.2 MA Theses 1971–2000

Engeli (1971): “Zur Problematik suprasegmentaler Merkmale der deutschen Sprache”

(Engl.: *On the difficulty of suprasegmental features in the German language*)

In his MA Thesis, Engeli (1971) provides the reader with an array of definitions of suprasegmental features, including intonation, pause, stress, and pitch range. In addition, he elaborately attends to the functions of prosody as well as its emotive facet. His Thesis mainly is theoretical in character with only few references to the intonational features of Swiss German. He does, however, mention the peculiar relationship between intensity and fundamental frequency of certain Swiss German dialects.

For Standard German, Engeli (1971) states that the higher the dynamic accent on a syllable, the higher the pitch, a co-occurrence which does not hold for all German dialects, however. Engeli reverts to Behaghel (1911) and Hirt (1925) who point out that the relationship between dynamic and pitch accents is reversed for Swabian and Lower Alemannic dialects, i.e. high dynamic accent – low pitch accent; low dynamic accent – high pitch accent. Swiss German is however not included in this group of Lower Alemannic dialects (Behagel 1911:74). Engeli (1971:34) argues that the contribution of pitch accents and dynamic accents to stress varies according to the language in question. He gives the example of German, for which dynamic gradation is more important – with the exception of Valais Swiss German – as opposed to French, where pitch accentuation plays a more significant role (*ibid.*). After discussing the problematic issues underlying the study of suprasegmentals in German, Engeli (1971) turns to an investigation of the articulation rate as one aspect of the suprasegmentals.

The final chapter of his MA Thesis can be understood as a plea for an investigation of Swiss German dialectal prosody. Engeli (1971) emphasizes that, despite numerous yet often short accounts on intonation in dialect grammars, a systematic description of Swiss dialectal intonation still remains a research desideratum. He highlights that albeit suprasegmental features were considered “an important and alluring task for future monographic research” by Hotzenköcherle (1962:240) in the introductory volume of the SDS, they were not included in the SDS, except for dynamic accents on the word and phrase level (quoted in Engeli 1971:101, translation AL). Engeli (1971) backs this up with a remark by Sonderegger (1968), who states the following, after having reviewed the currently existing literature on Swiss German dialects:

What needs to be worked on, apart from the elaboration of the historical and language-geographic overview, particularly concerns dialect typology in its narrow and wide sense: stylistics of the dialects, language histories of the

dialects or between dialect and written language, then [...] the clarification and explanation of the manifold questions regarding accents against the backdrop of the phonetic sciences.

(1968: 241, quoted in Engeli 1971: 101, translation by AL)

Spörri (1976): “Untersuchungen zur Satzintonation des Bergellischen und Oberengadiner-Romanischen”

(Engl.: *Investigations on sentence intonation of the Bergell and Upper-Engadin Romansh*)

Spörri's (1976) work on intonational features of the Bergell dialect and Upper-Engadin Romansh is pertinent to this study not because of the dialects in question, but because her work contributes to dialectal intonation studies in Switzerland.

The dialects investigated are spoken in the Southeast corner of the Canton of Grisons – close to the Italian border. The Bergell dialect is a Rhaeto-Roman dialect greatly influenced by Lombardian Italian. Spörri (1976) points out that in the time before her Thesis was written, mainly studies on the intonation systems of standard languages were conducted, while studies on dialectal intonation had been disregarded entirely. She maintains that there are no existing studies on the intonation of the Romance languages of Switzerland, much in contrast to accounts of Swiss German dialects such as the “Beiträge zur Schweizerdeutschen Grammatik”.

The experimental setting and the instruments Spörri (1976) used for her analyses can be considered advanced for that time period. She applied a pitch measurement device which via filtering eliminates harmonics and, as a result, measures only fundamental frequency. In addition to *f0* measurements, intensity measurements were also taken into account. Her subjects, mainly long-time residents, read aloud a text passage at their home. The sample texts were designed to elicit the melodic features of declaratives, questions, and exclamatory sentences.

Spörri (1976:47–48) discovers that Romansh speakers and Bergellers demonstrate a sentence-final, falling contour, regardless of sentence type. There are three exceptions to this rule. Firstly, in case of a phrase-final *, both dialect groups either remain on the same tone level or rise minimally. The second exception states that if the semantically most central word occurs in phrase-final position, the pitch may rise in declaratives. Thirdly, exclamations in Upper-Engadin Romansh can also show phrase-final rises. Furthermore, Spörri (1976) notes that Bergell dialect speakers seem to stress verbs more rigorously. The interplay between pitch movements and intensity – both contributing to “accent” – did not become clear from her data.²¹ In order to clarify this issue, further work needs to be done, so Spörri (1976).

21. It is not entirely clear how Spörri (1976:4) understands the term “stress” (German: “Betonung”), i.e. whether she considers it a concurrence of dynamic and musical accent.

Hegetschweiler (1978): “Comparing Native English Intonation and English Intonation of Swiss German”

Hegetschweiler’s (1978) Thesis was written in the context of second language learning. His paper aimed at establishing the degree to which English, spoken by non-native Swiss speakers, is characterized by L1 interferences on the intonational level. When reviewing the intonational features of Swiss German, Hegetschweiler (1978:24) as well mourned the fact that “[u]nfortunately there is still no study available of Swiss German intonation [...] [t]his field has been neglected in the past”. He did not seem to take into account, however, the numerous dialect grammars which actually do in many instances address suprasegmental features, even if only in passing. Hegetschweiler (1978: 24) specified that Swiss German intonation differed from that of Standard German in that it exhibits a “zick-zack melody” – a feature which apparently is also present in Southern German dialects. To be more specific, this means that, in Swiss German, we find a low pitch accent on the lexically prominent syllable but a high pitch accent on the adjacent lexically unstressed syllable, while in Standard German, stressed syllables normally carry a high pitch accent. This zick-zack melody is illustrated in Figure 5.4.

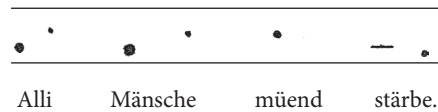


Figure 5.4. Hegetschweiler’s (1978: 25) “zick-zack melody” (German: *Alle Menschen müssen sterben*; Engl.: *All humans have to die*)

In Figure 5.4, the dots, framed by the f_0 baseline and topline, represent the f_0 of the corresponding syllable, whereas the bold dots stand for lexically stressed syllables. Hegetschweiler’s (1978) data consists of recordings of written English, read off by Swiss German Gymnasium students with varying competence levels of English. The recordings of a lecturer of the University of Zurich served as the “model” speech with the target intonation. The obtained data was processed with a fundamental frequency and intensity meter – the resulting oscillogram and the intensity curve were disregarded for the analyses. Yet the measured intonation contours were used merely to support Hegetschweiler’s (1978) auditive analysis. In a rudimentary way, he compared the differences in intonation and intensity between the target native English variety and the “Swiss English” articulation.

Hegetschweiler (1979) found several differences between the native and the Swiss English intonation. Most importantly, he noted that Swiss German speakers

tend to rise towards the end of continuing phrases (see Figure 5.5), whereas “English always has a fall” (1978:25).²²

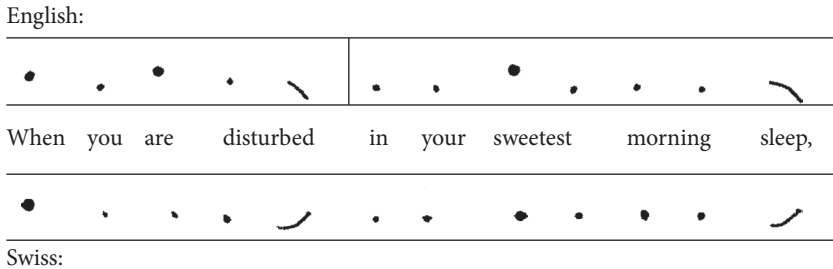


Figure 5.5. The phrase-final rises of Swiss German speakers in continuing phrases (1978:25)

Hegetschweiler (1978:50) ends his Thesis with a chapter on treatment, which refers to ways of helping Swiss German speakers acquire English intonation patterns. He explains that such treatment includes recurring exercises and practice as well as the exposition to a teacher who constantly addresses intonational deviations. Finally, he adds that being endowed with a musical ear also may make things easier (ibid.).

Radej (2000): “The English intonation of ‘bilingual’ speakers of English who are native speakers of Swiss German”

Like Hegetschweiler (1978), Radej (2000) set out to study the intonational interference for English spoken as a foreign language by Swiss German speakers. Her subjects, however, were advanced students of English at the English Institute of the University of Basel. Pilch’s (1977a, 1977b) work on Basel Swiss German as well as O’Connor & Arnold’s (1961) studies on Southern British English served as reference values. Radej’s (2000:25ff.) corpus consists of spontaneous English conversations between Basel Swiss German native speakers and Southern British English native speakers as well as a number of Basel Swiss German sentences. In her analyses, she primarily concentrated on statements and on open and closed questions. The elicited intonation contours were transcribed manually, based for the most part on audio input only.

Her findings regarding the intonational patterns of statements include that both Southern British English as well as Swiss German English (as spoken by her subjects) have a tendency to fall in pitch. Furthermore, Radej (2000:76) stresses

22. For a more detailed account of Hegetschweiler’s (1978) results, see pp. 45-46 of his MA Thesis.

that a great number of flat f_0 contours were found in Swiss English, which she attributes to the subjects' insecurities – paralleled by a low speech rate and several pauses and hesitations. She further found that Swiss-English speakers sometimes stressed certain particles, a feature which was not apparent in native speech (Radej 2000:76). This interesting phenomenon was explained by Radej as an L1 inference, since it is common in Swiss German to place high pitch movements on particles in spite of their relatively low value of semantic content (Radej 2000: 39; see Pilch 1977b).

5.3.3 Fitzpatrick's (1999) "The Alpine Intonation of Bern Swiss German"

Jennifer Fitzpatrick-Cole (1999) investigated the default pitch accent of Bern Swiss German in the context of the autosegmental framework (see Pierrehumbert 1980; Silverman 1992). From a typological standpoint, she set out to compare the default intonation contours of Northern Standard German and Bern Swiss German and stumbled across significant differences. While in Northern Standard German we find an H^*+L contour, i.e. a falling accent, Bern Swiss German has an L^*+H rising contour. This finding supports Sievers (1912:63–64) observation of such an inversion of tonal relations, which he made in the early 20th century for Southern German varieties. The following Figure provides an illustration of this tonal inversion.²³

L^* aligns with the stressed syllable and the f_0 does not rise for quite a lengthy part of the stressed syllable, frequently reaching its maximum only after the stressed syllable. Fitzpatrick (1999:943) argues that this tonal difference is not a realizational difference on the phonetic level, i.e. a late f_0 peak alignment, but is in fact phonological in nature. As a justification, she adds that such an analysis representing Bern Swiss German's default accent as H^*+L but phonetically realized with a very late f_0 peak alignment pushes our only interest in making an (unnecessary) link to Northern Standard German. Barker (2002), Gibbon (1998), and Kügler (2004) argue along similar lines. Gibbon (1998:93) labels the nuclear rise found in dialects along the Rhine valley as L^*+H , and Barker (2002, Chapter 6) also observes default nuclear rises in Tyrolean German, which he labels as L^*+H L^- , i.e. a rising nuclear accent followed by a low phrase tone for default declarative sentences. Similarly, Kügler (2004) provides evidence for L^*+H $L\%$ default accents in Swabian German (nuclear rise followed by a low boundary tone). He argues that this L^*+H pattern shares the same function between Bern Swiss German, Tyrolean German, and Swabian German (Kügler 2004:95).²⁴

23. Similar tonal inversions were also found for English dialects (see Grabe et al. 1998a).

24. For a discussion on the equivalence of dialectal intonation systems, see Peters (2006).

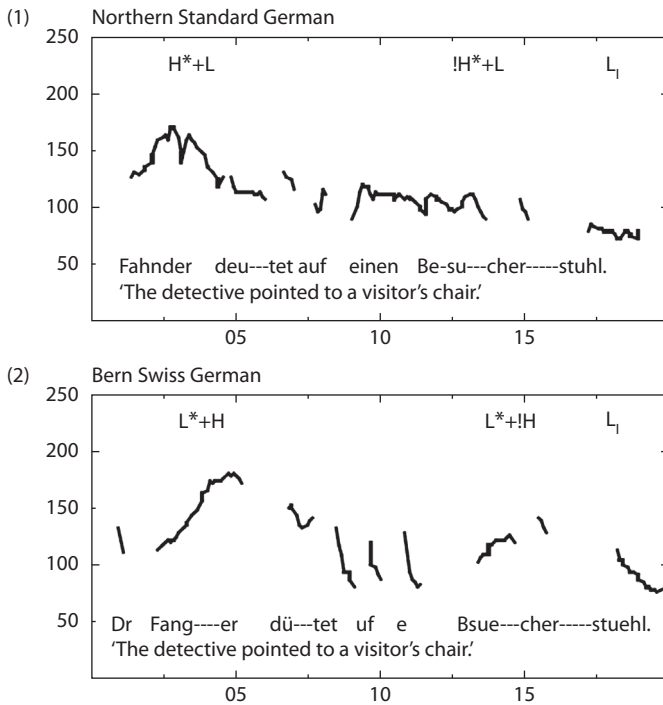


Figure 5.6. The H*+L falling default accent in Northern Standard German versus the L*+H rising default accent in Bern German (adopted from Fitzpatrick 1999:943)

5.3.4 Studies on Swiss Standard German

Albeit Swiss German speakers are very familiar with Standard German, their realization thereof is notoriously characterized by numerous features, which in Standard German are not present. This variety of Standard German, i.e. the kind of Standard German spoken in the Swiss German parts of Switzerland, is referred to as Swiss Standard German. The question remains, whether one can indeed speak of a Swiss Standard German variety (Mangold 2000: 1808; see Ammon 1995).

According to Hove, suprasegmentals are among the most important criteria to detect the identity of a Swiss German speaker conversing in Standard German (Hove 1999: 14). One of the most prominent characteristics in terms of lexical stress is that foreign words are usually stressed on the first syllable in Swiss German, but on the ultimate or penultimate in the standard, e.g. *Balkon*, *Kaffee*, *Telefon*, or *CD* (Lötscher 1983: 89; Siebs 2007). Meyer (1989: 26, translated by AL) points out the following differences between Swiss Standard German and Standard German intonation:

1. Compared to Standard German, the tunes (accents) of Standard Swiss German are more distinct, whereas the intensity distribution, in contrast, is generally more balanced.
2. The interplay between intensity and tone is regulated in a different way for each standard.
3. The speech rate of Swiss Standard German is generally slower.

Stock (2001: 171) comes to the same conclusions regarding the second point mentioned by Meyer, namely that the use of speech melody in Swiss Standard German often deviates from that of German and Austrian standards. He notes that Swiss Standard German is characterized by “lively melodic movements”, particularly in phrase-final position (Stock 2001: 171).

The intonation of Standard German as spoken by Swiss German speakers has received quite some attention, even if only treated marginally in some studies (see Meyer 1989; Ammon 1995; Hove 1999; Ulbrich 2005). The following pages will focus on the few yet most detailed accounts on Swiss Standard German intonation and suprasegmentals, which started emerging in the early 1980s.

Panizzolo (1982): “Die schweizerische Variante des Hochdeutschen”

(Engl.: *The Swiss variant of Standard German*)

Panizzolo (1982) provides a fairly exhaustive account of the segmental features of Swiss Standard German, yet dedicates only a minor chapter to suprasegmentals. In her analysis of intonation, she distinguishes between “Prätonie” (pretones) and “Tonie” (tones). The pretones begin at the first syllable of a sentence and end with the last syllable before the “last strong accent”, while the tones begin with the syllable with the last strong accent and end with the final syllable of the phrase (Panizzolo 1982: 40–41). She then introduces four types of tones: conclusive, interrogative, suspensive, and divisive. Panizzolo’s (1982) finding which is most pertinent to the present study, is that Swiss Standard German exhibits a rise/fall pattern for conclusive tones, as opposed to the falling pattern in Standard German.

Stock (2000): “Zur Intonation des Schweizerhochdeutschen”

(Engl.: *On the intonation of Swiss Standard German*)

In a pilot study, Stock (2000) investigated the Swiss Standard German intonation of the actors in Maximilian Schell’s movie adaptation of Friedrich Dürrenmatt’s novel “Der Richter und sein Henker” (1975). Subsequently, he compared his data with the data collected by Hove (1999), which was collected originally to investigate the segmental features of Swiss Standard German.

Stock (2000) focused on declaratives, requests, and WH-questions, performed an auditive and instrumental analysis, and subsequently compared the data to

their Standard German equivalents as spoken by selected German speakers (Stock 2000: 306). Here is an example of his comparison of the two standards:

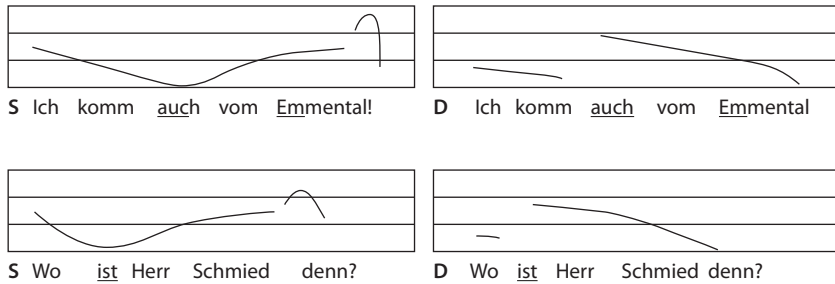


Figure 5.7. Stock's comparison between Swiss Standard German (S) and Standard German (D): The declarative sentence above is translated as *I too come from the Emmental!*, and the question means *Where might Mr. Schmied be?* (Stock 2000:306)

According to Stock (2000), the primary differences between Standard German and Swiss Standard German intonation lie in the post-nuclear contours: the Swiss steadily increase their f_0 and end with an abrupt fall. Additionally, Stock (2000: 307) observes that lexically stressed syllables bear a low f_0 , in contrast to the consecutive, unstressed syllables which increase in f_0 . Figure 5.7 further reveals that in Swiss Standard German, we find a high pitch onset, while the Standard German speakers start out on a much lower level. Stock (2000) then tested whether he would also find such typical Swiss Standard German f_0 contours, as retrieved from the actors' intonations, in Hove's (1999) corpus, while paying heed to external variables such as text, age, gender, and regional variation.

Stock (2000:313) concludes that the Swiss Standard German intonation contours he discovered in the piece occurred more frequently in spontaneous speech as opposed to prepared speech: the more comfortable the participant during the recording situation, the higher the number of these typical patterns. Interestingly enough, women seemed to apply the mentioned intonation contours more frequently in spontaneous speech than men (Stock 2000: 312).

Ulbrich (2005): "Phonetische Untersuchungen zur Prosodie der Standardvarietäten des Deutschen in der Bundesrepublik Deutschland, in der Schweiz, und in Österreich"

(Engl.: *Phonetic investigations of the prosody of German, Swiss, and Austrian Standard German varieties*)

In her Ph.D. Thesis, Ulbrich (2005) compares the standard varieties spoken in Germany, Austria, and Switzerland and gives extensive descriptions of their specific prosodic features. Her data consists of several news texts as well as one

folktale, read by 28 presenters of national broadcasting companies under the public law of Germany, Austria, and Switzerland. The data was then analyzed in three steps: several pretests, an auditive analysis of global and local prosodic features, and an acoustic analysis of global and local prosodic features.²⁵ The following subsection primarily focuses on Ulbrich's (2005) results regarding the intonation of Swiss Standard German.

Ulbrich (2005) discovered differences between the three standard varieties both on a local as well as on a global level. On a global level, the most pertinent results of the auditive analysis can be summed up as follows (Ulbrich 2005: 227–235):

1. The varieties are similar in terms of the f_0 interval between f_0 maxima and minima, yet the interval for the Germans is somewhat surprisingly larger. This is due to the fact that in declaratives, the f_0 maximum is already reached at phrase-initial f_0 peak, whereas the phrase-final lowering constitutes the f_0 minimum. In Austrian German, too, absolute f_0 maxima are reached in the first stressed syllable in phrase-initial position. For Swiss speakers, on the other hand, f_0 maxima occurred in phrase-medial positions as well, as can be observed in the following Figure:

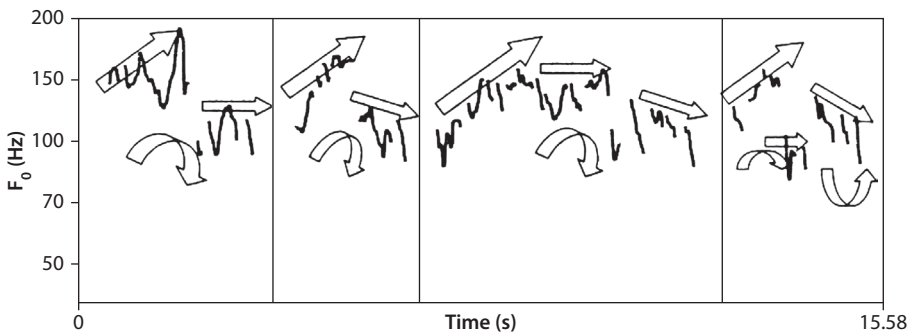


Figure 5.8. f_0 contours of a Swiss German newsreader. X-axis shows time, Y-axis exhibits frequency in Hz. The Swiss news reader realizes f_0 maxima in phrase-medial positions (adopted from Ulbrich 2005: 145)

2. Much in contrast to the Austrian or German standards, the declaratives of Swiss Standard German exhibit phrase-final rises or an unchanged f_0 level.

25. For further details on methodological issues, see Ulbrich (2005: 113–117).

The acoustic analysis, however, did not confirm these differences in the global f_0 contour (Ulbrich 2005:229). As for the analysis of the local intonation contour, Ulbrich (2005:117) introduced an inventory of five different intonation patterns: HH and LL (monotonal), HL and LH (bitonal), and the pattern LH_L for abrupt falls in f_0 after the pitch peak. In addition, Ulbrich (2005) applied the IViE labeling system introduced by Grabe et al. (2000) at Oxford University. The analysis showed the following results (Ulbrich 2005:230ff.):

1. Stressed syllables in the German and Austrian Standard are characterized predominantly by HL and LH_L patterns. The Swiss variety, on the other hand, exhibits a high percentage of low pitch stressed syllables, i.e. LH and LH_L contours.
2. Phrase-initial, stressed syllables are realized as LH* for Swiss German and Austrian, whereas the phrase-initial syllable is higher in f_0 for German.
3. Swiss and Austrian presenters have a more distinct f_0 interval in stressed syllables.
4. The f_0 contour of unstressed syllables that are preceded by high-pitched, stressed syllables falls for German and Austrian, but remains on the same f_0 level for Swiss German.
5. The phrase-final, stressed syllables in Swiss German and Austrian are characterized by a decrease in f_0 , followed by an unchanging or rising f_0 on the subsequent unstressed syllables. In German, on the other hand, the phrase-final position is marked by an absolute f_0 minimum (see Ulbrich 2002).

Apart from the above characteristics, Ulbrich (2005:219ff.) posits intensity as an additional parameter for prominence marking (see Ulbrich 2002; Hirschfeld & Ulbrich 2002). She explains that with regard to intensity, the acoustic analysis indicated that there is a rigorous distinction between stressed and unstressed syllables in German. In addition, she observed an increase in f_0 and dB in phrase-final positions for German, while the Swiss presenters remained at a level f_0 or intensity, or even showed an increase in both parameters (see Hirschfeld and Ulbrich 2002:69).

Ulbrich (2006) further looked at the f_0 declination of Standard German and Swiss German. For each variety, she analyzed the prepared, read speech of five TV and radio presenters (coherent text portions as well as sentences). She measured the f_0 on the syllable level in both initial and final stages of the periodic portion of the vowel in each syllable. These measurement points formed the basis for a linear regression line (Ulbrich 2005:167). Ulbrich (2005) discovered that, on an overall text level, the f_0 declination is less distinct for Swiss German speakers than for Standard German speakers, a finding which is of great interest for the present analysis. These differences were not significant, however (Ulbrich 2005:168). As a last point, as can be observed in Figure 5.8, the Swiss even rise in f_0 in text-medial

phrases, whereas the Germans exhibit a constant decline in f_0 in text-initial, text-medial, and text-final position (Ulbrich 2005: 170).

5.3.5 Results from speech synthesis research

Siebenhaar et al. (2004a), Siebenhaar (2004), and Häsler, Hove, and Siebenhaar (2005) showed that speech synthesis is a useful tool for the investigation of dialectal prosodic features. Due to physiological reasons, humans do not have the capacity to control prosodic components independently of one another. By means of speech synthesis, however, we can capture and isolate these prosodic components, which allows for a closer analysis of individual prosodic components such as timing, phrasing, or intonation. According to Häsler et al. (2005: 187), the benefit of generating dialectal models for speech synthesis lies in the gathering of results that are beyond a merely analytic approach “as the theory underlying speech synthesis in itself represents a functional model of language production” (translation AL). In addition, speech synthesis allows for a subsequent perceptual comparison of the generated models and thus an assessment of the model’s adequacy.²⁶

In 2001, a three year Swiss National Science Foundation project was launched, entitled “Elaboration of the fundamentals for the investigation of Swiss German prosody via synthetic modeling”.²⁷ This project investigated two Swiss German dialects, yet was not as such aimed at the description of their specific prosodic patterns. Rather, the aim was to create two dialectal speech syntheses and to thereby gain insights into dialectal prosody. The database consisted of spontaneous interviews with three Swiss German speakers: two Bernese speakers and one Zurich speaker. For each speaker, roughly 15 minutes of recording were analyzed, which amounts to about 8000–16000 segments per speaker (including pauses). The data was first segmented into individual sounds in Praat (2012) and these segments were then extracted and annotated on the segmental and syllable level (e.g. starting/end points of segments/syllables, duration of segments, type of nucleus etc.). The database was analyzed according to four prosodic parameters: pauses, phrasing, timing, and intonation. The following subsection will highlight the most significant findings, with a particular focus on the results of the intonation analyses.²⁸

26. For a detailed account of the functionality and implementation of this speech synthesis, see Häsler et al. (2005) and Siebenhaar et al. (2004).

27. SNF Project Number 1114-063702.00/1.

28. The following summary is based on the project’s final research report published in *Linguistik Online* (Häsler et al. 2005).

5.3.5.1 *Pauses*

Häsler et al. (2005:100) explain that pauses may (or may not) occur at any given point in a segmental string, which means that there is an “anything goes” principle when it comes to the occurrence of pauses in their corpus of spontaneous speech. Yet pauses are probable after syntactic boundaries, after conjunctions, and in the environment of emphatically accented words (Häsler et al. 2005:221).

5.3.5.2 *Phrasing*

For the analysis of phrases in the corpus, the following components were taken into account: pauses, lengthening, *f0* movements, intensity, syntactic structuring, and voice quality. Phrase boundaries were perceptually placed according to the above criteria.²⁹ Results show interaction effects between phrasing and pauses as well as between phrasing and timing. Segments in the environment of phrase boundaries are lengthened not only in phrase-final, but also in phrase-initial position (*ibid.*).

5.3.5.3 *Timing*

Timing is understood as the specific segment duration located on the same temporal axis as phrasing and pauses. Timing is difficult to investigate without taking into consideration larger linguistic constituents. The speakers show similar segment duration in their realization of * and long vowels, yet they differ significantly in terms of the length in short vowels. The short vowel of one of the Bernese speakers is shortest, followed by the second Bernese speaker. The Zurich speaker, on the other hand, produced the longest short vowels (Häsler et al. 2005:204). The location of the consonants, too, provided interesting results: consonants in the syllable coda are longer than those in the syllable onset (Häsler et al. 2005:206). Additionally, Häsler et al. (2005) observed phrase-final and phrase-initial lengthening in phrase boundary environment. Phrase-initial and phrase-final lengthening is relatively high for one of the two Bernese speakers as well as for the Zurich speakers, while the other Bernese speaker does not show such distinctive lengthening in either position. Finally, the variation for the three timing models has shown that for the Bernese speakers, the factor of intrinsic duration is more central, while for the Zurich speaker the surrounding segments seem to carry more weight (see Siebenhaar 2004).

29. The weight of each of these parameters with regard to phrase marking is not further specified.

5.3.5.4 *Intonation*

As for the intonation analysis, the Fujisaki Model (Fujisaki & Hirose 1982) was applied. Häsler et. al (2005:212) appoint PCs, which model global intonational contours, to phrases based on syntactic boundaries and pauses and ACs, which model fast intonational movements, to syllables in the text.

The first results concern the global intonation contour: PC amplitudes turned out to be fairly low for all three speakers when compared to read speech in Standard German and Swiss Standard German (see Mixdorff 2002a; Hirschfeld & Ulbrich 2002). Additionally, inter-group variation was discovered: the Zurich speaker indicates the highest PC amplitudes, one of the Bernese speakers the lowest. Finally, they correlate PC amplitudes with their frequency and show that PC amplitudes increase if the following PC is further away in the time domain (Häsler et al. 2005:216).

The findings regarding ACs are presented in detail, even though this analysis is based on the data of only two of the three speakers. In terms of the duration of the ACs, the Zurich speaker generally has longer ACs than the Bernese speaker, which the authors attribute to the faster speech rate of this specific Bernese speaker on the one hand, and to a different lengthening of syllables between the specific speakers, on the other. No significant differences were found regarding the temporal distance between the AC onset and syllable onset with which the AC is anchored. Neither are the AC amplitudes significantly different.

Furthermore, Häsler et al. (2005) ran a number of statistical tests for different linguistic variables that were believed to have an effect on the AC parameters. In terms of word class, it is assumed theoretically that ACs primarily fall on lexical words, while auxiliaries or grammatical words are less likely to be affected. The authors were able to show that this is indeed true: most ACs do indeed fall on lexical words. The AC amplitudes for word class do not differ significantly between the Bernese speakers but show significant differences for the Zurich speaker, with higher AC amplitudes on lexical words.³⁰ Syllable type (stressed, unstressed, or schwa syllable) was also extracted and tested for influence on AC amplitudes. The authors point out that ACs do not only link up with stressed syllables but also with unstressed or schwa syllables (Häsler et al. 2005:218). The Bernese speaker shows high amplitudes for unstressed and schwa syllables and lowest amplitudes for stressed syllables, contrary to the Zurich speaker, whose amplitudes are highest for stressed syllables, lower for unstressed, and lowest for schwa syllables. It could be argued, that the Bernese speaker's low AC amplitudes on stressed and higher

30. In their analysis, auxiliaries are not taken into account due to a small number of measurement points.

amplitudes on unstressed syllables parallels Fitzpatrick's (1999: 943) establishment of a L*+H default accent of the Bern variety.³¹

The final point of the discussion concerns the distance from the onset of the AC to the starting point of the syllable in relation to syllable length, as displayed in Figure 5.9.

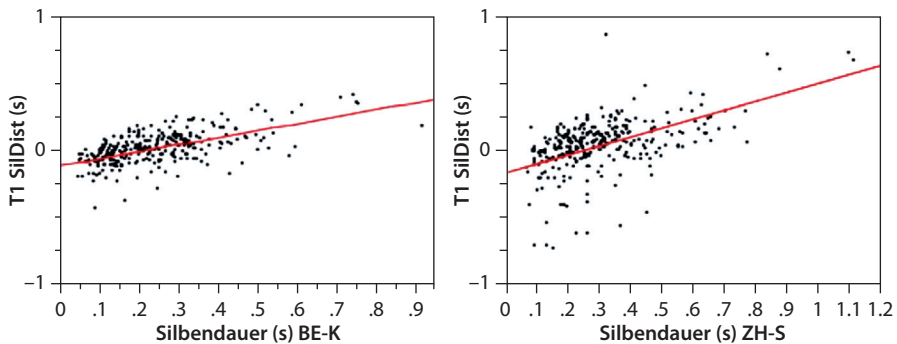


Figure 5.9. Temporal distance between AC onset and syllable onset in relation to syllable duration (adopted from Häsler et al. 2005: 220)

In Figure 5.9, negative T_1 silDist values indicate that the AC onset occurs before the syllable onset, while positive values denote the opposite. The Figure reveals a significant positive correlation for both speakers. The longer the syllable, the later the AC onset. Most importantly, however, this finding shows that “the position of the syllable peak is not a binary opposition, as suggested by ToBI annotation; instead it represents a continuum” (Häsler et al. 2005: 220, translation AL; see Atterer & Ladd 2004).

The authors conclude by confirming that there are obvious interaction effects between prosodic components, such as timing and intonation. They showed, for example, that syllable duration correlates with AC positioning. Such relations are given in the phonetic data and exist independently of a phonological interpretation. The obvious benefit of such an approach, so the authors, is that the phonetic detail of the data is preserved. The results are thus presented in milliseconds or Hertz rather than in categories such as H*+L or L*+H. This kind of research may be able to supplement existing work on intonation and prosody and “[f]or the prosodic comparison between speakers or varieties, phonetic differences, even if only sub-phonematic, may be perceptually relevant” (Häsler et al. 2005: 221, translation AL). In this respect, this project laid the groundwork for promising research in the future.

31. L* indicates the low pitch on the stressed syllable, H stands for the occasionally occurring pitch peak on the consecutive syllable.

5.3.6 Preliminary summary of previous work on Swiss German intonation

This section has reviewed the major existing works that have addressed Swiss German intonational features. Table 5.1–5.3 provide an overview of the most important findings. The tables are organized into three major directions: general observations on Swiss German intonation, dialect-specific intonational patterns, and Swiss Standard German intonation:

Table 5.1. Summary of general observations on Swiss German intonation

Author	General observations on Swiss German intonation
Stalder (1819)	Some Swiss German dialects are sung.
Mörikofer (1838)	Alpine varieties exhibit distinct rising and falling <i>f0</i> patterns.
Hegetschweiler (1978)	“zick-zack” melody: low <i>f0</i> on lexically prominent syllable – high <i>f0</i> on lexically unstressed syllable. Swiss German rise in continuation phrases.
Engeli (1971)	Peculiar relationship between stress and accent in Swiss German. VS: dynamic gradation is less important than accent placement.
Radej (2000)	Semantically insignificant words, such as particles, show high <i>f0</i> movements and stress.

Table 5.2. Summary of dialect-specific intonation patterns

Author	Dialect-specific intonational patterns
Weber (1987)	ZH: Stress and accent placement are very similar to Standard German (with a few exceptions). Calm and gradual <i>f0</i> movements; not a singing variety.
Fleischer and Schmid (2006)	ZH: low rising accents are default. Larger overall <i>f0</i> range than Standard German.
Baumgartner (1922), Marti (1985), Haldimann (1903)	BE: difference between stressed and unstressed syllables is less distinct than in Standard German. Not a singing variety (except for the Berner Oberland). Stress usually coincides with an increased <i>f0</i> .
Fitzpatrick (1999)	BE: L*+H default accent, where L* aligns with the stressed syllable. Phonologically different from Northern Standard German (H*+L).
Wipf (1910)	VS: Twin peak accents (both in stress and in accentuation via <i>f0</i>). Almost no difference in intensity between stressed/unstressed syllables. Vowels in final syllables carry strong dynamic accent. Singing variety. Stress and <i>f0</i> do not need to coincide but occur independently of each other. Sentence final rises in declaratives and falls in questions.
Brun (1918)	GR: dynamically stressed and unstressed syllables are not very different in intensity. Stress and <i>f0</i> do not coincide. Singing variety.

(Continued)

Table 5.2. Summary of dialect-specific intonation patterns (Continued)

Author	Dialect-specific intonational patterns
Meinherz (1920)	GR: clear distinction between dynamically stressed and unstressed syllables. Borderline case of a singing variety. Twin peaks in stress and in accent. <i>f0</i> and stress do not always coincide. Phrase accents usually in phrase-initial position.
Cavigelli (1969)	GR: certain words are stressed according to Rhaeto-Roman stress assignment, i.e. penultimate/ultimate syllable stress.
Siebenhaar et al. (2004a), Siebenhaar (2004), Häsler, Hove, Siebenhaar (2005)	BE and ZH: Low PC amplitudes compared to Standard German. PC amplitude increases if the following PC is further away in the time domain. Most ACs fall on lexical words. The longer the syllable, the later the AC onset. ZH: longer ACs than BE. ACs are more distinct for lexical words. BE: high ACs on unstressed and schwa syllables, low ACs on stressed syllables.

Table 5.3. Summary of Swiss Standard German intonation

Author	Swiss Standard German intonation
Meyer (1989)	Intensity distribution is more balanced. Tunes are more distinct than in Standard German. Interplay between tone and intensity is different than for Standard German.
Panizzolo (1982)	Conclusive phrases show a rise-fall pattern as opposed to a Standard German fall.
Stock (2000)	Post-nuclear contours are peculiar, with a rise-fall as opposed to a Standard German fall. Lexically stressed syllables show a low <i>f0</i> , contrasting unstressed syllables which may exhibit an increase in <i>f0</i> .
Ulbrich (2005, 2002, 2006) and Hirschfeld and Ulbrich (2002)	<i>f0</i> maxima may be reached in phrase-medial position (vs. phrase-initial for Standard German and Austrian German) and phrase-final rises in declaratives may occur. The Swiss variety mostly exhibits LH and LH_L contours on stressed syllables (vs. HL and LH_L for German and Austrian) and there is less difference in intensity between stressed and unstressed syllables. Less distinct global <i>f0</i> declination.

CHAPTER 6

Methods

This chapter addresses the methodologies applied in this study. First, the dialects and subjects under scrutiny are presented and contextualized in the framework of dialectology and language geography. The focus will then shift to the data collection procedure, keeping in mind that this approach is based on a corpus of spontaneous conversational speech with varying emotions. In a third subsection, the data preparation procedure is presented. Among other things, this section will explain why the syllabic level is viewed as the appropriate level of intonational analysis, as opposed to the segmental level or the word level. Finally, the data analysis procedure with the Command-Response model (Fujisaki & Hirose 1982) is introduced, followed by a portrayal of the modeling constraints.

6.1 Dialects chosen

The dialect regions considered in this study are: Bern [hereafter BE], Grisons [GR], Valais [VS], and Zurich [ZH] German. These regions were chosen based on the current dialectal situation in Switzerland: each of the dialects represents an opposing region in the East-West and North-South divide (see Hotzenköcherle 1961, 1984; Löttscher 1983). The North-South divide simultaneously allows for an analysis of intonational differences between Midland (BE, ZH) and Alpine dialects (VS, GR). The following Figure illustrates the location of the chosen dialect and the bold lines represent the general Swiss German geolinguistic structuring of the North-South as well as East-West divide (see Löttscher 1983: 137ff.).¹

1. This and subsequent maps of Switzerland are courtesy of *EDA* (Federal Department of Foreign Affairs), www.schweizerweltatlas.ch, accessed 16.04.2012.

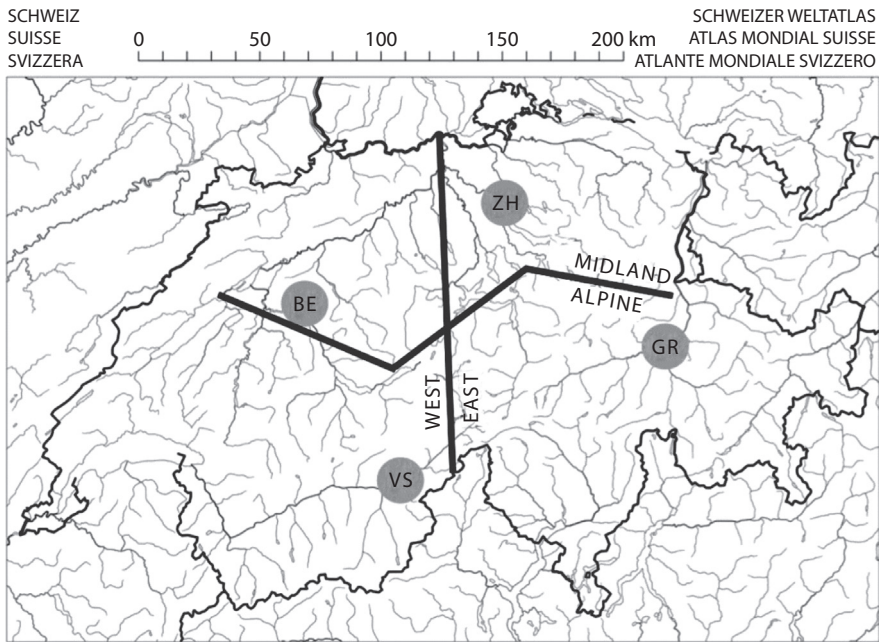


Figure 6.1. Dialects chosen for investigation: Bern (BE), Brig (VS), Winterthur (ZH), Chur (GR)

It should be noted at this point that the term *dialect* is inherently problematic. It is a highly polysemous expression, particularly in German, and different dialectological traditions, such as French, English, or German dialectological traditions, have developed their distinct definitions of the term (Ammon 1985:261). To make things worse, additional terms have surfaced and created terminological confusion in the German dialectological tradition. These terms include *Landschaftssprachenanalyse* (territorial language analysis), *Regionalsprachenanalyse* (regional language analysis), and *Gebietssprachenanalyse* (areal language analysis) (Stellmacher 1985: 189ff.). More importantly, questions about the definition of *region* in the context of dialectological studies have emerged (ibid.). Without a further discussion of existing approaches towards a definition of the term dialect, I will turn to the definition of dialect as perceived in the present study. This study adheres to a definition of dialect as proposed by Hotzenköcherle (1984) and to Siebenhaar's (2000) interpretation thereof. Hotzenköcherle (1984) calls the various dialectal regions of German-speaking Switzerland "Sprachlandschaften", i.e. *language areas* or, in a more narrow sense, *dialectal regions*. It should be pointed out that these dialect regions are neither geographically nor linguistically delineated clearly. Siebenhaar (1999:22ff.)

refers to these regions as *Grossraumdialekte* or *Grossraummundarten* (large area dialects). These linguistic areas reflect a lay person's knowledge of the dialect and its validity, which, so Siebenhaar, "is sustained by a number of linguistic criteria" (ibid., translation AL).

The decision as to which locations would be representative of the four chosen dialect regions was not an easy one. Dialects or dialect regions are inherently heterogeneous, a feature that does not exactly facilitate data elicitation and analysis (Oksaar 1985:216). Heterogeneity is even higher in larger cities and finding adequate representatives of the local dialect is impeded by migration movements that result in a melting pot of diverse dialectal variants (Löffler 2005: 138; Hotzenköcherle 1984:177). Due to the fact that secondary school students were chosen to be the subjects of the present study, recording sessions in larger cities were inevitable. Thus, the recordings were conducted in the following four cities, each over 10,000 inhabitants, and are considered to represent the larger dialect regions:

<i>Brig</i> ² :	representing the Western Alpine Valais dialect	(VS)
<i>Bern</i> :	representing the Western Midland Bernese dialect	(BE)
<i>Chur</i> :	representing the Eastern Alpine Grisons dialect	(GR)
<i>Winterthur</i> :	representing the Eastern Midland Zurich dialect	(ZH)

6.1.1 Brig - VS

Brig is the most important trade center and traffic junction of the Canton Valais. According to the Swiss National Census 2000, there are approximately 12,000 people living in the Brig-Glis district.³ Out of the four dialects, the VS dialect is the only Highest Alemannic dialect. The other three dialects belong to the High Alemannic dialects. The VS dialect is neighbor to two other languages: Italian in the South, and French in the West. VS Swiss German is generally spoken in the Upper Rhone valley (see Bohnenberger 1913:1). Hotzenköcherle (1984:177) further divides the VS dialect region into three major areas: firstly, from Lax to Oberwald, secondly, the area along the Rhone between Visp and the German/French language border, and thirdly, villages and valleys in between these two larger linguistic areas. The third area, particularly the Lötschental, behave differently on a case by case basis (ibid.).

2. The city of Brig encompasses the Brig-Glis district.

3. Retrieved from <http://www.brig-glis.ch/gemeinde/briginzahlen.php>, 16.04.2008.

6.1.2 Bern - BE

Bern, the capital of Switzerland, is not only the most important commercial center of Switzerland's largest Canton, but also acts as a traffic hub (see Baumgartner 1940: 101). In August 2008, its population was estimated at approximately 129,000 by the Swiss Federal Department of Statistics.⁴ Hotzenköcherle (1984: 193) notes that, of all the Swiss German dialects, no other variety features such a strong link between dialect and geographical conception of the Canton. Despite this seemingly rigid quality, BE German, too, is considerably heterogeneous. In fact, it is more heterogeneous than the dialectal variants of ZH (Keller 1961: 36). According to Hotzenköcherle (1984: 197), the BE variety can be divided into North and South: Northern Bernese (Laupen – Bern – Burgdorf in the middle), and Middle/Southern Bernese (Nidfluh – Faulensee – Brienz in the middle). Put differently, this corresponds to a division between the Midlands and the Alps. For instance, the Alpine speakers do not vocalize the *l* in words such as *Himmel* (Engl.: *heaven*), whereas Midland Bernese speakers do. The respective pronunciations are therefore /h'Im:U/ (Midland Bernese) and /h'Im:*l/ (Oberland Bernese) (Marti 1985: 24). An important characteristic for the present study is that the speech melody of the Alpine Bernese variety is said to have a distinct, singing quality (ibid.). In addition, the BE dialect spoken in the city of Bern is peculiar in that it has a long tradition of sociolects, distinctions which are primarily lexical in character and are still in use today, even if only in part (Siebenhaar & Stäheli 2000: 7).

6.1.3 Chur - GR

With approximately 33'000 inhabitants, Chur is the capital city of the Grisons Canton.⁵ It is the largest city of the Canton and is reputedly the oldest city in Switzerland, with first settlements as early as 11,000 BC.⁶ The Grisons Canton is often claimed to represent a Swiss linguistic microcosm: in a narrow space of only about 150 valleys, the languages spoken include Swiss German, Romansh, and Italian (Hotzenköcherle 1986: 152). The most important linguistic distinction in the GR dialect is that of Chur Rhine valley dialect and Walser dialect (see Meinherz 1920: 1; Hotzenköcherle 1986: 152). The Chur dialect borders with Romansh to the West (used in Domat/Ems), and the dialect to the North of Chur (down the

4. Retrived from http://www.bern.ch/leben_in_bern/stadt/statistik/in_kuerze/, 16.04.2008.

5. Retrieved from <http://www.gr.ch>, 16.04.2008.

6. Retrieved from <http://www.chur.ch>, 16.04.2008.

Rhine valley) diverges from the Chur dialect, increasingly exhibiting features of St. Gallen dialect (on the Swiss side of the Rhine) as well as features of Austrian German (Eckhardt 1991: 8).

6.1.4 Winterthur - ZH

Winterthur⁷ is the second largest city of the Canton of Zurich with a population of about 100,000 (July 2008).⁸ The general term Zurich German has a reputation of being a comparatively homogeneous dialect (see Keller 1961: 35). Nevertheless, three types of ZH German are usually discernable: Southern ZH German varieties, Northern ZH German varieties, and Northern borderline varieties, which are closer to North-Eastern Swiss German than to ZH Swiss German (Weber 1987: 20ff.). For the Northern group, it is differentiated between the Winterthur variety and the Unterland variety. The Southern part is more manifold, with a total of three subcategories: the Oberland variety (a more Alpine variety), the Ämtler variety, and the variety around Zurich lake, Zurich city, and the Limmat valley area (Weber 1987: 21–22). The Winterthur variety stands out amongst the other ZH varieties particularly because of its close and tense short vowels /i/, /u/, /y/, and /ø/. In other ZH dialects, these vowels are lax and open (Keller 1961: 35).

6.2 Subjects chosen

This study deals with issues of both phonetics (fundamental frequency modeling) as well as dialect geography/sociophonetics. The latter investigates the variability – the intonational variability to be more specific – of language under the dependence of varying geographic parameters. In other words, the different dialect regions act as independent parameters and language as the dependent parameter (König 1982: 471ff.). In such studies, it is common to control for the subjects' occupation, education, sex, and age (see Hagen & Boves 1994: 444). In studies on intonation, it is crucial to control for age because a person's *f0* may change throughout the lifetime (see Schötz 2006). In fact, the *f0* of females normally decreases until about the age of 50, where after it remains stable. The

7. Winterthur was preferred to Zurich because of favorable connection to a particular secondary school which offered to participate in this study.

8. Retrieved from <http://www.stadt.winterthur.ch>, 16.04.2008.

male *f*₀ exhibits the same initial pattern but then increases after the age of 50, sometimes reaching similar pitch heights as women (see Gerritsen 1985:80ff.; Schötz & Müller 2007). Other variables such as birthplace, current place of residence, the parent's origin, and so on are often elicited with a questionnaire (König 1982:471ff.).

As far as the selection of subjects is concerned, the most ambitious goal for empirical research is to achieve representativeness. The representativeness of a sample, i.e. the warranty that each element under observation has equal chances to be included in the sample, is one of the most ambitious goals of empirical social sciences (König 1982:473). Because of the heterogeneity in the dialects under scrutiny, it would of course be desirable to analyze a large sample. For a number of reasons, however, this is almost impossible (see Löffler 2005:138). One reason is the increased mobility on the part of the subjects, which very often entails contact with speakers of other dialects, impeding the geographical anchoring of the dialect in question. A survey of the Federal Department of Statistics shows that for the past 20 years, mobility has increased strongly from 29 kilometers in 1984 to 38 kilometers per person per day in 2005 (i.e. an increase of 30%).⁹ Christen (1998) examines deviations of present day dialectal speech from "Grundmundarten" (fundamental dialect regions) – which she bases on the SDS – on a largely segmental phonetic, phonological, and morphological level. She concludes that on the segmental phonetic level, the investigated dialects exhibit Swiss German-specific convergences, and that – at the same time – an accommodation towards Standard German is taking place. Whether or not this holds for the intonational domain is not addressed.

Other variables include the dialects of the parents or the subject's history of removals. Due to such factors, it is nearly impossible to assemble a large group of speakers that fulfills the targeted criteria of a homogeneous sample. Furthermore, studies in the field of acoustic phonetics often work with only a handful of subjects, since each speaker produces a vast amount of data in merely a few stretches of speech. One could of course opt to only analyze the speech of one authoritative representative per dialect. In other words, one would choose only one person who spent all of his/her life in a certain village and whose parents are from that same village (e.g. a teacher, pastor, or mayor) (Löffler 2003:41; König 1982:471ff.). This would provide a possible circumvention of the problem of finding authentic dialect speakers, and intra-dialectal variation could be alleviated. The present

9. Retrieved from <http://www.bfs.admin.ch/bfs/portal/de/index/themen/11/07/01/01/unterwegszeiten.html>, 16.04.2012.

study, however, aims to gain insights not only from inter-group but also from intra-group, cross-individual, intonational variation, which is why a larger sample was opted for.

The sample for the present study includes 10 speakers (5 male and 5 female) per dialect, which makes a total of 40 speakers. Swiss Gymnasia (grammar/secondary schools) were considered the most appropriate location to find subjects who would fulfill the above-mentioned criteria (similar occupation, education, and age). The following four Gymnasia were chosen for recordings:

<i>Brig:</i>	Kollegium Spiritus Sanctus
<i>Bern:</i>	Gymnasium Muristalden
<i>Chur:</i>	Bündner Kantonsschule
<i>Winterthur:</i>	Kantonsschule Rychenberg

The total of 96 subjects who were recorded represents a random sample of the total population of gymnasium students of all four gymnasia. Out of these 96 subjects, 40 subjects were selected randomly. Prior to the recordings, the subjects were asked to complete a short questionnaire regarding the following sociodemographic information: current place of residents and duration of residency, geographic origin of parents, and the city/village of kindergarten and elementary school they attended. In the form of an agreement, they gave their written approval that their recordings may be used for research purposes. They were further ensured anonymity, which was guaranteed in form of a code assigned to each subject. All of the above information is listed in Tables 6.1–6.4 and Figures 6.2–6.5 provide a geographical mapping of the subject's current place of residence, categorized according to dialect region. The commas in the figures between the subjects indicate that they reside in the same city/village, located closest to the leftmost subject. This grouping is particularly visible in Figure 6.3 for the BE and in Figure 6.4 for the Winterthur subjects, respectively (4 subjects live in Bern city, 8 in Winterthur city). Clearly visible, the subjects are more geographically dispersed in the Alpine varieties, particularly in the VS dialect region.¹⁰

10. The BE variety also entails three speakers from the Southwest of Bern (BE22m, BE21f, and BE04f).

Table 6.1. The codes assigned to the VS subjects and the sociodemographic information gathered in the questionnaire

Code	Current place of residence	Resident since	Geographical origin of father	Geographical origin of mother	Place where kindergarten was attended	Place where elementary school was attended
VS22m	Greich	1998	Lucerne, Andermatt, Brig	Eyholz	Riederalp, Montana	Montana, Mörel
VS23f	Glis	birth	Embd, Brig	Brig, Susten	Glis	Glis
VS25f	Randa	birth	Vals (GR)	Flums (SG)	Randa	Randa
VS28m	Mühlebach	birth	Ernen	Münster	Ernen	Ernen
VS29m	Naters	birth	Naters	Naters	Naters	St. Niklaus, Eyholz
VS41f	Guttet-Feschel	birth	Feschel	Guttet	Guttet-Feschel	Guttet-Feschel
VS44f	St. Niklaus	birth	St. Niklaus	St. Niklaus	St. Niklaus	St. Niklaus
VS46m	Ried-Brig	1990	Zurich	Holland	Ried-Brig	Ried-Brig
VS48m	Zermatt	birth	Zermatt	Susten	Zermatt	Zermatt
VS49m	Visp	birth	Visp	Visp	Visp	Visp

Table 6.2. The codes assigned to the BE subjects and the sociodemographic information gathered in the questionnaire

Code	Current place of residence	Resident since	Geographical origin of father	Geographical origin of mother	Place where kindergarten was attended	Place where elementary school was attended
BE02f	Hinterkappelen	birth	Bern	Bern	Hinterkappelen	Hinterkappelen
BE03m	Bern	birth	Hungary	Germany	Bern	Bern
BE04f	Meiringen	birth	Aarau	Meiringen	Meiringen	Hasliberg
BE05m	Bremgarten	birth	Muhen	Buchs (AG)	Bremgarten	Bremgarten
BE06m	Hinterkappelen	birth	Germany	Holland	Birma, Switzerland	Hinterkappelen
BE12f	Canton Bern	1991	Lebanon	Basel	Basel, Bern	Köniz, Schlieren
BE13m	Bern	birth	Bern	Lauterbrunnen	Münchenbuchsee	Bern
BE21f	Spiez	1993	Karlsruhe (Germany)	Schwäbisch-Hall (Germany)	Spiez	Spiez, Thun
BE22m	Münsingen	1993	Bern	Bern	Münsingen	Münsingen
BE23f	Wabern	1989	Bern	Bern	Wabern	Wabern

Table 6.3. The codes assigned to the ZH subjects and the sociodemographic information gathered in the questionnaire

Code	Current place of residence	Resident since	Geographical origin of father	Geographical origin of mother	Place where kindergarten was attended	Place where elementary school was attended
ZH01m	Wülflingen	1988	Valencia (Spain)	Wülflingen	Wülflingen	Wülflingen
ZH02f	Winterthur	1988	Switzerland	Switzerland	Winterthur	Winterthur
ZH03f	Winterthur	birth	Niederglatt	Winterthur	Winterthur	Winterthur
ZH22m	Winterthur	birth	Winterthur	Glarus	Winterthur	Winterthur
ZH23f	Winterthur	birth	Croatia,	Winterthur	Winterthur	Winterthur
ZH24m	Winterthur	birth	Graubünden	Denmark	Oberwinterthur	Oberwinterthur
ZH26f	Stadel	birth	Solothurn	Elgg	Stadel	Stadel
ZH31f	Winterthur	birth	Thun	Winterthur	Töss (Winterthur)	Töss (Winterthur)
ZH32m	Winterthur	birth	St. Gallen	Lucerne	Winterthur	Winterthur
ZH34m	Winterthur	1990	Herisau	Winterthur	Winterthur, Seen	Winterthur, Seen

Table 6.4. The codes assigned to the GR subjects and the sociodemographic information gathered in the questionnaire

Code	Current place of residence	Resident since	Geographical origin of father	Geographical origin of mother	Place where kindergarten was attended	Place where elementary school was attended
GR01m	Masein	1989	Masein	Zillis	Masein	Masein
GR03m	Chur	1988	Trin Digg	Zurich	Chur	Chur
GR04f	Felsberg	1988	Versam	Furna	Felsberg	Felsberg
GR06f	Araschgen	1988	Engadin	Passugg	Arascan	Arascan
GR24m	Domat Ems	1996	Tiefencastel	Zurich	Scharanz	Ems
GR30f	Chur	birth	Chur	Chur	Chur	Chur
GR31m	Chur	1989	Chur	Germany	Chur	Chur
GR33f	Untervaz	1989	Bad Ragaz, Sargans	Vaduz, Sargans	Untervaz	Untervaz
GR34m	Landquart	1995	Sils i.E, Samedan	Triessen (Lichtenstein)	Baden AG, Landquart	Landquart
GR35f	Chur	1989	Mels	Churwalden	Chur, Dalen	Chur, Dalen

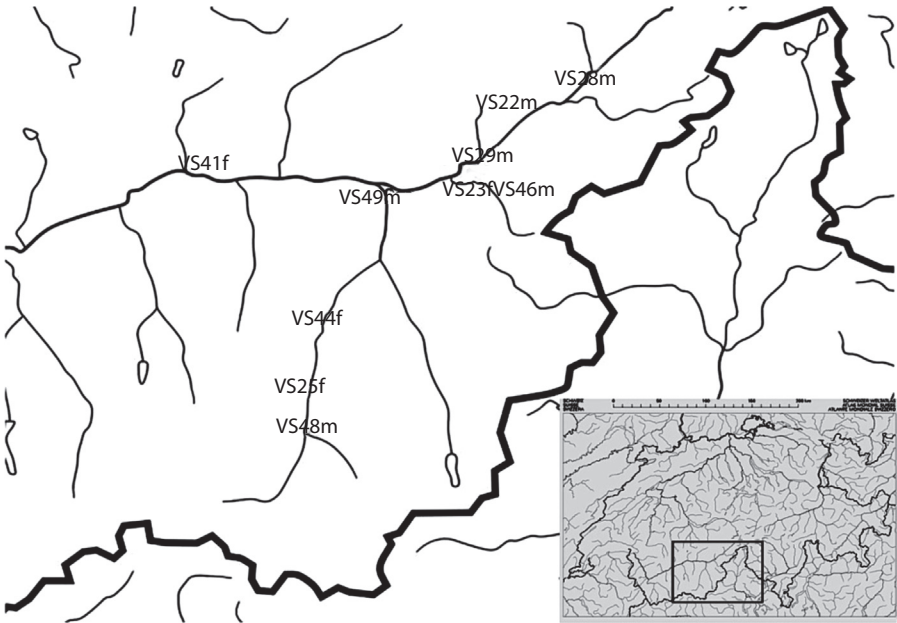


Figure 6.2. VS subjects' current place of residence

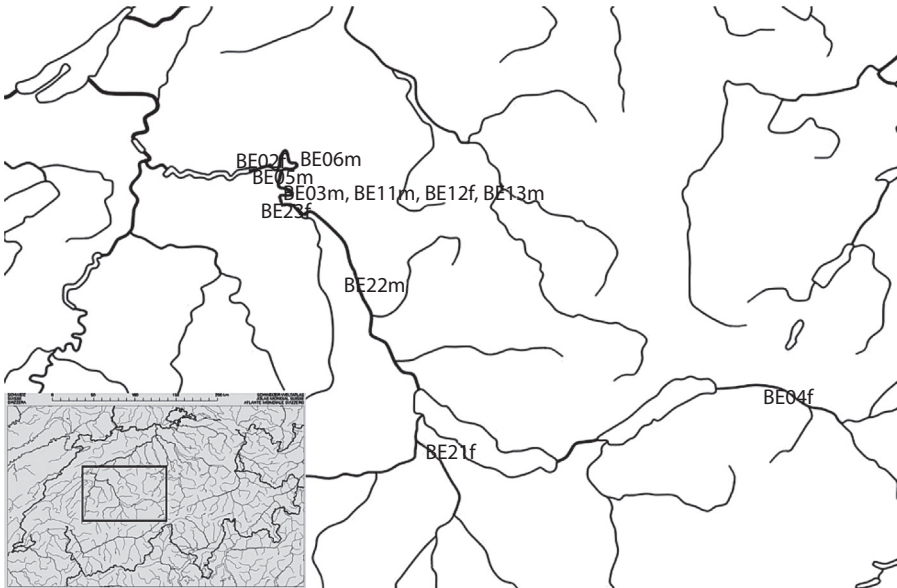


Figure 6.3. BE subjects' current place of residence



Figure 6.4. ZH subjects' current place of residence

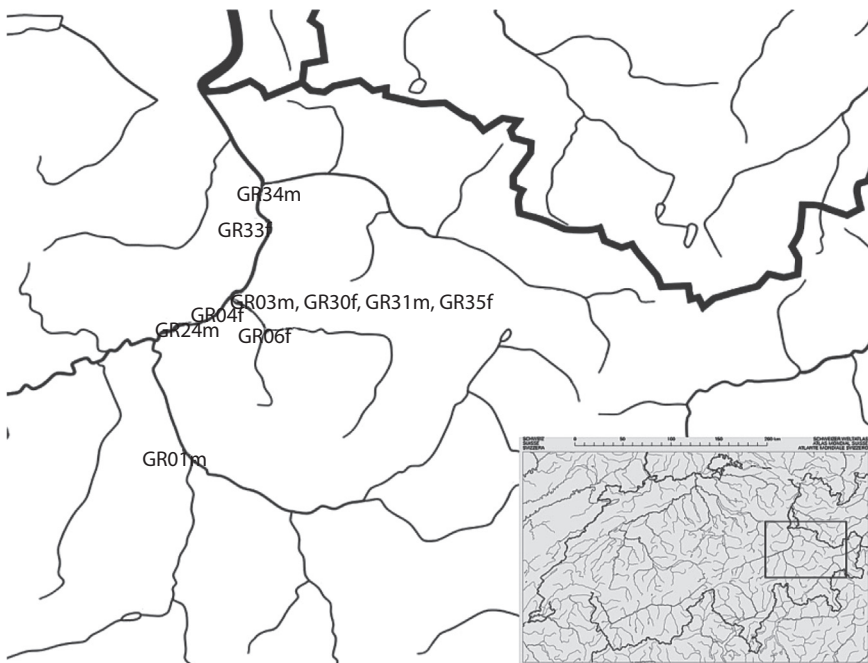


Figure 6.5. GR subjects' current place of residence

6.3 Data collection

The first recordings were made in Brig in the fall of 2005, followed by recordings in Bern, Winterthur, and Chur in 2006. The aim of the SNF project is to test for inter-dialectal intonational variation, eventually proposing an intonational geography of Swiss German. As mentioned in Section 5.3.5, the methodology of the current approach is based on Häsler, Hove, and Siebenhaar's (2005) work on Swiss German. Their study aimed at a description of the fine phonetic distinctions between the dialects, as minor phonetic differences may be perceptually relevant (Häsler et al. 2005:221). Therefore, their goal was to elicit as much regional variety as possible in order to test for fine phonetic, inter-dialectal differences. This is why they opted for the recording of spontaneous speech.

Similarly, the data for the present study was elicited via semi-structured spontaneous interviews (see Gilles 2005). Accordingly, the natural speech elicited in the present study is that of spontaneous interviews. Löffler (2003:49) describes this approach of data elicitation as follows:

The informant freely tells a story or an experience in front of the microphone. The interviewer is silent or merely backchannels via interested or affirmative behavior. (Löffler 2003:49, translation AL)

This choice of data elicitation brings with it two kinds of methodological issues. Firstly, as Schmidt (2001:27) observes, the field of conversational intonation research has not been able to put forth a methodological standard until today, namely a standard “which would allow for the decision whether intonation patterns of utterances with different complexities and different segmental-lexical bases are identical, similar, or not similar etc.” (translation AL). Secondly, in the light of the artificial conversational setting and the systematic observation of the subject's speech, it remains questionable whether one can still refer to the elicited data as “authentic speech”, a phenomenon first addressed by Labov as the *Observer's Paradox* (Labov 1972:209ff.). In the present study, however, natural, authentic speech is equated with appropriate speech, which means that speech is predictably perceived as suitable in specific contexts, regardless of the presence of formal or informal characteristics (see Wolfson 1997:124).

6.3.1 Recording devices

Different devices and microphones were used to record the subject's speech. These include two Edirol R-09 recorders as well as two Marantz PMD 671 recorders, one of which was generously provided by the Phonetics Laboratory at the University of Zurich. All devices are stereo audio recorders with a 24-bit,

44.1 kHz sampling rate. Further, one Sennheiser ME66 shotgun capsule microphone as well as three Sony ECM-MS907 microphones were employed. During the subsequent editing process, the bit size was decreased from 24-bit to 16-bit with Adobe Audition (2007) in order to enable further segmentation work with Praat (2012).¹¹ Furthermore, the signal was altered from stereo to mono and, in the case of low amplitudes, the signal was amplified in amplitude by 3–5 decibel in Adobe Audition (2007).

6.3.2 Interview setting and material

The interviews were set up in such a way that each interview with the subjects consisted of at least 15 to 25 minutes.¹² The informants were given standard explanations provided by the interviewers who had a short interview guide to hand. The interview was organized into four sections. In the first part, the actual spontaneous interview, the interviewee was asked specific questions designed to elicit long passages of free speech without disruption or breaking off. The topics covered the subjects' plans after graduation, their favorite past time activities, a description of prior vacation, and plans for a future vacation. The interviewer's task was solely to backchannel and if necessary provide new cues to stimulate the narrative. In the majority of cases, this technique alone already prompted roughly 15 minutes of recording per speaker. The second, third, and fourth part of the interview was designed for subjects who were uncooperative and did not provide the expected amount of free speech in part one of the interview. These backup techniques first involved the oral description of a favorite card or board game, a task based on an idea by Schlobinski (1996) which should elicit further passages of free speech. In case of continually poor participation on the part of the subjects, the interviewer resorted to picture naming tasks. The advantage of such tasks is, of course, that the picture acts as a stimulus for vocal production. The subjects were first asked to describe a picture devised for the *Boston Diagnostic Aphasia Examination* (Goodglass & Kaplan 1983), used originally to test speech production of potentially aphasic patients (see Figure 6.6). The fourth and last task involved a narration of Mercer-Mayer's (1980) *Frog Story*, while leafing through the pages of the booklet.

11. See http://www.fon.hum.uva.nl/praat/manual/Sound_files_3__Files_that_Praat_can_read.html, 16.04.2012.

12. Keller and Zellner Keller (2003) as well as Häsler, Hove, and Siebenhaar (2005) show that a data set of roughly 5 to 8 minutes per speaker is sufficient to create stable statistical models.

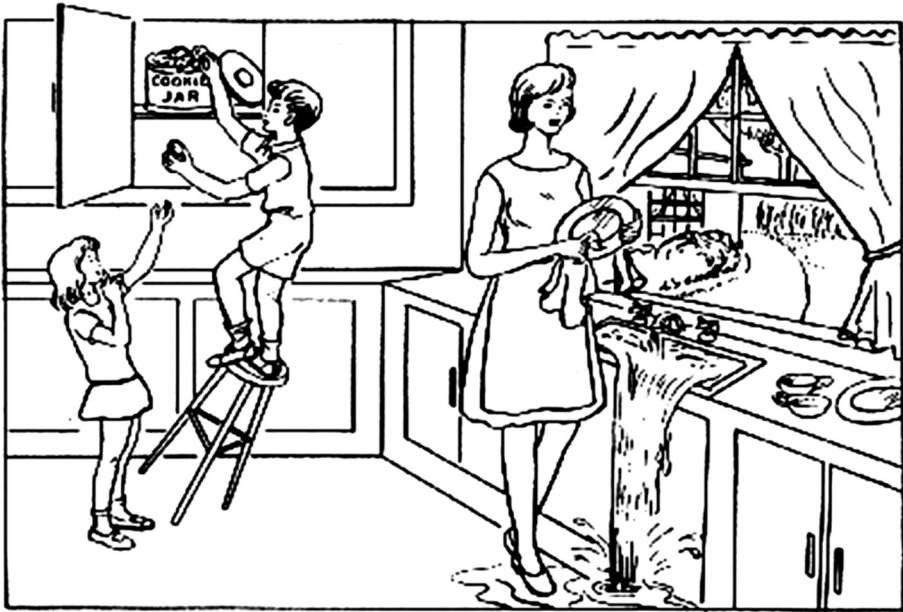


Figure 6.6. The cookie theft picture from the *Boston Diagnostic Aphasia Examination* (adopted from Goodglass & Kaplan 1983)

Roughly 90% of the present data was elicited throughout the spontaneous interview. Picture naming tasks were resorted to in only a few instances and there was no need to include any descriptions of the *Frog Story* (Mercer-Mayer 1980) into the present database.

6.3.3 Interview effects

The type of data and the chosen method of elicitation bear a number of problematic methodological issues presented in this section.

The first issue concerns the *interviewer-interviewee effects*. The way to ensure complete data validity is that every subject was interviewed under the same conditions (Werlen 1984:67). In the present project, this could not be achieved to such a degree. Among other reasons, this is due to the fact that the interviewers varied in age, gender, dialect, and other sociodemographics since all five SNF project members acted as interviewers. The sex of the interviewer, for example, is said to affect the interviewees' behavior, particularly in terms of social desirability, willingness to cooperate, and the desire for self-portrayal (Werlen 1984:86). According to Hyman (1954, quoted in Werlen 1984:86), interviewers of the opposite sex, for instance, are socially more desirable. Unfortunately, opposite-sex dyads were not upheld in principle in the present study. This may

in turn have the consequences that uncooperativeness – in the case of a same sex interview situation – may convert to limited vocal activity on the part of the subjects, which can in turn result in a lower mean length of utterances, i.e. shorter IPs (see Van Kleeck & Street 1982). Markel, Prebor, & Brandt (1972) have further shown that for both genders, vocal intensity increases if they talk to a member of the opposite sex.

Secondly, it can be assumed that the interview situation presents an *unfamiliar environment* for the interviewee. The interview settings in the present study are reminiscent of what Werlen (1984:81) refers to as the “school paradigm”. To be more precise, the subject is asked questions to which the interviewer may already know the answer. In this sense, the actual answers of the questions are not of genuine interest to the interviewer. The interview situation may thus be perceived as an interrogation rather than an interview. The direct translation of these factors into the elicited speech, particularly in terms of intonation, is not apparent. Possibly, the unnaturalness of the interview situation may inhibit the production of regional-specific linguistic forms (Werlen 1984:78). Additionally, due to the unnatural environment, the intonation contours of certain subjects may feature signs of an insecurity, fear or irritation, which according to Murray and Arnott (1993:1106) are characterized by a high pitch average and a wider pitch range (see 6.4.3).

A third phenomenon deals with *dialect accommodation effects* as well as *style accommodation effects* particularly significant for the VS dialect. It can be assumed that the VS speakers are aware of the unintelligibility of their dialect (see Ris 1992), which is why they are likely to accommodate to non-Valais German-speaking interviewers (see Schnidrig 1986; Werlen et al. 2002; Werlen 2005b; Clyne 1984). Giles’ (1973) accent mobility model suggested that subjects are likely to accommodate their speech styles to that of the interviewer if the interlocutor appears in a positive light (see Boves 1992). In case of a negative attitude towards the interviewer, the subjects may diverge from the interviewer’s style (see Bourhis & Giles 1977).

The fourth point of discussion addresses *task-specific effects on intonation*. As for the first task, the spontaneous interview, Siegman and Pope (1965) have shown that low specificity of interview questions can create a condition of informational uncertainty. This in turn may trigger uncertainty in the subjects, which in the end may translate into *f0* representations of fear or fear-like emotion, i.e. a high pitch average and a wider pitch range (Murray & Arnott 1993:1106). Even though all interviewers adhered to the interview guidelines, it is nevertheless possible that the specificity of the questions may have varied slightly. An obvious drawback of the second task, the game description technique, is that the interviewees are aware that the interviewer often knew the game in question, and thus knows how

it works. This may cause error awareness on the part of the subject, an insecurity that may cause a change in intonation, i.e. higher pitch average and wider pitch range. A significant drawback of the third task, the picture description, is the likelihood of phrase-final rises, as the items are narrated in a list like manner (see Selting 2001, 2007).¹³

Vocal quality effects should be mentioned as a fifth effect on intonation. Voice quality is highly multifunctional. For example, it may indicate the speaker's emotional and attitudinal state, or reveal personal characteristics, sociocultural indices, and allophonic variation. Most important to this study is that voice quality has interactional correlates (see Pittam 1994; Ní Chasaide & Gobl 1997). The type of phonation that poses a particular problem for the measurement of f_0 contours in Praat (2012) is creaky voice. This voice type, also referred to as vocal fry or pulse register phonation, causes errors in pitch determination algorithms (see Toshinori Ishi et al. 2008; see Mixdorff 1998: 62). In creaky voice, pitch is usually very low and f_0 of the glottal pulses is highly irregular (see Laver 1980: 122; Ní Chasaide & Gobl 1997: 450). An additional characteristic is low overall sub-glottal air pressure (Laver 1980: 124). Figure 6.7 gives an example of such measurement errors, here in the in phrase final syllables of *s *IntrEs*: 'i*r mi bejd*s ab*r iS r'ɔxt 'unt*rSIdl*x (German: *es interessiert mich beides aber ist recht unterschiedlich*; Engl.: *I'm interested in both but is fairly different*).

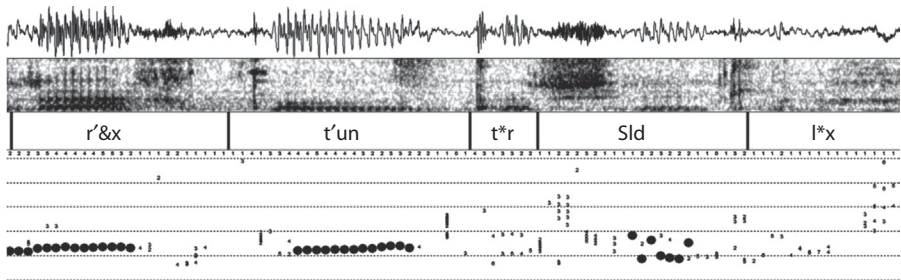


Figure 6.7. Measurement errors in final syllables of the second half of a phrase of a Bern Swiss German speaker

13. The analysis of the narration of the *Frog Story* would have proved additionally difficult as the story was originally designed for children. It was discovered that some of the subjects produced prosodic features typical of baby talk, such as higher pitch, slower tempo, exaggerated intonation, longer pauses, and shorter utterances (see Ferguson 1964; Fernald & Simon 1984).

The measurement errors occur predominantly in the sonorant components of the last two syllables *Sid* and *l*x*.¹⁴ One measure to counter such errors in pitch detection is to modify the pitch settings in Praat (2012). The lowering of the voicing threshold, however, did not improve the results. In contrast, increasing the measurement interval for *f*₀ from 100 (default) to 500 pitch points per second resulted in improved and more straightforward pitch measurements. The difficulty here is that Mixdorff's *FujiParaEditor* (2012), the program used for the intonational modelling in the present study did not respond well to such high frame durations. For this reason, *f*₀ measurements in Praat (2012) were deleted with incorrect measurements caused by unfavourable voice quality. Consequently, those syllables in question did not demonstrate any pitch measurements for the subsequent Fujisaki parametrization.¹⁵ If analyzed in the *FujiParaEditor*, the above semi-phrase looks as follows:

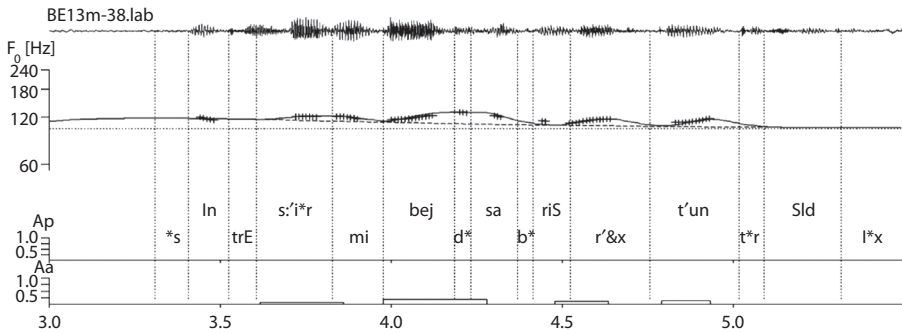


Figure 6.8. No *f*₀ measurement in phrase-final position – particularly affected are the syllables /*Sid*/ and /*l*x*/

In the context of the present study, the primary causes of immeasurable creaky voice contours are idiosyncratic voice quality, the interactional aspect of the interview, and the speakers' attitudes during the interview. From an interactional point of view, Kiessling et al. (1995) have shown that creaky voice in German often occurs in the proximity of morpheme boundaries. Laver and Eckert (1994: 70), too, note that sentence intonation in German declaratives usually falls and sub-glottal air pressure decreases, which is conceivably why many speakers

14. Naturally, the unvoiced segments, such as the syllable-initial /*S*/ in the last syllable, are deleted manually for a more accurate analysis (see Section 7.3.1).

15. The syllables affected by faulty measurements are of course not taken into consideration for the intonation analysis.

show vocal fry at the end of their conversational turns. In this sense, creaky voice can act as a signal to indicate the end of a turn since creaky voice often occurs at the end of a turn-constructive unit (see Sacks, Schegloff & Jefferson 1974). Such instances of creaky voice are also common in the present data, since the speech is elicited within a conversational setting. Laver and Eckert (1994: 153) pointed out a different voice type, the phenomenon of the “microphone voice”. This refers to the fact that the subjects adjust their voice slightly because they are aware of the microphone. In situations of long interviews, however, the subjects are said to adjust to the unnatural recording situation and, after a while, use their “normal” voices (*ibid.*). This in turn could suggest that creaks are more likely to occur towards the end of a 20-minute interview because the subjects start to feel more comfortable and the self-monitoring becomes less meticulous. On the attitudinal and emotional level, Grivicic and Nilep (2004: 8) discovered that creaky voice quality may indicate “passive reciprocity, a dispreference to continue the current topic, or a disalignment with the primary speaker”. On a final note, it was found that creaky voice is more likely to occur in male speech than female speech (Laver & Eckert 1994: 71).

While all subjects from the four different recording locations are subjects to the above-mentioned effects, not all recordings were conducted under the same circumstances. As for the recording site Brig, VS, all students seemed exceptionally cooperative and happy to be chosen as test subjects for this study. Further, the recording sessions were well integrated into the class schedule. In Bern, we encountered a somewhat different reaction: the subjects often appeared to be tense and under time pressure. In some instances their apparent unwillingness to co-operate was clearly communicated through their behaviour: they provided scarce answers, were obviously bored, and at times even clearly expressed annoyance by the interview situation or the interviewer. After some inquiries about their behaviour, it turned out that the recording sessions were preventing them from undertaking scholarly tasks. To be more precise, they were supposed to be writing mock essays for their final exams. Because of the recording time, they were of course robbed of 20 minutes worth of essay writing, which explains why some of the subjects seemed eager to return to class. The students in Zurich answered in a similar fashion as the students in Brig. They were mostly cooperative and did not seem to be under any time pressure. An interesting fact about the ZH students is that nearly all of them had a fairly precise idea of their future career plans, a feature which particularly for the BE students was not a given. Overall, in comparison to other participants, the ZH subjects were perceived to demonstrate a high level of self-confidence and extraversion. The GR subjects, on the other hand, were perceived as cooperative but less cooperative than the VS or ZH participants. Since intonation is highly sensitive to emotional

and psychological states, the condition of the speakers and the circumstance of the recording session are likely to affect f_0 contours. Cooperative, at points even seemingly happy, interviewees, such as those primarily encountered in the VS, demonstrate more vocal activity. This results in longer mean lengths of utterances, complex phrases, and a higher proportion of assertive phrases¹⁶ (Van Kleeck & Street 1982). So-called happy speakers additionally reach a much wider pitch range and a higher pitch average, as well as smooth upward pitch inflections (Murray & Arnott 1993: 1106). Reticent or irritated interviewees, on the other hand, are likely to behave oppositely: they exhibit less vocal activity, shorter mean lengths of utterances, and less complex phrasing (Van Kleeck & Street 1982). Effects of bored or annoyed speech include lower pitch averages, somewhat wider pitch ranges, and wide, downward terminal inflections (Murray & Arnott 1993: 1106). As for the perceived dominance and extraversion of the ZH speakers, Giles and Street (1994: 105) mention that the relationship between pitch and dominance is confusing, as studies have shown correlations with high pitch, low pitch, and pitch range. As for intensity, Trimboli (1973) found a positive correlation between extraversion and vocal intensity.

6.4 Data preparation

6.4.1 Transcription

In a first step, each of the 15 to 25-minute recordings were cut into chunks of speech of about 5–12 seconds with Adobe Audition (2007). Each of these cuts encompasses roughly 150–200 waveforms per speaker and each waveform contains at least one intonational phrase. Most waveforms, however, contain several IPs, as the recordings were perceptually cut into sections at points where pauses occurred.

The data was then transcribed by all project members, using a 7-bit printable ASCII Swiss German SAMPA¹⁷ (*Speech Assessment Methods Phonetic Alphabet*), originally created at the *Laboratoire d'Analyse Informatique de la Parole (LAIP)* at the University of Lausanne (see Siebenhaar, Zellner Keller & Keller 2002). The obvious advantage of using SAMPA as opposed to IPA is the convenience of inputting the phonemic transcript with the computer keyboard: each sound is encoded in a single, easy to read character. Except for a number of symbols, the Swiss German

16. Assertive phrases have a falling intonation by default, except in VS Swiss German, where statements are marked by rises (see Wipf 1910).

17. For more details on the Swiss German dialectal SAMPA used, see the reference at the beginning of this study.

SAMPA resembles the SAMPA devised for the basic six languages (Danish, Dutch, English, French, German, and Italian) (see Wells 1997).¹⁸ Additionally, long and short variants of vowels are treated as different segments in the transcription, lexical stress is understood as a feature of vowels, and syllabic consonants are treated separately from schwa plus consonant.¹⁹ In the current study, the transcribers enriched the notations with (+) and (–) in word-final position to distinguish between lexical (+) and grammatical (–) words. This transcriptive annotation is crucial as it can be expected that the difference in parts of speech affects intonation (see for example Möbius 1993a:136ff.). If a word is categorized as lexical it automatically also carries stress (‘). Grammatical words can also carry stress but only if focused by the speaker. Lexical stress, as marked by the transcribers, is transcribed as it occurs in the underlying lexicon entry for reasons explained in Section 6.4.3. Primary stress assignment is marked by means of /’/.

Table 6.5 gives an example of a Swiss High German SAMPA transcription, along with its IPA (phonetic) transcription and the Standard German equivalent (figure adopted from Keller-Flückiger 2008:38). The equation sign (=) indicates a syllabic consonant, which in the present data encompasses /=r/, /=m/, and /=l/.

Table 6.5. Transcription sample of Swiss High German SAMPA, IPA, and Standard German (Engl.: *And that I find almost most important about the job*) (adopted from Keller-Flückiger 2008:38)

Swiss German SAMPA	U- das- f’lnd+ 1– f’aS+ =m- v’lxtigSt*+ am – br’u*f+
IPA	[ʊ das f’lnd ɪ faʃ m̥ v’lxtiɡʃtə am bʁ’ʊʁf]
Standard German	,Und das finde ich fast am wichtigsten am Beruf’

Evidently, the phonemic transcription should be as objective as possible. However, the very practice of phonetic transcription is based on the principle that the actual deduction of speech production of a speaker happens via one’s own perception. In other words, the transcription ultimately relies on a comparison of the speech of researcher and participant. In the words of Almeida & Braun (1986), the deduction occurs via the imitation of what is heard by the proprio-receptive adjustment of the articulatory organs. Thus, it is difficult to name a measure of what is correct or incorrect in a transcription. Due to the fact that different people

18. See the SAMPA reference at the beginning of this study for further information about the symbols used in the Swiss German SAMPA.

19. Because the LAIP project was particularly interested in timing features of a Swiss High German TTS synthesis, it was crucial to distinguish between vowels in stressed or unstressed positions, as stress would affect duration. Hence, stress is viewed as a segmental feature of vowels (Siebenhaar, Zellner Keller, Keller 2002:166).

with different Swiss German dialects worked on the transcriptions of the present data, the transcripts originally contained much variation. The only method to approximate different ways of notation is via inter-subjectivity comparisons, i.e. comparisons of the different notations of the transcribers (Almeida Braun 1986: 160). This comparison does not offer an objective measure of the correctness of a transcription, but an inter-subjective measure of reliability. Consequently, transcripts were continually checked for their reliability during the initial stages of the transcription process.

6.4.2 Segmentation

After the transcription phase, a semi-automatic segmentation was performed with an MBROLA-based text-to-speech [hereafter TTS] aligner, created at LAIP at the University of Lausanne (see Dutoit 1997). This tool aligns phonemes (.txt file) with the sound file (.wav file) and generates a phonetic/phonological output file. This step of automatic labeling saves approximately one third of the time needed for an exclusively manually segmented corpus. After the automatic segmentation, a manual adjustment of the labels is performed in which the pre-segmented files are corrected and adjusted. Each sound file is segmented into the corresponding phonemes with the help of the oscillograms and spectrograms in Praat (2012).

In the current study, the labelers primarily adhered to Schwab et al.'s (1998) *Conventions de segmentation pour la construction de diphones* established at LAIP, as well as to Ellbogen's (2005) *Conventions for Segmentation*. The following is a summary of the most relevant labeling conventions:

1. In case of a gradual transition period between two phonemes (for example /s/ and /S/ as well as *vowel + vowel* combinations), the label is set into the middle of adjacent segments, as illustrated in Figure 6.9 (adopted from Schwab et al. 1998: 2).

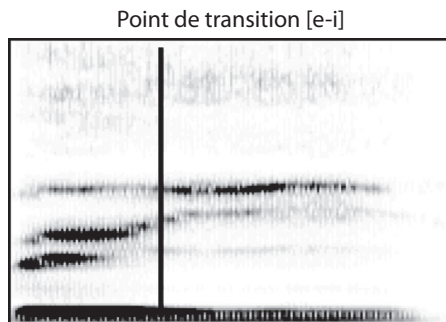


Figure 6.9. Transition period between the vowels /e/ and /i/. In such instances, the label is placed into the middle of the transition period between the two neighboring phonemes (adopted from Schwab et al. 1998: 2).

2. The label is always placed according to spectrogram and oscillogram and is always placed in the positive 0 crossing on the oscillogram (see Ellbogen 2005: 3).
3. For voiced sounds, the label is placed at the point where the first periodic element begins (see Ellbogen 2005: 4).
4. As is illustrated in Figure 6.10, plosives – *lenis* and *fortis* in Swiss German (see Willi 1993) – are labeled according to their three stages: stop, occlusion (/Vg/ and /Vd/ for lenis plosives and /_k/ and /_t/ for fortis plosives), and burst. If the plosive is preceded by a nasal, the occlusion phase may be omitted as the nasal itself implies oral closure. In case of difficulties regarding the distinction of fortis or lenis, it is primarily the extended length of fortis plosives after sonorants (see Willi 1993) as well as the extended length of the occlusion that is indicative for fortis plosives (see Heike 1964).

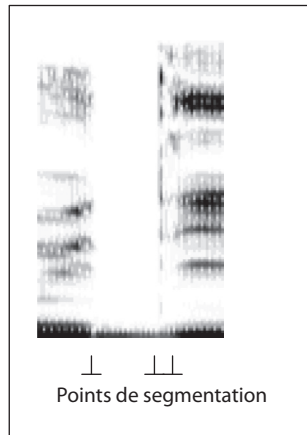


Figure 6.10. Plosives are labeled according to their three phases: stop, occlusion, and burst (adopted from Schwab et al. 1998: 4)

5. In case plosives clash, the first phoneme of the second word is normally labeled. For example, /hɛt- d*-/ is labeled as /h/ /ɛ/ /Vd<_ttVd/ /d/ /*/ (German: *hat den*; Engl.: *has the*). (<) indicates the underlying phonological phoneme string (see Ellbogen 2005: 4).
6. On the same note, the underlying phonological form of a segment is added should a speaker phonetically deviate from the dialect-specific phonological entry or reduce certain forms, including de-accentuation in lexical words.²⁰

20. Why this phonological labeling is necessary for the intonation analysis is shown in Section 6.4.3 on annotation.

This phonological completion is again marked by the (<) sign. For instance, variations of the word *eigentlich* (Engl.: *actually*) are vast in Swiss German, which may lead to realizations such as /'ejVgg*x/ or /*x/. In the latter case, for instance, the transcription is given as /*x<'ejVgg*n_ttlIx/.

7. Silence measurements exceeding 200ms are treated as pauses and are labeled (_).
8. Filled pauses are labelled with a pound sign (#). The pound sign precedes every phoneme that is part of a larger constituent of a pause-filler.

Point 6 of this list of criteria for sound segmentation is in need of further qualification. The phonological labeling is straightforward in connection with reductions, as illustrated in the example of *eigentlich*, realized as /*x/ and phonologically completed as /'ejVgg*n_ttlIx/. However, for syllabic consonants, this completion is somewhat more delicate. The syllabic, consonantal realization /=r/, for example, does not need to be phonologically completed further if the underlying form is /*r/, as in /od=r/ (German: *oder*; Engl.: *or*). If the phonetic realization is /=r/, as for example in /v=rS'inli/ (German: *wahrscheinlich*; Engl.: *probably*), the /=r/ is phonologically completed with /ar/, as the underlying form of the realized syllabic consonant is not /*r/. Contrarily, in cases with words that have more than one lexical entry in the dialects in question, the labeling procedure is somewhat different. The Swiss German word *also* (German: *also*; Engl.: *well*), for example, demonstrates a number of different, non-reduced realizations, such as /aso-/ , /also-/ , and /awso/. In the VS dialect, for example, the female personal pronoun *sie* (Engl.: *she*) can be realized as both /Si/ or /si/. In the case of such stylistic variants, the phonetic realization was regarded as the phonological counterpart, i.e. no phonological completion was performed.

The procedure of data segmentation is not unproblematic as the articulatory process is continuous. Articulatory-constant sounds are inexistent as, in speech, we do not produce single sounds but sound sequences in which sounds overlap. In some instances, it can be exactly this overlap in the sonogram that makes up for the main part of a sound (see Machelett 1994). Heike (1982:641) notes that accuracy of the labeling of a continuous speech stream depends on two variables. On the one hand, the size of the phonetic contrast of adjacent sounds: the greater the contrast the easier the segmentation. On the other hand, it depends on the articulation rate of the speaker: the higher the rate the more difficult the segmentation. This difficulty in sound segmentation becomes most evident when labelers of the same file compare their placement of the labels, as shown in Figure 6.11.

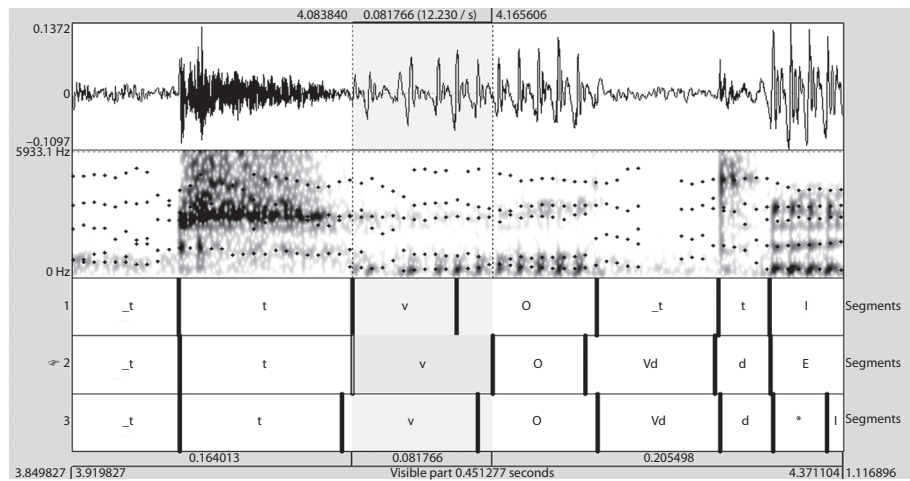


Figure 6.11. Labeling variation: transitions between voiced labio-dental fricatives and adjacent vowels, in particular, are a common source of labeling variation

Figure 6.11 illustrates how the labeling of the end point of /v/ differs for each of the three researchers who worked on this segment. Such difficulties were encountered in particular for labio-dental voiced fricatives, vowel-vowel transitions, and glottal fricatives. Because of such difficulties of manual segmentation, the researchers were frequently asked to label the same files in order to cross-check their transcriptions, discuss problems, and minimize variation. Generally, a tolerance level of 10ms for the placement of the labels was adhered to.

6.4.3 Annotation

Annotation is a means to establish how the model parameters for the four dialects vary with regard to the various linguistic, paralinguistic, and non-linguistic properties of intonation, which were introduced in 2.6. This means that the data is annotated with each of these variables and they are assigned a separate column in a Microsoft Excel sheet. Statistical analyses are then carried out with JMP (2005). The annotation was performed on a syllabic level, which will be justified below. Subsequently, I give an overview of the variables that were annotated for, the levels established for each variable, and the manner in which this annotation was performed. The variables can be broadly categorized into either linguistic, paralinguistic, or non-linguistic properties. The categorization follows the distinction proposed by Fujisaki (2004) (see Chapter 2).

Linguistic variables: The variables *stress* and *word class* are assigned to this category.

Paralinguistic variables: This includes the variables *focus*, *phrase type*, and *prosodic paraphrasing*.

Non-linguistic variables: The variables that belong to this category are *articulation rate*, *emotion*, and *sex*.

6.4.3.1 Annotation on the syllabic level

This study adheres to the definition of the syllable in an acoustic sense, meaning that the segments that constitute a syllable are structured in such a way that there is a change in sonority from a less sonorant edge to a more sonorant nucleus and back to a less sonorant edge (see Allen 1973: 38). From an articulatory perspective, this translates into an opening and a closing of the oral cavity. From the onset edge to the nucleus, consonants follow the *sonority hierarchy* from the least to the most sonorant sound – plosives, fricatives, nasals, liquids, approximants, and vowels (see Sievers 1881). Towards the syllable coda, on the other hand, the consonants align in reversed order. It should be pointed out that it is by no means uncontroversial to treat the syllable as a phonological constituent, as there are a number of unresolved issues, including the syllabification of consonants clusters, particularly in German (Ulbrich 2004: 50), and the syllabification of VCV strings (see Blevins 1995: 227; Auer 1994). These concerns aside, the annotation was performed on a syllabic level. The syllable was chosen as the appropriate level of analysis because it corresponds to the lowest level on which regular prosodic parameter variation is found. For example, the syllable is the smallest unit that can receive stress, which is why it is directly related to intonational representation. Thus, the syllable represents a structural anchor point for abstract prosodic features, such as tone or stress (Günther 1999: 171). Vaissière (1983: 61), too, notes that “prosodic phenomena such as tone, stress, and accent are customarily considered qualities of the syllable”, and that “[t]he syllable is the minimal unit for *f₀*, duration and intensity contrasts.”

The syllabification in the present study is performed against the background of Sievers’ *sonority hierarchy* (1881). Syllable boundaries are placed in such a way that the sound with least sonority is always located in onset initial position, whereby extra-syllabic material is assembled in the coda. In other words, the onset must always strictly adhere to the sonority hierarchy. Plosives are handled in such a way that the occlusion is part of the coda of the previous syllable, while the burst forms the onset of the syllable containing the plosive. The consonant /h/ is always in the first position of the onset, except if it is part of a participle, e.g. in the phrase **r het hose* a kha*: (German: *er hat Hosen angehabt*; Engl.: *he wore pants*). In this case, the /h/ does not occur in onset-initial position because the syllable *kha*, in which /h/ occurs, is part of the participle of the verb *aha* (German: *anhaben*; Engl.: *to wear*).²¹

21. For a cross-dialectal comparison of syllabification in Swiss German, see Keller-Flückiger (2008).

6.4.3.2 Linguistic variables

Stress: *Stress* is annotated for because stressed syllables usually show an increased f_0 (see 2.4). *Stress* is also understood as a feature of lexical words. This distinction between lexical and grammatical words with regard to stress assignment is justified in so far as, in connected speech, most grammatical words generally occur in unstressed forms and reduced vowels.²² Thus, the metrically strongest syllables in lexical words are labeled as stressed, the rest are marked as unstressed (see Siebenhaar, Zellner Keller, Keller 2002: 166). There are two special cases to this labeling:

- a. For some dialects, different stress patterns are common for the same word, such as in VS German *Militär* (Engl.: *military*), which may either carry first syllable or last syllable stress. In these instances, stress is assigned depending on the speaker's phonetic realization.
- b. In spontaneous speech, speakers commonly reduce forms, which in some instances may lead to de-accentuation. In these instances, the underlying phonological form is reconstructed and the original metrical stress is annotated for. The rationale behind working with the underlying phonological form and not with the actual phonetic realization is two-fold. Firstly, the assessment of the contribution of *metrical stress* to local f_0 contours in Swiss German as represented in the speaker's lexicon is exactly one of the many goals of this study. Secondly, an auditory and visual labeling of *stress* based on the actual phonetic realization might turn out to be circular: chances are that f_0 prominent, yet unstressed syllables are accompanied by distinct high f_0 movements, which is why these syllables are at risk of being falsely perceived and labeled as stressed (see Hirst 1983).

First stressed syllable in an AC: The first stressed syllable that occurs in an AC is marked with an (x) in order to test whether a different positioning of the first stressed syllable in the AC bears any effects on AC amplitude or timing.²³

Word class: This variable is annotated for because three studies which also apply the Fujisaki model have reported effects of word class on intonation (Möbius 1993a; Mixdorff 1998; as well as Fujisaki & Ohno 1995). Möbius (1993a:144), for instance, shows that AC amplitudes are 10% higher for nouns than for other parts of speech. In the current approach, only lexical words carry stress, while

22. In their citation forms, however, each word has at least one stress (Cruttenden 1986: 21).

23. Stressed syllables that occur after the first stressed syllable are not annotated and, therefore, not statistically analyzed.

grammatical words are viewed as largely unstressed. Grammatical words can, however, carry stress if they are in focus. Each syllable of a grammatical or lexical word is labeled separately either as (–), indicating a syllable of a grammatical word, or (+), representing a syllable of a lexical word. In the present study, grammatical words largely encompass the following parts of speech: auxiliaries, modals, articles, personal pronouns, possessive pronouns, relative pronouns, conjunctions, and prepositions. Lexical words are principally verbs, nouns, adjectives, and adverbs.

The distinction between lexical and grammatical words is not always an easy one. Generally, the transcribers bore in mind that grammatical elements occupy a content-structuring function. In some cases, however, the distinction between lexical and grammatical category is less straightforward. The word *nach* (Engl.: *after*), for example, is normally marked as grammatical in the present data, such as in *je nach dem* (Engl.: *depending on*). In cases in which *nach* occurs in verbal constructions, however, such as in *er dachte nach* (Engl.: *he was contemplating*, literally: *he was thinking about*), the word *nach* is lexical and carries stress. A frequently occurring lexical item that can be both grammatical or lexical is *ja* (Engl.: *yes*). *Ja* can function as pause filler and modal particle or gradation particle, but may also constitute an answer to a yes/no question. In the case of *ja* as a filled pause, an accurate labeling is not important as filled pauses are treated separately anyway. If it functions as a modal or gradation particle, it is generally labeled grammatical, whereas if it constitutes an answer to a yes/no question, it is lexical and thus carries stress.

6.4.3.3 Paralinguistic variables

Focus: *Focus* is annotated for because straightforward *f0* correlates have been established for focused constituents as well as for pre- and post-focus elements in IPs that carry narrow foci (see Batliner & Nöth 1989; Botinis et al. 2001; Xu et al. 2004 etc.). The speech data was assessed for *narrow focus* by means of taking into consideration aspects of givenness of information, contrasting information, and emphasizing information during the course of the interview.

In the case of IPs that contain a narrow focused constituent, which is the case in approximately 50% of all IPs in the current study, all syllables of the focused word(s) are marked with (x). If any of the syllables of the focused word are spanned by an AC – which is normally the case given that *f0* movements are closely associated with focus marking – this particular AC is labeled as *focus* AC. If an IP contains several ACs, the ACs preceding the *focus* AC are labeled as *no focus* ACs, the AC that immediately precedes the *focus* AC is labeled as *pre-focus* AC, while all ACs that follow the *focus* AC are labeled *post-focus* ACs

(see Figure 6.12). Seldom, it is also possible that entire IPs are focused, in which case all focused syllables are marked with (x). Note that it is also possible that more than one focused constituent is present in different locations in the IP. This phenomenon is rare, however, and only occurs in distinctly long IPs. In this case, ACs between *focus* ACs occupy both a *post-focus* and *pre-focus* position at the same time, priority is given to the first *focus* AC in the IP. The AC is thus labeled as *post-focus* AC of the first AC. Furthermore, a small number of focused syllables were found (on average about 25 per speaker), which do not fall under an AC. In these exceptions, focus is obviously not marked by *f0* movements but, rather, by other acoustical means, such as duration and/or intensity. These exceptions are not included in these analyses.

In IPs with narrow focused constituents, IPs were further labeled with regard to the *location of the focused constituent* in the phrase. Mixdorff (1998a: 96) has shown that PC magnitude behaves differently depending on the location of the focused constituent. For this categorization, the exceptions described above are taken into consideration. Now, depending on the number of ACs and exceptions in an IP, it is distinguished between four types of IPs. Firstly, we find *first position focus*, i.e. the focus AC or the exception is the first of at least 2 ACs in the IP (or an exception and an AC). Secondly, we find *medial position focus*, for which at least 3 ACs or 2 ACs and one exception are required. Then there is the *ultimate position*, which must feature at least 2 ACs or 1 AC and one exception, whereas the final category consists of IPs which carry only one single focused constituent.²⁴ This only applies to focus constituents that are spanned by ACs. In the case of exceptions, i.e. words are in focus but not spanned by ACs, those ACs that span focused syllables are given priority. If an IP contains an exception focus in phrase-medial position and a regular focus AC in phrase-final position, the IP is labeled as *ultimate position focus*. Figure 6.12 illustrates a phrase with medial position focus as well as the annotation for focus marking in the present data.

In the case of broad focus IPs, i.e. normal stress, a syntactically determined phrase accent may be conceivable, yet was not further taken into consideration and thus not labeled in the present study. On the one hand, this was not conducted because the question of whether or not phrase accents are necessary in German is yet to be answered (see Möbius 1993a: 22; Grønnum Thorsen 1989: 9) and secondly, it would go beyond the scope of this study.

24. Note, again, that in rare occasions and only in distinctly long IPs, the occurrence of more than one focused constituent is possible. In these cases, the first focus AC in the IP is given priority.

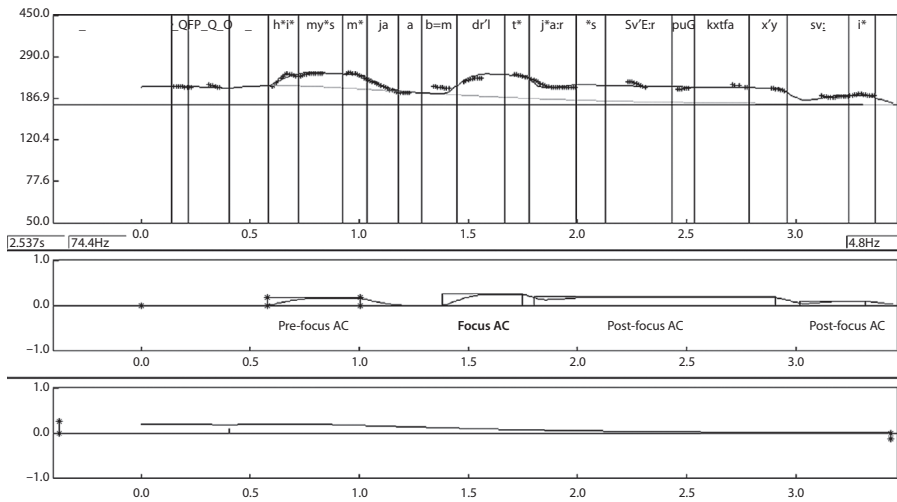


Figure 6.12. Focus marking of an IP with phrase-medial focus

Phrase type: Intonation is a means of signaling the end or the continuation of a conversational turn (see 2.7.2). Gilles (2005:21ff.) has shown that, in the vast majority of the German dialects he investigated, prototypical turn-endings are characterized with falling contours in nuclear position. Continuations, on the other hand, are used to indicate that a speaker intends to hold the floor and are principally realized with a phrase-final rise (ibid.; see Cruttenden 1971). Consequently, in the present study, IPs are labeled according to their conversation-structuring functions, i.e. they are labeled as marking continuity or finality. The criteria according to which phrases were labeled as terminating or as continuing are loosely based on Gilles' (2005) framework of labeling the conversational function of IPs.

In his studies on *f0* contours of continuing and terminating phrases of German dialects, Gilles (2005:45) postulates four fundamental categories for IP labeling: *continuation*, *potential continuation*, *termination*, and *potential termination*. For an IP to be labeled terminated, it has to be finished both syntactically as well as pragmatically. Syntactically finished entails that the structural syntactic positions must be occupied, meaning that the given syntactic structure does not require a further constituent in order for it to be perceived as syntactically complete (Gilles 2005:48). The criteria for pragmatic completion are more difficult to establish. Gilles (2005:50) argues that the context as well as the given turn must clearly indicate the end of the conversational activity. However, the most evident indicators for finality are turn-changes. Mere backchannelling on the part of the interviewer is not considered a turn change.

In the present study, the categorization of *continuing phrases* [C phrases] and *terminating phrases* [T phrases] is performed according to turn-related and syntactic criteria in the style of the method suggested by Gilles (2005).²⁵ I follow Goodwin (1981) in defining a *turn* to be “the talk of one party bounded by the talk of others” and *turn-taking* to be “the process through which the party doing the talk of the moment is changed.” Figure 6.3 shows the criteria according to which phrases were labeled as T or C *phrases*.

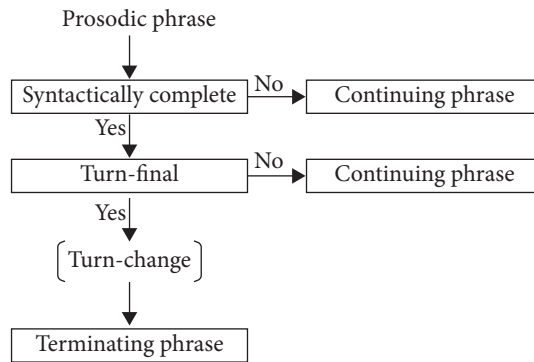


Figure 6.13. Flow-chart of the determination of IP functions as applied in the present study

For a prosodic phrase to be labeled *terminating* it must be syntactically complete and must also suggest finality in the context of a given turn. As Gilles (2005: 50) suggests, turn-finality is understood in the sense that the conversational context must clearly imply an end of a conversational activity, which in the present study translates into an end of a narration or a sub-topic. A turn-take can, but need not, occur after a prosodic phrase that fulfills these syntactic and turn-related criteria.

Prosodic paragraphing: Intonation also occupies a *paragraphing function* in fluent speech (see 2.7.2, Tseng et al. 2005; Tseng & Su 2008). The best way to study the effects of prosodic paragraphing on the *f0* contour would entail a labeling according to Tseng et al.’s (2005) *multi-phrase hierarchy*, in which IPs form breath groups and breath groups in turn form entire prosodic groups. It can also be argued that prosodic groups further form major discourse units, which,

25. While Gilles differentiates between continuation, potential continuation, potential termination, and termination, the present study only distinguishes between continuation and termination, for the sake of simplicity. In a number of instances in this study, phrases were also labeled as Q, i.e. questions. These questions came about if the interviewee misunderstood the interviewer’s input or utterance. Question phrases are not further considered in this study.

consecutively, contribute to an entire discourse (see Fujisaki 2008a). However, since such a labeling is highly time-consuming and a rather difficult process for spontaneous speech data, and because prosodic paragraphing is not the main focus of the present study, a simpler method is suggested. PC magnitude and PC duration are regressed on the following three independent variables:

- a. PC magnitude of previous IP
- b. PC duration of previous IP
- c. Strength of break (i.e. pause length) preceding the IP

These regressions also shed some light on prosodic paragraphing. Mixdorff (1998a: 142ff.), for example, has shown that the longer the previous IP, the higher the PC magnitude of the IP investigated. In other words, the longer the elapsed time since the start of the previous IP, the more the declination line is adjusted again in the actual IP. If *PC magnitude of the previous IP* is a significant predictor of PC magnitude, for example, we can conclude that the magnitude of the previous has a sequential effect on the actual IP. These relationships would not, of course, provide information of paragraphing effects on *f₀* contours in specific IP positions in the prosodic group, but they would underline the close-knit relationship between PC magnitude and duration of preceding IP and actual IP. As for the final variable, pauses frequently trigger a reset in the declination line where the degree of the reset is indicative of the importance of the boundary (Vaissière 1983: 57). Pitch resets thus place prominence, which is why they carry a direct communicative function (Fox 2000: 309). Tseng et al. (2005), too, observe long pauses before and after speech paragraphs. Hence, pause duration is very well considered a means of investigating paragraphing structure.

6.4.3.4 *Non-linguistic variables*

Articulation rate: Fujisaki and Hirose (1984) have shown that in both PC magnitude as well as in AC amplitude, we find a decrease for faster *articulation rates*. Mixdorff (1998), too, has shown that an increased speaking rate leads to a reduction in AC amplitude, in the number of prosodic phrases, and in the duration of pauses. Moreover, a faster speaking rate leads to AC sandhi. Therefore, articulation rate was also considered for the analyses.

One way of measuring *articulation rate* is to consider the number of syllables produced per second.²⁶ It was decided to work with three classes of speakers for every dialect: *fast*, *normal*, and *slow* speakers (as applied by Mixdorff 1998 and

26. For a discussion of other means of measuring speaking and articulation rate, see Roach (1998).

Ohno et al. 1998). In order to form these categories, the span of the *articulation rate* (including filled pauses) was calculated for each dialect, i.e. the rate of the fastest minus that of the slowest speaker. Subsequently, this number was divided by three and, for each dialect group, those speakers in the bottom third category were labeled as *slow*, speakers in the middle group as *normal*, and the ones in the top third as *fast*.

Emotion: Section 2.6.3 has highlighted the role intonation plays in emotion expression, particularly in spontaneous speech (see Murray & Arnott 1993). Thus, it is interesting, in general, to examine the effects of emotion on *f0* contours, which an annotation for emotion would enable. As has been pointed out earlier, the subjects were not always exposed to the same recording conditions. Hence, in this particular case, it is especially pertinent to investigate these effects since the relative proportions of emotion phrases per dialect seem to be skewed. For the emotion decoding process, different categories of emotions had to be established, a rather difficult task given the fluidity of the concept of *emotion*. *Anxiety*, for example, a relevant emotion in the context of the present study, is not clearly defined and delimited: some studies categorize it as *fear*, some as *nervousness*, and some possibly assign it to yet another taxonomy (Frick 1985:420). Thus, observed emotions are frequently contrasted to a *neutral* emotional state, which facilitates the categorizations of different emotions by comparing deviations from this norm (Kranich 2003:619).²⁷ This is also the technique applied in the present study.

Five different emotion categories were established. Nearly all of the emotion categories presented in Murray and Arnott's (1993) review of existing literature on the vocal expression of emotion were applied. *Anger* was not included as none of the speech data seemed to be spurred by *anger*. As a point of measurement contrast, the *neutral* category was added for reasons explained above. Only a handful of emotion categories were used since a larger quantity of annotation types would render subsequent analyses difficult by impeding the detection of systematic patterns (Stibbard 2001:205). The emotion categories as well as their implications in the context of the present study are shown in Table 6.6.

Emotions were decoded on the level of the IP despite the fact that in authentic conversations vocal emotions can change rapidly on even lower prosodic levels (see Kranich 2003:62). This measure was undertaken for the sake of simplicity and because the emotional aspect is not at the core of the present study. More detailed labeling and analyses are worth pursuing in controlled follow-up experiments. After initial cross-comparisons between different labelers, the annotation

27. This approach is met with criticism, as "neutral" emotions are, too, always emotionally laden to some degree (e.g. bored, un-cooperative etc.). For more information on this subject, see Kranich (2003).

for *emotion* for all speakers was performed only by one labeler. This measure is justified since studies have shown that people are able to assess emotions fairly accurately based on vocal parameters (see Scherer 1982; Banse & Scherer 1996).

Table 6.6. Primary, human vocal emotions (loosely based on Murray & Arnott 1993) and the interpretation thereof for the present study

Human vocal emotions (loosely based on Murray & Arnott 1993)	Interpretation in the present study
Neutral	Speaker shows normal cooperation; does not seem to be particularly happy, bored, anxious, or annoyed.
Happiness	Speaker shows extraordinarily cooperative behavior; happy to talk to interviewer; laughter.
Bored	Speaker is bored; not very cooperative.
Fear	Speaker is anxious to talk to interviewer; does not feel comfortable talking to the interviewer.
Disgust	Speaker is annoyed to have to talk to the interviewer; speaker seems under time pressure and wants to get out of the interview situation.

Sex: *Sex* is of course investigated further as an extralinguistic variable. A number of f_0 correlates of sex have thus far been observed: McConnell-Ginet (1978) shows that females exhibit a comparatively wider pitch range, and Henton (1989) observed female speech as high-pitched and “over-emotional”. Daly and Warren (2001) confirm speaker-sex differences for New Zealand English, in which females exhibit higher pitch ranges and demonstrate more pitch-dynamic speech, i.e. their rate-of-change of the f_0 contour is more distinct than in male speech. Cruttenden (1986:136), too, notes that females demonstrate more “expressive” intonation which, here, refers to a more frequent application of rises and a wider pitch range.

Application of the Fujisaki model

This section is concerned with the linguistic interpretation of the Fujisaki model components, the selected model parameter configuration, a presentation of the applied modeling procedure, as well as a discussion of modeling difficulties and their ramifications on the f_0 analyses and statistical analyses. To the author's present knowledge, no study has to this date attempted a Fujisaki modeling of corpus-based, spontaneous speech f_0 contours. Hence, it should be noted that the modeling procedure bears explorative character and is by no means carved in stone.

7.1 Linguistic interpretation of the model components

7.1.1 Fb

The logarithm of *fundamental frequency* is the sum of a constant, namely Fb (or *base f_0*), and the two time-varying parameters *phrase component* and *accent component* (see Section 4.2). Linguistically and physiologically, Fujisaki and co-workers have modified their definition of Fb over the years, incorporating either one or both of the notions of speaker-specificity and utterance-specificity. Both conceptualizations are justifiable. Mixdorff (1998a:58), a proponent of the speaker-specific definition, argues that a modeling framework which understands Fb as utterance-specific forfeits cross-comparisons between the magnitudes of utterances of different sentence types, such a high a Fb in yes/no questions against a low Fb in statements. This is due to the varying Fb which affects both PC and AC parameters extensively due to the additive nature of the model (see Chapter 4). It is not entirely accurate to say, however, that in this framework cross-comparisons of the magnitudes of different sentences modes are difficult to draw. Rather, what is required is a declaration of an utterance-specific Fb in the presentation of the results. In this sense, utterance-specific Fbs also bear linguistic significance in that yes/no questions, for instance, exhibit higher Fbs than declaratives (see Möbius 1993a:159). Conversely, in advocating the utterance-specific interpretation of Fb , Möbius (1993a:78) points out that a speaker-specific conceptualization of

Fb would entail serious measurement effort and difficulties, if a speaker's lowest-possible *f0* ought to be established.

In the present study, *Fb* is understood as “the lower limit of fundamental frequency below which vocal fold vibration cannot be sustained in the glottis of a speaker” (Fujisaki 1981:3). In other words, *Fb* is understood in a physiological sense as being *speaker-specific*.¹ The reason for choosing an *Fb* that is constant and speaker-specific, as opposed to utterance-specific, was to reduce the number of model parameters, especially since the current study already evaluates a high number of linguistic, paralinguistic, and extralinguistic variables.² Hence, *Fb* is understood as the speaker-specific, lowest asymptotic voice fundamental frequency value onto which the phrase and accent components are superposed.

7.1.2 Phrase component

The phrase component consists of a quick rise followed by a gradually falling slope towards the asymptotic *Fb* and serves as the base contour onto which local *f0* movements are superposed. As suggested by Fujisaki and co-workers, this contour can sensibly be applied for a description of the global declination tendency of *f0*. This strategy is also pursued in the current study. Since declination is directly linked with a reset in pitch, each new phrase impulse is understood as a reset of the declination line. The benefit of using the phrase component for a modeling of declination is flexibility: the initial starting value as well as the declination rate can be varied, while the overall effect of declination is nevertheless maintained (Taylor 1994: 17). A boost in PC magnitude translates into a steeper declination, which equally increases perceived global pitch range (Taylor 1994: 40).³

The linguistic unit in which declination is observed in the present approach is the IP (or simply the *phrase*). An IP is the most significant production unit of intonation and serves as a means of chunking information into sensible units (see 2.2). In the current study, each PC placed refers to one IP. A PC may include one syllable only or may span entire syntactic phrases and clauses.

1. The method used in this study to establish this lowest *f0* value will be illustrated in 7.2.1.

2. Admittedly, an utterance-specific *Fb* modeling framework would have been just as reasonable since both approaches are valid.

3. Whether or not declination is caused by thyroid translation as suggested by Fujisaki (1987) or triggered by the decreasing transglottal air pressure (see Collier 1975) lies beyond the scope of this discussion.

Consequently, an utterance can include one or several PCs. PCs are delimited by a bundle of different prosodic features, including f_0 movements (PC initially or finally), pausing, lengthening of syllables, syntactic structure, and so on. These features may mark PC boundaries either by occurring together or independently.

7.1.3 Accent component

In the present study, the accent component is understood acoustically as a device for marking prominence through f_0 on the local level. Prominence can be altered in three different ways: firstly, it can be boosted by increasing the *amplitude* of the accent component (or by suppressing the amplitude if neighboring constituents have high amplitudes), secondly, it may be lengthened by increasing the *duration* of the AC, and thirdly, it can be *timed* differently depending on the location of T_1 and T_2 relative to segment onset and offset. The short rising time of the accent component makes it possible to capture quick f_0 movements, i.e. local rises and falls. Linguistically, ACs have been associated with different phonological concepts: Fujisaki (1987) relates ACs to prosodic words (i.e. word accents), Möbius (1993a) to stress groups, and Mixdorff (1998) to intonemes (i.e. tone switches at accented syllables). The present study, however, bypasses such a direct link between local f_0 movements and a phonological concept for reasons explained below.

Despite the fact that the literature on the correspondence between word stress and the acoustic properties thereof struggles with a number of unresolved issues and ambiguities, it is nevertheless commonly assumed that word stress is a function of f_0 (see Kochanski et al. 2005: 1039). As a natural consequence, ACs that model local rises and falls in intonational studies are anchored with stressed syllables, stress groups, or tone switches at accented syllables. In the present study, however, word stress and f_0 movements are treated as independent variables, which is why ACs are not anchored with stressed syllables. Instead, they are linked with exactly those syllables they span, i.e. the syllables that are made more prominent by f_0 marking. In other words, the modeling of local accents with ACs happens blindly, not taking into account phonological information on the segmental level. The reason for choosing this method of analysis is threefold:

1. The role of f_0 in stress marking is not yet fully comprehended.
2. There is no study on the acoustic features of different Swiss German dialects' realizations of word stress that confirms that the primary acoustic correlates of stressed syllables are indeed f_0 movements. Hence, as to the author's knowledge, there is not yet sufficient evidence for Swiss German to justify an anchoring of all ACs on the level of stressed syllables.

3. Earlier studies on Swiss German have pointed out incongruence between the placement of word stress and high f_0 movements. Wipf's (1910:22) observations on the intonation of VS Swiss German are as follows:

“[t]he general rule which applies to all of the Swiss German dialects that dynamic accent and musical accent coincide and that unstressed syllables carry a lower tone does not hold for [the dialect spoken in] Visperterminen”.

(1910:22, translation AL)

She continues to say that the distribution of high and low pitch accents on high dynamic and low dynamic accent syllables is entirely free (ibid.). As a means of illustration, an example from the current corpus is given in the following Figure. It shows the f_0 contour of a VS speaker articulating the declarative phrase **m, j'a:, m'I litEr* (German: *mh, ja, Militär*; Engl.: *mh, yes, military*). In VS Swiss German, the first syllable of *m'I litEr* can, and in this example does, bear word stress, while in High German and other Swiss German dialects, the word stress is normally on the last syllable.

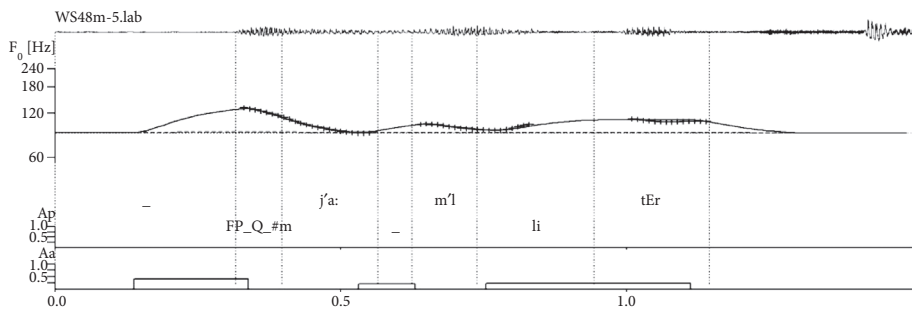


Figure 7.1. VS speaker's f_0 contour of the phrase **m, j'a:, m'I litEr* (German: *mh, ja, Militär*; Engl.: *uhm, yes, military*), with a minor f_0 fall on the lexically stressed syllable *m'I*, followed by a rise on *litEr*

In the phrase shown in Figure 7.1, the stressed syllable *m'I* is modeled with a falling AC, while the subsequent, unstressed syllables *litEr* are modeled with a rising AC. Informal measurements in the syllable nuclei in Praat (2012), indicate an increase in pitch from *m'I* (92 Hz) to *li* (101 Hz) and to *tEr* (106 Hz), while the amount of intensity remains fairly stable throughout the whole word: *m'I* (62 dB) *li* (52 dB) *tEr* (60 dB). Wipf (1910:21), too, discovered such an even distribution of intensity on nearly every syllable in VS German. In this example, unstressed syllables are marked more prominently with f_0 than the lexically stressed syllable. If ACs are anchored with precisely the syllables they span, one can examine whether VS German is, for example, particularly characterized by f_0 movements

on unstressed or schwa syllables. If ACs were only anchored with stressed syllables in the previous example, then the AC under *litEr* would have to be anchored with the syllable *m'I*, which is difficult since a). *m'I* already carries its own AC and b). the f_0 movement clearly occurs on the unstressed syllables *litEr*, reaching its peak in the ultimate syllable *tEr*. Hence, by treating f_0 movement and word stress as independent parameters, the contribution of word stress on the representation of f_0 movement can be explored in an unbiased manner.

Based on this argumentation, the linguistic understanding of the accent component in this study is given in principal by the component's acoustic interpretation. The speaker-specific, asymptotic F_b provides the necessary pitch values to position the phrase component, resulting in a speaker-specific declination line. Based on the phrase component, the accent components are added to model local prominence markings through f_0 . In other words, the ACs are employed to model local f_0 movements that are marked by rises from and falls onto the declination line. Figure 7.2 visualizes this addition of F_b , *phrase component*, and *accent components*.

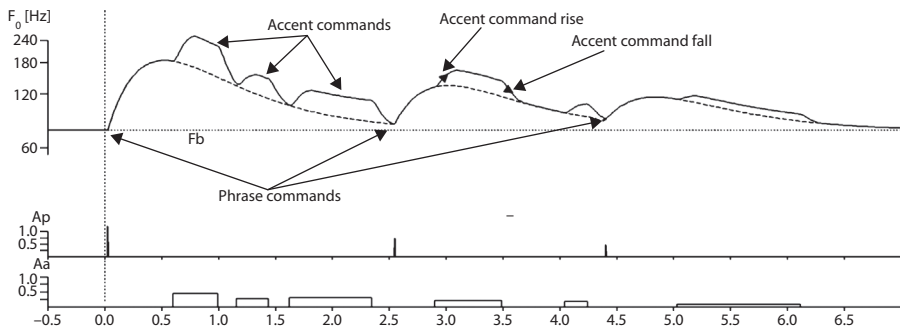


Figure 7.2. The speaker-specific F_b serves as the basis onto which phrase components are superposed. The phrase component, in turn, provides the basis for local f_0 movement modeling

7.2 Parameter configuration

The advantage of the application and optimization of the Fujisaki parameters is that it renders a high number of parameters constant. As will be shown in this section, merely the model parameters *PC magnitude*, *AC amplitude*, and *AC timing* (T_1 and T_2) vary, while the parameters F_b , T_0 , α , and β are kept constant. This allows for a more straightforward interpretation of the modeled f_0 contours. Females and males can be modeled in an identical manner, without having to

apply a normalization procedure of any kind. Phrase as well as accent command parameters are relative f_0 excursions superposed onto Fb , the latter of which is a speaker-specific value.

7.2.1 Fb

Fb is understood as the *speaker-specific zero level frequency* and is held constant for every speaker. In the words of Ladd (1996:267), Fb is conceived of as an f_0 level “largely unaffected by raising or lowering the voice or by other within-speaker range changes”. In Praat (2012), the value for Fb is established by locating the lowest f_0 measurement in all speech data provided by one speaker, which on average approximates 130 phrases per speaker. A majority of f_0 minima was found in utterance-final or utterance-medial positions. In the process of determining a speaker-specific Fb , special attention was paid to passages of vocal fry, which as a general rule generate lower yet inaccurate measurements. Figure 7.3 provides an example of the Fb measured for a male Grisons speaker.

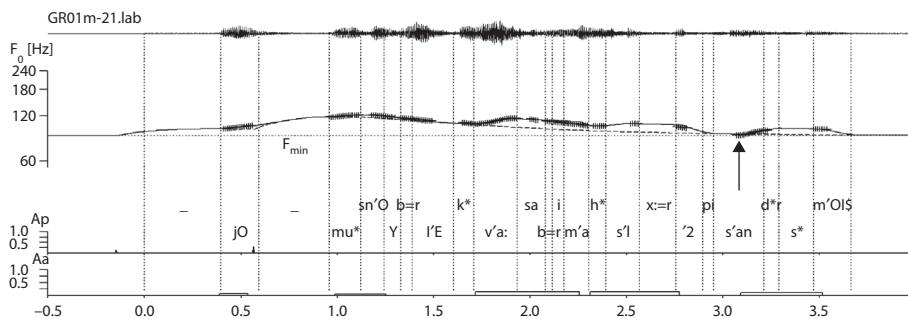


Figure 7.3. An arrow indicates the sentence-final Fb of the speaker GR01m

While Möbius (1993a:80) points out that establishing the lowest speaker-specific f_0 is fairly problematic from a point of view of measurement, it was perceived in this study that the examination of such a high number of phrases per speaker allows for the establishment of a fairly accurate zero level f_0 that is unaffected by intentional raising or lowering of the voice.

7.2.2 Phrase component

The *phrase component* includes the parameters *PC magnitude*, its onset time T_0 , as well as α (the time constant of the input signal). Further, this study examines the values of *PC duration*, which denotes the length of one IP. While the parameters *PC magnitude* and T_0 , which convey linguistic information, vary

according to the information content of the utterance (Fujisaki, Hirose & Ohta 1979:169), the time constant α was shown to remain nearly constant across utterances and across speakers of different languages given that it reflects the underlying physiological mechanism of the larynx (Fujisaki & Hirose 1982). In their publications, Fujisaki et al. use on average an α of 3.1.

In this study, α is kept constant for all utterances and speakers. This decision is primarily based on the above findings on α variation, which is minimal across utterances and speakers. Secondly, a phrase-to-phrase, utterance-to-utterance, or speaker-to-speaker variation of α significantly affects the magnitude value, on the one hand, and the positioning of T_0 relative to phrase onset, on the other hand. In addition, α variation can cause different positioning and frequency of ACs, which poses considerable problems to subsequent data interpretation. The following Figure illustrates the effects of a varying α on T_0 , PC magnitude, and the frequency and positioning of ACs.

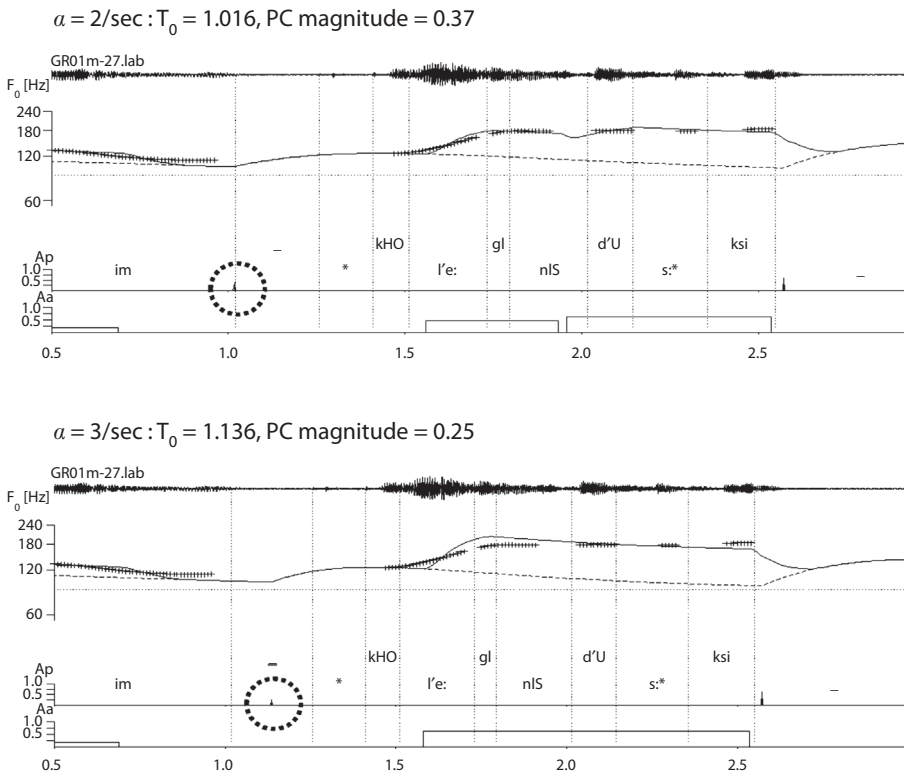


Figure 7.4. Varying α and its effects on T_0 , PC magnitude, and the frequency and positioning of ACs

Figure 7.4 shows an $\alpha = 2/\text{sec}$ (top panel) and $\alpha = 3/\text{sec}$ (bottom panel) modeling of the phrase **khOl'e:gIn IS d'Us:ksi* (German: *eine Kollegin war draussen*; Engl.: *a friend was out there*). As can be observed in the illustration, a change of $1/\text{sec}$ in the time constant critically affects a number of model parameters. Firstly, T_0 shifts forward, which – with a modeling condition that requires the PC peak to coincide with the utterance onset – is due to the relation of T_0 and α being $T_0 = 1/\alpha$ (see Pätzold 1991). Secondly, the PC magnitude decreases due to the different impulse responses of the phrase control mechanism with the varying α . In the above example, the time constant further has an effect on the modeling of local accents: for $\alpha = 3/\text{sec}$, one AC suffices, whereas for the $\alpha = 2/\text{sec}$ condition, two ACs are needed for appropriate modeling. In the SNF project that preceded the present study, a varying α was applied. First, the parameter was varied from utterance to utterance and, at a later stage, from speaker to speaker (see 5.3.5; Häslér, Hove & Siebenhaar 2005). Unfortunately, the effects of the varying α on the other model parameters were not elaborated on in their presentation of the results.

While it appears sensible to maintain α constant, the question remains as to the exact value at which it should be set since, in the words of Fujisaki and Hirose (1982:67), “there may be languages which may involve laryngeal controls not formulated in our present model”. In order to achieve a numerical optimization of α , Hansjörg Mixdorff’s (2000) program for estimating Fujisaki parameters was applied. The Fujisaki model parameters are estimated from ESPS-waves-based f_0 contours (Mixdorff 2009:2), a procedure which involves a number of steps. In the final stages of the estimation procedure, the PC parameters are optimized by “incrementally varying alpha, A_p and T_0 , for all phrase commands and calculating the distance between the resulting phrase component and the low frequency contributions [LFC] before and after each variation of each parameter of every individual phrase command” (Mixdorff 2009:2). The best possible α can be determined by feeding α values between 0.2 and 3.0 into the estimation procedure. One can thus retrieve the one α with the best-fitting result, i.e. the α that exhibits the smallest distance between the resulting f_0 contour and the original LFC contour.

This procedure was conducted for ten BE speakers, ten VS speakers, five ZH speakers, and three GR speakers. Ten waves were tested per person, and each wave was approximated with an α of 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4, 2.6, 2.8, and 3.0. Subsequently, a search of the lowest mean delta between the resulting phrase component and the original f_0 contour was conducted for each wave. For 15 different alphas (0.2–3.0, 0.2 incremental steps) in ten waves for each of the 28 speakers, this accounts to a total of 4200 comparisons. This procedure was not performed for all speakers because, firstly, it is highly time-consuming

and, secondly, not all speech data of each speaker had been segmented and annotated by the time α was optimized. Overall, the mean optimal α turned out to be 1.6.⁴ This finding is interesting in so far as, according to Fujisaki et al.'s publications, an optimal α for Japanese is approximately 3.0/sec, while for High German, the applied α values are 2.0/sec (Mixdorff 1998) and 3.1/sec (Möbius 1993a).⁵ Despite this interesting finding of such a comparatively low α value in Swiss German, a constant α value of 2.0/sec was chosen for the current study, based on three reasons. Firstly, as suggested by Fujisaki (1987:174), the ratio between the two natural angular frequencies for rotation (AC) and translation (PC) of the thyroid is estimated at 3:1 (Fujisaki 1987:174). In order to maintain, or at least approach a decent physiological interpretation of the model, α cannot be set at such a low value, since this would entail a β of only 4.8, which would be non-sensical considering that the accent component needs a quick rising time to adequately capture local rises and falls.⁶ Secondly, if the modeling condition holds that the peak of the phrase component is reached at utterance onset – which entails $T_0 = 1/\alpha$, as is the case in the present study – the temporal distance between T_0 and phrase onset would amount to as much as 625 milliseconds (see Pätzold 1991). This figure, however, does not appear to present an accurate reflection of the activation time of the thyroid cartilage, since the time for the translation of the thyroid cartilage is usually estimated at 200ms (Fujisaki 1987:174). Thirdly, it is desirable, especially for future comparative analyses, to employ an α value identical (or at least similar) to other studies on German, especially Mixdorff 1998 ($\alpha = 2.0/\text{sec}$) and Möbius 1993a ($\alpha = 3.1/\text{sec}$), and preferably an α which is also comparable to Fujisaki parameter analyses of languages other than German. Since α represents such a critical model parameter, the application of the same α value greatly facilitates cross-comparisons of the results of studies using the Fujisaki model. For these reasons, the time constant α was set constant for all phrases, utterances, and speakers at 2.0/sec.

The amount of variation in the α parameter has a direct impact on the value of the second parameter of the phrase component, namely T_0 . If α is kept constant, as is the case in the present study, then T_0 is fixed at $1/\alpha$ (see Pätzold, quoted in

4. For the BE speakers, the mean α value was 1.7, while for the other dialects it was 1.5.

5. At this point, it can only be speculated about the meaning of such comparatively low α values for Swiss German. The finding suggests that the Swiss German speakers globally rise and fall significantly more slowly in f_0 since α determines the steepness of rise and fall of the global component. This finding is without doubt in need of further research.

6. Yet, in Fujisaki et al.'s analyses and publications, the ratio of the damping factors β and α frequently is at 7:1 (β at approximately 20/sec and α at approximately 3.1/sec).

Möbius 1993a: 105). Thus the application of a constant α value of 2.0/sec in the present study results in a fixed T_0 at 500ms (1/2). In other words, the timing of the phrase component (T_0 minus utterance onset time) is constant for utterance-initial phrases and need no longer be examined in the data. The modeling of the phrase component and the setting and the roles of the parameters discussed in this section will become clearer in the next sections on modeling techniques.

7.2.3 Accent component

The accent component includes the parameters T_{1p} , T_{2p} , A_{ap} , and β and describes the local and relatively quick f_0 movements, superposed onto the phrase component. Further, this study examines the values of *AC duration*, which denotes the length of an AC, demarcated by T_1 (beginning) and T_2 (end). The timing parameters T_1 and T_2 , as well as the parameter *AC amplitude* which marks the degree of prominence through f_0 , vary from sentence to sentence because they reflect the lexical information of the words. In contrast, the parameter that characterizes the speaker's physiological mechanism, β in the case of ACs, is assumed to be constant from utterance to utterance as well as between speakers (Fujisaki 1981: 11) since "the mechanical properties of the human larynx should be approximately the same regardless of language" (Fujisaki 1992a: 172). This assumption of consistency was tested by Möbius (1993a: 111ff.), who conducted a study on whether β is dependent on the position of the accent group in the utterance or whether it is speaker-dependent. In both instances, the variables did not seem to correlate. Based on these arguments, the damping factor β of the accent component is kept constant for all utterances and all speakers in the present study.

The values for β and α were not determined in the same fashion: α was optimized, whereas β was not. The reason for this is that the great majority of works employing the Fujisaki model use a default β of 20/sec. Fujisaki et al.'s earlier publications suggest a β between 20.5/sec and 26/sec (Fujisaki 1981), and, in later publications, an β of 20/sec is regularly applied. Mixdorff (1998, 2002) as well as other authors (e.g. O'Reilly et al. 2008) also use a β of 20/sec, while Möbius (1993a) calculates an optimal β of 16/sec for his data. For this reason, a β of 20/sec was applied for all utterances and for all speakers. The relative ceiling level γ of accent components, which guarantees that the accent component will reach its maximum in a finite time, was set at 0.9, the default suggested in Fujisaki et al.'s publications.

Clearly, the applied parameter settings of β and α do not live up to the ratio between the two natural angular frequencies for rotation (AC) and translation (PC) of the thyroid estimated at 3:1 (Fujisaki 1987: 174). Instead, a ratio of 10:1 is established if a β of 20/sec and an α of 2.0/sec are employed. In Fujisaki et al.'s publications, this ratio is commonly at 7:1, yet Fujisaki (1987: 174) remarks that it

is to be determined “whether or not the 7 to 1 ratio commonly observed for β and α in the analysis of actual f_0 contours can be fully accounted for by the suggested mechanism”.

7.3 Modeling

7.3.1 Pre-processing

The pre-processing of the wave files involved a number of steps, all of which are explained in detail in this section. The following Figure provides an overview of these steps for pre-processing:

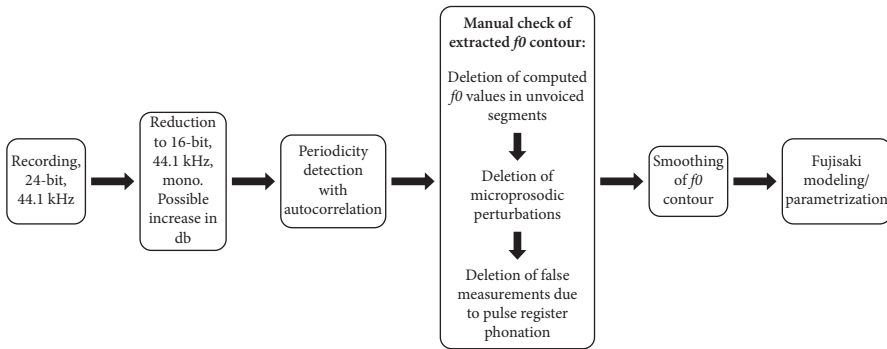


Figure 7.5. Stages of pre-processing, from initial recordings to Fujisaki parametrization

The first processing step was the recording of the speech data with a 24-bit, 44.1 kHz sampling rate. In a second step, the bit size was decreased from 24-bit to 16-bit with Adobe Audition (2007) in order to enable further segmentation work with Praat (2012). The signal was further altered from stereo to mono and, in cases of very low amplitude, was amplified by 3-5 decibel in Adobe Audition (2007). The third step concerns the generation of a pitch object file from a sound object. This pitch file is required for the modeling with Mixdorff’s *FujiParaEditor* (2009) and, in the current study, was retrieved through Praat (2012). The applied algorithm for f_0 extraction in Praat (2012) performs a periodicity detection based on an autocorrelation method (Boersma 1993). The frame duration was set at the default threshold value of 0.0, which means that per second, a computation of 100 pitch values with a pitch floor of 75 Hz was performed. The pitch ceiling values were set at 300 Hz for males and 600 Hz for females. Voicing thresholds and occasions of silence were used in their default settings. The fourth processing step involves manual checking of the extracted

f_0 contour. Manual checking is necessary since automatic periodicity detection is not always reliable and since microprosodic variation needs to be accounted for. While it is debated if microprosodic effects are perceptually relevant (see Lehiste 1970; Kompe 1997), it was decided in the present study to manually minimize microprosodic effects with the justification that microprosodic modeling is not intended with the Fujisaki model (Fujisaki & Hirose 1982). The manual correction in Praat (2012) consisted of the deletion of estimated f_0 values in unvoiced segments and the deletion of microprosodic effects. The latter includes effects such as the increase in f_0 in the transition phase of voiceless obstruents (fortis in Swiss German) and vowels, or the decrease in f_0 in the transition phase of voiced obstruents (lenis in Swiss German) and vowel f_0 . These effects were deleted in such a way that the estimated f_0 candidates of the respective transition phases were deleted as minimally as possible, since subsequent LPC (linear predictive coding) analysis-re-synthesis relies on the pitch points present in the corrected pitch file. The most reliable portions of the f_0 contour typically were vowels and nasals. Microprosodic perturbations and estimating f_0 values in unvoiced segments are not the only reason why the f_0 contour was manually corrected, however. As shown at 6.3.3, creaky voice, which frequently results in errors in pitch determination algorithms such as octave jumps (see Toshinori Ishi et al. 2008), posed a particular problem for the measurement of f_0 contours in the present study. As pointed out earlier, no solution was achieved by modifying the pitch settings in Praat (2012) (i.e. lowering the voicing threshold or increasing the time steps of estimating f_0). Hence, incorrect f_0 estimations for syllables with unfavourable voice quality were deleted entirely in Praat (2012), which meant that these deleted pitch measurements were not available for subsequent Fujisaki parametrization. In a fifth step, f_0 smoothing of the extracted f_0 contours at a bandwidth of 10 Hz was performed in Praat (2012) in order to allow for a more straightforward visual and auditory analysis/resynthesis during the actual last step of the data processing, the modeling process of the f_0 contours.

7.3.2 Modeling procedure

This section addresses the different stages of the modeling/parameterization of intonation contours with the Command-Response model. The core of f_0 modeling consists of the *analysis-by-synthesis* method, which enables a decomposition of a given f_0 contour into its constituents (Bell et al. 1961). Bell et al. (1961:1727) describes this procedure as follows:

The term *analysis by synthesis* is used to refer to an active process that can be applied to signals that are produced by a generator whose properties are known.

The heart of an analysis-by-synthesis system is a signal generator capable of synthesizing all and only the signals to be analyzed. The signals synthesized by the generator are compared with the signals to be analyzed, and a measure of error is computed. Different signals are generated until one is found that causes the error to reach some smallest value, at which time the analyzer indicates the properties of the internally generated signal (1961:1727).

To put it in other words, acoustic representation of f_0 is described by means of a synthetic representation generated from the Fujisaki parameters. In this synthesized version, different values of the components, such as magnitudes, amplitudes, and timing, can be estimated. This process is depicted in the below Figure (Fujisaki 1987: 166).

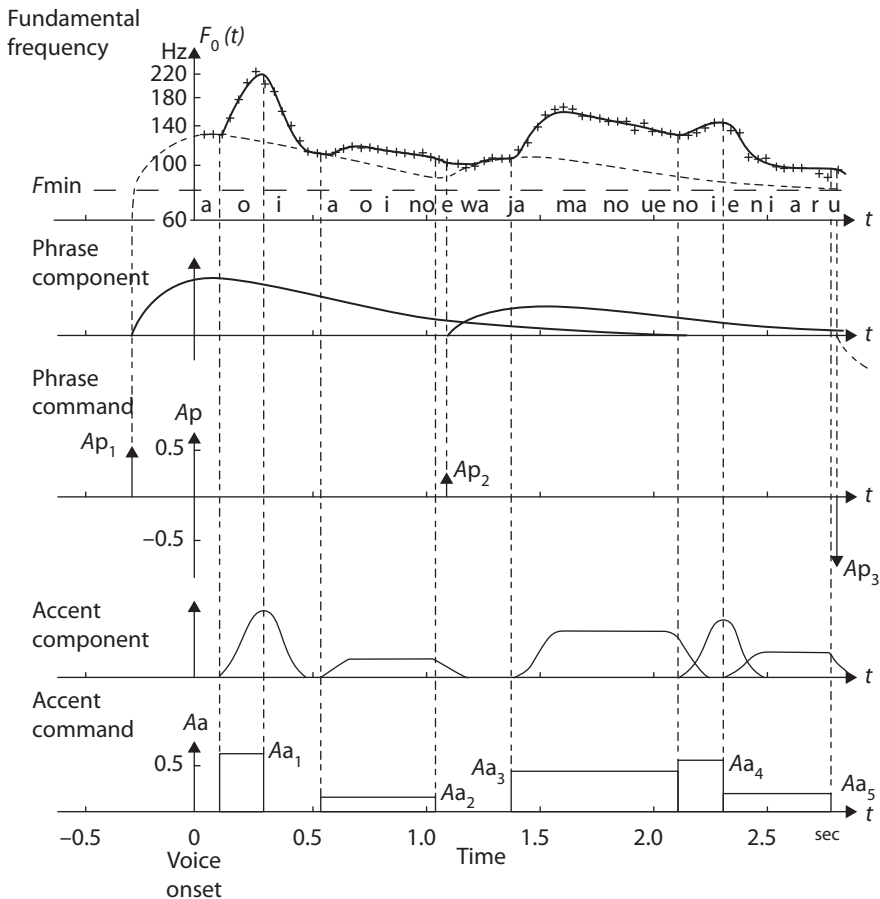


Figure 7.6. Example of *analysis-by-synthesis* of a Japanese declarative sentence for which the optimum decomposition into AC and PC is achieved (according to Fujisaki 1987: 168)

Similar to the data pre-processing procedure, the modeling procedure consists of different stages, depicted in the following Figure.

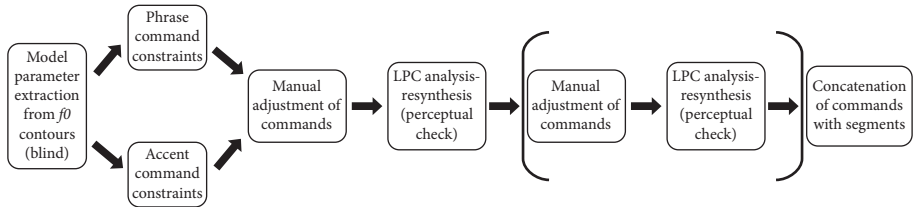


Figure 7.7. Modeling stages from parameter extraction to concatenation of the parameters with the segments

The first stage is that of model parameter extraction from f_0 contours. Mixdorff developed a rather useful program for the extraction of such parameters, the latest version of which (2009) was applied in the present study. The Fujisaki model parameters are estimated from ESPS-waves-based f_0 contours (Mixdorff 2009:2). The estimation consists of a number of steps, including a quadratic spline stylization, high-pass filtering, component separation, and Fujisaki model command initialization (see Mixdorff 2000, 2009). Technically, the Command-Response model can generate any f_0 contour. In order to ensure a linguistically meaningful interpretation of model parameters, however, modeling constraints for PCs and ACs need to be introduced, since otherwise the interpretation of the retrieved parameters becomes increasingly difficult (see Möbius 1993a:79). Since the analysis method is not formally defined in the papers presented by Fujisaki and co-workers, particularly not with regard to spontaneous speech, modeling constraints needed to be postulated.⁷ Due to the importance of modeling constraints, the following section will discuss these constraints for both PCs as well as ACs in detail.

7.3.2.1 Modeling constraints for PCs

PCs are the parameters used to model the global intonation contour of an IP. In the present study, each IP is modeled with one separate PC. As discussed earlier, IPs are delimited by a bundle of features on different linguistic levels, which can either occur together or independently to mark phrase boundaries. The most important cues for IP (and thus PC) demarcation are pitch rests, pauses, syntactic completion, laryngealization, intensity changes, breathing, changes in topic, and performance errors. The major concern regarding the establishment of PC modeling constraints

7. For a tentative modeling of spontaneous speech with the Fujisaki model, see Mixdorff and Pfitzinger 2005 or Mixdorff 2008.

is the ambiguity as to the weight of the individual factors for IP demarcation, which is even more prevalent in spontaneous speech: a division between IPs in spontaneous speech is often not apparent at first since the above-mentioned features for IP demarcation can occur independently of each other. In order to shed light on the weight of the individual parameters for PC placement and to enable a linguistically meaningful interpretation of the analysis results, *systematic modeling constraints* regarding PC placement need to be established.

In the present study, the most critical, systematic cue for the placement of a new PC is a reset in pitch. If no apparent pitch reset is perceived either auditorily or visually in *FujiParaEditor* (2012), a potential IP boundary is examined for the following physiological, linguistic, pragmatic, or syntactic features, which may serve as further cues for an IP boundary.

The following constraints are presented in order of importance:

- Constraint 1. Pausing/breathing – before IP, after IP
- Constraint 2. Phrase-final lengthening, laryngealization
- Constraint 3. Performance errors, filled pauses
- Constraint 4. Syntactic completion

Constraint 1. In the course of an utterance, pitch resets clearly are the most salient signal for a placement of a PC. However, breathing and pausing typically also trigger a reset in pitch (Vaissière 1983:57), which is why Peters et al. (2005) stress that pausing serves as a reliable cue for IP boundary marking. A textbook example of PC placement according to pitch reset triggered by pausing is shown in Figure 7.8.

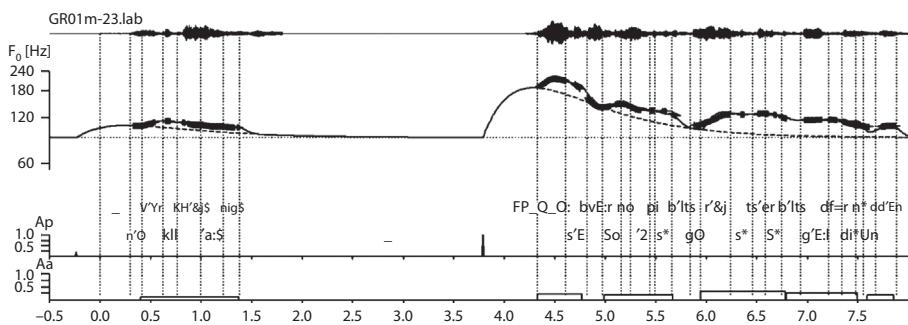


Figure 7.8. Pitch reset after a pause of 3 seconds

The phrase in Figure 7.8 reads as follows: *n'O v'Yrkli kH&S 'a:nig. jO: s'Eb v'E:r So no '2pIs * b'its gO r'&js* ts'erS * b'its g'E:ld f=rdi*n* Und d'E'n* (German: *noch wirklich keine Ahnung. Ja das wäre schon noch was ein wenig reisen gehen zuerst ein*

wenig Geld verdienen; Engl.: *really don't have a clue yet. Yes, it would be something to go traveling first earn some money*). While this is a good example to illustrate the importance of pauses and breathing for a reset of the declination line, resetting may also occur in the absence of pauses (see Fujisaki et al. 1979), particularly in the case of spontaneous speech, see Figure 7.9.

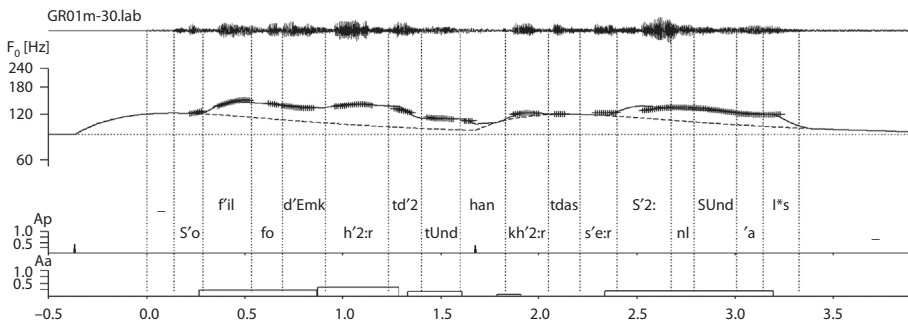


Figure 7.9. Pitch reset after the syllable *han*

The utterance reads *S'o f'il fo d'em kh2':rt d'2t Und han kh2':rt das s'e:r S'2:n IS Und 'al*s* (German: *schon viel von dem gehört dort und habe gehört dass es sehr schön ist und alles*; Engl.: *have heard much about that and have heard that it is very beautiful and everything*). Here, the reset occurs after *han* (Engl.: *have*), where no pause is located.

Constraint 2. Phrase-final lengthening, frequently accompanied by pulse phonation, is a further indicator for phrase-finality and thus IP demarcation. Figure 7.10 gives an example of this phenomenon.

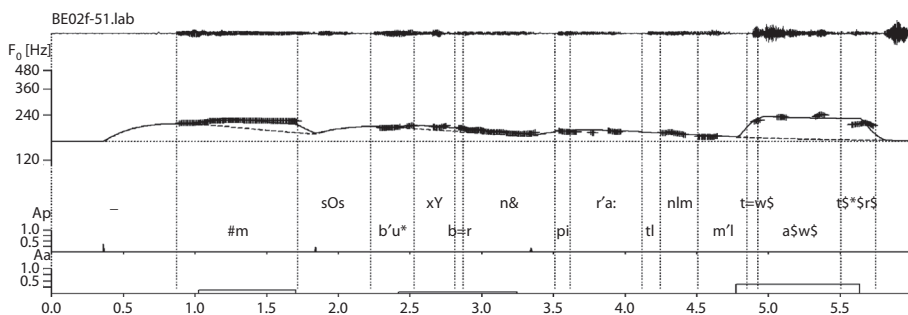


Figure 7.10. Phrase-final lengthening in the syllable *ne*, followed by a new PC.

The phrase in Figure 7.10 reads **m: sOs b'u*x Yb=r ne pir'a:tIn im m'It=wawt*r* (German: *mh so ein Buch über eine Piratin im Mittelalter*; Engl.: *um such a book*

about a female pirate in the Middle Ages). The first PC models the filled pause *m, the following PC the phrase sO s b'u*x Yb=r nɛ, while the final PC generates pir'a:tIn im m'It=wawt*r. The point of interest lies in the syllable nɛ, which is 507ms long and appears in phrase-final position, thereby triggering a subsequent pitch reset modeled with a new PC.

Constraint 3. Performance errors are very much a characteristic of spontaneous speech and more often than not trigger a new PC, as shown in Figure 7.11.

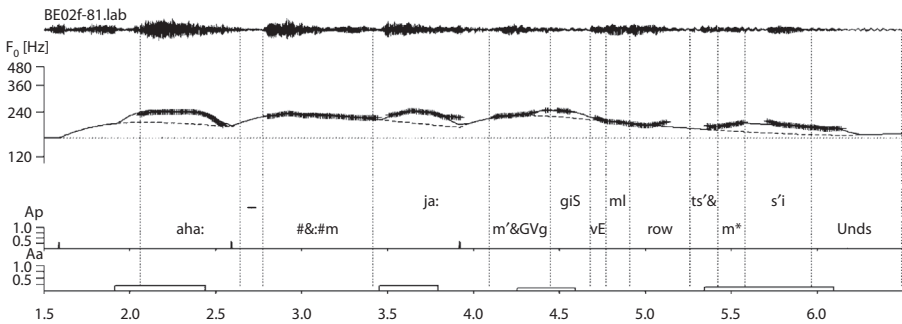


Figure 7.11. Pitch resets after the disfluencies *aha:* and *ɛ:m ja:*

The phrase reads *aha: ɛ:m ja: m'GgiS vE mI r ow ts'ɛm* s'I Unds* (German: *aha ähm ja manchmal wenn wir alle zusammen sind und es*; Engl.: *I see um yes sometimes when we are all together and it*). The first PC models *aha:*, the second PC *ɛ:m ja:*, the third *m'GgiS vE mI r ow ts'ɛm* s'I Unds*.

Constraint 4. Syntactic completion is used as a cue for IP boundary marking only if it did not become clear from the previously mentioned cues whether or not a new PC should be placed. Syntax, i.e. syntactic completion of a phrase, can trigger a reset in pitch, as shown in Figure 7.12.

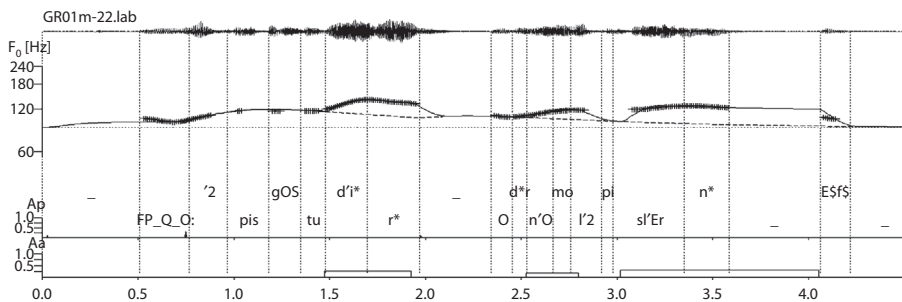


Figure 7.12. Syntactic completion may mark an IP boundary

The phrase of the Grisons speaker reads *jO: '2pis gO Stud'i*r*, Od*r n'Omol '2pis l'Erne* (German: *ja etwas studieren gehen oder nochmals etwas lernen*; Engl.: *yes go study something or learn another thing*). The first phrase *jO:* is a filled pause, the second and third phrases are syntactically complete phrases. In addition, the third phrase is clearly demarcated by the preceding pause. In spontaneous speech, syntactic cues for IP demarcation are comparatively weak, since syntactic phrase and prosodic phrase frequently do not coincide, as shown in the next Figure.

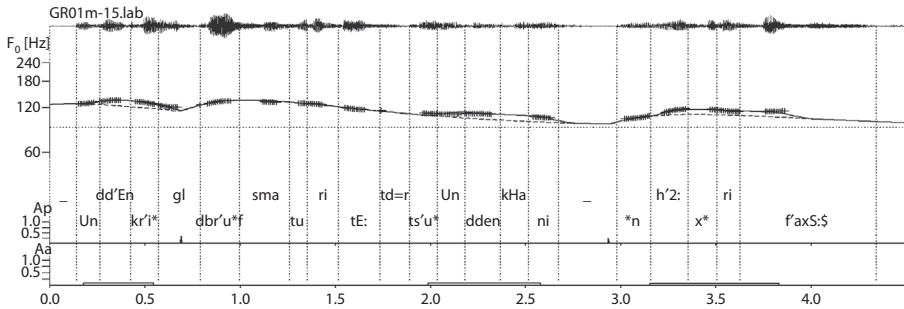


Figure 7.13. Syntactic phrases and prosodic phrases in spontaneous speech may not coincide

The phrase illustrated in Figure 7.13 reads *Und d'En kr'i*g I d br'u*fsmaturitE:t d=rt's'u* Und den kHan i *n h'2:x*ri faxS:-* (German: *und dann kriege ich die Berufsmaturität dazu und dann kann ich eine höhere Fachsch*; Engl.: *and then I get the vocational baccalaureate with it and then I can (attend) a higher college*). The first PC models subject and verb without the complement *Und d'En kr'i*g i*, the second PC models the complement of the previous phrase and the subject and verb of the following phrase *d br'u*fsmaturitE:t d=rt's'u* Und den kHan i*, while the third PC again models the complement of the previous phrase **n h'2:x*ri fax:-*.

Now that the *modeling constraints* for PCs have been established, the actual *placement of PC* can be addressed. If α is held constant, the placement of PCs involves two model parameters: T_0 and *PC magnitude*. As for T_0 , PCs are placed in such a way that, if possible, the peak of the PC is reached at phrase onset (see principles below). As for *PC magnitude*, PCs are placed in such a way that they touch the measured f_0 values at least once yet never overshoot any f_0 value. In other words, the PC should approximate the f_0 contour maximally and touch the f_0 contour at a maximal number of measured f_0 points, while never overshooting any given f_0 values.

These modeling principles were adhered to for two reasons:

Principle a.

Placing the peak of the PC at segment onset allows for the actual modeling of the declination trend in the IP, since the global component can only decrease once the peak has been reached.

Principle b.

PCs serve as baselines onto which local rises and falls are superposed. Since negative ACs are not applied in the present study, no measured f_0 value of the IP is allowed to go below the baseline.

As mentioned earlier, the underlying principle of choosing an optimal parameter configuration is to maintain as many parameters constant as possible in order to enable a more straightforward interpretation of the modeled f_0 contours. If α is kept constant, as it is the case in the present study, and (a) is adhered to, then the distance of T_0 is a fixed constant at 500ms before utterance onset. Thus, T_0 need no longer be examined as a varying parameter, at least not in utterance-initial IPs (see Möbius 1993a: 105). Figure 7.14 illustrates this modeling principle on the same utterance depicted in Figure 7.8.

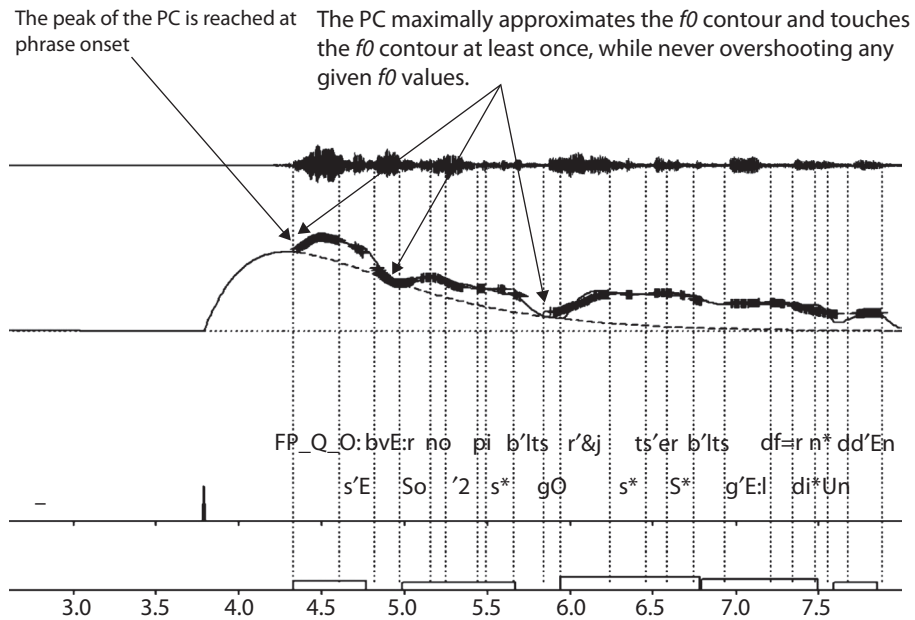


Figure 7.14. Modeling principles of PCs

While this procedure applies to the majority of utterance-initial phrases, which are normally preceded by a pause of > 500 ms, such idealistic placement of PCs often does not apply for utterance-medial or utterance-final phrases. Phrases that occur later in the utterance frequently feature significantly shorter pre-phrasal pauses than utterance-initial phrases, or feature no pauses at all. For these phrases, the same modeling criteria as for utterance-initial phrases apply in principle. Since, however, the temporal distance between T_0 and phrase onset is less than 500ms, the peak of the PC cannot be reached at phrase onset. Therefore, in order

to approximate the criterion of reaching the PC peak at phrase onset, the temporal distance between T_0 and phrase onset in these phrases is kept at a maximum without interfering with the modeling of the last syllable of the previous phrase. Figure 7.15 illustrates this modeling principle on the same utterance depicted in Figure 7.13 above.

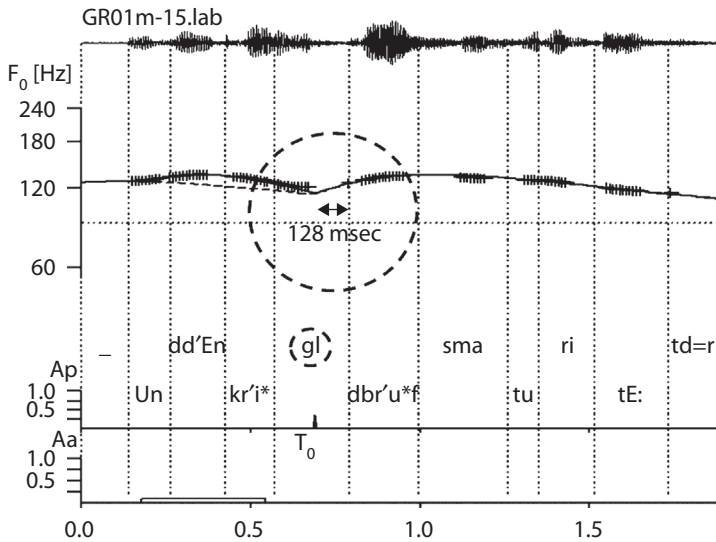


Figure 7.15. Placement of non-utterance-initial PCs: the temporal distance between T_0 and phrase onset is kept at a maximum (here 128ms) without interfering with the modeling of the last syllable of the previous phrase (*gl*)

7.3.2.2 Modeling constraints for ACs

ACs are understood as a means of marking local prominence through f_0 . The local f_0 movements generally either rise from or fall onto the baseline (phrase component) and are modeled accordingly by either rising or falling ACs. *Falling ACs* model accents with falling f_0 movements only and the falling contour is captured by the falling part of the AC. *Rising ACs*, on the other hand, apply to all of the remaining local f_0 movements. As a general rule of thumb, *rising ACs* are applied if there is a local f_0 rise from the baseline, while *falling ACs* are placed for local f_0 that falls onto the baseline.

The present study distinguishes between six types of local f_0 movements, five of which are modeled with rising ACs and one with a falling AC. Note that for all of these six accent types, pitch interruptions, i.e. gaps in the measured f_0 contour, may or may not occur. These interruptions are due to voiceless sounds, deleted microprosodic influence, unfavorable voice quality, technical measurement

errors, etc. It is important to understand that the overall movement of the f_0 contour is, however, not affected by such pitch interruptions. The following Figure 7.16 provides examples of how rising and falling ACs model the six types of accents:

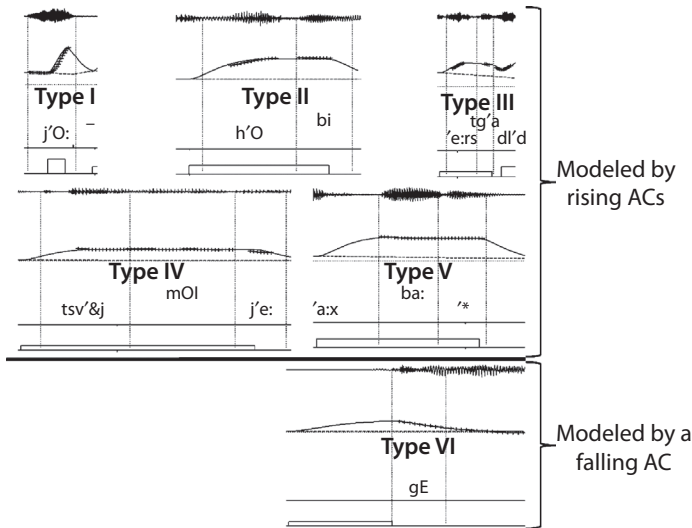


Figure 7.16. The six accent types, modeled by rising and falling ACs

The f_0 movements of the six accent types are characterized as follows (note that types I to V are modeled by rising ACs, while only type VI is modeled by a falling AC):

- Type I: a singular rise
- Type II: a rise and a subsequently maintained pitch level
- Type III: a rise (possibly, with a subsequent, equal level pitch) and a successive fall
- Type IV: a flat contour followed by a fall
- Type V: a flat contour, neither preceded by a rise or a fall
- Type VI: a falling f_0 only

In the present study, the fundamental principle of AC modeling is to model all of the perceptually relevant f_0 movements (not already captured by the baseline) with a *minimal number* of ACs. As mentioned earlier, the Command-Response model is able to generate any f_0 contour accurately by the application of an infinite number of commands. Thus, in order to avoid linguistically insensible modeling and to guarantee systematicity, it is necessary to adhere to the condition of minimalistic AC modeling, while keeping the perceptual discrepancy between original and resynthesis at a minimum.

The modeling of these six accent types is conducted according to the criteria of the *AC amplitude*, T_1 , and T_2 setting listed below. The analysis is conducted auditorily and visually in Mixdorff's *FujiParaEditor* (2012).

Amplitude:

Types I-VI: *AC amplitude* is adjusted in such a way that it captures a maximal number of the extracted f_0 measurement points of the corresponding accent.

T_1 :

Types I, II, and III: T_1 is given by the location at which the AC begins to rise.

Types IV and V: T_1 is placed in such a way that the AC reaches its peak at contour onset in order to optimally capture the flat contour.

Type VI: T_1 is less relevant, since falling ACs are modeled with the falling part of the AC only, rendering T_2 more critical. T_1 is adjusted in such a way that the temporal distance between T_1 and T_2 is kept at a minimum while the falling part of the AC is nevertheless able to capture as many measured f_0 points of the fall as possible.

T_2 :

Type I: T_2 is placed at the point where the rise reaches its maximum.

Type II: T_2 is placed a). at the last measured point of f_0 before the discontinuation of measured f_0 points or b). at the point where a rise in f_0 occurs, which is in turn modeled by a subsequent AC.

Types III and IV: T_2 in rise-fall or flat-fall ACs is placed in such a way that the last f_0 measurement of the fall of the contour is captured by the AC.

Type V: T_2 is placed at the point with the last measured pitch points.

Type VI: T_2 is placed at the first measured pitch point of the falling contour.

Figure 7.17 illustrates the modeling of local accents in a sample of a multi-phrase utterance.

The phrase in Figure 7.17 reads *j'O: d'as IS 'erSt g'ad letSti ks'i: In mal'O:rka, mIt min*r b'eSt* kOl'e:gin * v'Ux** (German: *ja das war erst gerade kürzlich in Mallorca, mit meiner besten Kollegin eine Woche*; Engl.: *yes that was just recently in Mallorca, with my best friend a week*). The phrase exhibits all six AC types defined above.

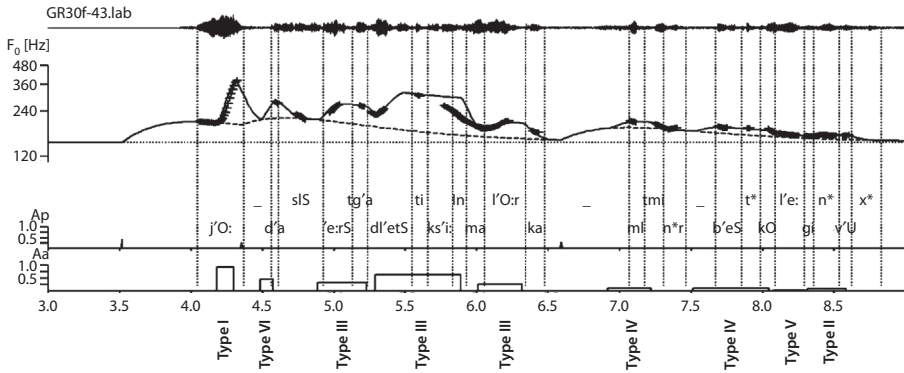


Figure 7.17. Sample modeling of a multi-phrase utterance

Since the present study associates ACs with exactly the syllables they span and not with stressed syllables exclusively, the timing of the ACs are expressed relative to the segment onset of the first syllable the AC spans and the segment offset of the last syllable the AC spans. In other words, the timing of the ACs is not simply associated with the nearest stressed syllable. Three different timing values are extracted:

Two timing values of rising ACs:

T1dist_rise: the temporal distance between AC onset and segment onset (T_1 -segment onset). *T1dist_rise* represents the time taken to prepare the vocal folds for the voicing of local accents, resulting in a negative value if T_1 occurs before segment onset (early) and a positive value if T_1 occurs after segment onset of the accent (late).

T2dist_rise: the temporal distance between AC offset and segment offset (T_2 -segment offset). If the temporal distance between T_2 and segment offset is distinct, the f_0 falls early relative to segment offset (early). If the temporal distance between T_2 and segment offset approximates 0, the f_0 of the accent falls shortly before or at segment offset (late). This timing value is not discussed in the subsequent statistics sections since its linguistic implications are not particularly decisive.

The timing value of falling ACs:

T2dist_fall: the temporal distance between AC offset and segment offset (T_2 -segment offset). Falling ACs are only modeled with the falling part of the AC, hence, the only timing value of interest is T_2 . Again, if the temporal distance between T_2 and segment offset is distinct, the f_0 falls early relative to segment offset (early), whereas if the temporal distance between T_2 and segment offset approximates 0, the f_0 of the accent falls shortly before or at segment offset (late). Note, too, that this timing value is not discussed further in the statistics sections

given its lack of linguistic importance and the low number of falling ACs (13% of all ACs in the present corpus).

Figure 7.18 illustrates such early and late rises of rising ACs relative to segment onset ($T1dist_rise$), early and late falls of rising ACs relative to segment offset ($T2dist_rise$), as well as early and late falls of falling ACs relative to segment offset ($T2dist_fall$).

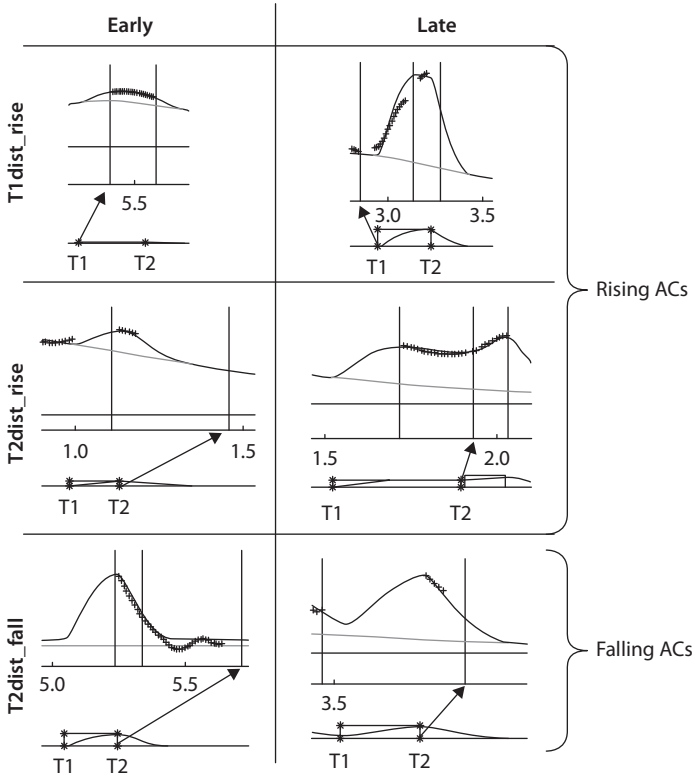


Figure 7.18. Extraction of the timing parameters $T1dist_rise$, $T2dist_rise$, and $T2dist_fall$

The most critical parameter is $T1dist_rise$ because it largely determines whether the AC shows a rising or a falling trend. Thus, when speaking of AC *timing* in the subsequent statistics chapters, we mean $T1dist_rise$ only. If the accent starts with a relatively high f_0 value (e.g. Type IV or V accents, generally ACs with a falling tendency), T_1 occurs early relative to segment onset so as to render the larynx muscle tense in order to begin at a higher f_0 . If the accent starts with a low f_0 value (e.g. Type I, II, and III, generally rises), the time to set the muscles at the right position is not required and voicing can begin after segment onset.

7.3.2.3 LPC-resynthesis

During and after the modeling stage described above, resynthesis on the syllable level as well as on the phrase level was applied repeatedly. Resynthesis is a highly beneficial derivative of the analysis-by-synthesis approach, by means of which the agreement between measured f_0 contour and the generated synthetic counterpart can be evaluated. Mixdorff's *FujiParaEditor* (2012) is able to perform a resynthesis based on LPC (linear predictive coding) analysis. A waveform is produced from the f_0 contour generated with the current parameter values of the Fujisaki model (see Mixdorff 1998:74 ff.). Such resynthesis allows for a systematic change in one parameter at a time, so that the contribution of each parameter towards the given f_0 contour can be evaluated individually. If the synthesized variant of the f_0 contour is not satisfactory, the commands can be adjusted accordingly. The aim is to find a balance between placing a minimal number of ACs while keeping the perceptual difference between original and resynthesized contour to a minimum. As was shown in Figure 7.6, the LPC-resynthesis may be applied repeatedly during the modeling process until the modeled f_0 contours are satisfactory.

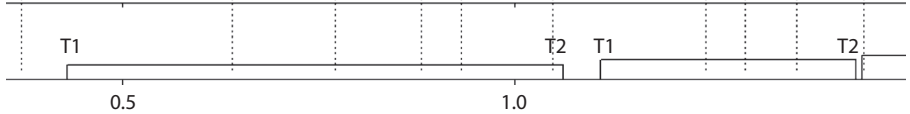
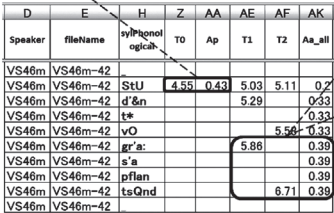
7.3.2.4 Concatenation of commands with segments

The final stage of the modeling process involved the concatenation of the model parameters T_0 , PC magnitude, AC amplitude, T_1 , and T_2 with the segments on the syllabic level. This final step is illustrated in Figure 7.19. It is critical to bear in mind that ACs are not anchored with stressed syllables but, instead, are linked with exactly the syllables they span, i.e. the syllables that are made more prominent via f_0 marking (see 2.4). The link between local f_0 accents with the segments happens blindly, which means that phonological information on the segmental level is not taken into account.

The phrase in Figure 7.19 reads *StUd'ɛnt* vO gr'a:s 'apflants=nd* (German: *Studenten die Gras anpflanzen*; Engl.: *students who grow weed*). The PC is anchored with the syllable *StU*, the ACs are linked with the corresponding segments, e.g. *gr'a:s 'apflants=nd*.

The concatenation of AC model parameters with the segments was conducted along the principle that each segment can maximally be associated with one AC.⁸ In a few instances, however, it is possible that the fall of an AC of type III, IV, or VI, or the flat part of a Type II or Type V AC and the rising part of a subsequent Type I, II, or III command or the flat part of a Type II or Type V AC occur in the same syllable. Figure 7.20 provides an illustration of this special case:

8. Theoretically, it is possible to have a f_0 twin peak accent within one syllable, as described by Wipf (1910:22) for VS Swiss German.



of the subsequent AC. In such a case, the more dominant f_0 movement of a syllable governs the concatenation with the segments. In the example in Figure 7.20, the syllable *ne* is thus linked with the first AC since the falling f_0 movement exhibits slightly more measurement points than the rising contour. Consequently, the T_i of the subsequent AC is assigned to the following syllable *b**.

7.4 Modeling difficulties

The process of the f_0 contour modeling and analysis in the current study gave rise to a number of modeling difficulties. This concerns primarily the modeling of slow-rises, low accents, and nearly flat sections at relatively high and low f_0 contours. In this section, examples from the present study are presented, and the gravity and ramifications of these modeling difficulties are discussed.

7.4.1 Flat contours

A common feature of spontaneous speech is the occurrence of nearly flat sections of relatively low or relatively high f_0 values, since declination is often present only to a minimal degree. For low f_0 values, Fujisaki et al. (1998) suggest that such sections be modeled with low-magnitude PCs and a sequence of ACs with increasing amplitudes towards the end of a phrase. For flat, high-level f_0 movements, they propose a high-magnitude PC followed by a sequence of small PCs in short intervals. From a linguistic point of view, both of these approaches are dubious. However, since the Command-Response model does not provide further modeling freedom one must decide between either of these approaches.⁹ For the modeling of a high or low flat f_0 contour, the present study chose to employ one PC followed by a sequence of ACs that rise in amplitude. Such systematic, linguistically motivated placement of PCs was preferred to an arbitrary placement of a sequence of PCs because, in the Fujisaki model hierarchy, AC placement depends largely on prior placement of PCs. This approach, however, comes at a severe price

9. An application of both approaches, as suggested by Fujisaki et al. (1998), is not reasonable since in one instance, the ACs are placed in such a way that they can no longer be interpreted sensibly, while in the second approach the same can be said for PCs. In opting for one of these suboptimal approaches, at least all contours are modeled equally, which means that the possible bias regarding the interpretation of one parameter value (either AC or PC) only concerns one of the parameters, not both. Preferably, a third component should be applied, which allows for the modeling of such contours, the implementation of which is discussed in the next two subsections.

in the analysis of spontaneous speech: AC position in the IP correlates positively with AC amplitude, as shown in Figure 7.21.

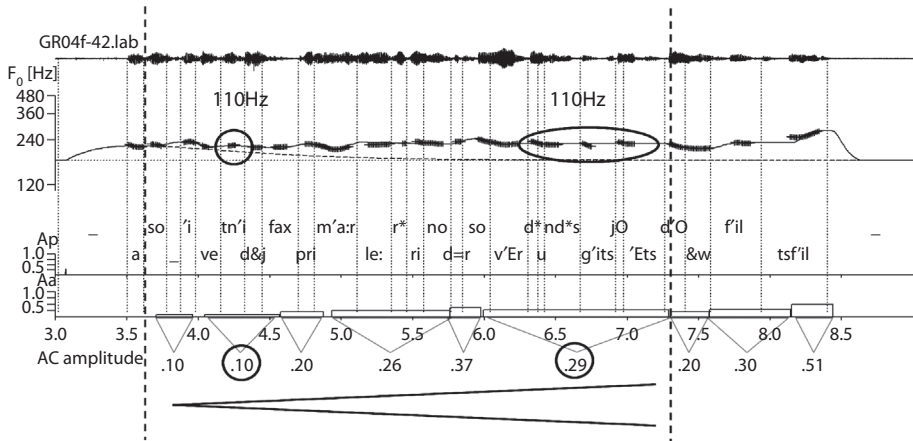


Figure 7.21. The modeling of flat f_0 contours with one linguistically plausible PC and a sequence of subsequent ACs with increasing amplitude towards phrase-medial and phrase-final position. The increased amplitude does not correspond to a proportionate increase in Hertz

The one-PC phrase in Figure 7.21 reads *aso 'I vet n'id & jfax prim'a:rl:r*rin od=r so v'Er d* und *s g'its jo 'Ets d'o & w f'il tsf'il* (German: *also ich will nicht einfach Primarlehrerin oder so werden und es gibt ja jetzt da auch viel zu viele*; Engl.: *well I don't want to simply become an elementary school teacher and there are way too many of them now*). The ACs of interest in the above Figure are demarcated between the dotted lines and AC amplitudes are indicated below the triangles. We find that AC amplitude increases from phrase beginning to phrase middle and end, from .10 to .29 amplitudes, interrupted by a local accent on the syllables *d=r so* (AC of .37). Despite the fact that the AC amplitude increases from *tn'i* (.10) to *nd*s g'its jo 'Ets* (.29), an informal measurement of the f_0 of these two segments in Praat (2012) reveals a nearly identical mean of 110 Hz. Hence, the shown increase in AC amplitude in phrase-medial and phrase-final position is solely a measure to compensate for the falling slope of the PC when modeling flat f_0 contours.¹⁰ In this case, an increase in AC amplitude does not translate into the expected effect, namely local f_0 prominence. Instead, AC amplitude correlates positively with the position of the AC in the IP for all dialects investigated in the present study, particularly for the GR dialects: the later the AC in the IP, the higher its amplitude.¹

10. Possibly, this effect may be lessened by application of the optimized α of 1.6/sec: the global contour may exhibit a less rapid approximation towards the F_b , which is likely to affect the placement of ACs.

7.4.2 Slow-rising phrases

In spontaneous speech, slow global rises in phrase-medial position towards phrase endings are common, particularly for continuing phrases. High-rises signal that the speaker intends to hold the floor. As pointed out at 4.5.3, slow-rises cannot be adequately modeled with the current components of the Fujisaki model, since the phrase component rises too rapidly to capture slow-rises. Fujisaki et al. (1998) again advise to model slow-rises by means of a sequence of small-magnitude PCs. This method, however, undermines the linguistic interpretation of such commands, which is why the present study again opted for a modeling with a sequence of ACs that increase in amplitude towards the end of the phrase. An example phrase is shown in Figure 7.22.

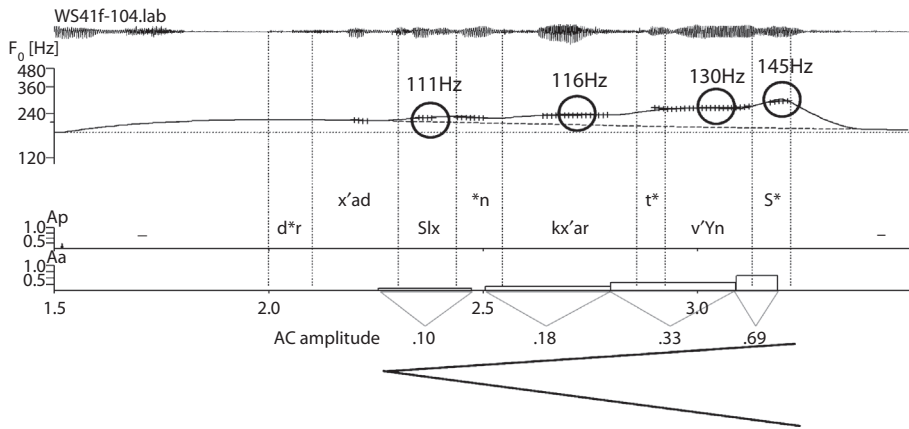


Figure 7.22. The modeling of slow-rises: a linguistically plausible PC, followed by a sequence of ACs that increase in amplitude distinctly to model the gradual increase in *f*₀

The phrase in Figure 7.22 reads *d*r x'ad Slx *n kx'ar t* v'Yn S** (German: *der(jenige) kann sich eine Karte wünschen*; Engl.: *that person can ask for a card*). In the Figure, AC amplitudes for each of the ACs are given below the triangles. To adequately capture the slow-rise, the AC amplitudes are characterized by a distinct increase from .10 to .69 towards the end of the phrase. Again, such a systematic, linguistically motivated placement of PCs was preferred to an arbitrary placement of a sequence of PCs because of the model's hierarchical structure: AC placement depends largely on prior placement of PCs.

This approach of modeling slow-rises is not optimal, however. The placement of ACs suggests that the slow-rises represent locally controlled *f*₀ movements. On the contrary, however, slow-rises are understood as global *f*₀ movements, affecting large portions of the IP, and should therefore be modeled as signaling continuation or incompleteness globally. The undesired effect of such modeling is of course the

fact that the positive correlation between AC position in IP and AC amplitude is heightened to an even more distinct degree.

Mixdorff, too, observes such slow-rises in the data sets of his Ph.D. and concludes that this type of intonational control is not included in the Command-Response model, which originally was geared towards Japanese utterances. Consequently, Mixdorff introduces a new *slow-rise component* into the model.

7.4.2.1 *Slow-rise component*

The slow-rise component is additively superposed onto the model and, according to Mixdorff (1998:98ff.), aims at fulfilling the following conditions:

- a. Compensation of the falling contour of the phrase component in order to generate a gradual linear rise (with a time constant similar to α).
- b. A slow, initial rise of the component (unlike the onset characteristics of the phrase component) to establish a concave global rise.

The slow-rise component generally shares the characteristics of the accent component, except for the absence of γ (the relative ceiling value) and the inclusion of a time constant similar to that of the phrase component (Mixdorff 1998:98ff.) In the Fujisaki equation, the term takes on the following form, with $T3$ and $T4$ denoting the starting and end points of the command:

$$+ \sum_{k=1}^K A s_k [G s(t - T3_k) - G s(t - T4_k)]$$

where

$$G s(t) = \begin{cases} 1 - (1 + \delta t) \exp(-\delta t), & \text{for } t \geq 0, \\ 0, & \text{for } t < 0. \end{cases}$$

Regrettably, Mixdorff did not integrate the slow-rise component into the *FujiParaEditor* (2012) program. This gap was bridged by Jürg Bigler, who implemented the slow-rise component into the *FujiParaEditor* (2012) in the framework of the SNF project on which the present study is based. Informal modeling with this new version of the editor was successful for the modeling of the following contours:

- a. *slow-rises*: global rises are captured most optimally with this new component.
- b. the modeling of *low flat* and *high flat* contours: the falling trend of the phrase component can be compensated for by the application of one slow-rise

command, thereby generating a flat contour. Again, however, since the actual f_0 contour does not rise slowly but is flat, the inherent meaning of slow-rise is not captured in this approach. The slow-rise is only applied to make up for the falling tendency of the PC. Nevertheless, such an implementation seems more sensible than applying a series of ACs gradually increasing in amplitude. The latter modeling strategy should only be applied to model local f_0 prominence.

Unfortunately, the slow-rise component could not be implemented quantitatively in the present study for the following reasons:

1. Unlike for phrase and accent components, a physiological correlate of the slow-rise component has not been scientifically proven thus far. Still, Mixdorff (1998:99) notes that “speakers are capable of consciously raising their pitch at a slow rate (portamento)”. Fujisaki (1995), as well, observes that languages other than Japanese or English may require further controls.
2. Jürg Bigler’s implementation of the additional command in the *FujiParaEditor* (2012) is not yet refined enough at this point to allow for an efficient drag and drop style modeling. On the contrary, the modeling with the new command is an exceedingly slow and tedious process. In light of the vast amount of the data on which this study is based, it seemed much more reasonable to only work with the original model parameters.
3. As noted earlier, taking into consideration the numerous variables investigated in the present study, including linguistic, paralinguistic, and non-linguistic variables, one of the aims of this study is to reduce the number of varying Fujisaki parameters to a minimum, enabling a more straightforward interpretation of the modeled f_0 contours. An implementation of this slow-rise component would require further analyses of the model parameter’s amplitude, T_3 and T_4 as well as justification for the selection of the time constant delta.

Nonetheless, an implementation of this additional component appears to be very promising for future quantitative analyses of f_0 contours. To allow this, however, the latest version of the *FujiParaEditor* (2012) would need to undergo refinement.

7.4.3 Slow-rising local accents

As mentioned at 4.5.3, another drawback of the Fujisaki model is its incapability of generating certain slow-rising accentual contours (Taylor 1994). In slow-rising accents, the rise in f_0 usually takes longer than the fall, yet the opposite may also be the case, i.e. that the rising time is much shorter than the falling time. Visual examples of both slow-rising and slow-falling accents are depicted in Figure 7.23.

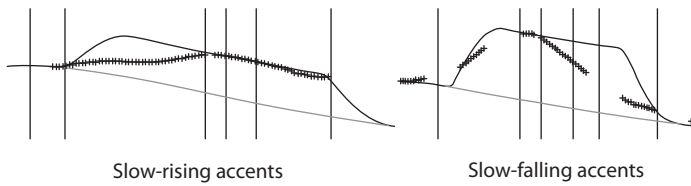


Figure 7.23. Slow-rising and slow-falling accents

Slow-rising accents were observed for the BE dialect in particular, while slow-falling accent seem highly prevalent in the GR variety. The problem here is that both the rising and falling parts of one AC are generated with the same time constant (here 20/sec). As a consequence, the modeling of such slow-rising accents remains fairly inaccurate.

Now that the theoretical bases as well as the methodology applied in the present study have been presented, we now proceed to the overall results of the analysis of the five chosen parameters.

CHAPTER 8

Overall results

The aim of this section is to give the reader an overall impression of the dialects' f_0 behavior. For PCs, these parameters include *PC magnitude* and *PC duration*, and for ACs, they are composed of *AC amplitude*, *duration*, and *timing* (*T1dist_rise*). Note that in this chapter only the most relevant results are reported. Before we turn to the presentation of the overall results, let me briefly touch on the methods of data transformation, the figure details, and the visual presentation of the statistical results. Statistical information on the various tests performed can be found in the endnotes.

8.1 Statistical preliminaries

8.1.1 Data transformation

It is desirable to work with parametric tests in order to be able to conveniently analyze a given data set. One assumption of parametric statistical tests is that the data follow a near Gaussian distribution. Following West, Finch, and Curran (1995), data that show skewness > 2 and kurtosis > 7 deviate strongly from a normal distribution. The measurement parameters of the present study along with their skewness and kurtosis values are illustrated in the following Table.

Table 8.1. Skewness and kurtosis of the five parameter

	Skewness	Kurtosis
PC magnitude:	0.93	1.4
PC duration:	1.8	7.9
AC amplitude:	1.3	2.7
AC duration:	1.8	5.4
T1dist_rise:	1	1.1

While both the kurtosis and the skewness values for AC duration and PC duration are comparatively high, the values for PC magnitude and AC amplitudes,

too, are distinct. Hence, a data transformation of PC magnitude and AC amplitude data was conducted in order to reliably work with parametric tests. This decision is based on two main reasons: firstly, because of the high degree of skewness and kurtosis of the two most critical model parameters, AC amplitude and PC magnitude, and, secondly, because all of the investigated linguistic, paralinguistic, and non-linguistic variables in this study involve calculations with either PC magnitude and/or AC amplitude values, yet not necessarily with the three other parameter values.

In a transformation, the data is converted to a new scale to fit the condition of parametric tests more optimally (see Osborne 2002a). Each observation undergoes a mathematical operation, which then serves as the basis for subsequent statistical tests. Because all observations are treated in the same way, subsequent comparisons remain valid. The type of data transformation depends on how the data is distributed, and with a large set of observations it is worthwhile to compare the effects of the different types of transformations. Hence, the AC amplitudes as well as PC magnitudes were transformed by a number of mathematical functions, including logarithmic, square root, cube root, and reciprocal transformations. For the current data, a conversion with square root proved to be the most powerful operation so as to overcome the violation of the assumptions of parametric tests. A feature of square root transformations is that numbers above 1.00 behave differently than numbers between 0.00 and 0.99 in that the square root for data above 1.00 becomes smaller, whereas the square root of data below 1.00 becomes larger (Osborne 2002a).

Once the data have been transformed, the effect of the transformation can be tested with normal quantile plots. In a normal quantile plot, the data is compared to a normal distribution, whereby the degree of normality depends on how well the data points cluster around a straight line. Figures 8.1 and 8.2 provide examples of the distributions and normal quantile plots of the original PC magnitudes and AC amplitudes and the corresponding values after the square root transformation.

It becomes apparent that the transformed data on the right-hand-side cluster more closely around the straight line representing the normal distribution. For the transformation of the AC amplitudes, the obtained distribution is less normal than for PC magnitudes. Yet, ANOVAs are not very responsive to moderate deviations from normality, which is why subsequent analyses are legitimate.

8.1.2 Figure details

The statistical analyses of the present study were performed with JMP (Release 7.0, 2008). For illustration purposes, data are commonly illustrated with *mean*

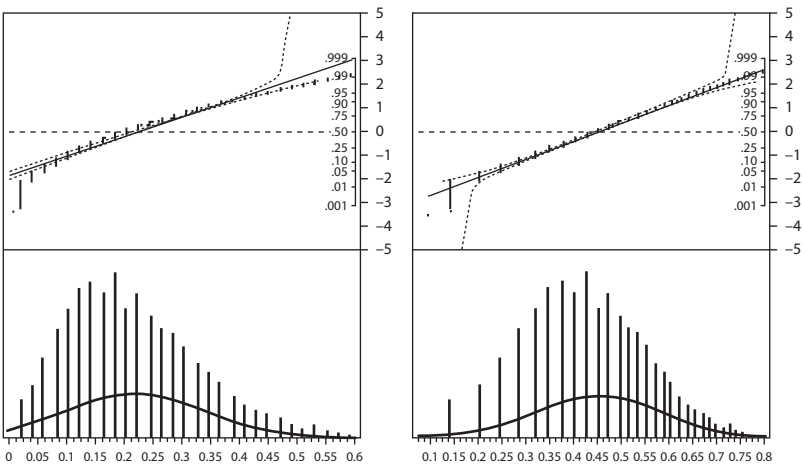


Figure 8.1. PC magnitude before (left) and after (right) square root transformation

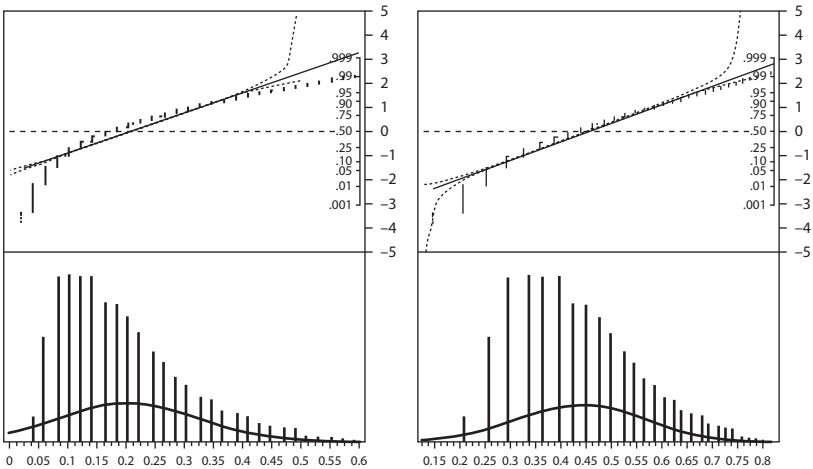


Figure 8.2. AC amplitude before (left) and after (right) square root transformation

diamonds plots. In order to accurately interpret these plots, this section will provide some information on the mean diamonds plots. As an example, see Figure 8.3, which shows a mean diamonds graph for the nominal independent variable *dialect* on the x-axis, and the numeric, dependent variable PC magnitude on the y-axis. The diamonds plot illustrates the total response sample mean and the group means (the line across each diamond). Additionally, the vertical endpoints of each diamond indicate the 95% confidence intervals for each group's mean.

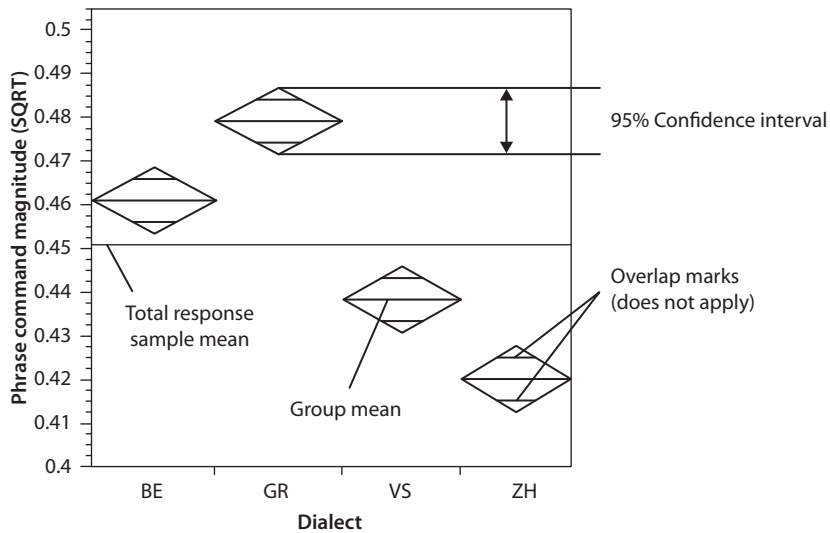


Figure 8.3. Figure details as used in the present study (see Sall et al. 2005: 203)

If the group means are similar to the mean of response, they will be close to the grand mean. Furthermore, if the confidence intervals (the end points of the diamonds) do not overlap, the differences between the corresponding means of the groups are significant (yet the reverse is not necessarily true) (Sall et al. 2005: 172). In case of balanced data where the means are composed of the same number of observations, the overlap marks are also graphically indicative of statistical significance, in which two means are significantly different when their overlap marks do not overlap. Yet, this graphical device is not applicable to this study since the present data does not yield the exact same number of observations of a variable between the groups. While mean diamonds plots are the most applied and relevant means of visual display in this study, other figures are also resorted to. These figures will be explained when employed.

8.1.3 Presentation of statistics

For ANOVAs as well as linear and logistic regressions, statistics are presented as follows:

$$\text{e.g. } F(3, 1893) = 112, p < .0001$$

where

$$F(\text{degrees of freedom between groups, degrees of freedom within groups/error}).$$

For the statistical analyses, an alpha level of .05 is used in all instances. If not further specified, two-sided t-tests are applied as opposed to one-sided tests. T-tests are also applied for post-hoc testing. Regressions with R^2 values $< .01$ are generally not reported even if the corresponding p values suggest statistical significance: small p values in regressions are common particularly in large data sets. In chi-square tests of independence, the *Pearson* chi-square coefficient is reported along with the test p value.

8.2 Summary of analyzed data

The current study is based on data of 40 speakers. For each speaker, an average of 3:09 minutes of spontaneous speech material were analyzed (empty pauses¹ excluded), which equals 1020 syllables on average (including filled pauses). However, 7.5% of this data – an average of approximately 77 syllables per speaker – was discarded for further analyses due to poor f_0 measurements, mostly caused by creaky or whispery voice. Table 8.2 provides detailed information for each speaker: the individual articulation duration (top cell), the total number of syllables including filled pauses (medial cell), as well as the percentage of syllables that were discarded because of unfavorable measurements (bottom cell). The percentage of unusable data ranges from 0% (e.g. GR03m) to 35% (e.g. GR31m) and is based on the number of total syllables of the respective speaker.

8.3 *Fb*

The first model parameter to consider is the base frequency for each of the 40 speakers, summarized in Table 8.3. On average, the base frequencies for the BE males are at 88 Hz, for BE females 167 Hz, for GR males 80 Hz, for GR females 172 Hz, for VS males 93 Hz, for VS females 174 Hz, and for ZH males and females 80 Hz and 168 Hz, respectively. For both, males and females, the standard deviation amounts to 12.

1. Silences (including occlusion phases of plosives) beyond 200ms were labeled as pauses.

Table 8.2. Overview of analyzed data per speaker: total articulation duration (top cell), number of syllables produced (medial cell), and percentage of syllables discarded from of the total of syllables due to inaccurate *f0* measurements (bottom cell)

BE02f	BE03m	BE04f	BE05m	BE06m
04:37	03:37	03:13	04:05	02:48
1137	923	1003	1154	920
6%	15%	4%	2%	9%
BE12f	BE13m	BE21f	BE22m	BE23f
03:59	03:06	02:28	03:10	02:41
1071	953	837	1040	871
18%	22%	1%	7%	7%
GR01m	GR03m	GR04f	GR06f	GR24m
03:01	03:23	03:29	02:56	03:29
944	1042	1054	934	1071
4%	0%	7%	4%	1%
GR30f	GR31m	GR33f	GR34m	GR35f
02:48	03:20	03:04	03:19	03:01
1008	1015	989	995	926
1%	35%	12%	3%	1%
VS22m	VS23f	VS25f	VS28m	VS29m
03:02	03:02	02:52	03:17	02:53
1076	1113	923	1059	1136
24%	12%	2%	5%	0%
VS41f	VS44f	VS46m	VS48m	VS49m
03:47	03:57	02:47	02:40	02:27
1184	1338	919	945	927
8%	11%	4%	5%	0%
ZH01m	ZH02f	ZH03f	ZH22m	ZH23f
03:32	02:59	03:01	02:39	03:21
1048	1067	992	1072	1121
7%	6%	0%	1%	5%
ZH24m	ZH26f	ZH31f	ZH32m	ZH34m
02:44	02:39	02:54	03:10	03:00
1030	817	1054	1033	1037
15%	5%	9%	17%	5%

Table 8.3. *Fb* for all 40 speakers

BE02f	BE03m	BE04f	BE05m	BE06m
161	101	171	76	81
BE12f	BE13m	BE21f	BE22m	BE23f
179	102	158	81	167
GR01m	GR03m	GR04f	GR06f	GR24m
89	75	175	180	76
GR30f	GR31m	GR33f	GR34m	GR35f
148	79	189	84	172
VS22m	VS23f	VS25f	VS28m	VS29m
105	162	184	116	93
VS41f	VS44f	VS46m	VS48m	VS49m
178	175	75	88	83
ZH01m	ZH02f	ZH03f	ZH22m	ZH23f
78	151	185	77	151
ZH24m	ZH26f	ZH31f	ZH32m	ZH34m
75	188	168	78	94

8.3.1 Effects with other model parameters

To test whether *Fb* is a predictor of any of the investigated parameters for males and females, linear as well as logistic regressions were calculated. The linear regressions of PC duration, AC amplitude, AC duration, and AC timing on *Fb* showed $R^2s < .01$. The only $R^2 > .01$ is found in the female group, when PC magnitude is regressed on *Fb*. In this case, *Fb* explains 6% of the variance in PC magnitude in that the higher the *Fb* of the female speaker, the lower the PC magnitude.² This trend is valid for all dialects.

8.4 Phrase component

8.4.1 PC magnitude

A one-way analysis of variance was calculated on the PC magnitudes of the dialects. It turned out to be significant.³ For a comparison among the levels, a

two-sided student's t-test revealed that all levels differ significantly.⁴ All dialects exhibit nearly identical standard deviations. Overall, the GR speakers show the most distinct PC magnitude values, while the lowest magnitude values are found for ZH speakers. The following table gives an overview of the respective means (SQRT) and standard deviations.

Table 8.4. Measures of dispersion and means for overall PC magnitude

	Mean (SQRT)	Standard deviation
BE	0.46	0.13
GR	0.48	0.13
VS	0.44	0.13
ZH	0.42	0.13

8.4.2 PC duration

The duration of IPs ranges between .2 seconds to 10.8 seconds, with an overall average of 1.47 sec. Descriptive statistics of IP length reveal that the duration of IPs are longest for ZH speakers ($M = 1.56$, $SD = 1.02$), followed by BE ($M = 1.49$, $SD = .99$), VS ($M = 1.45$, $SD = .88$), and GR ($M = 1.39$, $SD = 1$). A one-way ANOVA shows significant effects.⁵ A two-tailed t-test indicates that the differences between GR and BE, ZH and GR, and ZH and VS are significant.⁶

The number of syllables in an IP ranges from 1 to 38 syllables. Descriptive statistics of the number of syllables per IP demonstrate that on average, the ZH produce the highest number of syllables in IPs ($M = 8.7$, $SD = 6.2$), followed by VS speakers ($M = 8.3$, $SD = 5.6$), BE speakers ($M = 7.29$, $SD = 5.2$), and GR speakers ($M = 7.2$, $SD = 5.5$). Again, the ANOVA confirms significant effects.⁷ T-tests of the inter-level differences between VS and BE, VS and GR, ZH and BE, as well as ZH and GR turned out to be significant.⁸

The number of ACs in an IP in the present data ranges from 0 to 16. Descriptive statistics of the number of ACs in an IP demonstrate that the ZH and VS speakers produce the highest number of ACs in an IP (ZH: $M = 2.5$, $SD = 1.8$; VS: $M = 2.4$, $SD = 1.8$), followed by the GR and BE speakers for whom the number of ACs in an IP is lower (GR: $M = 2.2$, $SD = 1.8$; BE: $M = 2.1$, $SD = 1.5$). A one-way ANOVA on the dialects' frequencies of ACs in an IP confirms significant effects.⁹ A t-test indicates that the differences between VS and BE, VS and GR, ZH and BE, and ZH and GR are significant.¹⁰

8.4.3 Effects with other model parameters

To test whether PC magnitude and PC duration predict any of the other six investigated parameters, linear regressions were performed. A regression of AC amplitude on PC magnitude shows statistically significant results. PC magnitude accounts for 1.3% of the variance in AC amplitude, where the higher the PC magnitude, the higher AC amplitude.¹¹ Again, this trend is valid for all dialects.

With regard to PC duration, a regression of AC amplitude on PC duration confirms significant results: the longer the IP the higher the AC amplitude. PC duration explains 2% of the variance in AC amplitude.¹² Moreover, a regression of AC duration on IP duration shows significant effects. The duration of the IP accounts for 2% of the duration in the ACs: the longer the IP duration, the longer the AC duration.¹³ Both of these findings are valid for all four dialects. A logistic regression was used to estimate the probability of falling ACs from PC duration. As it turns out, PC duration is a significant predictor for falling ACs with a predictive value of 2%. The probability of falling ACs increases with the length of the IP.¹⁴ This trend is again valid for all four dialects. To test whether PC duration predicts PC magnitude, PC magnitude was regressed on PC duration. The regression shows that duration of IP is not a predictor of PC magnitude (overall and in all dialects $R^2 > .01$).

8.5 Accent component

8.5.1 AC amplitude

A one-way ANOVA was calculated on the dialects' AC amplitudes. Table 8.5 gives the overall AC amplitudes for all speakers. Table 8.5 shows means (SQRT) as well as standard deviations. The analysis was significant.¹⁵ For a comparison among the

Table 8.5. Measures of dispersion and means for overall AC amplitude

	Mean (SQRT)	Standard deviation
BE	0.42	0.12
GR	0.44	0.14
VS	0.43	0.12
ZH	0.43	0.12

levels, a student's t-test yielded significant differences between nearly all levels.¹⁶ The only difference which is not significant is ZH with BE.¹⁷ On a local level, too, the GR speakers demonstrate the most distinct amplitude values, while the AC amplitude values of BE speakers are least distinct.

8.5.2 AC duration

The duration of the ACs in the present data ranges between .1 and 4.4 seconds, with an overall average of .47 seconds. Descriptive statistics of the AC length demonstrate that AC durations of the BE speakers are longest ($M = .49$, $SD = .29$), followed by GR ($M = .47$, $SD = .29$), ZH ($M = .46$, $SD = .27$), and VS ($M = .46$, $SD = .27$). A one-way ANOVA on the differences between the groups is significant.¹⁸ A two-tailed t-test shows that the differences between GR and BE, VS and BE, and ZH and BE are significant.¹⁹

In the present data, the number of syllables per AC ranges from 1 to 34, with an overall average of 2.5 syllables per AC. Descriptive statistics of the number of syllables per AC reveal that the ZH speakers produce the most syllables per AC ($M = 2.6$, $SD = 1.6$), followed by VS ($M = 2.6$, $SD = 1.6$), GR ($M = 2.4$, $SD = 1.7$), and BE ($M = 2.4$, $SD = 1.5$). A one-way ANOVA on the dialects' frequencies of syllables per AC is significant.²⁰ A t-test shows that the differences between VS and BE, VS and GR, ZH and BE, as well as ZH and GR are significant.²¹

8.5.3 AC timing

The one-way ANOVA, calculated on the dialects' *T1dist_rise* values, turned out to be significant.²² For a comparison among the levels, a two-sided student's t-test shows significant differences on nearly all levels.²³ The only insignificant differences were found between ZH ($M = -.5$, $SD = 85$) and VS ($M = -4$, $SD = 79$) as well as between VS and GR ($M = -6$, $SD = 89$). The BE ($M = 13$, $SD = 91$) speakers clearly exhibit the latest rises relative to segment onset, namely 13ms after segment onset, while the rises in all other dialects start before the segment onset, particularly for GR dialect (-6ms).

8.5.4 Effects with other model parameters

To test whether each of the values of the different AC parameters are correlated with other AC parameter values, correlations as well as logistic regressions were performed. A logistic regression was calculated to estimate the probability of falling ACs from AC duration. AC duration was found to be a significant predictor for falling ACs, with a predictive value of nearly 20%: the shorter the AC, the more

probable it is a falling AC.²⁴ Furthermore, all of the timing parameter values were correlated with AC duration. The correlations were significant in all instances. With regard to rising ACs, the correlation indicates that the longer the AC, the later the rise (relative to segment onset) and the earlier the fall (relative to segment offset).²⁵ In terms of falling ACs, the correlation reveals that the longer the AC, the earlier the fall (relative to segment offset).²⁶

CHAPTER 9

Linguistic variables

9.1 Stress

For the variable stress, two different types of analyses were performed:¹ firstly, analyses on the number of stressed syllables in the ACs, and, secondly, analyses on the position of the first stressed syllable. For these analyses, only ACs are considered.

9.1.1 Number of stressed syllables in AC

For the majority of the present analyses, the interest lies in establishing the differences between ACs with *0 stressed syllables*, *1 stressed syllable*, and *2 or more stressed syllables*. For this reason, three categories are established and the respective ACs are grouped together, whereas all ACs with *2 or more stressed syllables* are pooled together into one group. Generally, the *number of stressed syllables in an AC* is understood as a measurement variable in nature, since the interval between the measurements is meaningful and constant (1 syllable). In the current analysis, however, due to the inclusion of *0 stressed syllables* and the pooling of ACs with *2 or more stressed syllables*, the interval between the measurements of this variable is no longer meaningful and constant and is thus treated as ordinal.

In order to test whether there is a difference in the relative proportions of ACs with 0, 1, and 2 or more stressed syllables between the four dialects, a chi-square test of independence was performed. The test confirmed significant differences between the dialects.²⁷ The percentage of ACs with 0 stressed syllables for the BE group is 36%, for GR it is 32%, for VS 35%, and for the ZH dialect only 31%. As for ACs with only 1 stressed syllable, the BE group exhibits 50%, the GR 52%, the VS only 46%, and the ZH 51%. ACs with 2 or more stressed syllables have the following distributions: BE 14%, GR 16%, VS 19%, and ZH 18%.

1. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

9.1.1.1 AC amplitude

A correlation of the specific number of stressed syllables in the AC and AC amplitude shows the following overall trend: the more stressed syllables in the AC, the higher the amplitude.²⁸ In order to test whether AC amplitudes differ significantly between ACs with 0, 1, or 2 or more stressed syllables for each dialect, ANOVAs were calculated, see Figure 9.1.

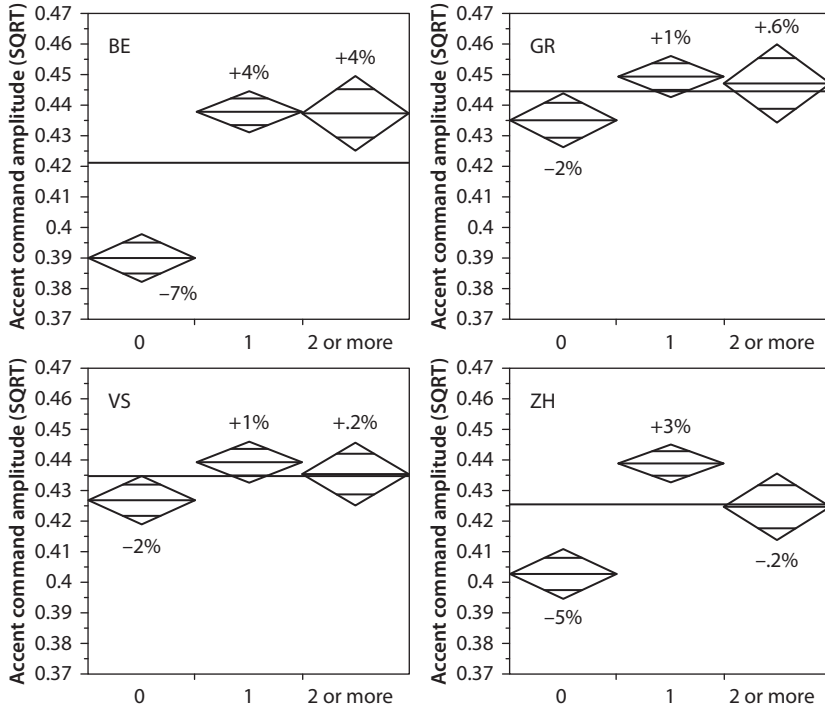


Figure 9.1. Mean diamonds plots showing the amplitudes of the different AC types according to dialect

All ANOVAs showed significant effects.²⁹ In the BE dialect, ACs with 0 stressed syllables clearly show the lowest amplitudes (-7%), while ACs with 1 and 2 or more stressed syllables indicate equally high amplitudes.³⁰ In the GR dialect, ACs with 0 stressed syllables show the lowest amplitudes (-2%), while ACs with 1 and 2 or more stressed syllables show slightly higher amplitudes (+1% and +.6%).³¹ A nearly identical pattern is also found in the VS variety, for which the amplitudes of the different AC types do not differ extensively.³² The ZH group exhibits the greatest discrepancies in amplitude between the different AC types. The inter-level tests confirm significant differences in all instances.³³

Cross-dialectal differences

In order to test whether the dialects differ among each other with regard to the amplitudes of the three different AC types, further ANOVAs were calculated. This time, the ANOVAs concern the dialects' amplitudes according to different AC types, see Figure 9.2.

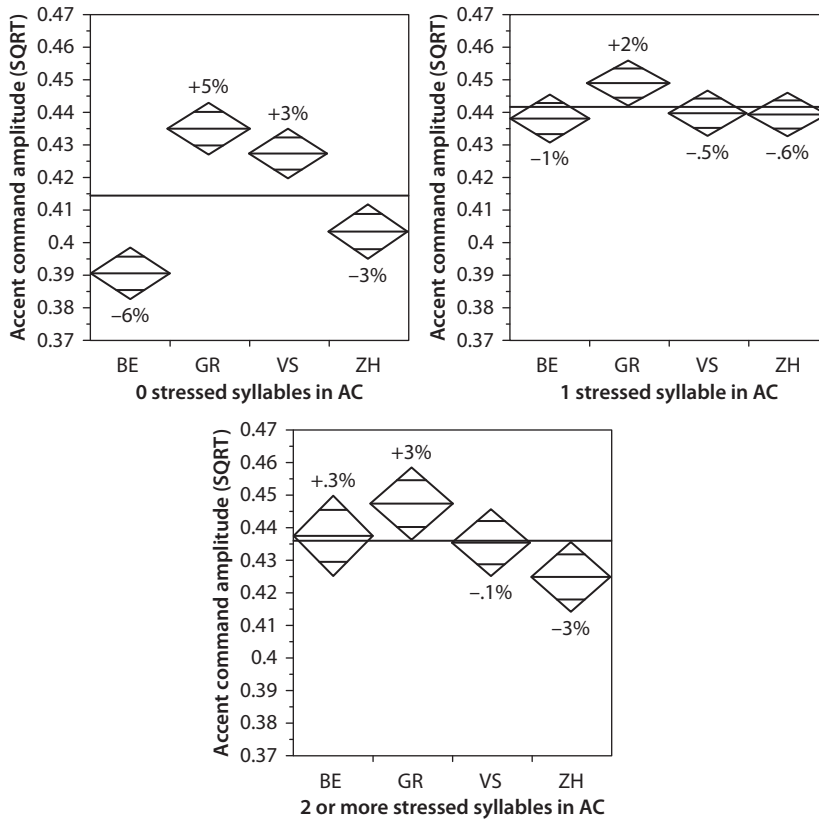


Figure 9.2. Mean diamonds plots showing the dialects' AC amplitudes, according to different AC types. The deviations indicated in percentages are calculated based on the mean of response

The ANOVAs on the dialects' amplitudes of ACs with 0 or 2 or more stressed syllables show significant effects.³⁴ For ACs with 0 stressed syllables, the GR indicate the highest (+5%), the BE by far the lowest amplitudes (-6%). All inter-dialectal differences are significant, except those between VS and GR.³⁵ In ACs with only 1 stressed syllable, the amplitudes is again highest for the GR (+2%) and lowest for the BE group (-1%). For ACs with 2 or more stressed syllables, we find a slightly different pattern: here, the amplitudes are highest for the GR (+3%), second highest for the BE (+.3%), and lowest for the ZH (-3%). Only the differences between GR and ZH are significant.³⁶

9.1.1.2 AC duration

Duration positively correlates with the number of stressed syllables in the AC for each dialect. This means that the more stressed syllables in the AC, the longer the AC.³⁷ Consequently, the different AC types' durations according to dialect are not further investigated. However, the dialects' AC durations according to type of ACs are of further interest, since this will reveal cross-dialectal differences, see Figure 9.3.

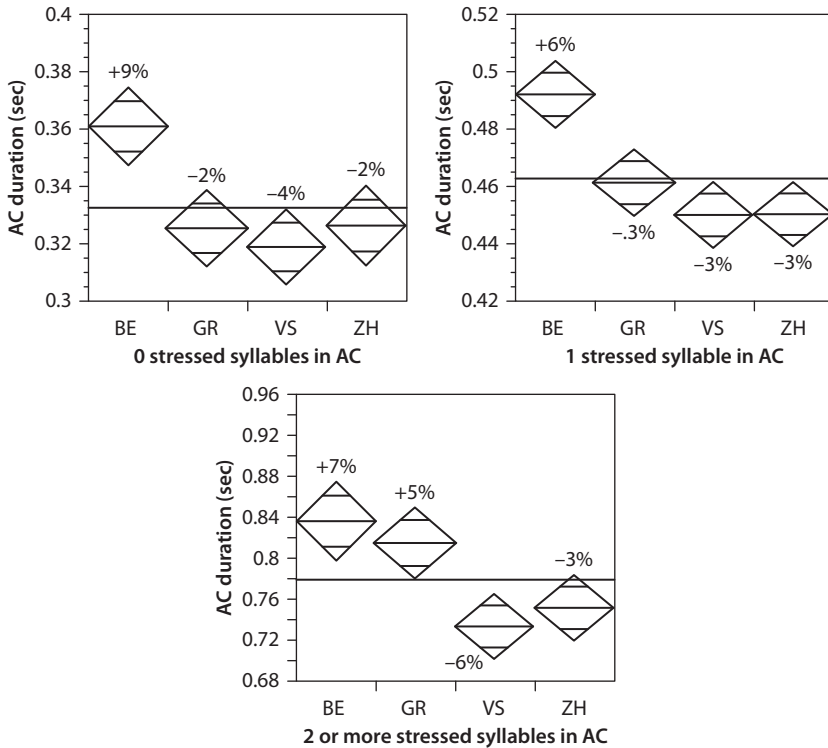


Figure 9.3. Mean diamonds plots showing the dialects' AC durations according to different AC types

All ANOVAs were significant.³⁸ In ACs with 0 stressed syllables, the BE show the longest (+9%), the VS the shortest durations (-4%). The BE group behaves significantly different from all other dialects.³⁹ In ACs with 1 stressed syllable, we find a similar cluster as in ACs with 0 stressed syllables. Here, however, the VS and ZH indicate nearly equally short AC durations.⁴⁰ In ACs with 2 or more ACs, we find a somewhat different grouping: the BE and GR indicate fairly long AC durations (+7%, +5%), while it is the VS again which exhibits the shortest durations (-6%).⁴¹

9.1.1.3 AC timing

Again, if the specific number of stressed syllables per AC is used for the analyses for a *Pearson's correlation* between *T1dist_rise* and the number of stressed syllables in the AC for each dialect, a positive correlation is established.⁴² The results suggest that the more stressed syllables in the AC, the later the rise of the AC relative to segment onset. Given this tendency, the *T1dist_rise* values of different AC types according to the dialects are not further investigated. However, the dialects' *T1dist_rise* values according to the different AC types are of interest. In order to test whether the dialects differ with regard to the *T1dist_rise* values of the three different AC types, further ANOVAs were calculated, see Figure 9.4.

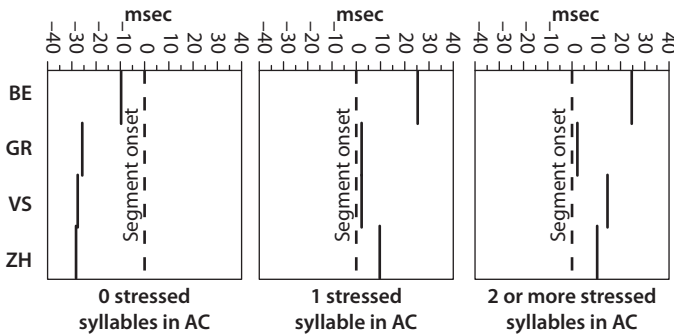


Figure 9.4. Illustration of the dialects' *T1dist_rise* values according to AC types

All ANOVAs were again significant.⁴³ For all AC types, the BE exhibit the latest rises. In ACs with 0 stressed syllables, the GR, VS, and ZH group all behave nearly identically, while for ACs with 1 stressed syllable, only the GR and VS dialects behave similarly (+2ms). In ACs with 2 or more stressed syllables, the GR rise earliest (+2ms), the BE latest (+24ms).⁴⁴

9.1.1.4 Summary and discussion

Distribution

Variable: Most ACs contain only 1 stressed syllable and about 15% of all ACs incorporate 2 or more stressed syllables. Interestingly, however, more than a third of all ACs do not contain any stressed syllable at all.

Firstly, this finding reflects some of the insights put forth by Kochanski et al. (2005) and Silipo and Greenberg (1999) for British English, Irish English, and American English. They found that a great number of stressed syllables in their corpus of spontaneous speech are marked with distinct *f0* movements. Such *f0* movements were also found, however, for an equally great number of unstressed

syllables. They therefore conclude that stressed syllables are louder and longer than surrounding syllables but that f_0 movements alone are weak predictors of stress. In this study, too, f_0 excursions do not necessarily involve stressed syllables since ACs consisting only of unstressed syllables show local f_0 excursions as well. This finding corroborates the meaningfulness of treating f_0 and stress as separate variables, as applied in the present study (see Section 6.4.3).

Secondly, the above result may further serve as evidence of what is frequently found in the literature on both Swiss German (see Hegetschweiler 1978:24) as well as Swiss High German intonation (Ulbrich 2005:230): Swiss German default accents often demonstrate a low f_0 in an otherwise stressed syllable, and a high f_0 in subsequent, otherwise unstressed syllables. This incongruence between stress and pitch has been observed particularly for the Alpine varieties.² The first AC in the following Figure provides an example of this phenomenon:

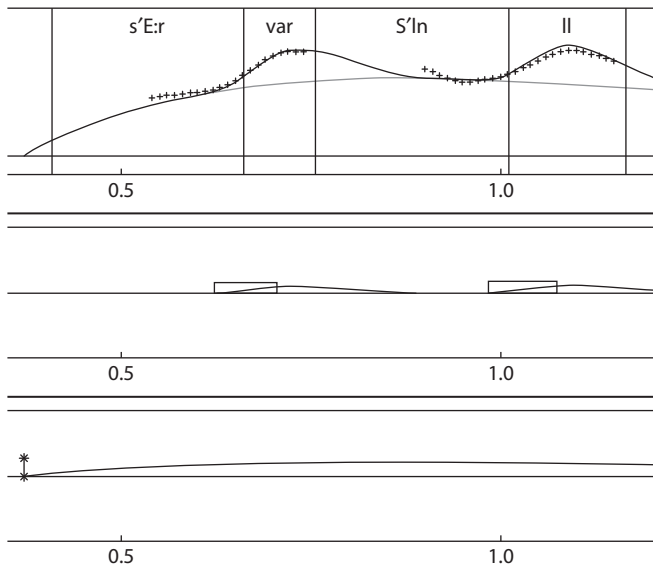


Figure 9.5. f_0 peaks on metrically weak syllables in the VS variety

The excerpt reads *s'E:r var S'In li* (German: *sehr wahrscheinlich*; Engl.: *very likely*). The first stressed syllable *s'E:r* is comparatively low in pitch, while the following unstressed syllable *var* carries a high f_0 . This syllable is followed by a low pitch, stressed syllable, which again is followed by a high pitch unstressed

2. For the VS dialect, see Baumgartner (1922:23) and Wipf (1910:22), for the GR dialect see Brun (1918:28) and Meinherz (1920:38).

syllable *li*. While the T_1 of the first AC is linked with *s'Er* – because the rise begins at the very end of the syllable *s'E:r* – the T_1 of the second AC is linked with *li*, since all the measured f_0 points in the syllable *S'In* are captured by the phrase component. The second AC is thus an example of an AC without any stressed syllables.

AC amplitude

Variable: Overall, the highest amplitudes are found in ACs that contain 1 or more stressed syllables. If the AC does not contain any stressed syllables, it is generally lower in amplitude.

This finding underlines that in the stream of speech, metrical stress can cause higher f_0 excursions, a result that was anticipated given the vast amount of literature on acoustic correlates of stress (see Isačenko & Schädlich 1970; Vaissière 1983; Lieberman 1960 etc.). If we take into consideration the above findings that more than a third of all ACs do not contain any stressed syllables, yet are modeled by means of ACs, we can conclude that even though f_0 excursions *may* be caused by stressed syllables, this *need not* necessarily be the case. Thus, the results of this study concur with Kochanski et al.'s (2005) findings that f_0 is a weak predictor of stress. What is characteristic about f_0 excursions caused by stressed syllables, however, is that they are higher in amplitude than f_0 excursions for ACs without stress. In sum, the current findings suggest that metrical stress does not have to be accompanied by local movement (ACs), yet in case it is, stress seems to cause distinctly higher AC amplitudes.

Dialect behavior: The Midland varieties, and BE in particular, strongly distinguish between low amplitude ACs with 0 stressed syllables and high amplitude ACs with 1 stressed syllable. The Alpine varieties, on the other hand, make much less of a distinction in amplitude between the three AC types.

While all dialects show approximately the same proportions of ACs with 0 stressed syllables, we find that the differences in amplitude between ACs with 0 stress and ACs with 1 or more stressed syllables are more distinct for the Midland variety than the Alpine variety. We find a striking North-South divide with the Alpine showing similar amplitudes for all AC types, regardless of whether the AC contains 0 or several stressed syllables. This ties in with Wipf's (1910:22) observation that unstressed syllables in VS Swiss German can also carry higher tones, and Meinherz' (1920:38ff.) remark that weak syllables in the Grison dialect often carry higher pitch accents than highly dynamic ones. In contrast, we find a distinct difference between 0 stress ACs (low amplitude ACs) and ACs with 1 or more stressed syllables (much higher amplitudes) for the Midland varieties. In other words, the

contribution of metrical stress to AC amplitude seems to play a more critical role in the Midland varieties, in particular, in the BE group (see Chapter 13).

AC duration

Dialect behavior: We find the same picture in nearly all AC types: the BE show by far the longest durations, the VS and ZH the shortest.

This finding suggests that increased AC duration is a typical feature of the BE dialect. It is very likely that this characteristic has to do with the overall slow articulation rate of this dialect group, which will be discussed at a later stage (see Section 11.1). The BE's slow articulation rate can be attributed to longer mean duration of vowels as well as more distinct phrase-final lengthening (see Leemann & Siebenhaar 2007). Long segments are likely to mirror more distinct AC duration.

AC timing

Variable: Overall, we find that the more stressed syllables in the AC, the later the rise relative to segment onset.

A possible interpretation may include the comparative saliency of late rises. If a speaker intends to mark a certain AC as more salient, s/he is not restricted to merely applying amplitude properties but can also make use of the timing of AC onset relative to the segment. As Gussenhoven (2002: 53) argues, later rises, which often reach their peaks late in the first syllable or as late as in the adjacent syllable, are more prominent and elicit more "unusual occurrence" interpretations. Moreover, late peaks are commonly perceived as if they were higher (*ibid.*). Accordingly, the more stressed syllables in the AC, the more marked the timing of the AC compared to other AC types.

Secondly, this finding may involve the positive correlation of *T1dist_rise* with AC duration: ACs are generally longer the more stressed syllables they contain. Thus, we can reformulate the above condition as follows: the *longer the AC*, the later the rise relative to segment onset. Now, the argumentation for a link between T_1 and AC duration goes as follows: from the viewpoint of execution time, long ACs provide the speaker with sufficient time available to realize a fully-fledged *f0* contour *and* reach the pitch peak in time, even with a rise that starts after the segment onset of the first syllable in the AC. In shorter ACs, on the other hand, which typically happen to be ACs with 0 stressed syllables, less time is available which may cause a forward shift of T_1 in the temporal domain to compensate for the comparatively short time span given to realize the *f0* contour and reach the pitch peak in a given segment.

Dialect behavior: In all instances, the BE demonstrate much later rises than the other dialects, particularly the GR dialect.

The comparatively late T_1 s relative to segment onset may be a correlate of long segment durations in the BE group (see Leemann & Siebenhaar 2007): even if T_1 occurs after segment onset, a pitch peak realization within a long target segment is still possible. However, if the segments are shorter, the larynx needs to anticipate this duration and adjust accordingly (earlier T_1) so as to guarantee a peak realization on the target segment.

9.1.2 Position of first stressed syllable in AC

For the majority of the present analyses, the interest lies in establishing the differences between ACs with stress in *0 syllable position*³ (no stress), *first syllable position*, and *second or later syllable position*.⁴ For this reason three categories are established and the respective ACs are grouped together; all ACs with stress on the second or later syllable are all pooled together into one group. Generally, the *position of first stressed syllable in an AC* is understood as a measurement variable in nature, since the interval between the measurements is meaningful and constant (1 syllable). In the current analysis, however, due to the inclusion of 0 position stress and the pooling of ACs with stress on the second or later syllable, this variable is treated as ordinal.

In order to test whether there is a difference between the four dialects in the relative proportions of ACs without stress, ACs with stress in first syllable position, or a stressed syllable in second or later syllable position, a chi-square test of independence was performed. The test confirmed significant differences between the dialects.⁴⁵ As for ACs with no stressed syllable, the BE exhibit 36%, the VS group 35%, GR 34%, and ZH 31%. The percentage of ACs with first syllable stress for the ZH group is 47%, BE and GR 46%, and VS only 42%. What is interesting is the distribution in ACs with post-first syllable stress: the BE show the least ACs (18%) and the VS the most (23%). GR (20%) and ZH (22%), on the other hand, are located between these two extremes.

3. Note that the first level of this variable, *0 position stress*, is collinear with the level *0 stressed syllables in the AC*.

4. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

9.1.2.1 AC amplitude

In order to test whether, in the different dialects, the AC amplitudes of the above mentioned AC types differ significantly, ANOVAs were calculated, see 9.6.

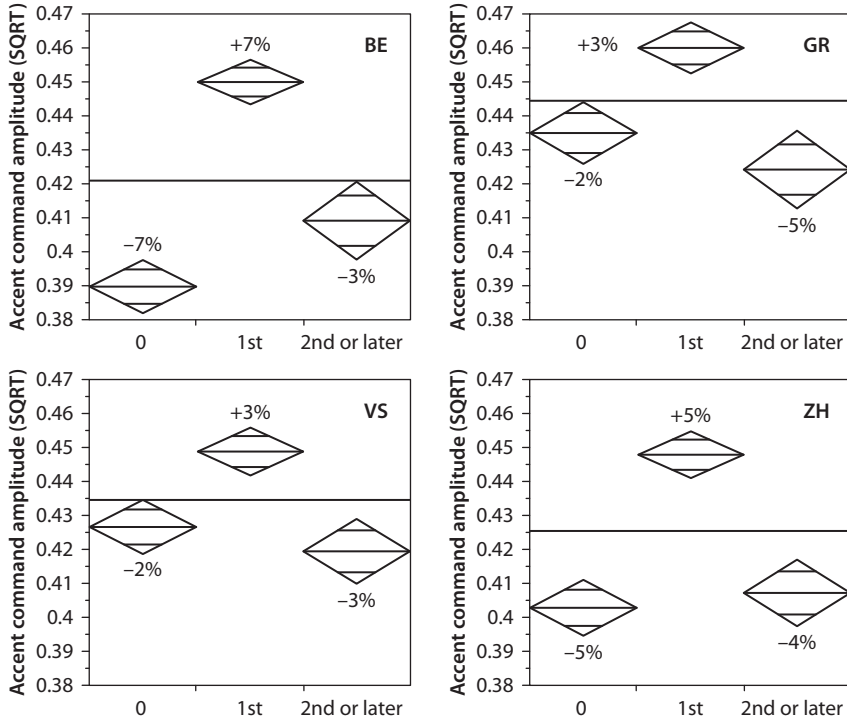


Figure 9.6. Mean diamonds plots showing the different AC types' amplitudes according to dialect. The percentages indicate the in-/decreases compared to the corresponding means of response

All ANOVAs were significant.⁴⁶ In the BE group, ACs with first syllable stress show by far the highest amplitudes (+7% from mean of response), while ACs with 0 stressed syllables are by far the lowest (-7%).⁴⁷ A similar trend can be observed for the ZH group, for which not all inter-level differences turned out to be significant, however.⁴⁸ The GR and VS speakers demonstrate a slightly different pattern: here, while the highest amplitudes are also found in ACs with first syllable stress, the lowest amplitudes fall on ACs with stress in second or later syllable positions.⁴⁹

Cross-dialectal differences

In order to test whether the dialects differ among each other with regard to the amplitudes of the three different AC types, further ANOVAs were calculated. This

time, the ANOVAs are run with regard to the dialects' amplitudes according to different AC type, see Figure 9.7.

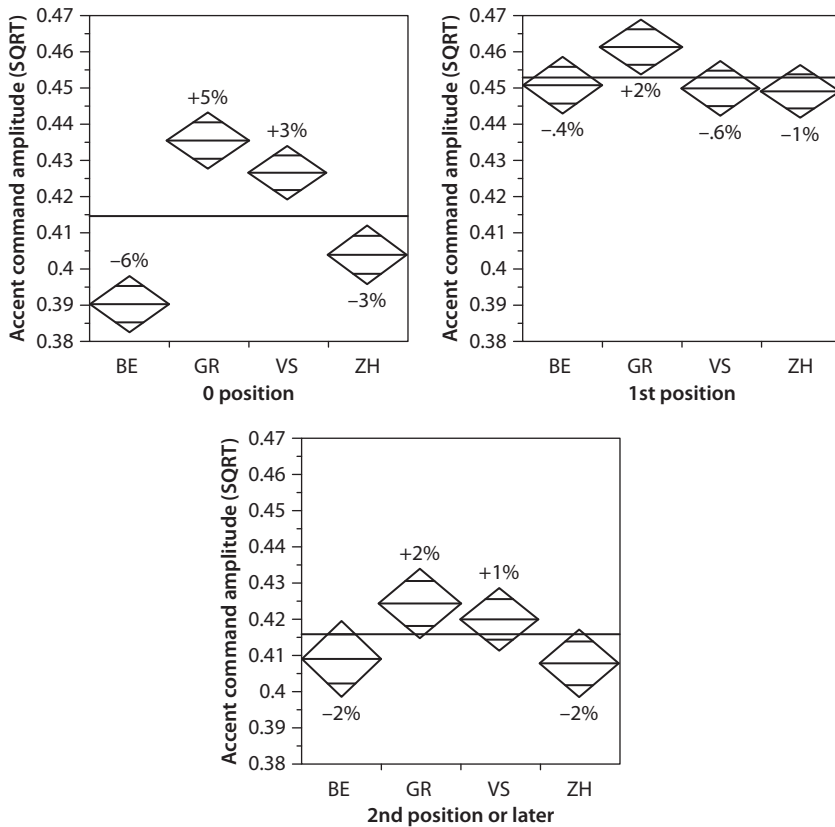


Figure 9.7. Mean diamonds plots showing the dialects' AC amplitudes according to different AC types. The percentages stand for the in-/decrease when compared to the corresponding means of response

The ANOVAs show significant effects for the amplitudes of ACs with 0 position stress as well as for ACs with post-first syllable stress.⁵⁰ In ACs with 0 stressed syllable position, the GR demonstrate the highest (+5%) and the BE the lowest amplitudes (-6%). All inter-dialectal differences are significant, except for the ones between VS and GR.⁵¹ In ACs with first syllable stress, the GR show the highest (+2%) and the ZH the lowest amplitudes (-1%), yet, overall, all dialects show fairly similar amplitudes. In ACs with stress in second syllable position or later, again the GR exhibit the highest (+2%), the ZH and BE groups the lowest amplitudes (-2%). The differences between GR and BE as well as between ZH and GR are significant.⁵²

9.1.2.2 Summary and discussion

Distribution

Variable: The position of stress in most ACs containing stress, is on the first syllable (approx. 48%), while about one third of all ACs do not contain any stressed syllables at all. About 21% of the ACs carry stress on the second syllable position or later.

These findings show that ACs were modeled, i.e. f_0 movements occurred, whether or not there was an underlying stress and whether or not the stress occurred in any certain position. However, for most ACs found in this corpus, there is stress and it seems to be placed on the first syllable in the AC. Hence, stress on the first syllable seems to trigger f_0 movements more than no stress or stress in a different position in the AC. On a separate note, it should be analyzed further whether the above finding may be due to frequency effects, considering that Swiss German is a first syllable stress language, more so than High German (see Löttscher 1983: 89).

AC amplitude

Variable: Overall, ACs with stress on the first syllable exhibit by far the highest amplitudes, while the amplitudes for the other two AC types are both fairly low.

As was already hypothesized above, first syllable stress seems to trigger distinctively high f_0 movements. The high AC amplitudes on ACs with first syllable stress may be due to general phonetic/phonological properties. In order to examine whether such intrinsic properties are indeed the main reason for the increased f_0 amplitude, one would also have to consider language-specific stress patterns as well as semantic/pragmatic prominence placement. In other words, one would have to consider all the specific segments concerned.

Dialect behavior: All dialects show distinctly high amplitudes in ACs with stress on the first syllable. However, this holds true especially for the Midland varieties. Furthermore, the Midland varieties show very low amplitudes in ACs with 0 stress, while the Alpine varieties exhibit higher amplitudes.

This finding ties in with the interpretation regarding the relationship between AC amplitude and the number of stressed syllables in an AC (0, 1, 2 or more). We found that the differences in amplitude between ACs with 0 stress, on the one hand, and ACs with 1 or more stressed syllables, on the other hand, are distinct for the Midland variety, yet very low for the Alpine variety. In the same way, it seems that the differences regarding the position of stress are somewhat more distinct for the Midland variety, yet lower for the Alpine variety. This supports the assumption that, for Alpine varieties, stress seems to have a more moderate effect on local f_0 movements (see Chapter 13).

9.2 Word class

9.2.1 Number of lexical syllables in AC

Words were either labeled as lexical or grammatical.⁵ For the majority of the present analyses, the interest lies in establishing the differences between ACs with *0 lexical syllables*, *1 lexical syllable*, and *2 or more lexical syllables*. For this reason, three categories are established and the respective ACs are grouped together, whereas all ACs with 2 or more lexical syllables are pooled together into one group. Generally, the *number of lexical syllables in an AC* is understood as a measurement variable in nature, since the interval between the measurements is meaningful and constant (1 syllable). In the current analysis, however, due to the inclusion of 0 lexical syllables and the pooling of ACs with 2 or more lexical syllables, the interval between the measurements of this variable is no longer meaningful and constant and is thus treated as ordinal.

In order to test whether there is a difference between the dialects regarding the relative proportions of ACs with 0, 1, and 2 or more lexical syllables, a chi-square test of independence was performed. This test confirmed significant differences between the dialects.⁵³ The percentages for ACs with 0 lexical syllables are as follows: 21% for BE, 21% for GR, only 17% for VS, and 20% for ZH. As for ACs with only 1 lexical syllable, the BE group exhibits 35%, the GR 33%, the VS only 32%, and the ZH 30%. For ACs with 2 or more lexical syllables, we find the following distributions: BE 44%, GR 46%, VS 51%, and ZH 50%.

9.2.1.1 AC amplitude

In order to test whether, overall, AC amplitudes differ significantly for ACs with 0, 1, and 2 or more lexical syllables for each dialect, ANOVAs were calculated, see Figure 9.8.

All ANOVAs confirmed significant effects.⁵⁴ In all dialects, the AC amplitudes are clearly lowest in ACs with 0 lexical syllables. For the BE, GR, and VS, the highest amplitudes are found in ACs with 2 or more syllables. The ZH group, on the other hand, has the highest amplitudes in those ACs with exactly 1 lexical syllable (+3% from mean of response). Several inter-level differences are significant: for the BE and the VS groups, all inter-level t-tests yielded significant effects.⁵⁵

5. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

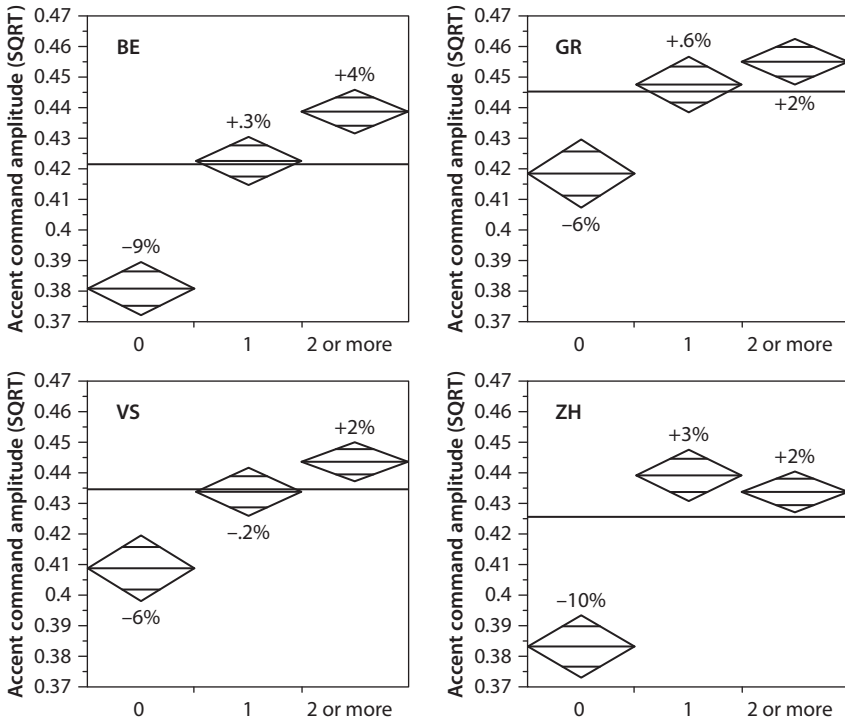


Figure 9.8. Mean diamonds plots showing the different AC types' amplitudes according to dialect

Cross-dialectal differences

In order to test whether the dialects differ with regard to the amplitudes of the three different AC types, further ANOVAs were calculated, see Figure 9.9.

All ANOVAs are significant.⁵⁶ In ACs with 0 lexical syllables, the GR and VS speakers show the highest (+5%, +3%), the BE and ZH clearly the lowest amplitudes (-4%).⁵⁷ In ACs with 1 lexical syllable, the GR again show the highest amplitudes (+3%), the BE, once more, the lowest amplitudes (-3%).⁵⁸ In ACs with 2 or more lexical syllables, the GR again exhibit the highest amplitudes (+3%), this time, however, the amplitudes are lowest for the ZH group (-2%).⁵⁹

9.2.1.2 Summary and discussion

Distribution

Variable: About half of all ACs contain 2 or more lexical syllables, 30% contain only one lexical syllable, and approx. 20% consist of only grammatical syllables.

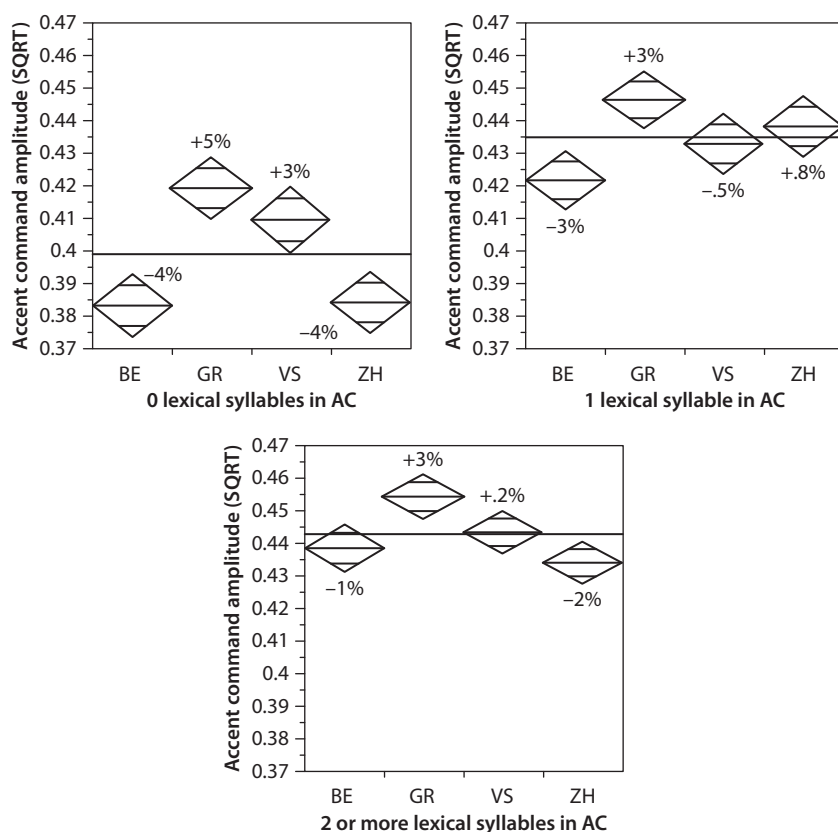


Figure 9.9. Mean diamonds plots showing the dialects' AC amplitudes according to different AC types

This distribution indicates that ACs can generally fall on any syllables, regardless of whether the syllables are part of lexical or grammatical words. Yet, in 80% of all ACs, at least one lexical syllable is present. The finding that lexical words, mainly proper nouns, are particularly marked with ACs has been confirmed also for German (Möbius 1993; Mixdorff 1998) and British English (Fujisaki et al. 1995). Clearly, frequency effects play a certain role in this distribution since, overall, almost two thirds of all syllables produced in the present corpus are lexical syllables. Nevertheless, it is rather remarkable that every fifth AC would fall on syllables that only belong to grammatical words, which are inherently unstressed, except if in focus. It may be the case, of course, that, in these instances, the speaker placed a narrow focus on these constituents, rendering them highly *f₀* prominent.

There are two ways of explaining these distributions. On the one hand, the dialects under scrutiny may not make a rigorous distinction between lexical and

grammatical syllables in terms of AC placement. On the other hand, the speakers may have placed a great number of focal accents on grammatical words. The second hypothesis has to be rejected since, overall, only 7% of the ACs that do not contain any lexical syllables are in focus. Hence, it seems more plausible to assume that, while lexical syllables generally attract ACs, the dialects under scrutiny also place *f*₀ prominence on grammatical syllables. This finding ties in nicely with Radej's (2000) and Pilch's (1977b) observations of an intonational peculiarity of Swiss German, namely that one often encounters high *f*₀ movements on particles, despite their comparatively low semantic content.

AC amplitude

Variable: The overall trend is as follows: the more lexical syllables in an AC, the higher the amplitude.

This finding underlines the more distinct semantic and communicative weight of lexical words, which is signaled by higher *f*₀ movements. As noted earlier, the same tendencies have been found for German (Möbius 1993; Mixdorff 1998) and English (Fujisaki et al. 1995).

Dialect behavior: All dialects follow the above trend. The differences in amplitude are more distinct for the Midland varieties, however, in particular for the ZH dialect. The amplitude of ACs with 0 lexical syllables is significantly lower than the amplitude of ACs with 1 or 2 or more lexical syllables. In a cross-dialect comparison, the VS and GR dialects stand out in that they exhibit much higher amplitudes in ACs with 0 lexical syllables when compared to the Midland varieties.

While all dialects distinguish between high amplitude ACs with 1 or more lexical syllables and low amplitude ACs with only grammatical syllables, the distinctions are less elaborate in the Alpine varieties. Once more, we encounter an Alpine – Midland divide. As was the case with stress, it seems that word class, too, seems to have less of an effect on local *f*₀ contours in the Alpine dialects. Accordingly, the Alpine varieties exhibit much higher amplitudes than the Midland varieties for ACs with grammatical syllables only. Both Wipf (1910:22, for VS) and Meinherz (1920:38ff, for GR) observed that these varieties commonly show high tones in unstressed syllables which, if placed in the context of the present study, explains the high amplitudes in ACs with only grammatical syllables: grammatical words are by assumption unstressed.

Paralinguistic variables

10.1 Focus

10.1.1 Accent component

Descriptive statistics

As mentioned in the annotation section (see 6.4.3), of all the 4823 IPs, 51.2% are narrow focus IPs, while 48.8% are broad focus IPs. In the narrow focus condition, approximately 25 syllables per speaker were perceived as focused even though they were not assigned an AC.¹ Of all ACs in narrow focus IPs, 13% were labeled as no focus ACs (phrase-initial position), 17% as pre-focus ACs (which immediately precede the focused constituent), 47% as focus ACs, and 23% as post-focus ACs. All dialects show comparatively similar distributions. For the following analyses, broad focus IPs, i.e. normal-stress IPs, were suppressed: all results are based on IPs that contain narrow focus ACs only.

10.1.1.1 AC amplitudes

To test whether the means of AC amplitudes differ between *no focus*, *pre-focus*, *focus*, and *post-focus* ACs, ANOVAs were performed for each dialect, see Figure 10.1. It is crucial to note that these calculations are made regardless of the position of the focus AC in the IP and the number of ACs in an IP.² That is, if an IP only has three ACs and the first one is the focus AC, then the final two are labeled *post-focus*. If an IP has five ACs and the last one is the focus AC, then the first three ACs are labeled *no focus*, the fourth *pre-focus*, and the last one is labeled *focus* AC.

1. These exceptions are not included in the following narrow focus analysis results. See the annotation chapter for further insight into the methods applied for IP and AC focus labeling.

2. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

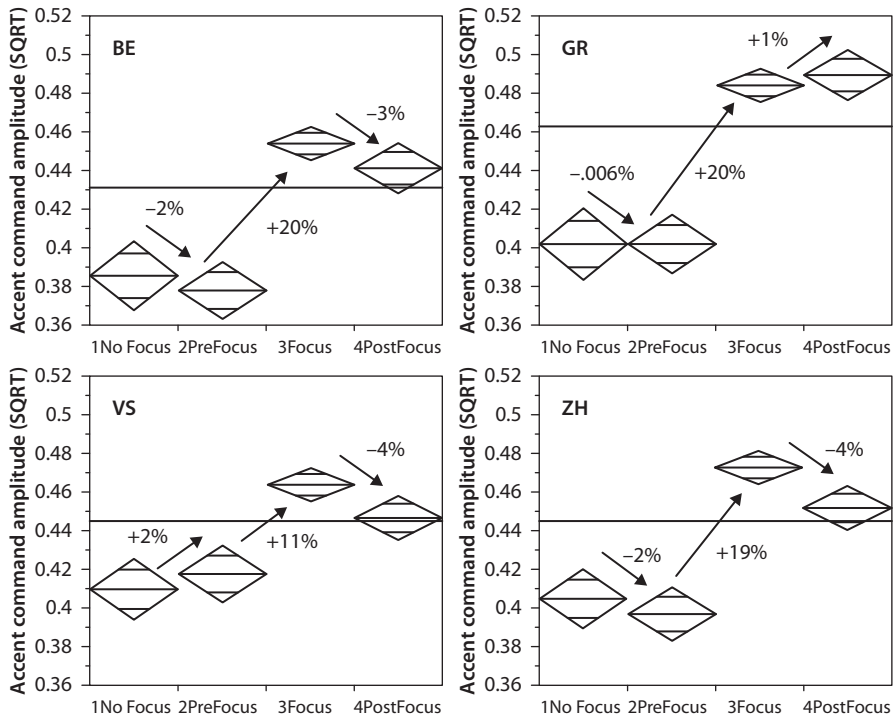


Figure 10.1. Mean diamonds plots showing the amplitudes of the different AC focus types per dialect. The percentages and arrows show the relative increase/decrease in amplitude from position to position, whereas the previous position serves as base

The ANOVAs in the above Figure show significant effects for every group.⁶⁰ In the BE group, we find a fall-rise-fall pattern for AC amplitudes: pre-focus ACs are 2% lower than no focus ACs, the increase from pre-focus to focus ACs is 20%, and then, the amplitude decreased again by 3% from focus to post-focus ACs.⁶¹ In the GR dialect, we find a fall-rise-rise pattern from one focus position to another.⁶² The lowest amplitudes are also found in pre-focus position, the highest, however, occur in post-focus position (+1%), a feature that stands out in comparison to all of the other dialects. No focus and pre-focus ACs show nearly identical amplitudes and the increase in amplitude from pre-focus to focus is +20%, like in the BE group. In the VS dialect, the lowest amplitudes are found in no focus ACs, the highest in focus ACs – unlike the other dialects. The VS behavior for the transition from pre-focus, to focus condition is also striking: they show a less distinct increase of only +11%. Then, for post-focal ACs, the amplitude decreases again (–4%).⁶³ The ZH group exhibits a fall-rise-fall pattern very similar to that of the BE group: the amplitudes in pre-focus ACs are 2% lower than in focus, then they rise

as much as +19%, after which point they decrease again from focus to post-focus AC (−4), just as observed for the VS group.⁶⁴

Cross-dialectal differences

ANOVAs were calculated on the dialects' AC amplitudes according to the different AC focus types. This examines in which AC focus type the dialects vary most in amplitude, see Figure 10.2.

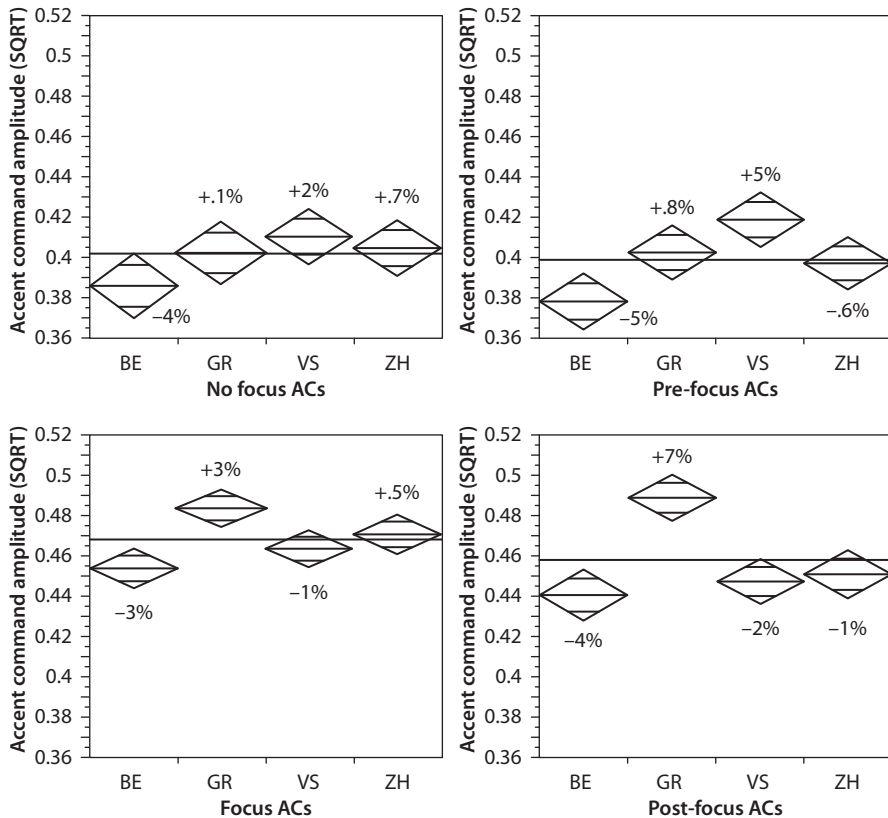


Figure 10.2. Mean diamonds plots showing the dialects' AC amplitudes according to different AC focus types. The variation is calculated based on the mean of response for each AC type

The effects of the pre-focus, focus, and post-focus AC amplitudes in the corresponding ANOVAs are significant, whereas the effects for the no focus condition are not.⁶⁵ In no focus ACs, the VS show the greatest amplitudes (+2%), the BE the lowest (−4%). In the pre-focus position, we find a similar pattern as in the no focus position; here, however, the ANOVA showed significant effects. The VS demonstrate the highest (+5%), the BE the lowest amplitudes (−5%), while

the values for GR and ZH approximate the mean of response. The differences between GR and BE, VS and BE, ZH and BE, as well as ZH and VS are significant.⁶⁶ In focus ACs, the GR demonstrate the greatest (+3%), ZH the second greatest (+5%), VS and BE the lowest amplitudes (particularly BE: -3%). The differences between GR and BE, VS and GR, and ZH and BE are significant.⁶⁷ As for post-focus ACs, the GR exhibit again the highest (+7%), the BE again the lowest amplitudes (-4%). The values for the VS and ZH fall close to the overall response mean, similar to the values for focus ACs. For this instance, the differences between GR and all other dialects are significant.⁶⁸

10.1.1.2 Narrow focus durations

To test whether there is a difference in the means of AC durations for no focus, pre-focus, focus, and post-focus ACs for each dialect, ANOVAs were performed. We find significant effects for every group, as can be observed in Figure 10.3.⁶⁹

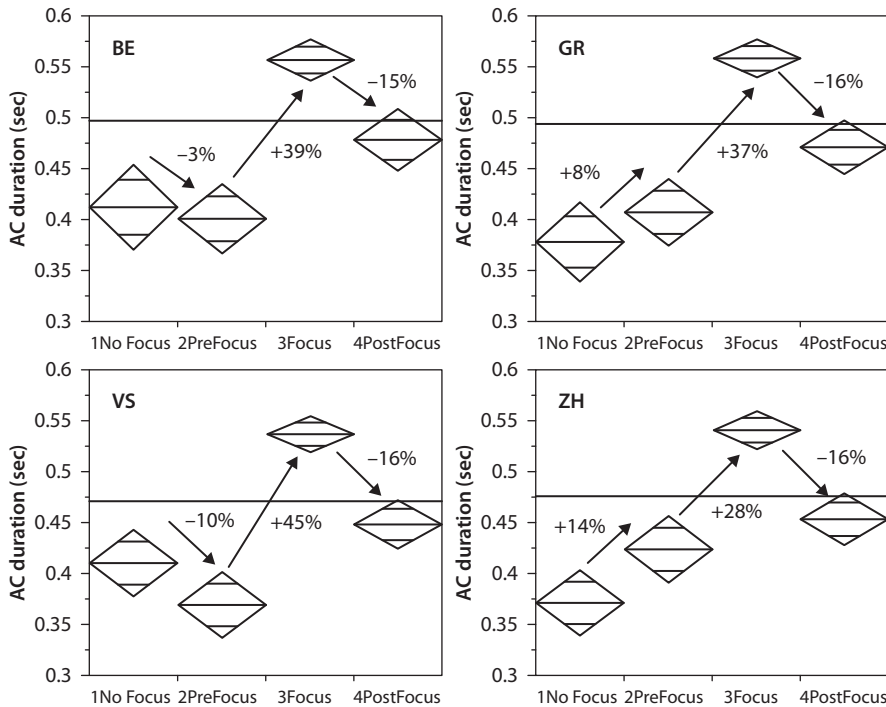


Figure 10.3. Mean diamonds plots showing the different AC focus types' durations by dialect. The percentages and arrows represent the relative increase/decrease in duration from position to position (previous position serves as base)

In the BE group, we find a decrease-increase-decrease pattern in AC duration: pre-focus ACs are 3% shorter than no focus ACs, focus ACs are 39% longer than pre-focus ACs and post-focus ACs are again 15% shorter than focus ACs.⁷⁰ In the GR dialect, we find an increase-increase-decrease pattern from one focus position to another.⁷¹ The shortest durations are found in no focus and the highest in focus ACs, while no focus and pre-focus ACs show a difference in duration of +8%. The increase in duration from pre-focus to focus (+37%) is similar to that of the BE group. In the VS dialect, the shortest durations are found in pre-focus, the highest in focus ACs. The decrease in duration in the first transition is -10%. Most remarkably, however, the increase from pre-focus to focus ACs amounts to +45%. Post-focal ACs, on the other hand, exhibit shorter durations again (-16%).⁷² Finally, in the ZH dialect, we encounter an increase-increase-decrease pattern similar to that observed in the GR variety. The increase in amplitude from no focus to pre-focus ACs is 14%, the highest increase when compared to the other three dialects. In contrast, the increase in duration from pre-focus to focus ACs is smallest with only +28%. Once more, we observed a decrease in duration from focus to post-focus ACs of -16%.⁷³

Cross-dialectal differences

ANOVAs were calculated on the AC durations of the different dialects according to focus types in order to reveal cross-dialectal differences, see Figure 10.4.

The ANOVAs on AC durations show significant effects only for the pre-focus position.⁷⁴ In no focus ACs, the BE and VS, as well as the GR and ZH form similar groups with equally long AC durations. In pre-focus ACs, the ZH clearly exhibit the longest durations (+6%), while the VS show the shortest durations (-8%). The differences between these two dialects are significant.⁷⁵ In focus as well as post-focus ACs, all dialects behave very similarly with regard to AC duration. For post-focus, the durations for the BE (+4) is somewhat longer than for the VS group (-3%), which demonstrate the shortest durations.

10.1.2 Summary and discussion

10.1.2.1 Accent component

AC amplitude

Variable: In narrow focus ACs, we observe the overall tendency that pre-focus AC amplitude is suppressed compared to the amplitude of preceding no focus ACs. Focus ACs, on the other hand, are clearly marked with very distinct amplitudes. Post-focal accents show somewhat lower amplitudes than focal accents, yet they are distinctively higher than no focus and pre-focus accents.

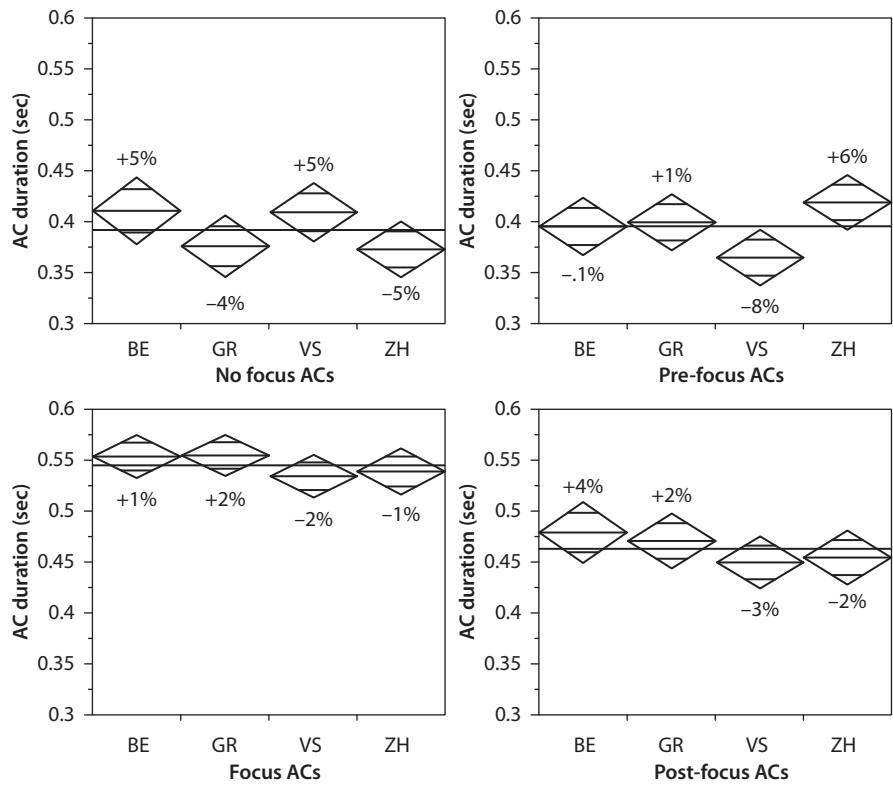


Figure 10.4. Mean diamonds plots showing the dialects' AC durations according to different AC focus types. The deviations in percent are calculated according to the mean of response for each focus type

The result that focal accents show distinctively high AC amplitudes was anticipated and has been observed in other studies on focus in the framework of the Command-Response model (e.g. Mixdorff 1998; Fujisaki 1992). Compressed pre-focal accents, too, were confirmed in other studies (e.g. Botinis et al. 2001). The highly significant difference in amplitude between a slightly compressed pre-focal accent (when compared to preceding no focus accents) and a subsequent, high-pitched focal accent adds even more weight to the saliency of the focal accent. Post-focal accents have been reported to be flattened and low in f_0 (see Xu et al. 2004; Botinis et al. 2001; Cooper et al. 1985). While this finding, too, is partly confirmed by the present analysis, it is interesting to observe that post-focal accents are only shown to be slightly lower than focal accents (in two dialects the differences are not significant). That is, compared to no focus ACs, focus greatly boosts the f_0 contours in post-focus ACs.

Dialect behavior: Differences between focus AC types: The AC amplitudes of the four AC types differ significantly in all dialects. The VS stand out in that their pre-focal accents are higher in amplitude than the no focus accents. Furthermore, the VS show only a minor difference in amplitude for pre-focus and focus accents. Interestingly, the post-focal accents of the GR group are higher in amplitude than the focal accents.

Cross-dialectal differences: In all AC types, the dialects show different AC amplitudes, except for no focus ACs. The BE show particularly low pre-focal, focal, and post-focal accents. On the contrary, the VS exhibit especially high pre-focal accents. The GR stand out in that they exhibit particularly high post-focal accents.

The VS is the only dialect in which pre-focal accents are not suppressed but, on the contrary, are boosted by approximately 2% if compared to no focus accents (base). Consequently, the differences in amplitude between pre-focus and focus ACs are only minor in the VS group. As mentioned earlier, in the VS group declination, too, seems to occupy a crucial role for focus marking, which may be a reason why the different ACs in narrow focus IPs differ less than in the BE, the ZH, or the GR groups. A further interesting fact is that focus ACs in the GR variety seem to have a particularly boosting effect on the amplitudes of post-focal accents.

AC duration

Variable: Overall, focus ACs are longer than no focus, pre-focus, and post-focus ACs. Post-focus ACs are the second longest ACs.

The result of an increased duration in focal accents, as well as the lengthening of subsequent segments, has already been established for High German (see Mixdorff 1998a: 92ff.). Also, Heldner (1996) and Bauer et al. (2004) emphasize the role of duration in focus marking for Swedish and Hong Kong Cantonese and assign only a minor role to *f₀*. As shown in the above subsection, focus greatly boosts the AC amplitude of post-focal accents. This also applies to AC duration, yet to a lesser extent. As for the reduction in the duration of pre-focal accents: this phenomenon occurs only in two dialects (the Western dialects). The Eastern varieties behave conversely: the focused constituent boosts the duration of the pre-focus AC.

The highly significant ANOVAs on the AC durations of each dialect's different AC focus types, as well as the inter-level comparisons with the t-tests, suggest that duration plays a critical role for focus marking in Swiss German. As shown above, the same calculations were also highly significant for AC amplitudes behavior, yet only 18 of the inter-level t-tests were significant (for duration, 20 inter-level t-tests confirmed significant results). Whether or not this suggests that duration is a more critical parameter for focus marking is certainly an interesting question for future research on focus in Swiss German.

Dialect behavior: Differences between focus AC types: we find highly significant differences between the four AC types in all dialects. The VS stand out in that they strongly differentiate between no focus AC and a much shorter pre-focus AC. The pre-focus AC is again followed by a quite distinctly longer focus AC.

Cross-dialectal differences: It is striking how both Eastern varieties show a decrease-increase-decrease pattern for AC duration in the course of a focus IP, while the Western varieties clearly show an increase-increase-decrease pattern. The only ACs in which the dialects differ are pre-focus ACs. The VS show by far the shortest, the ZH the longest durations.

We find that the pre-focal accents in the VS exhibit a special behavior. It seems as though the primary means for marking focus in the VS dialect is AC duration: no other dialect exhibits such distinctly different AC durations for focus and pre-focus, as well as for no focus and pre-focus AC durations. These results on focus and AC duration further the support the notion of an East-West divide, see Chapter 13.

10.2 Phrase type

Descriptive statistics

The descriptive statistics show that, overall, 66% of all phrases are *continuing phrases* [*C phrases*] and 33% are *terminating phrases* [*T phrases*], and only 1% of the phrases are labeled as *question phrases* [*Q phrases*].³ Q-phrases usually came about if the interviewee was unclear as to the question asked by the interviewer. However, with only 38 phrases (out of a total of 4829 analyzed IPs), Q-phrases were not included in the following statistical analyses.

If we look at the phrase type distributions per dialect, we obtain the following frequencies:

BE:	69%	continuing
	30.6%	terminating
	0.4%	questions
GR:	64.1%	continuing
	35%	terminating
	0.9%	questions
VS:	66%	continuing
	32.2%	terminating
	0.8%	questions

3. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

ZH: 66% continuing
 33.3% terminating
 0.7% questions

These distributions confirm that there is comparatively little variation among the different dialects regarding the relative proportions of phrase types.

10.2.1 Phrase component

10.2.1.1 PC magnitude

To test whether the PC magnitudes of C and T phrases were significantly different for each dialect, student's t-tests were calculated.⁴ In all dialects, the PC magnitudes for C phrases are higher than for T phrases. In every dialect except the GR, the differences in PC magnitude according to phrase type are significant.⁷⁶ In Figure 10.5, the differences in PC magnitudes for the different dialects are indicated and the PC magnitudes of the C phrases are compared to those of the T phrases (base).

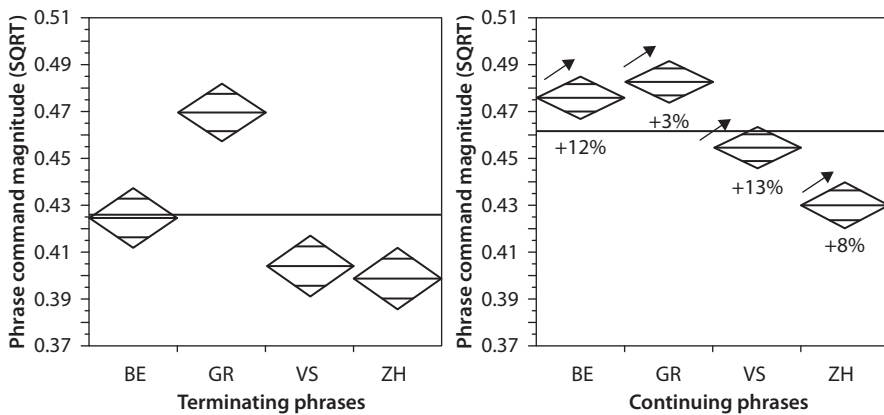


Figure 10.5. Comparison of PC magnitudes for C and T phrases per dialect. The percentages on the right indicate the increase in PC magnitude for C phrases as compared to the T phrase condition (base)

As can be observed in the above Figure, the greatest increases in magnitude are found for the BE (+12%) and the VS dialect (+13%). This means that increase

4. In these calculations, the PC magnitudes of IPs with varying numbers of ACs were compared, which means that the magnitudes were compared regardless of the number of ACs in the IPs. The number of ACs for a single IP in the current study ranges from 1 to 16, but on average, an IP consists of 2 to 3 ACs. 28% of all IPs contain only 1 AC, 25% contain 2, 21% contain 4 or more, 16% contain 3, and 10% contain no AC.

in declination for these two dialects in C phrases is 12% and 13%, respectively. The GR make almost no distinction between the two conditions (+3%), and the ZH exhibit an 8% increase in magnitude.⁷⁷

Furthermore, the dialects' PC magnitudes can be compared according to C and T phrases in order to reveal the cross-dialectal differences in declination, see Figure 10.6.

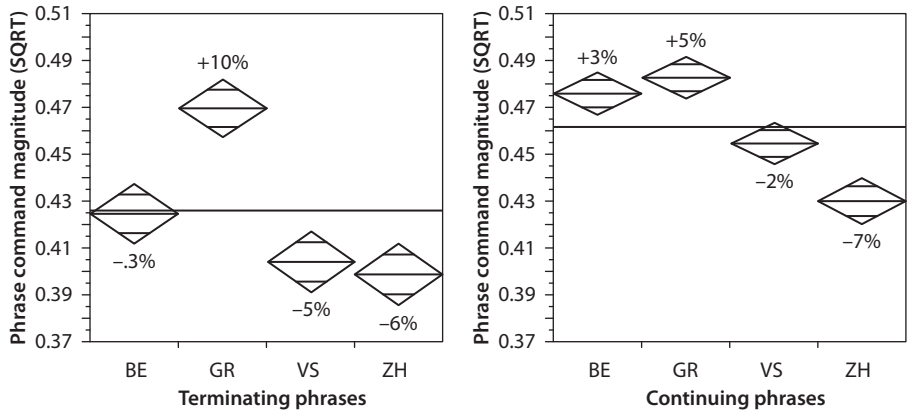


Figure 10.6. Mean diamonds plots showing the dialects' PC magnitudes according to phrase type. The percentages indicate the deviations between the mean of response (base) and the corresponding dialect

Both ANOVAs are significant.⁷⁸ Compared to the mean of response for the C phrase condition, the GR exhibit the highest amplitude values (+5%), followed by BE (+3%) and VS (-2%), while the ZH group shows the lowest PC magnitude values (-7%). The differences between GR and ZH, BE and ZH, GR and VS, VS and ZH, as well as between BE and VS are significant.⁷⁹ As for the T condition, the GR have by far the highest magnitudes (+10%), BE is just about at the overall mean (-3%), and the VS (-5%) and ZH (-6%) groups exhibit the lowest magnitudes. The differences between GR and BE, VS and BE, VS and GR, ZH and BE, and ZH and GR are significant.⁸⁰

10.2.1.2 PC duration

To test whether the IP durations of C and T phrases differ for the dialects, student's t-tests were again calculated.⁵ For all dialects, the IPs are longer for T phrases than for C phrases, yet the differences are significant in only two of the dialects, namely

5. Again note that in these calculations, PC durations of IPs with varying numbers of ACs were compared, meaning that the magnitudes were compared regardless of the number of ACs in the IPs.

BE and VS. For GR and ZH, the differences are not significant.⁸¹ Figure 10.7 shows this comparison with arrows and percentages – where C phrase PC duration is compared to T phrase PC duration (base) – for every dialect.

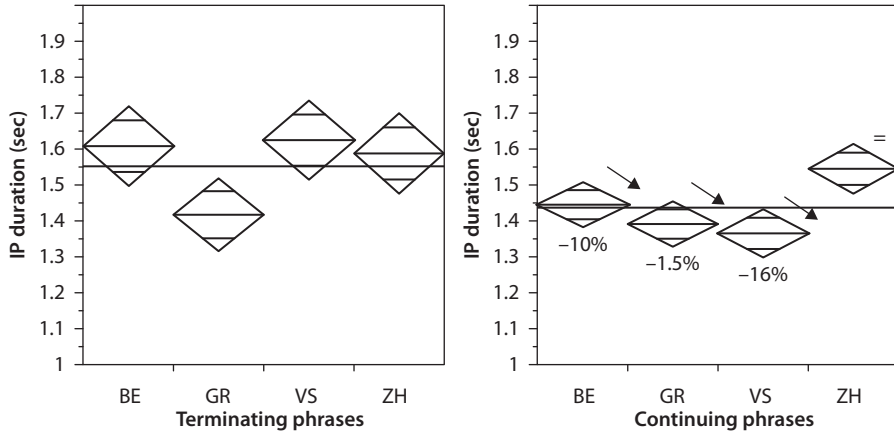


Figure 10.7. Mean diamonds plots showing the dialect's IP durations according to phrase type. The percentages indicate the decrease of IP duration for C condition if compared to the T condition (base) ('=' is used for differences < 3%)

Figure 10.7 indicates that, for C phrases, the greatest decrease in duration if compared with T phrases as a basis can be found in the VS dialect (-16%). The ZH group makes almost no distinction between the two conditions (< 3%), meaning that they exhibit almost identical durations for both C phrase and T phrase IPs.⁸²

The IP durations according to C and T phrases, if compared between the dialects, are as follows:

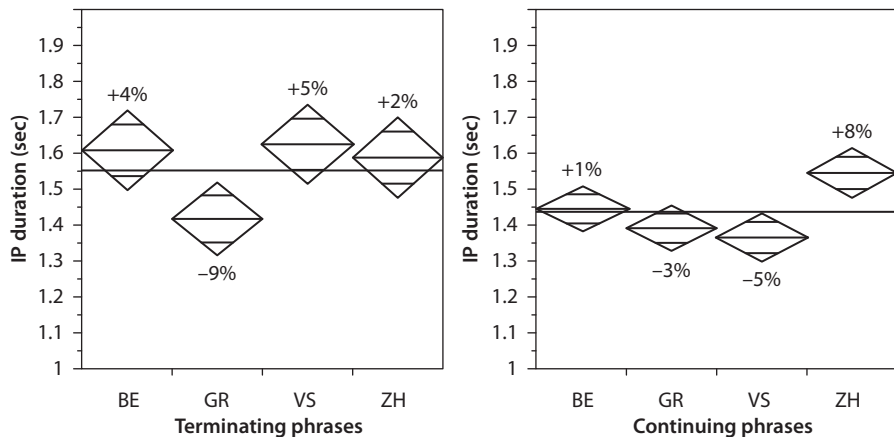


Figure 10.8. Mean diamonds plots showing the dialects' PC durations according to phrase type. The percentages indicate the deviations of the dialect compared to the mean of response (base)

Both ANOVAs show significant effects.⁸³ In the C condition, the IPs are longest for the ZH group (+8%) and shortest for the VS (−5%). The differences between ZH and BE, ZH and GR, as well as ZH and VS are significant.⁸⁴ In the T condition, on the other hand, the longest IP durations are found in the VS group (+5%) and the GR group has the shortest durations (−9%). The differences between GR and BE, GR and VS, and GR and ZH are significant.⁸⁵

10.2.2 Accent component

10.2.2.1 AC amplitude

Overall

To test whether AC amplitudes in C and T phrases are significantly different for the different dialects, t-tests were applied. These tests reveal that the overall AC amplitudes are significantly lower for T phrases than for C phrases, as can be observed in Figure 10.9.⁸⁶

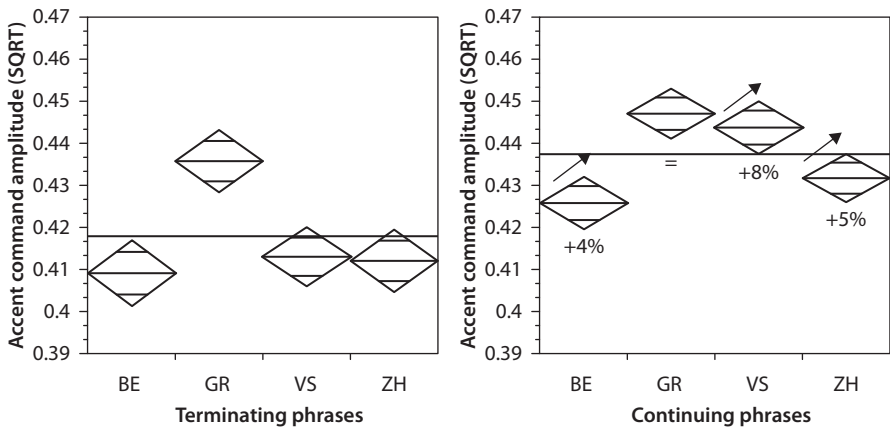


Figure 10.9. Comparison of C phrase and T phrase amplitudes of ACs per dialect. The percentages indicate the increases in amplitude from the T condition (base) to the C condition (“=” is used for differences < 3%)

As the above Figure indicates, the greatest increase in amplitude occurs in the VS group (+8%), while the increase in the GR group amounts to less than 3%. The AC amplitudes for the BE (+4%) and ZH (+5%) group increase only moderately if C phrases are compared to T phrases.⁸⁷

The cross-dialectal differences in prominence marking through *f0* for C and T phrases are displayed in Figure 10.10. For all four dialects, the calculated ANOVAs show the AC amplitudes according to phrase type.

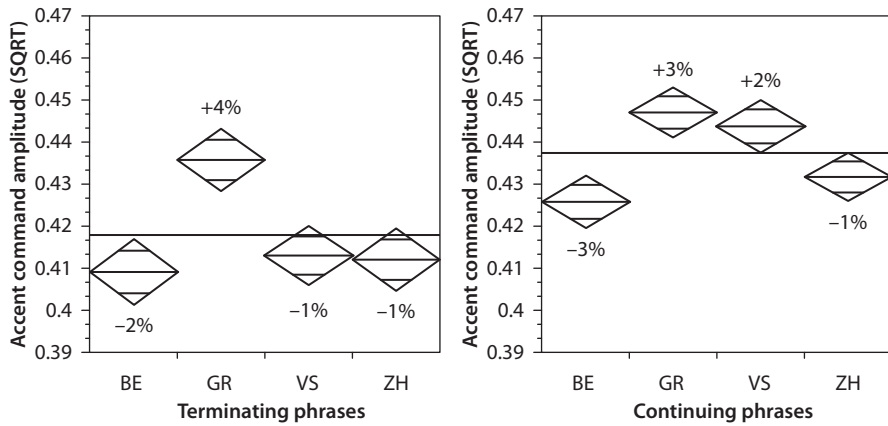


Figure 10.10. Mean diamonds plots showing the dialects' AC amplitudes according to phrase type. The percentages indicate the deviations between mean of response (base) and the corresponding dialect

Both ANOVAs are significant.⁸⁸ In the C phrase condition, the GR demonstrate the highest amplitudes (+3%) and BE the lowest (-3%). The differences between GR and BE, VS and BE, and GR and ZH are significant.⁸⁹ In the T condition, the GR group clearly stands out with a deviation of +4% from the overall response mean, whereas the other three dialects are located at a similar position, slightly below the average mean. The differences between GR and all of the other dialects are significant.⁹⁰

AC sequence

In order to perform a fine-grained analysis of local *f0* movements, ACs in the IP were labeled as *first*, *medial*, *penultimate*, *final* and *only*. This annotation allows for a fine-grained analysis of *f0* representation in various positions within the IP. Particularly the penultimate and ultimate AC positions are of interest when it comes to the interactive function of intonation, indicating continuity or finality (cf. Gilles 2005; Cruttenden 1971). In an IP with only 2 ACs, the first AC is labeled *first* the second *ultimate*. In case of 3 ACs, the first remains *first*, the second is labeled *penultimate*, and the third *ultimate*. In case of 4 ACs in an IP, all four labels *first*, *medial*, *penultimate*, and *ultimate* are assigned. Each additional AC increases the number of *medial* ACs in an IP. Note that IPs on average only contain 2–3 ACs, some IPs contain as few as one, others as many as ten. The following calculations are made regardless of the number of ACs in the IP, that is, all *first* ACs were pooled together, all *medial* ACs were pooled together etc. ANOVAs

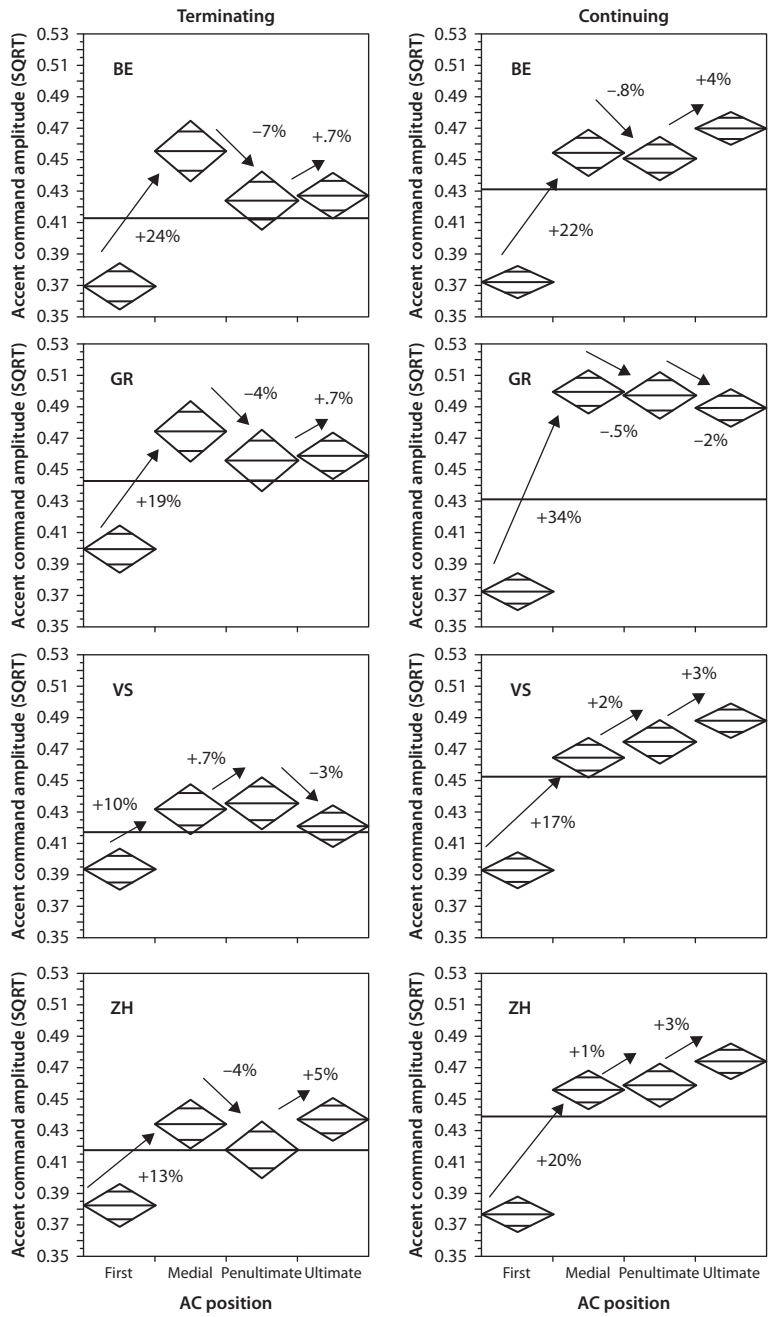


Figure 10.11. Mean diamonds plots showing the different AC positions' amplitudes according to C and T phrases and dialects. The percentages represent the increases or decreases in AC amplitude from position to position, the previous position serving as base

were performed to test for the difference in the amplitude means of first, medial, penultimate and ultimate ACs according to the two different phrase types and four dialects.⁶

For C as well as T phrases, significant effects were found for all dialects by means of ANOVAs on the amplitudes of ACs in different sequential positions,⁹¹ see Figure 10.11, along with the increases or decreases in AC amplitude from position to position: the previous position serves as the base for the calculations.⁹²

As for the T phrases in Figure 10.11, all dialects make a stark distinction between first and medial AC amplitude.⁹³ The BE increase the most (+24%), while the VS increase the least (+10%). The subsequent transition in amplitude from the medial to the penultimate AC is negative for three of the four dialects, increasing only slightly for the VS group (7%).⁹⁴ The transition from penultimate to ultimate phrase is marked by an increase in amplitude for all dialects except the VS group, which demonstrates a decrease of 3%.⁹⁵

In terms of the C phrases, the GR group stands out because of its high jump from the rather low, IP-initial amplitude to the strikingly high, IP-medial AC amplitude. The VS make the least distinct transition with an increase of only 17% from first to medial AC.⁹⁶ As for the transition from medial to penultimate AC, the BE and the GR groups decrease in amplitude, while the VS and the ZH behave oppositely.⁹⁷ The transition from penultimate to ultimate AC in C phrases is characterized by an increase in all dialects except for GR, which is marked by a decrease of 2%.⁹⁸

Comparing T and C phrases

Figure 10.12 provides a comparison of the amplitudes of ACs with regard to their position in the IP. The AC amplitudes for C phrase ACs in various positions are compared to the corresponding T phrase amplitudes, sorted according to dialects. Such a comparison reveals whether, for C and T phrases, a certain dialect exhibits differences in amplitude for the *first* AC of an IP, for instance.

With regard to IP-initial AC amplitudes, three of the four dialects do not exhibit significant differences between C and T phrases. For the GR dialect, the first AC amplitudes for C phrases are 7% lower than for T phrases. For medial ACs, the only dialect with identical amplitude value for C and T phrases is the BE group. The other three dialect groups demonstrate increasing amplitudes: GR +4%, VS +18%, and ZH +5%. The amplitudes of penultimate ACs in all dialect

6. For the calculations in this subchapter, IPs with only one AC were not taken into consideration.

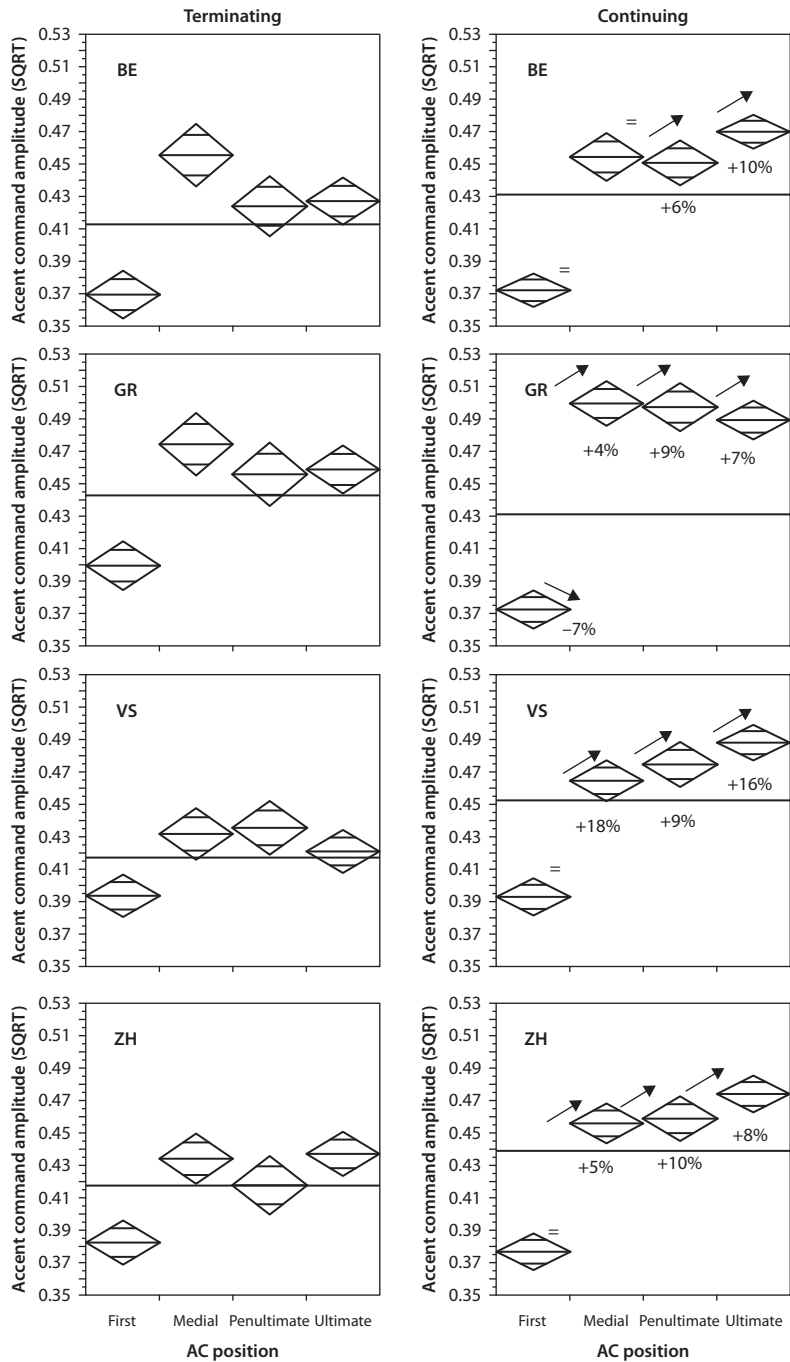


Figure 10.12. The amplitudes of ACs in different positions by dialect and phrase type. Percentages and equal signs indicate the difference in amplitude in C phrases, based on the values measured for T phrases. Differences < 3% = "="

groups are significantly higher for C than for T phrases: BE +6%, GR +9%, VS +9%, and ZH +10%. In the same vein, the ultimate ACs for C phrases are also significantly higher than for T phrases, particularly in the VS group: BE +10%, GR +7%, VS +16%, ZH +8%.⁹⁹

Cross-dialectal differences

Finally, ANOVAs were run on the AC amplitudes of the different dialects according to phrase type and AC position to show which AC position yields the most amplitude variation among the dialects, see Figure 10.13. Seven out of eight ANOVAs showed significant effects.¹⁰⁰ The only result that is not significant for the different dialects is the ANOVA on the amplitudes of ultimate ACs in C phrases. The differences among the levels, i.e. between the dialects, are significant in a number of cases.¹⁰¹

In the T condition, the GR group shows the highest amplitudes for every AC position (+3.3% for first, +6.2% for medial, +5.2% for penultimate, +5.4% for ultimate AC position). Interestingly, the VS group demonstrates the second highest AC amplitudes for ACs in first position (+1.6%), yet scores the lowest amplitudes for ACs in medial position (−3.5%) as well as ultimate position (−3.7%).

In the C condition, the AC amplitudes for ACs in first position are by far the highest for the VS group (+4%), while for ACs in medial and penultimate positions, the GR group are highest again (+6.3% and +5.6%, respectively). Furthermore, it is interesting to note that for ACs in ultimate position, the values for both mountain varieties as well as the two Midland varieties are nearly identical.

10.2.2.2 *AC timing*

Overall

To test whether, in each dialect, *T1dist_rise* is significantly different for C and T phrases, t-tests were applied, which yielded significant results, see Figure 10.14.

Figure 10.14 indicates that the greatest increase in *T1dist_rise* values is found in the ZH dialect. For the T condition, the rise occurs before segment onset, while for the C condition, it takes place after segment onset with a difference of +14ms. The BE behave similarly, only here, for both the C and T condition, the rises take place after segment onset. For the GR and VS dialects, the differences between the two phrase types are only minor.¹⁰²

In a next step, the dialects' *T1dist_rise* values for C and T phrases are compared in order to reveal cross-dialectal differences, as shown in Figure 10.15.

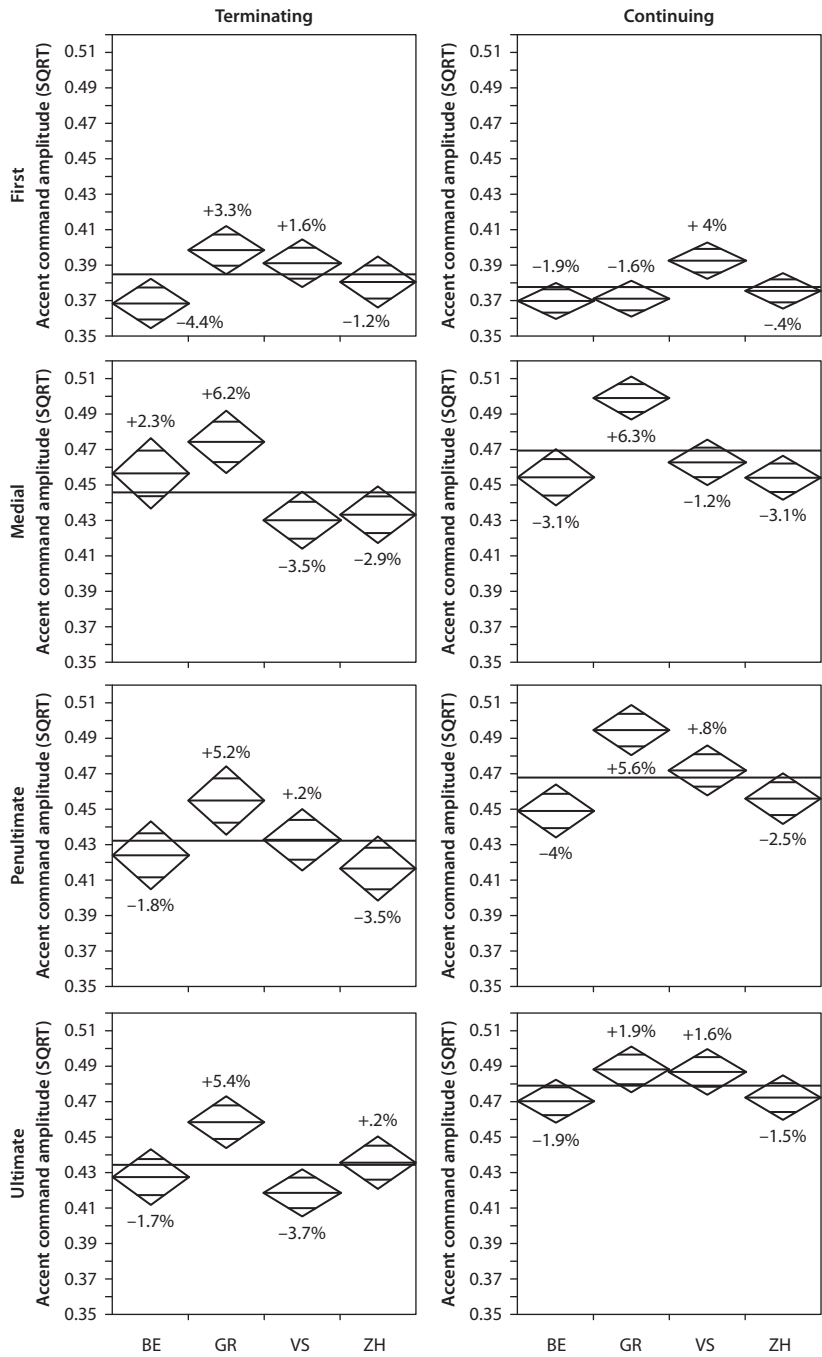


Figure 10.13. Mean diamonds plots showing the dialects' AC amplitudes, sorted by AC position and phrase type. The percentages show the deviations from the mean of response (mean of response serves as base)

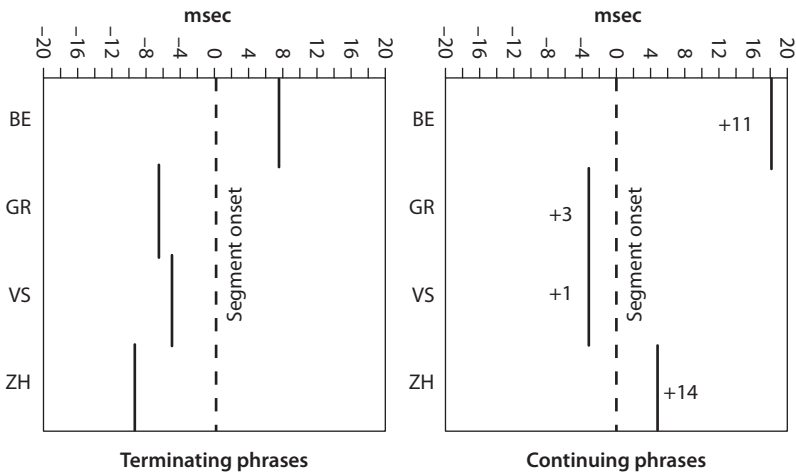


Figure 10.14. Comparison of T phrase *T1dist_rise* (left) with C phrase *T1dist_rise* (right) according to dialect. The numbers indicate the increases in ms from the T condition (base) to the C condition

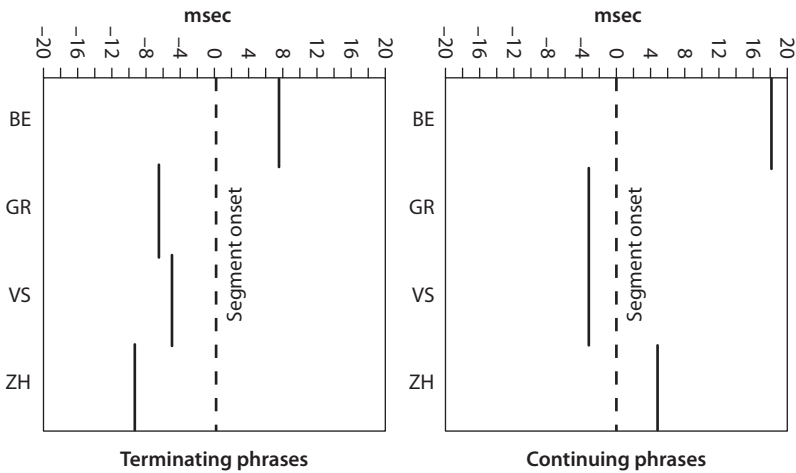


Figure 10.15. Illustration of the dialects' *T1dist_rise* values according to phrase type

Both ANOVAs are significant.¹⁰³ In the C condition, the BE rise latest at 18ms after segment onset. The ZH also show rises after segment onset (+5ms), while the GR and VS speakers rise earliest: both rise 3ms before segment onset. The differences between GR and BE, VS and BE, ZH and BE, ZH and GR, as well as ZH and VS are significant.¹⁰⁴ In the T condition, the BE group again demonstrates the latest rises (+7ms after segment onset). This behavior differs significantly from the other dialects, which all show similar early rises.¹⁰⁵

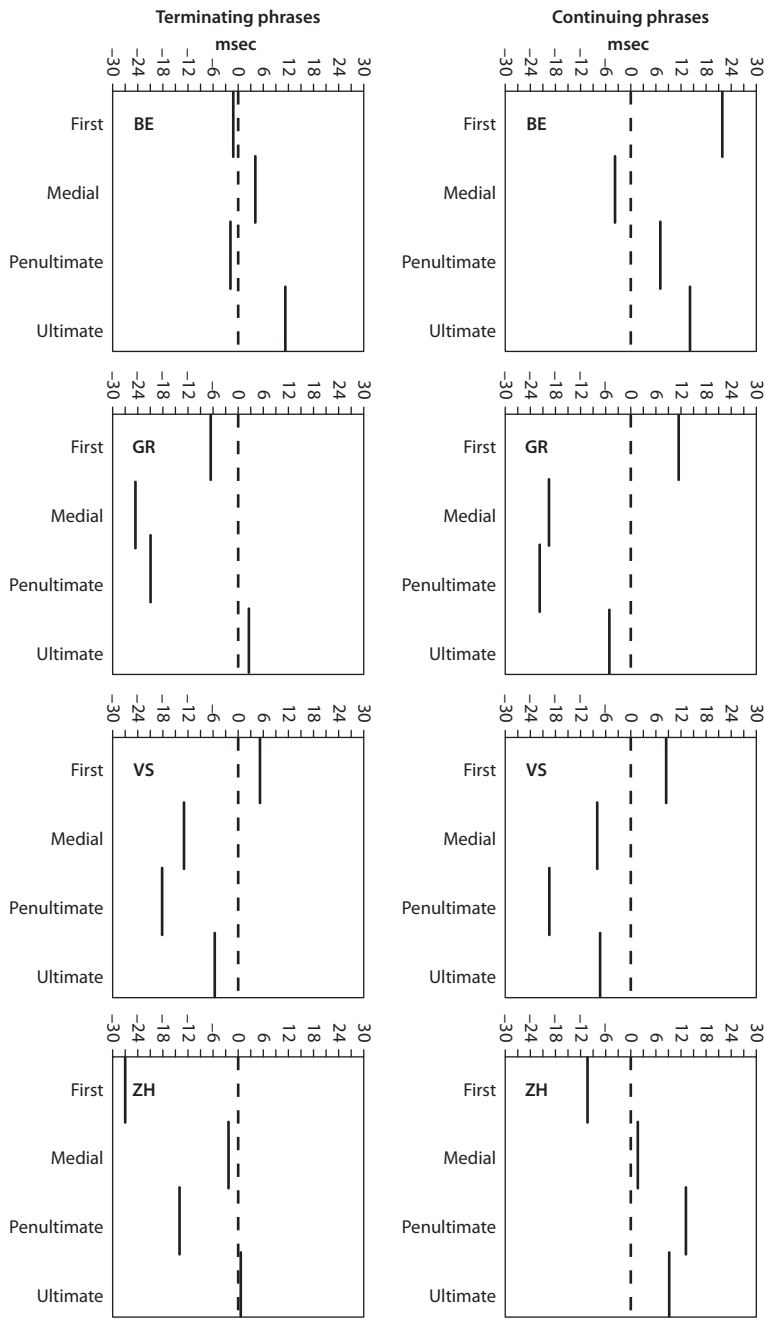


Figure 10.16. Illustration of the different AC positions' $T1dist_rise$ values according to C phrases (left) and T phrases (right), as well as dialects. The dotted line stands for segment onset at 0 sec

AC sequence

ANOVAs were performed to test whether there is a difference in the means of *T1dist_rise* of first, medial, penultimate and ultimate ACs for the different dialects and phrase types.⁷ The results are significant for all groups, except for the T condition in the BE dialect.¹⁰⁶ Figure 10.16 illustrates these data.¹⁰⁷

Regarding T phrases, the BE show a rise after segment onset in medial as well as ultimate position, while the penultimate ACs rise earliest (−3ms). In the GR group, only ultimate ACs occur after segment onset (+1ms), while medial ACs, in particular, occur early (−25ms).¹⁰⁸ In the VS group, the only ACs that start after segment onset are ACs in first position, while penultimate ACs start earliest (−19ms).¹⁰⁹ In the ZH group, the first ACs rise earliest (−26ms), the ultimate ACs latest (+1ms). In the ZH group, ultimate ACs are the only ACs to rise after segment onset.¹¹⁰

In C phrases, the BE rise after segment onset in all positions except for the medial position: the latest rises are found for ACs in first position (+21ms).¹¹¹ In the GR group, too, the latest rises happen in first AC position, the only position in which the rise occurs after segment onset (11ms). The earliest rises, on the other hand, occur in the penultimate AC position (−22ms).¹¹² The VS group behaves similar to the GR: first ACs mark the latest rises at +9ms, the only rises after segment onset, while the rises for the remaining positions start before segment onset. The rises are particularly early for penultimate ACs (−19ms).¹¹³ In the ZH group, we find a similar pattern as observed for BE: for three positions the rise occurs after segment onset. However, in the ZH group the rise before segment onset happens on the first ACs (−10ms), not the medial ACs like BE.

Comparing C and T phrases

In a next step, the *T1dist_rise* values for different AC positions in both C and T phrases are compared for each dialect. This reveals whether there is, for example, a difference in the *T1dist_rise* of first ACs in C as opposed to T phrases in a specific dialect. These data are presented in Figure 10.17.

The greatest divergence of *T1dist_rise* values in the BE group is found for ACs in first position (+22ms). For the BE group, only this difference between T and C phrases turned out to be significant.¹¹⁴ We also find the greatest difference in first AC position for the GR group: in this case, the ACs start 18ms earlier in the C condition, a difference that also is significant.¹¹⁵ In the VS group, none of the *T1dist_rise* values are significantly different between C and T phrases, whereas

7. For the calculations in this subchapter, IPs with only one AC were not taken into consideration.

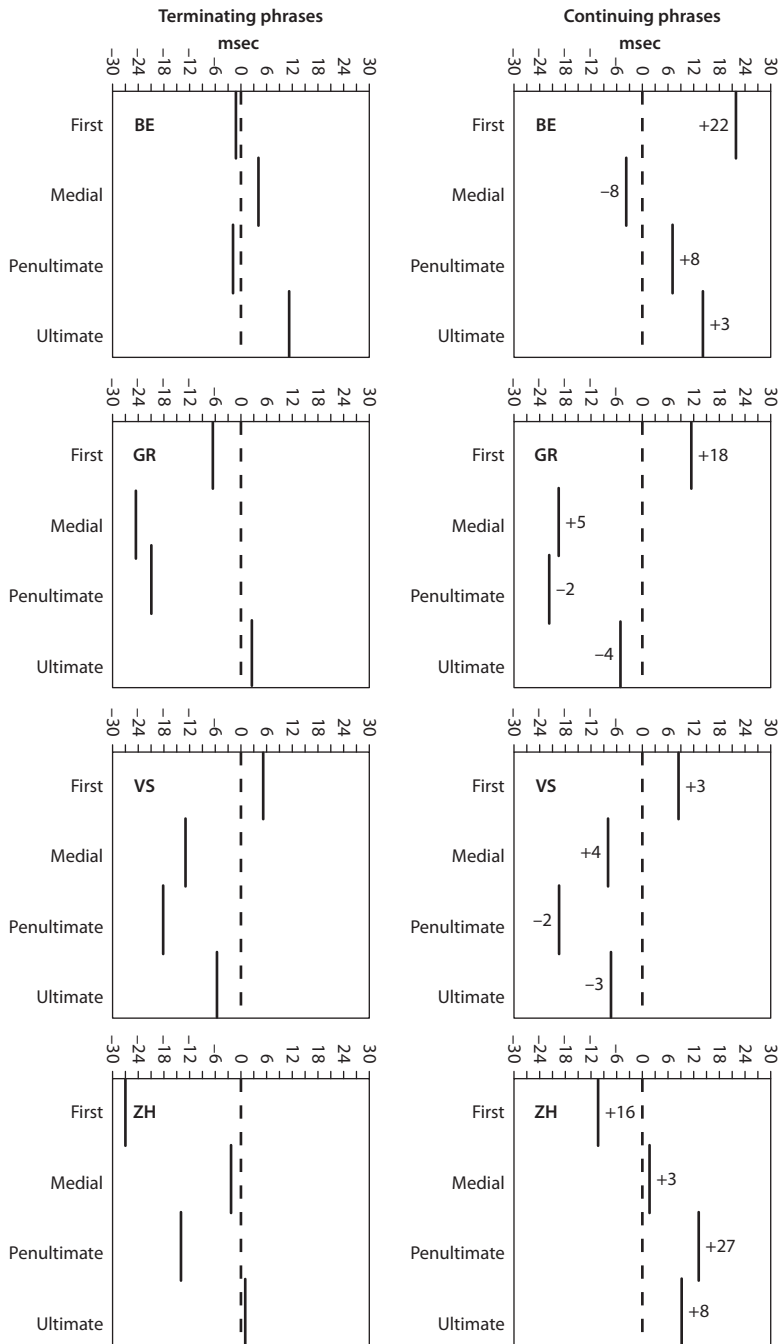


Figure 10.17. *T1dist_rise* values for ACs in different positions by dialect and phrase type. Values in the C condition indicate the temporal difference between C and T phrases (ms). The C phrases are measured by the T phrases

in the ZH group, statistical tests confirm significant differences between C and T phrases for ACs in first positions.¹¹⁶ The difference for ACs in penultimate position turned out to be significant as well, showing significantly later rises in C phrases (+27ms).¹¹⁷

Cross-dialectal differences

Finally, ANOVAs were run on the dialects' *T1dist_rise* values according to phrase type and AC position in order to establish which AC position exhibits the most variation for C and T condition between the dialects, see Figure 10.18. All ANOVAs except the penultimate and ultimate AC positions in T phrases show significant effects.¹¹⁸

In the first AC position of the T condition, the ZH show the earliest rises (−27ms), while the VS is the only group that rises after segment onset (+6ms). The BE, on the other hand, rises right at segment onset.¹¹⁹ As for ACs in medial position, the rises are earliest for the GR (−24ms) and latest for the BE (+4ms). The BE is the only group that begins to rise after segment onset.¹²⁰ In penultimate ACs, all dialects rise before segment onset, the earliest being GR (−20ms), the latest being BE (−2ms). In ultimate position, BE, GR, and ZH rise after segment onset (BE being the latest with 11ms), and only the GR group shows a rise before segment onset (−5ms).

In the C condition, all dialects show rises after segment onset, particularly the BE group (+21ms). The only exception is the ZH dialect with early onsets at −10ms.¹²¹ In the medial position, the above trend is reversed: all dialects show rises before segment onset, especially the GR group (−20ms), and the ZH group is the only group that rises after segment onset (+2ms).¹²² In penultimate position, the two midland dialects show rises after segment onset, while the VS and GR dialects demonstrate particularly early rises. The GR dialect rises as early as 22ms before segment onset.¹²³ In ultimate position, the same pattern is found as in the penultimate position: the BE and the ZH form one group, while the GR and VS behave similarly as well. The BE show the latest (+14ms), the VS the earliest rises (−7ms).¹²⁴

10.2.3 Summary and discussion

10.2.3.1 *Phrase component*

PC magnitude

Variable: Overall, C phrases exhibit higher PC magnitudes than T phrases.

The high magnitude in C phrases confirms the notion of prosodic paragraphing. When answering a question, the informant starts at a higher *f0* in anticipation of the series of IPs to come. From a communicative point of view, high

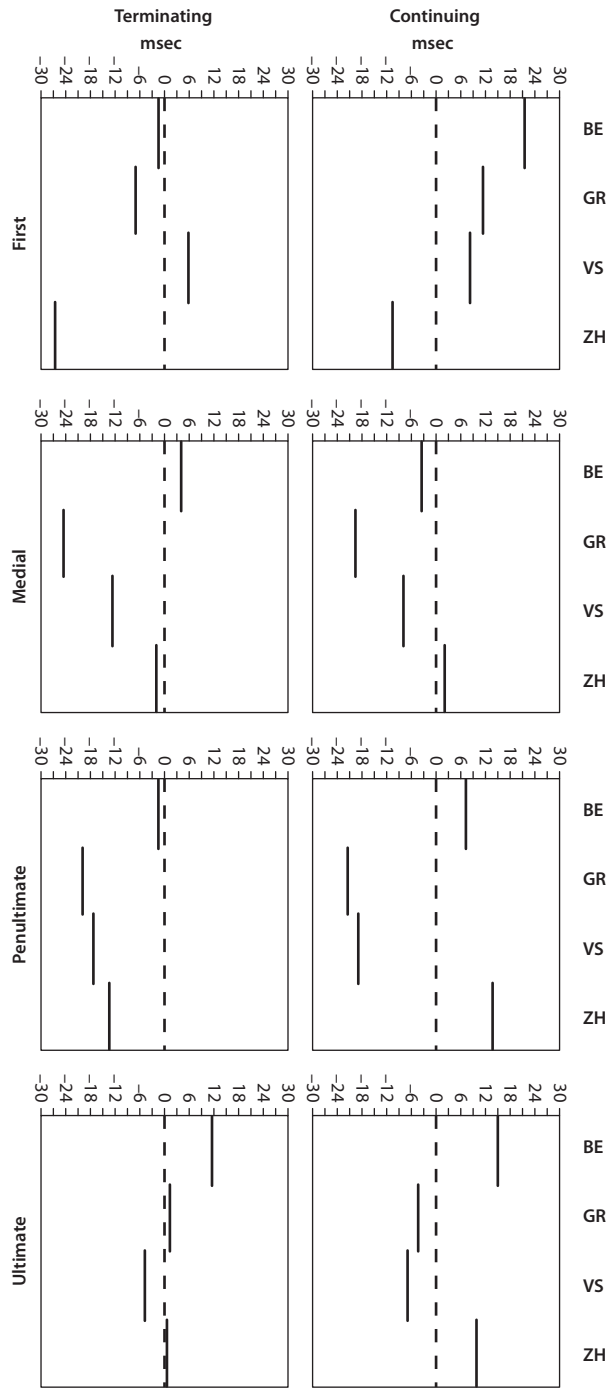


Figure 10.18. Illustration of the dialects' *T1dist_rise* values, sorted by AC position and phrase type

beginnings mark new topics and are produced in order to signal a speaker's communicative intention (Gussenhoven 2002: 52). This phenomenon was also observed by Fujisaki (1987) as well as by Mixdorff (1998: 143), who also established a negative correlation between PC magnitude and phrase count from the head of the utterance. This explains why all but the final PC of the prosodic group must maintain a high level magnitude so as to signal continuity in the course of speaking.

Dialect behavior: Differences between C and T phrases: The BE and VS dialects stand out in that they make a strong distinction in PC magnitude between C and T phrases. The ZH make nearly no distinction.

Cross-dialectal differences: The differences between the dialects are quite distinct, particularly in C phrases: here, the ZH speakers show by far the lowest, the BE and GR speakers the highest PC magnitudes. In T phrases, on the other hand, the highest magnitudes are found by far in the GR and the VS, while the ZH group exhibits by far the lowest magnitudes.

The literature on declination patterns of Swiss German dialectal speech is largely inexistent, as has become evident in the literature review. Most of the existing research is focused on local accents. The only account on PC magnitude is found in Häsler, Hove, and Siebenhaar (2005), which showed inter-group variation in PC magnitude. Of the one Zurich and the two Bernese speakers they analyzed in their study, the Zurich speaker indicated the highest, one of the Bernese speakers the lowest PC amplitudes. These calculations did not, however, take into consideration phrase function. The results of the present study attest quite the opposite: the ZH speakers were found to have the lowest declination in both C and T phrases.

As for a comparison of Swiss German and Standard German, it seems that PC magnitudes are generally lower for speakers of Swiss German than for Standard German speakers. Ulbrich (2006) showed that, on an overall text level, the f_0 declination of speakers of Swiss High German is less distinct than of speakers of Standard German. In their study, Häsler, Hove, and Siebenhaar (2005), too, confirmed this tendency. The current study can neither confirm nor confute this tendency.

PC duration

Variable: Overall, C phrases show shorter durations than T phrases.

This finding could possibly be explained from a speech production point of view: online speech planning of longer overall messages (C phrases) might trigger comparatively shorter prosodic phrases. A longer message is obviously

delivered in more than one utterance, which might well include a larger number of pauses for breath-intake and further speech planning than found in T phrases. Additionally, speech production errors that are short in nature, such as slips of the tongue, speech repairs, false starts, as well as other disfluencies, are often labeled as C phrases and frequently modeled with single PCs.

Dialect behavior: The BE and VS make again a strong difference in PC duration between C and T phrases, while the ZH show almost no distinction.

With regard to the analysis of PC magnitude as well as PC duration, the BE and VS group behave alike in that they mark C and T phrases rigorously for both variables. The ZH speakers, in contrast, make almost no distinction between the two phrase types for both declination and duration.

Cross-dialectal differences: When the dialects are compared, we find that the PC duration differences among the dialects are not as distinct as was observed for PC magnitude. The differences are more elaborate in T phrases, in which the GR speakers demonstrate the longest durations. In C phrases on the other hand, it is the ZH speakers that exhibit the longest durations.

10.2.3.2 Accent component

AC amplitude

Variable: Overall, ACs in C phrases have a higher amplitude than ACs in T phrases. The AC amplitudes in C phrases tend to increase gradually from first to ultimate AC position. The pattern found in T phrases is quite similar: we find an increase in amplitude from first to medial AC, a decrease in amplitude from medial to penultimate AC, and again an increase in amplitude from penultimate to ultimate AC. The differences between C and T phrases are particularly distinct in penultimate and ultimate ACs, where C phrase ACs show much higher amplitudes.

Higher amplitudes in C phrases were expected (see Gilles 2005). The speaker intends to hold the floor and signals non-finality by a comparatively high f_0 in PC-penultimate and PC-ultimate ACs. This explains why the difference in amplitude between C and T phrases is most distinct in penultimate and ultimate ACs. This trend is illustrated in Figure 10.19.

The C phrase in the above Figure reads *also hEt m* dip IS m* diplOm'i*rt* (German: *also hat man Dip ist man diplomiert*; Engl.: *well one has the dip one is licensed*), and the T phrase reads *mI r'ejtst das and*rs ts ksè:* (German: *mich reizt das anders zu sehen*; Engl.: *I am tempted to see the other side of things*). The C phrase displays the prototypical increase in AC amplitude from first to ultimate AC. Note, however, that the increase in AC amplitude may partly be enhanced by

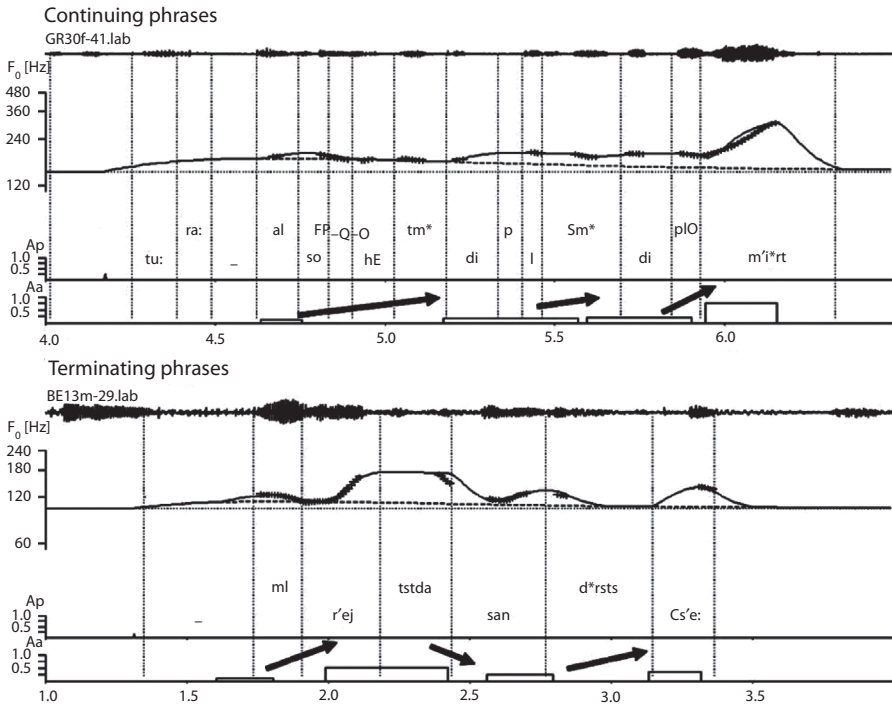


Figure 10.19. Prototypical behavior of C and T phrase AC amplitudes, based on overall results

suboptimal modeling of high flat f_0 contours, as frequently found in spontaneous speech C phrases. This is the case in the second and third AC, in which the absolute pitch remains the same, but the AC amplitude nevertheless increases to compensate for the falling PC slope. In the T phrase, we find a low first AC amplitude, followed by a higher amplitude in the medial AC and a subsequent decrease in amplitude before the amplitude eventually rises again in the end.

This finding reflects what is frequently mentioned in the literature on Swiss High German intonation: the interference of Swiss German phrase-final f_0 contours on Swiss High German pronunciation. Stock (2001: 171) mentions that for Swiss High German, we often find “lively melodic movements” in phrase-final position, which are characterized by an increase in f_0 , followed by a quick, abrupt fall. Panizzolo (1982: 41), too, noted that in “conclusive tones” (i.e. statements), she found rise-fall patterns in Swiss High German but falling patterns in Standard German. Ulbrich (2005: 227ff) also observed that, in Swiss High German declaratives, the Swiss either show phrase-final rises or remain on the same level, much in contrast to Austrian or German speakers.

Dialect behavior: AC sequence: In T phrases, the BE stand out because they make a stark distinction between first and medial ACs and a stark distinction between medial and penultimate ACs. The VS group is salient because of the comparatively small difference in the amplitude of first and medial AC. The VS group is further salient because it is the only group without a rise-fall-rise pattern in T phrases. Instead, they exhibit a rise-rise-fall pattern. In C phrases, the GR stand out because they make a very strong distinction between first and medial ACs. Furthermore, the GR are special in that they exhibit a rise-fall-fall pattern, while the BE show a rise-fall-rise, and the VS and ZH a rise-rise-rise patterns.

Differences between C and T phrases: When C and T phrase ACs are compared, the GR are the only group in which the first AC is lower in the C than the T condition. The VS group exhibits an interesting behavior for ACs in ultimate position: the amplitudes are much higher in C as opposed to T phrases.

Cross-dialectal differences: Ultimate ACs in C phrases are very similar in amplitude in all of the dialects. In all other positions, however, both T and C phrase ACs show much more variance. This is particularly true for penultimate ACs in C phrases and ultimate ACs in T phrases. In the first instance, the GR exhibit the highest, the BE by far the lowest amplitudes. In ultimate ACs of T phrases, the GR show again by far the highest, the VS the lowest amplitudes.

In these findings, the local f_0 patterns of T phrases in the VS group are of particular interest. The established pattern of AC amplitude variation for T phrases is a rise-rise-fall, as is illustrated in Figure 10.20.

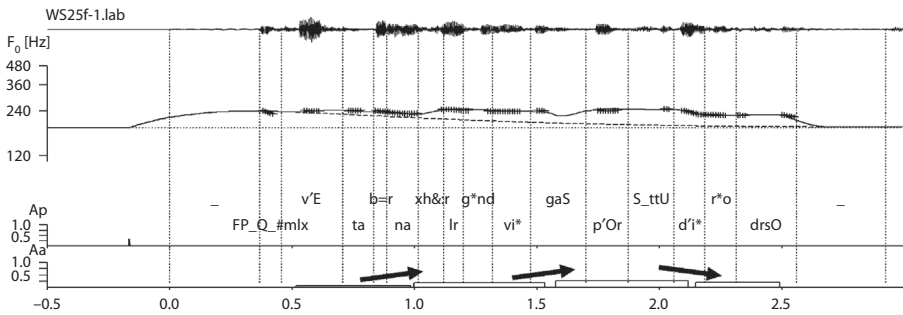


Figure 10.20. Prototypical f_0 contour of a multi-AC T phrase, as produced by a VS speaker

The phrase in Figure 10.20 reads *mix v'Et ab=r naxh&:r Irg*ndvi* ga Sp'Ort StUd'i*r* odr sO* (German: *ich will aber nachher irgendwie Sport studieren gehen oder so*; Engl.: *but afterwards I want to go study sports or something*). This phrase depicts a prototypical T phrase f_0 pattern of a speaker of the VS dialect. We find

a minor rise in amplitude from first to medial AC, followed by another rise, before the amplitude eventually decreases again from penultimate to ultimate AC. This pattern stands in contrast to what earlier research has shown. Wipf (1910: 22), for instance, observed that the predominant contour in sentence final position of declaratives is a rising tone, an observation which was not made for any other Swiss German dialect. Hence, one would have expected to find high AC amplitudes in the phrase-final position of T phrases. On the contrary, however, the VS is the only dialect whose phrase-final AC amplitudes are *lower* than the amplitudes of medial and penultimate ACs. In fact, a cross-dialectal analysis of AC amplitudes in T phrases reveals that, out of all the dialects, the amplitudes of ultimate ACs in the VS dialect are by far the lowest. What is most interesting, however, is that the VS dialect is the only dialect that marks an increase in AC amplitude from medial to penultimate AC. This increase in f_0 in IP-medial/IP-final position in declaratives is a very distinct feature that separates the VS group from the other dialects.

Of further interest is Haldimann's (1903: 298) observation, that declaratives in the BE dialect usually have a falling intonation. The results of the present study do not support this notion, however, at least not from the point of view of AC amplitudes. Conversely, in the BE dialect, the amplitudes of ultimate ACs are higher than those of penultimate ACs. What is worth noting, however, is that both penultimate and ultimate ACs are comparatively low compared to medial ACs, a difference which is not found to such a degree in the other three dialects. Possibly, this is what Haldiman (*ibid.*) perceived as a generally falling tendency of BE declaratives: the strong contrast in f_0 between medial ACs with a high f_0 and penultimate and ultimate ACs with a lower f_0 . On the other hand, Haldimann's (*ibid.*) perception might also have to do with the timing of the fall, i.e. T_1 and T_2 interpreted below.

AC timing

Variable: In T phrases, virtually all dialects rise late (i.e. after segment onset) in ultimate AC position. With regard to the differences between C and T phrases, C phrase ACs rise later relative to segment onset than T phrases. This is the case especially for ACs in first position, while ACs in medial and penultimate positions generally differ least.

The finding that T_1 s in ultimate ACs of T phrases are overall placed after segment onset suggests parallels to what Gilles (2005) and Fitzpatrick (1999) have hypothesized – and in part confirmed – for Swiss German dialects. They claim that a default L*+H contour is prevalent in the nuclear accent position of T as well as C phrases. In what follows, I attempt to draw a tentative comparison between

these insights and the present findings, bearing in mind the different underlying assumptions and theoretical underpinnings.⁸

Despite methodological differences, it is possible to perform a tentative extraction of ACs in ultimate AC position in T phrases that carry stress in first syllable position (on average, about 50% of all T phrase ACs). For these particular ACs, T_1 is anchored with the first metrically strong syllable, while T_2 is not, unless the AC spans over exactly only this syllable. This procedure is performed for all dialects. Results show that in all dialects, T_1 occurs distinctly after segment onset in the stressed syllable, that is, $T1dist_rise$ is of positive polarity. Mixdorff (2002b: 4) records an average $T1dist_rise$ value of 132ms for L^*+H accents, while H^*+L accents score a negative mean of -60ms. Now in Swiss German, the corresponding ACs are likely to be of Type I, II, and III since they frequently exhibit onsets that occur after segment onset. For ACs of types IV and V, on the other hand, the onset usually happens before segment onset because they already start with a high flat f_0 rise. In other words, the corresponding contours in Swiss German are late rising contours, occurring in ultimate ACs of T phrases, as suggested by Gilles (2005) and established for BE German by Fitzpatrick (1999). An example of such a late rising ACs that starts with a stressed syllable is shown in Figure 10.21.

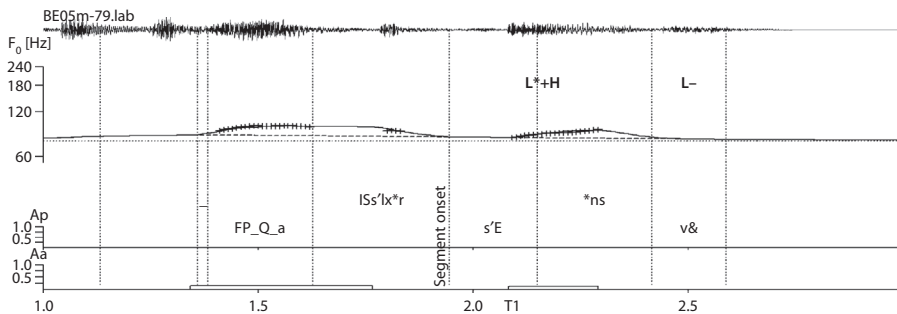


Figure 10.21. L^*+H L^- accents in phrase-final positions of T phrases

8. Both Gilles (2005) and Fitzpatrick (1999) applied an autosegmental-metrical framework, which impedes a direct comparison of their results and the findings of the current study. Although it is possible to compare ToBI labels with Command-Response model commands (see Mixdorff 2002b), such a comparison is difficult to perform given the parametrization used in this study: ACs are not anchored with metrically strong syllables. This characteristic, however, constitutes an important prerequisite if ToBI labels are to be compared with Command-Response model commands.

The phrase above reads *ja: IS s'lx*r s'E*nsvɛʁt* (German: *ja ist sicher sehenswert*; Engl.: *yes is certainly worth seeing*). L^* aligns with the stressed syllable while the f_0 peak is reached late in the unstressed syllable **ns*. Note that this finding is only valid for those 50% of ACs in T phrases that start with a stressed syllable in ultimate position. The other 50% of T phrase ACs in the different dialects are made up of ACs without any stressed syllable (20%) and ACs with stress on the second syllable or later (30%).

As to the differences between C and T phrases, the finding that C phrases generally rise later ties in nicely with Kohler's (1991b) observations on peak alignment. He found that the relocation of the f_0 peak relative to the stressed vowel from early to late results in a categorical change in meaning (Kohler 1991b:125 ff.). Kohler (1991b) notes that medial or late rises relative to segment onset, as found in C phrases in the present study, entail new information open for discussion or the introduction of a new argument. The generally earlier rises in T phrases, on the other hand, imply established facts that leave no room for discussion (ibid.). While these insights apply to nuclear accents in phrase-final position, in particular, it seems valid to tentatively translate these findings to the overall means of *T1dist_rise* values found in this study.

With regard to global movements, Vaissière (1983:59) notes that rising f_0 movements are generally associated with the notion of a beginning, while falling f_0 movements tend to be associated with an end. Possibly, these findings also hold on a local level, which would mean that clearly rising f_0 movements (i.e. late T_{1s}) mark beginnings, and flat or flat-falling f_0 movements (early T_{1s}) denote finality. Mixdorff (1998a:96), too, found late onset times in phrase-final C phrase ACs, as opposed to earlier onsets in the corresponding ACs of T phrases.

Finally, it should be pointed out that this finding might partly be due to the more distinct durations of ACs in C phrases. After all, AC duration correlates with *T1dist_rise* in that the longer the AC, the later the rise relative to segment onset. The fact that AC durations in C phrases are longer may in turn be a consequence of suboptimal modeling. Since the PC magnitude in C phrases is shown to be higher, the declination is comparatively steep, which means that the longer AC durations are likely to constitute a compensation measure for the modeling of high, flat f_0 contours.

Dialect behavior: AC position patterns: The BE show particularly late rises in ultimate ACs of T phrases. If only ultimate T phrase ACs with first syllable stress are considered, the BE demonstrate the latest rises (+39ms), followed by the ZH (+34ms), the GR (+22ms), and the VS (+15ms). The greatest difference in T1dist_rise values between the different AC positions in T phrases is found for the ZH group. In C phrases, the least differences are found for the BE, the greatest differences for the GR.

Differences between C and T phrases: All dialects, except the VS variety, exhibit significantly later rises in first ACs of C phrases. In the ZH group, also penultimate ACs rise significantly later for C than for T phrases. In the VS group, no significant differences between C and T phrases are found for any AC position.

Cross-dialectal differences: The dialects differ in all AC positions except the penultimate and ultimate ACs of T phrases. In C phrases, the latest rises for nearly all ACs are observed in the BE group. Also in penultimate and ultimate ACs of C phrases, the Midland varieties rise late, while the Alpine varieties rise early.

Interestingly, non-significant cross-dialectal differences were established for *T1dist_rise* values in penultimate and ultimate AC positions in T phrases. This underlines the previously discussed finding on late rises in T phrases in all dialects. If only first-syllable-stress, ultimate ACs of T are considered, however, we observe that the BE show the latest rises (+39ms after segment onset). This corroborates Fitzpatrick's (1999) claim that in the BE variety, we predominantly find a L*+H accents in nuclear position in T phrases, in which the *f0* peak is often reached in the following unstressed syllable. In the VS variety, this rise occurs earlier, yet still after segment onset (+15ms). It is salient that in ACs with stress on the first syllable, both Alpine varieties rise comparatively early, while the Midland varieties, particularly BE, exhibit late rises. The same is true for C phrases: we encounter a distinction between early-rising Alpine varieties and later-rising Midland varieties.

10.3 Prosodic paragraphing

For the analysis of prosodic paragraphing, only PCs are taken into consideration.⁹ As noted in the annotation section, the variables to be considered are *pre-phrase strength of break* (i.e. pause length before an IP), *duration of previous IP*, as well as *magnitude of previous IP*.

10.3.1 PC magnitude

10.3.1.1 *Strength of break*

A linear regression of PC magnitude on the strength of break was performed, see Figure 10.22.

9. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

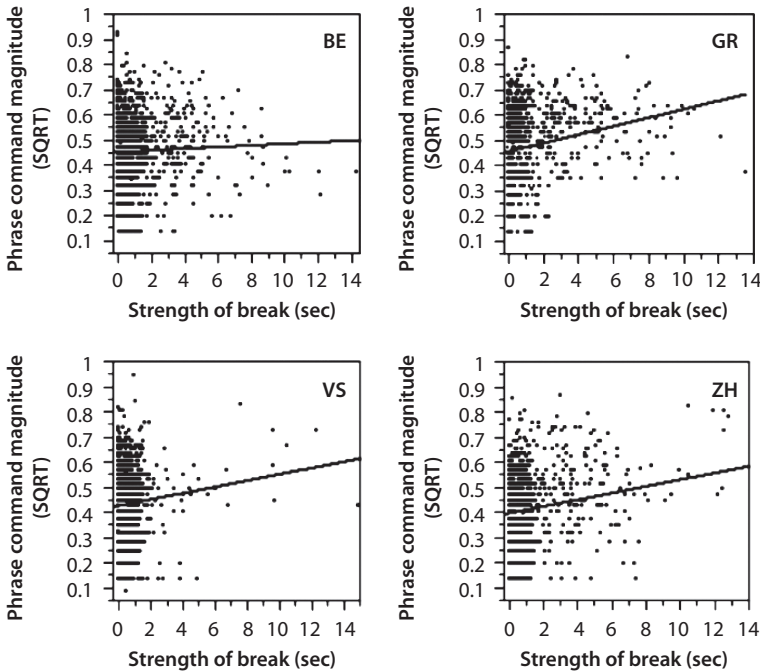


Figure 10.22. Cross-plots of PC magnitude as a function of strength of break by dialect

All of the regressions were statistically significant, except for the BE data.¹²⁵ The results suggest that the more distinct the pre-phrase break, the higher PC magnitude. The low R^2 values indicate, however, that strength of break is a rather weak predictor for PC magnitude.

10.3.1.2 *Duration of previous phrase*

A regression of PC magnitude on duration of the previous IP only confirms statistical significance in the VS group, yet also with a low R^2 value.¹²⁶ This means that duration of previous IP is a poor predictor of PC magnitude. Still, there is a general trend which posits that the longer the previous IP, the higher the PC magnitude of the actual IP.

10.3.1.3 *Magnitude of previous phrase*

A linear regression of PC magnitude on the PC magnitude of the previous IP is significant for all dialects, see Figure 10.23.

The finding is that the PC magnitude of the previous IP predicts the PC magnitude of the actual IP with R^2 s between .07 and .11.¹²⁷

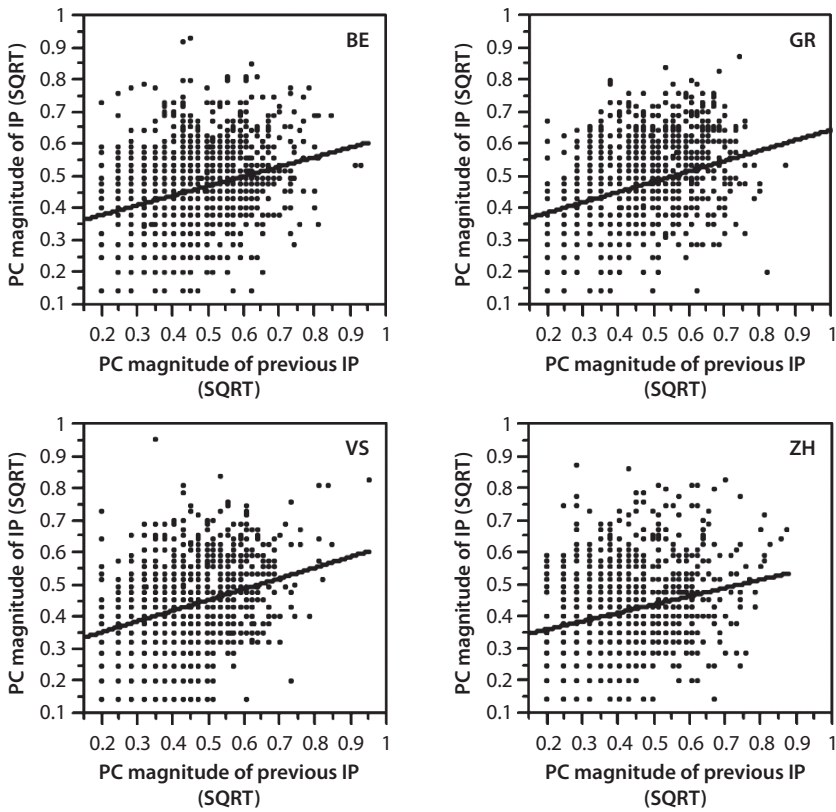


Figure 10.23. Cross-plots of PC magnitude as a function of PC magnitude of previous IP by dialect

10.3.2 PC duration

10.3.2.1 *Strength of break*

A linear regression of PC duration on the strength of break was performed for all dialects. The trend holds that the stronger the break, the shorter the duration of the IP. Yet this regression is only statistically significant for the VS group.¹²⁸ Hence, strength of break is not considered a significant predictor for IP duration.

10.3.2.2 *Duration of previous phrase*

A linear regression of PC duration on the duration of the previous phrase was performed for all dialects. Three dialects exhibit the tendency that the longer the previous IP, the longer the actual IP. For the VS dialect the opposite trend is established. The analyses show that the regression is significant only for the ZH

dialect.¹²⁹ Hence, duration of previous phrase is not considered a significant predictor for IP duration.

10.3.2.3 *Magnitude of previous phrase*

A linear regression of PC duration on magnitude of the previous IP was performed for all dialects. For the BE, GR, and VS, the tendency was found that the higher the declination of the previous phrase, the shorter the duration of the actual IP. This trend is significant only in the BE group, however.¹³⁰ The VS speakers behave conversely. Hence, magnitude of the previous IP, too, is a poor predictor of IP duration.

10.3.2.4 *Summary and discussion*

PC magnitude

Variable: All tested variables are relatively weak predictors of PC magnitude. The best predictor is the PC magnitude of the previous IP, followed by the strength of break. The regressions show that the higher the magnitude of the previous IP, the higher the magnitude of the actual PC. Furthermore, it was found that the longer the pause preceding the IP, the higher the magnitude.

The finding regarding the predictive power of the previous IP magnitude was anticipated. This result confirms that the declination of the previous IP has a sequential effect on the actual IP, as illustrated in Figure 10.24.

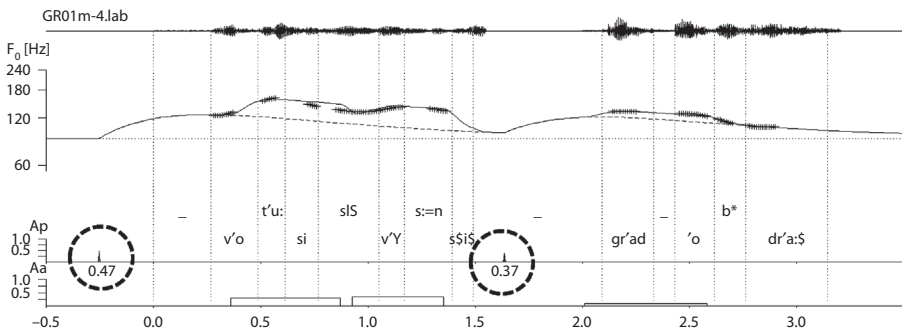


Figure 10.24. The PC magnitude of an IP predicts the PC magnitude of a subsequent IP

In this example, the PC magnitude of the first IP is .47, the magnitude for the following IP is 0.37. Now, with regard to paragraphing effects on f_0 contours, this finding may not provide information on the role of IP position within a prosodic group, it does, however, suggest a dependency between the magnitude of an IP and the magnitude of the previous IP. The interaction of PC magnitudes may be

interpreted as a result of an ergonomic and economical way of producing intonation: the less discrepancy in f_0 the less effort and the more seamless the flow of speech. In fact, it would be very uneconomical for adjacent IPs to behave conversely when it comes to PC magnitude, unless for purposes of special marking. In other words, constantly abrupt discrepancies in f_0 would be strenuous to produce, and they would be a source of distraction and confusion for the listener. Rather, proportional behavior is much more natural since it contributes to an intergradient sequence of IPs. In the light of this finding, we can further draw the tentative conclusion that since the IPs of an utterance affect each other in certain ways, they contribute to a greater purpose: they form a whole, a bigger unit called a prosodic paragraph.

As for the strength of break, it seems as though pause length plays a critical role in predicting PC magnitude. This finding can be illustrated by the following example in Figure 10.25:

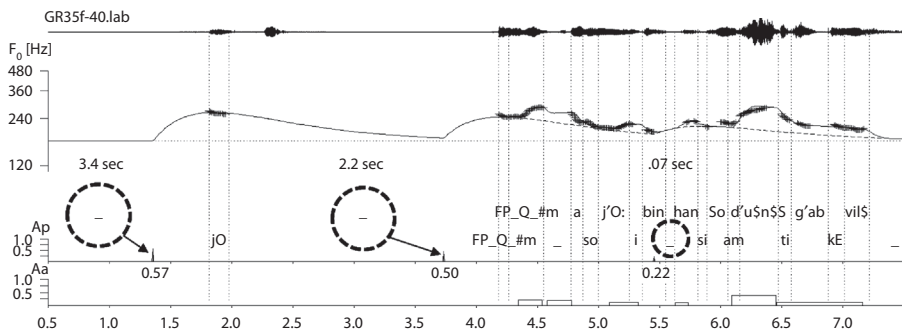


Figure 10.25. Strength of break, i.e. pause length, predicts PC magnitude of the subsequent IP

In this three-phase utterance, the first phrase is preceded by a 3.4-second pause and modeled with a PC magnitude of .57. The second phrase is preceded by a slightly shorter 2.2-second pause and is modeled with a magnitude of .5, while the third phrase is preceded by a .07-second pause and modeled with a .22 magnitude PC. This is an excellent example of the positive correlation between duration of pre-phrase pause and PC magnitude. As noted by Vaissière, pauses can trigger a reset in the declination line, where the strength of the reset mirrors the importance of the boundary (Vaissière 1983:57). Accordingly, strong boundaries are marked with longer pauses, which has a direct effect on the degree to which the declination line is reset (see Tseng et al. 2005).

Cross-dialectal variation: For all dialects, the PC magnitude of a previous IP is a similarly strong predictor of the magnitude of the actual PC. For the strength of break, on the other hand, we find some variation: its predictive power of PC magnitude is higher for the GR than the BE group.

PC duration

Variable: None of the investigated variables are significant predictors of PC duration.

What is striking in these and the above results is the low predictive power of the performed regressions. Possibly, this is due to the highly spontaneous nature of the present data. Note that Tseng et al. (2005: 86) steered clear of using spontaneous speech when establishing the multi-phrase hierarchy so as “to reduce rapid shift and planning strategies”. Spontaneous speech is, after all, a stream of speech without punctuation that is filled with pauses, repairs, and false starts. Shriberg notes that disfluencies frequently influence one third of utterances (2005). All of these typical features of natural speech may impede a straightforward paragraphing structure since repairs, fillers, and false starts are frequently modeled with separate PCs.

Non-linguistic variables

11.1 Articulation rate

Descriptive statistics

Table 11.1 summarizes the average articulation rates for the four dialects, with and without filled pauses (left and right, respectively).¹

Table 11.1. Summary of the articulation rates by dialect (with and without filled pauses)

	Rate (w/FP)	Rate (w/o FP)
BE	4.99	4.98
GR	5.23	5.24
VS	5.79	5.8
ZH	5.74	5.76

The analysis will be based on three classes of speakers for each dialect: *slow*, *normal*, and *fast* speakers.² These three categories were formed as follows. Firstly, the span of the articulation rate (including filled pauses), i.e. the fastest speaker's rate minus the slowest speaker's rate, was calculated for each dialect. Subsequently, this number was divided by three. The speakers of the bottom third category were then labeled as *slow* speakers, the speakers of the medium third as *normal* speakers, and the ones in the top third as *fast* speakers. This leads to the categorization shown in the below Table 11.2.

1. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3

2. This kind of classification was also applied by Mixdorff (1998) and Ohno et al. (1998).

Table 11.2. Articulation rate categories per dialect

Speaker	Syll. / sec.	Speaker	Syll. / sec.	Speaker	Syll. / sec.	Speaker	Syll. / sec.
BE02f	4.1	GR04f	5	VS41f	5.2	ZH01m	4.9
BE03m	4.3	GR34m	5	VS25f	5.4	ZH26f	5.1
BE12f	4.5	GR03m	5.1	VS28m	5.4	ZH32m	5.4
BE05m	4.7	GR24m	5.1	VS46m	5.5	ZH03f	5.5
BE13m	5.1	GR31m	5.1	VS44f	5.6	ZH23f	5.6
BE04f	5.2	GR35f	5.1	VS22m	5.9	ZH34m	5.8
BE23f	5.4	GR01m	5.2	VS48m	5.9	ZH02f	6
BE06m	5.5	GR06f	5.3	VS23f	6.1	ZH31f	6.1
BE22m	5.5	GR33f	5.4	VS49m	6.3	ZH24m	6.3
BE21f	5.7	GR30f	6	VS29m	6.6	ZH22m	6.8
Range:	1.6	Range:	1	Range:	1.4	Range:	1.9
SD:	0.56	SD:	0.3	SD:	0.45	SD:	0.57
Fast	Normal	Slow					

Since the relative proportions of articulation rate types between the dialects are very different, the overall means for each type cannot be used for calculations. Therefore, only the data of one dialect was used. The ZH dialect seemed most suitable for analysis since it exhibits the greatest standard deviation, which means that the articulation rates are spread out over a distinctive range of values. For this reason, the ZH dialect was chosen for the following analyses.

11.1.1 Phrase component

11.1.1.1 PC duration

A one-way ANOVA on the PC durations of the three articulation rates shows significant differences.¹³¹ The data are presented in Figure 11.1.

Fast speakers produce the longest IPs (+23% from mean of response) followed by slow speakers (−2%), while speakers with normal articulation produce the shortest IPs (−6%). The differences between fast speakers and normal and slow speakers are significant.¹³²

11.1.1.2 Summary and discussion

Distribution

Variable: The results show that, if filled pauses are included, the VS and ZH speakers are the overall fastest speakers (around 5.8 syll/sec), followed by the GR (5.2 syll/sec), while the BE are the slowest (5.0 syll/sec).

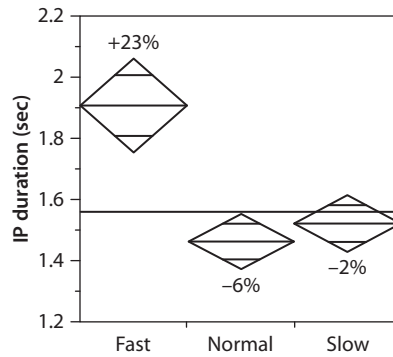


Figure 11.1. Mean diamonds plot showing PC durations of speaking-rate types

This finding confirms two prevalent stereotypes about the ZH and BE dialects, namely that the BE talk ‘slowly’ and the ZH ‘rapidly’ (see Ris 1992:756). Leemann and Siebenhaar (2007) have shown that the slow articulation rate in the BE variety is primarily due to distinctly longer mean durations of vowels. The BE dialect, however, was also shown to exhibit distinct phrase-final lengthening.

PC duration

Variable: Fast speakers show the longest IP durations, normal speakers the shortest. Slow speakers fall in between.

The same tendency is also established for the other three dialects, yet the differences between slow and fast speakers are significant in the ZH group only. Once again, this finding is difficult to interpret. It can be said, however, that any deviation from the normal articulation rate, regardless of whether slower or faster, seems to result in an increase in IP duration. Since higher articulation rate can lead to a reduction of phrase boundaries (see Mixdorff 1998a:176, Ladd et al. 1998), one could argue that such a reduction in PCs may result in overall longer IPs. The IPs of slow speakers, on the other hand, are probably longer because their ACs are found to be longer as well as.¹³³

11.2 Emotion

In terms of the distribution of emotional categories, 81% of all the phrases were labeled as *neutral*, 8% as *bored*, 6% as *happy*, 4% as *disgust*, and 1% as *fear*.³ It is

3. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

important to bear in mind that these emotions are assigned to *phrases*, i.e. entire IPs. The first point of interest is the difference in the distribution of emotional categories for the different dialects, especially since the subjects were recorded under different emotional/psychological circumstances. On average, a total of 1209 IPs were analyzed per dialect. All groups, except for the BE dialect, show similar cell sizes. Figure 11.2 illustrates the bar chart of dialects and type of emotion, whereas Table 11.3 gives the corresponding contingency table.

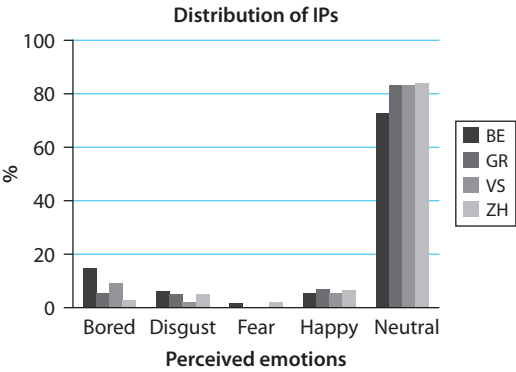


Figure 11.2. Column chart of perceived emotions

Table 11.3. Contingency table of the dialects according to type of emotion

Count Row %	Bored	Disgust	Fear	Happy	Neutral	Total
BE	183 14.76	72 5.81	20 1.61	65 5.24	900 72.58	1240
GR	68 5.28	60 4.65	3 0.23	86 6.67	1072 83.17	1289
VS	110 9.19	25 2.09	2 0.17	65 5.43	995 83.12	1197
ZH	30 2.71	54 4.87	22 1.98	71 6.4	932 84.04	1109
Total	391	211	47	287	3899	4835

A chi-square test of independence was performed, which showed significant differences in the relative proportions of types of emotion for the different dialects.¹³⁴ The most salient differences are found in the percentages of bored phrases: for the GR, VS, and ZH, these amount to roughly 3–9%, whereas for the BE group, 15% were labeled as bored. A similar trend can be observed for the disgust category: the percentage value for the BE is nearly 6%, for the GR,

ZH, and particularly the VS group, the number of disgust phrases is lower. As for happy phrases, the distribution is nearly the same for all dialects. The same holds true also for neutral emotion phrases. For neutral phrases, the BE indicate only 73%, however, while the remaining three dialects produced around 83.5% of neutral phrases.

Since the relative proportions of emotion types between the dialects are significantly different, overall means of the different emotion types cannot be used for calculations. Therefore, only the data of one dialect is further analyzed. The values of the BE speakers were considered as suitable for the subsequent analyses since the cell sizes of bored and disgust emotions are comparatively large in this group, and thus suitable for statistical analyses. The category of *fear* is suppressed, however, since there are only 20 counts in the cell.

11.2.1 Phrase component

11.2.1.1 PC duration

To test for differences in the PC durations of the different emotion types, an ANOVA was performed. The overall PC durations for the different emotion types is illustrated in Figure 11.3.

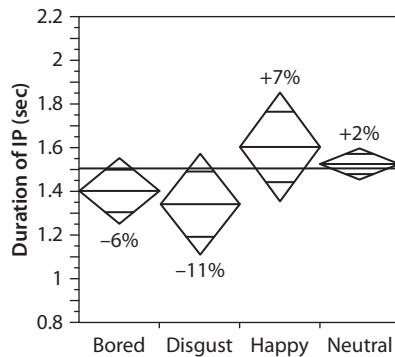


Figure 11.3. Mean diamonds plot showing the PC durations of different emotion types. The percentages indicate the deviation from the mean of response

This analysis did not yield significant results.¹³⁵ Happy phrases show the longest durations (+7%), while disgust emotion phrases are shortest (-11%).

Different IP constellations

The PC durations of IPs with a varying number of ACs for different emotion types is not examined any further since, as mentioned earlier, the frequency of ACs in an IP correlates positively with IP duration. However, the PC

durations of the different emotion types can be further examined according to different IP types (IPs with less than 3 ACs, or 3 or more ACs). In IPs with 3 or more ACs, the emotion types show significant differences in IP length, see Figure 11.4.¹³⁶

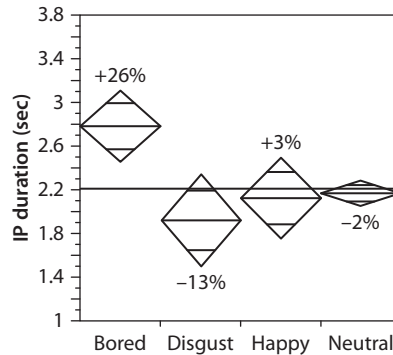


Figure 11.4. Mean diamonds plot showing the PC durations of different emotion types. Only PCs with 3 or more ACs are considered. The percentages indicate the deviation from the mean of response

In IPs with more than 3 ACs, bored phrases show the longest durations (+26%), while disgust phrases exhibit the shortest durations (−13%). The IPs for happy (+3%) and neutral (−2%) emotion behave fairly similarly.¹³⁷

11.2.1.2 Summary and discussion

Distribution

Variable: The vast majority of the labeled phrases are neutral phrases, followed by bored and happy emotion phrases. Disgust and fear phrases are proportionately irrelevant. The BE exhibit a comparatively high number of bored phrases.

The relative proportions of emotion phrases for the different dialects are fairly similar, with exception of the BE speakers. With regard to an interpretation, it is important to keep in mind that the BE speakers were recorded under special conditions: at the time of the recordings, they were supposed to be writing mock essays for their final exams. Because of the recording sessions, however, each subject was cut 20 minutes short of finishing their essays, explaining why some of the subjects seemed eager to bring the interview to an end and get back to class. This would explain why the subjects seemed comparatively tense and appeared to be under time pressure. Their unwillingness to cooperate is mirrored in the unusually high number of bored phrases.

The overall scarce occurrence of fear, i.e. apprehension, phrases (only 1%) is somewhat surprising. After all, the interview setting might well have featured certain characteristics that are generally believed to foster apprehensiveness and/or uncertainty on the part of the subjects, such as a low specificity of questions on the part of the interviewers (Siegman and Pope) or the failure to apply opposite-sex dyads (see Hyman 1954). For a more detailed study of emotion, as well as articulation rate, a refined methodological concept should be developed and applied in follow-up experiments.

PC duration

Variable: Overall, the emotions investigated did not yield any significant differences in IP duration. In IPs with 3 or more ACs, bored phrases are distinctly longer than other phrases.

If the same analysis is conducted, we find similar trends in the other dialects for the overall durations. If only IPs with 3 or more ACs are considered, we find the same trend as found in the BE dialect in the ZH variety, as well. Bored phrases are second longest in the VS variety, and third longest in the GR group.

In IPs with 3 or more ACs, bored emotion phrases are found to be longer than other phrases, a finding that is particularly distinct for the ZH and the BE group. Van Kleeck and Street (1982) have shown that reticent or annoyed interviewees, a label suitable for many of the BE interviewees, are likely to show less vocal activity, less complex phrasing and a shorter mean length of utterances. In our data, however, it is exactly the BE speakers which, in their bored phrases, exhibit the longest PC duration. Possibly, this is related to their overall slow articulation rate: longer segments lead to longer ACs, while longer ACs result in longer PC durations.

11.3 Sex

11.3.1 Phrase component

11.3.1.1 *PC magnitude*

To test whether there are differences in the PC magnitudes of males and females, a pooled t-test was calculated.⁴ The data are illustrated in Figure 11.5.

4. For detailed information about the labeling/annotation methods applied in this subsection, see Section 6.4.3.

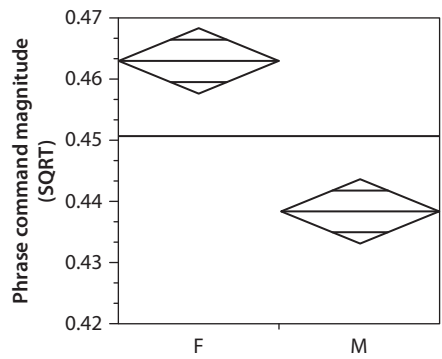


Figure 11.5. Mean diamonds plot showing the PC magnitude of males and females

The analysis was significant.¹³⁸ Females show significantly higher PC magnitudes than men.

Different IP constellations

As a next step, the PC magnitudes of different IP types, meaning IPs with a different number of ACs, were compared.⁵ Figure 11.6 shows the data.

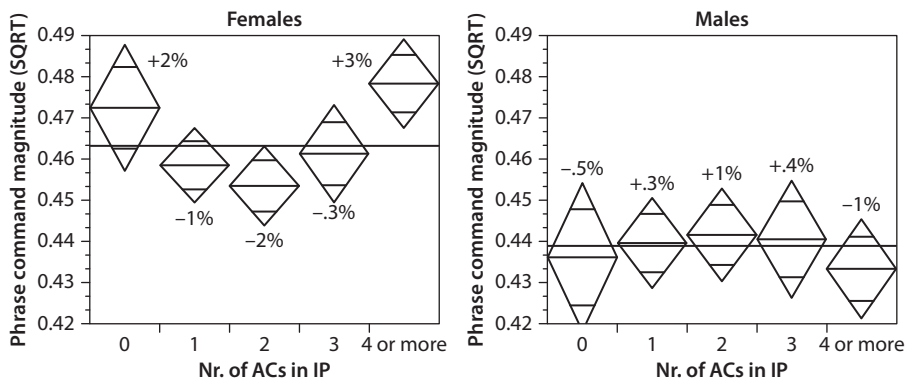


Figure 11.6. Mean diamonds plots showing the PC magnitudes of different IP types, sorted by sex. The percentages indicate the in-/decreases from the means of response

The ANOVAs on female speakers showed significant effects.¹³⁹ The pattern for female speakers is interesting in so far as IPs with 0 ACs and IPs with 4 or more ACs show nearly the same declination (+2%, +3%), while IPs with only 1 and IPs

5. As pointed out earlier, an average IP in the current analysis consists of 2 to 3 ACs, yet the maximum range of ACs that was found is 1 to 16.

with 2 ACs show a much lower declination (-1% , -2%).¹⁴⁰ Male speech exhibits converse tendencies: the highest declination is found in IPs with 2 ACs ($+1\%$), the lowest declination for IPs with 4 or more ACs (-1%).

Cross-sex comparisons

In order to find out whether the PC magnitudes of the different IP types differ between the two sexes, pooled t-tests were calculated. Figure 11.7 indicates the data.

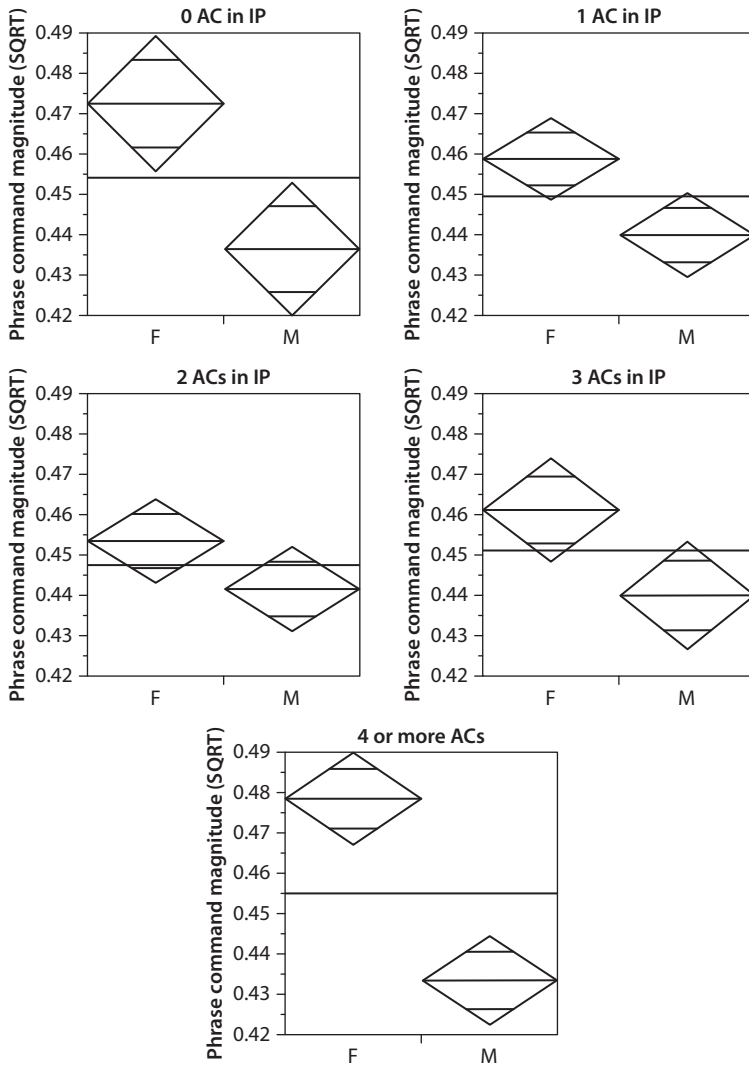


Figure 11.7. Mean diamonds plots showing the PC magnitudes of males and females according to IP type

All of the five t-tests showed significant differences between male and female speech except for IPs with 2 ACs.¹⁴¹ The females demonstrate higher magnitude values in all IP types. The difference between the two sexes is particularly distinct for IPs with 0 and 4 or more ACs.

11.3.2 Accent component

11.3.2.1 AC amplitude

In order to detect whether males and females differ overall with regard to AC amplitude, a pooled t-test was calculated. The AC amplitudes for the different sexes are illustrated in Figure 11.8.

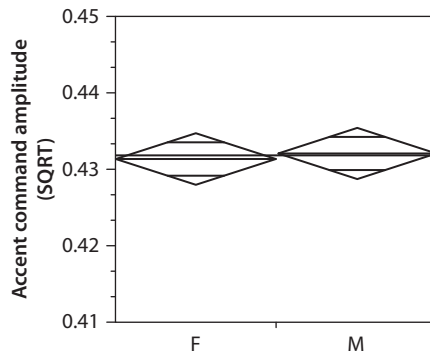


Figure 11.8. Mean diamonds plot showing AC amplitudes according to sex

As can be easily guessed from Figure 11.8, the analysis is not significant.¹⁴² A closer look at the dialects shows that all dialect groups exhibit the same pattern: the differences in AC amplitude are not significant between males and females. The only exception is the BE group, in which males show significantly higher amplitudes.¹⁴³

AC sequence

To examine whether there is a difference in the means of AC amplitudes of first, medial, penultimate, and ultimate ACs between male and female speech,¹⁴⁴ ANOVAs were performed. Figure 11.9 shows the data.

Both ANOVAs showed significant effects.¹⁴⁵ Females increase by 24% from first to medial ACs (the first AC serving as base). Males, too, show a distinct increase, yet only by 18%. The pattern that follows is similar in both groups: AC amplitude falls (–1%) from medial to penultimate AC, and rises again (females +2%, males +1%). Several inter-level differences are significant in both groups.¹⁴⁶

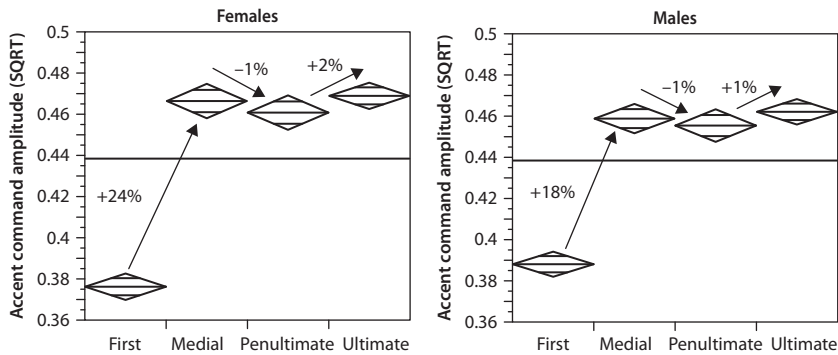


Figure 11.9. Mean diamonds plots showing the different AC positions' amplitudes, according to sex. The arrows and percentages show the in-/decreases in AC amplitude from position to position

Cross-sex comparison

Lastly, t-tests on the AC amplitudes of the different sexes were run according to AC position in the IP. This would show in which AC position males and females vary the most in amplitude. The data is presented in Figure 11.10.

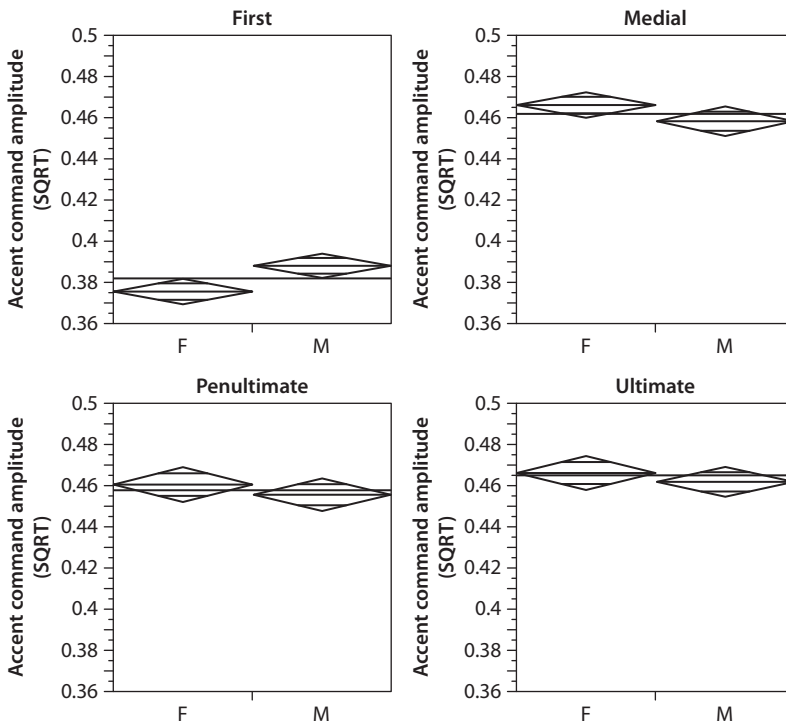


Figure 11.10. Mean diamonds plots showing the AC amplitudes of males and females, sorted by AC position in IP

Only the t-test in first ACs indicated significant differences. Here, the AC amplitudes of the women are significantly lower than the amplitudes of the men.¹⁴⁷ In all other AC positions, females indicate slightly higher AC amplitudes than men.

11.3.2.2 *Summary and discussion*

PC magnitude

Variable: Overall, females exhibit significantly higher declination than males. This is particularly the case in distinctly long and short ACs.

An increased PC magnitude translates into an increased perceived global pitch range as well as, at the same time, a steeper declination (Taylor 1994:40). Therefore, the above findings tie in nicely with what has been observed by McConnell-Ginet (1978), Daly and Warren (2001), and Cruttenden (1986), namely that females exhibit a comparatively wider pitch range. Why this is particularly so in distinctly short and distinctly long ACs is difficult to interpret. McConnell-Ginet (1978:82) argues that the higher and more dynamic pitch range produced by women is a means of attracting and holding the listener's attention. She explains that females make use of this strategy more extensively than men due to their comparative powerlessness. In her view, "dynamic rendition of the text is invaluable in holding the listener's attention if one lacks the authority to require that attention" (McConnell-Ginet's 1978:82). Furthermore, she attributes the higher pitch range of women's speech to more frequent contact with younger children. Verbal stimuli produced with an increased pitch range might better attract the children's attention (see also Fernald & Simon 1984).

AC amplitude

Variable: Overall, no difference in AC amplitude was found between females and males. However, females were shown to make a greater distinction in amplitude between first and medial ACs. Accordingly, females exhibit lower amplitudes in first ACs if compared to male speech.

This result provides support for the hypothesis postulated in the previous paragraph. The more distinct AC amplitudes between first and medial position ACs in female speech create a more distinct intonational contrast. This contrast, combined with their more distinct pitch range, contributes to intonational prominence, and renders their speech somewhat more salient and expressive than male speech (see Cruttenden 1986:136).

CHAPTER 12

Linear models

This final chapter of the statistics section is devoted to the dialect-specific, statistical regression models for the prediction of fundamental frequency behavior. Such a model was created for each dialect and for each of the investigated five parameters.

Linear regression models allow for a distillation of the relative contribution of independent variables in the explanation of f_0 variability. The modeling is performed in order to answer questions about the predictive power of the different linguistic, paralinguistic, or non-linguistic factors. In other words, for each given parameter in all of the four dialects, the models expose those variables which explain most variability (best predictors) or least variability (weakest predictors). In the end, this will allow a comparison of the relative weight of predictors and will shed light on the differences and similarities between the different dialects.

In a first subsection, the choice for and assumptions of the statistical methods will be explained, followed by a brief and general discussion as to how independent variables were selected and how the relative importance of the variables was determined. Finally, a short introduction to the visual presentation of the models is given. The main part of the section revolves around the individual statistical models.

12.1 Preliminaries

12.1.1 Multiple linear regressions

Direct multiple linear regression models [henceforth MLR] are applied in the present study because, even though the response variables are continuous, the explanatory variables fall into all of the three different categories of scales: continuous, nominal, and ordinal. JMP (release 7.0) automatically represents non-continuous variables with dummy variables in linear regressions.

There are four core assumptions that are not robust to violation when it comes to the relationship between dependent and independent variables that

underlie MLR. All of the created models adhere to these assumptions (based on Osborne 2002b):

Linearity, between the dependent variable and the independent variables. This was tested by examining the residual plots¹ where the residuals are regressed on the predicted values. The points are symmetrically distributed around the horizontal line at 0.

Homoscedasticity, i.e. equal variances of residuals. Again, this was tested by observation of the scatterplots when the residuals are regressed on the predicted values. The residuals are randomly scattered around 0.

Normality of residuals, i.e. the residuals are distributed normally. This was tested with normal quantile plots. In all instances, the residuals follow normal or near normal distributions.

Multicollinearity, i.e. a strong relationship between two or more predictors, should not be present as it renders variable-specific regression coefficients instable and obstructs subsequent inference processes. Multicollinearity was tested with the variance inflation factor [henceforth VIF]. A VIF of > 10 is indicative of a serious degree of multicollinearity for a variable. Collinear variables are treated accordingly, but more on that in the section on the actual models.

The assumptions of a logistic regression are less strenuous than those of an MLR. In a binary logistic regression – in this case rising and falling ACs as response – the dependent variable must be dichotomous and each case can only occur once and in either one or the other group. Multicollinearity, too, should not be present. Both of those assumptions are met in the created, nominal logistic regression models.

12.1.2 Selection of independent variables

The models created in the present study are not geared at capitalizing on the *coefficient of determination* (R^2), i.e. maximizing the degree to which a given model is able to predict a certain phenomenon. Therefore, the aim is not to create models which include as many potential predictors as possible. Rather, the models aim at a (hopefully) linguistic explanation of the investigated parameters. Hence, model-internal effects, as described in the overall results section, are not included in the statistical models although they would additionally boost the R^2 and thus increase the predictive power of the model. Instead, the analyses include only linguistic, paralinguistic, and non-linguistic predictors.

There are, however, further reasons which support this decision:

Circularity

We have already established model-internal interactions, such as for instance the interaction between *T1dist_rise* and AC duration (see Chapter 8). Such effects

1. Residuals are the unexplained variations that occur once the regression model has been fit.

are not included in the statistical models since they may create circularity. If AC duration were used as an explanatory variable for *T1dist_rise* (the longer the AC, the later the rise relative to segment onset), then it is equally legitimate to use *T1dist_rise* as an explanatory variable for AC duration (the later the rise relative to segment onset, the longer the AC). In other words, an independent variable X that explains variance in a response variable Y is not to be determined itself by Y.

Suboptimal modeling

As an example, it was shown earlier (see Chapter 8) that AC position in the IP correlates with AC amplitude, or that PC magnitude is a significant predictor of AC magnitude and duration. Such effects are largely attributed to the suboptimal modeling of flat *f0* contours typical of spontaneous speech. Thus, these effects are not included in the statistical models since they are largely due to suboptimal modeling.

The downside of not including model-internal effects is that the predictive power of the models will be rather low. The upside, on the other hand, is that sensible linguistic interpretation is facilitated since the contribution of each investigated linguistic, paralinguistic, and non-linguistic variable towards explaining *f0* variance can be assessed individually for each dialect.

The models were built as follows:

1. Each potential predictor was tested for statistically significant association with the corresponding response in bivariate tests, which for the most part amounts to the statistics conducted in the previous sections.² Additional, dialect-specific, tests were run for the predictors *emotion*, *articulation rate*, and *sex*: the previous analyses of *emotion* only considered the BE dialect, the analyses of *articulation rate* only the ZH dialect, while for *sex*, the data of all dialects were pooled together. Those variables that showed significant effects with the dependent variable were considered for use in the linear models.
2. In a second step, the MLRs and nominal logistic regressions were performed. Since no *a priori* hypotheses were postulated with regard to a specific order of entry of the predictor variables, a direct, simultaneous method was applied. The regressions were first conducted with all predictors that confirmed significant effects in the bivariate tests. Subsequently, the effect sizes of all factors within the models were assessed with standard output effect tests, as well as with likelihood-ratio tests for nominal logistic regressions in JMP.

2. For the statistical models, Q phrases as well as phrases labeled as *fear* phrases are suppressed due to low frequencies. The applied predictors are those variables used in the previous statistics sections, i.e. the variables largely contain the same levels. In case levels of a variable were pooled together and/or nested, further explanations are provided.

3. In case of non-significant effects, the predictor with the lowest non-significant effect was removed and the model was run again. This procedure was repeated until the model contained only significant effects. The aim was to work with parsimonious models that contain only the most critical explanatory variables. The reason for this is the fact that statistical models become more stable if they include only those predictors that, in the model, are associated significantly with the response.

It is crucial to note, at this point, that the weight of the variables in the models of the different parameters do not directly translate to the findings on the same variables given in the statistics section earlier (although they are likely to). It needs to be borne in mind that the ranking of the variables is based solely on *calculations within one parameter and within one dialect*. In other words, if the VS dialect exhibits a special behavior with regard to a certain variable and, thus, stands out particularly in the statistics section because of distinct cross-dialectal differences, this does not mean that this variable is automatically a predictor in the VS models. This is because the variables in the models are not weighed against absolute values of the *same variables in other dialects*, but are weighed against *other variables within the same dialect*. For example, if a dialect stands out particularly with its very strong distinction in PC magnitude between C and T phrases, it is likely that the variable *phrase type* will in fact be a predictor in the model of that dialect, probably a predictor which is ranked higher than in the corresponding models of other dialects. However, this need *not* be the case, since each model lists only the *relative weight* of a predictor compared to the weight of other predictors in the same dialect. It may be the case that the dialect in question exhibits other (and possibly more important) predictors than *phrase type* when it comes to explaining PC magnitude variance, for example.

12.1.3 Determining relative importance of explanatory variables

Determining the relative importance of factors in an MLR or decomposing R^2 into its component parts is quite thorny, and there are several methods to choose from. In the present study, Type III sums of squares (also referred to as *partial sums of squares*) are used for this estimation. Type III sums of squares are calculated in that the full model is compared to the full model without the factor of interest. In other words, the Type III sums of squares of a variable provide the additional variance that is explained when the variable of interest is added to the model. The relative importance of each individual factor is determined by its contribution in percent to the total Type III sums of squares of the significant effects in the model, i.e:

$$\text{Relative importance of an effect} = \left(\frac{\text{Type III sums of squares of the effect}}{\text{Sum of the Type III sums of squares of all significant effects}} \right) * 100$$

With regard to determining the relative importance of effects in logistic regressions, a similar procedure was chosen. For all factors, likelihood-ratio tests were applied. Likelihood-ratio chi-square tests “are calculated as twice the difference of the log-likelihoods between the full model and the model constrained by the hypothesis to be tested (the model without the effect)” (JMP: Statistics and Graphics Guide 2008). Consequently, the likelihood-ratio chi-square value of each variable is understood to provide the additional variability that is explained when this effect is added to the model. Thus, the relative importance of an effect in a logistic regression is defined as likelihood-ratio chi-square values of the effect divided by the sum of the likelihood-ratio chi-square values of all significant effects, times one hundred. The weight of the variables is reported in the same fashion as ANOVAs, yet, instead of reporting the degrees of freedom within groups, the sum of squares or the chi-square test values are reported.

12.1.4 Visualization of statistical models

The results of the statistical models are visualized with radar charts, a highly useful technique to illustrate multivariate data. Figure 12.1 explains the illustration of the statistical model of AC amplitude for the VS dialect.

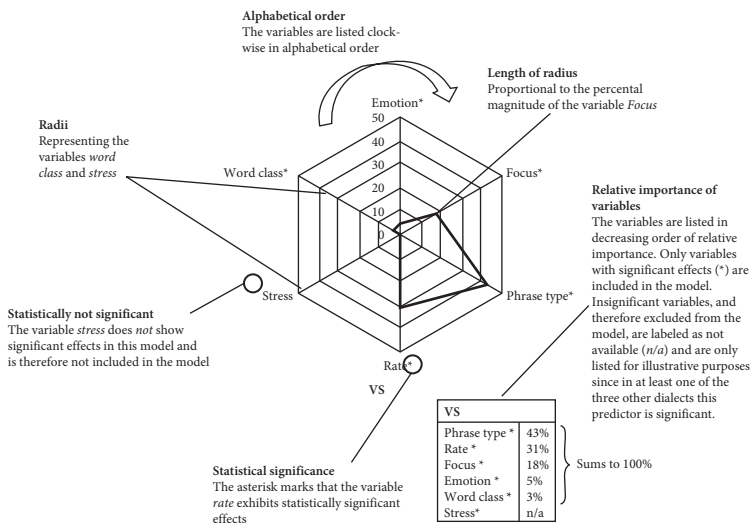


Figure 12.1. Explanation of the applied radar charts

12.2 Phrase component

12.2.1 PC magnitude

The effects used to create the PC magnitude models are fairly easy to follow. It should be noted, however, that the variable *position of focus* is nested in the factor *focus* (two levels: narrow focus/broad focus). That is, the effect of whether the focused constituent in a narrow focus IP occurs in phrase-initial, medial, final or only position is fitted in the main effect *focus*. Bivariate tests showed that *emotion* does not have a statistically significant association with the dependent variable in any of the dialects. For this reason, *emotion* is not included as a predictor in any of the models. The radar charts illustrating the models and the corresponding relative importance values within the models are shown in Figure 12.2 and Table 12.1, respectively.

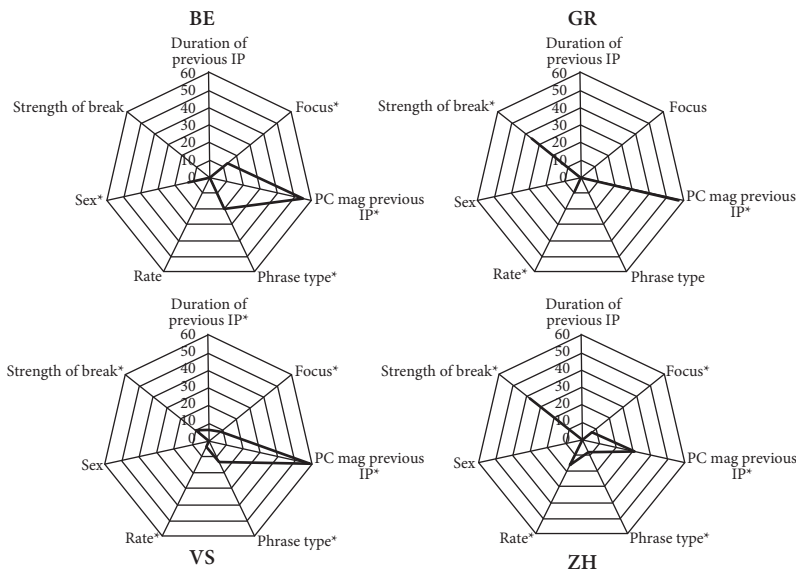


Figure 12.2. Radar charts illustrating the MLRs of PC magnitude for all four dialects

In the BE dialect, the four explanatory variables produced an adjusted R^2 of .14 for the prediction of PC magnitude.¹⁴⁸ In other words, the model explains 14% of the variance in PC magnitude. The single best predictor is the *PC magnitude of previous IP* (55%), followed by *phrase type* (20%), *focus* (13%), and *sex* (12%).¹⁴⁹

In the GR dialect, six predictors show an adjusted R^2 of .17 for the prediction of PC magnitude.¹⁵⁰ This means that the model explains 17% of the variation in

Table 12.1. Relative importance of effects within the PC magnitude models

BE		GR	
PC mag previous IP*	55%	PC mag previous IP*	56%
Phrase type*	20%	Strength of break*	35%
Focus*	13%	Rate*	9%
Sex*	12%	Duration of previous IP	n/a
Duration of previous IP	n/a	Focus	n/a
Rate	n/a	Phrase type	n/a
Strength of break	n/a	Sex	n/a
VS		ZH	
PC mag previous IP*	58%	Strength of break*	37%
Phrase type*	14%	PC mag previous IP*	31%
Strength of break*	9%	Rate*	17%
Focus*	8%	Phrase type*	9%
Duration of previous IP*	6%	Focus*	6%
Rate*	4%	Duration of previous IP	n/a
Sex	n/a	Sex	n/a

PC magnitude. The strongest effects are again found for *PC magnitude of previous IP* (56%), followed by *strength of break* (35%), and *rate* (9%).¹⁵¹

In the VS dialect, the six predictors *PC magnitude of previous IP*, *phrase type*, *strength of break*, *focus*, *duration of previous IP*, and *articulation rate* explain 16% of the variance in PC magnitude, i.e. they produce an adjusted R^2 of .16.¹⁵² The best predictor of PC magnitude is by far *the magnitude of previous IP* (58%), followed by *phrase type* (14%), *strength of break* (9%), *focus* (8%), *duration of previous IP* (6%), and *rate* (4%).¹⁵³

In the ZH variety, the five predictors showed an adjusted R^2 of .18, which means that the linear model explains 18% of the variance in PC magnitude.¹⁵⁴ In contrast to the other three dialects, the best predictor for PC magnitude in the ZH variety is the *strength of the break* prior to the IP (37%), followed by *PC magnitude of previous IP* (31%), the *articulation rate* (17%), *phrase type* (9%), and *focus* (6%).¹⁵⁵

12.2.2 PC duration

Like the effects for PC magnitude, the effects used to create the PC duration models are also straightforward. However, for PC duration, the variable *focus* (two

levels: narrow focus/broad focus) no longer contains the nested variable *position of focus* since, as mentioned earlier, a phrase is evidently longer if it contains a medial position focus because a preceding as well as a following AC are needed in order to label an AC as *medial-position focus*. Thus, these phrases are inherently longer. The radar charts illustrating the models as well as the corresponding relative importance values within the models are shown in Figure 12.3 and Table 12.2 respectively.

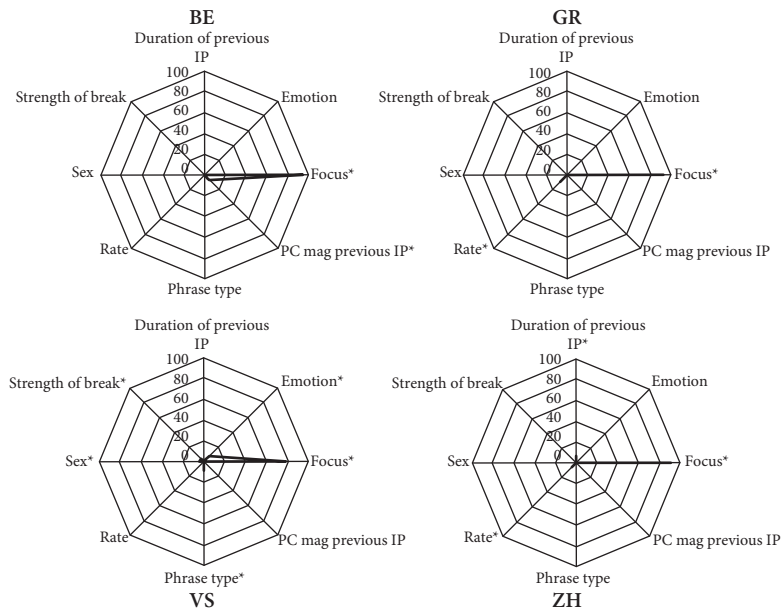


Figure 12.3. Radar charts illustrating the MLRs of PC duration for all four dialects

In the BE dialect, the two explanatory variables show an adjusted R^2 of .07 for the prediction of PC duration.¹⁵⁶ The model thus explains 7% of the variance in PC duration. The best predictor is by far *focus* (94%) followed by the other predictor *PC magnitude of previous IP* (6%).¹⁵⁷

In the GR dialect, there are also only two significant predictors for PC duration: *focus* and *articulation rate*, which produce an adjusted R^2 of .13.¹⁵⁸ The model explains 13% of the variation in PC duration. *Focus*, too, is by far the best predictor of PC duration (91%), followed by *rate* (9%).¹⁵⁹

The VS dialect exhibits by far the greatest number of significant predictors for PC duration: *focus*, *phrase type*, *emotion*, *sex*, and *strength of break*.¹⁶⁰ They explain 16% of the variance in PC duration, i.e. they produce an adjusted R^2 of .16. The best predictor of PC magnitude is again *focus* (78%), followed by *phrase type* (8%), *emotion* (7%), *sex* (4%), and *strength of break* (4%).¹⁶¹

Table 12.2. Relative importance of effects within the PC duration models

BE		GR	
Focus*	94%	Focus*	91%
PC mag previous IP*	6%	Rate*	9%
Duration of previous IP	n/a	Duration of previous IP	n/a
Emotion	n/a	Emotion	n/a
Phrase type	n/a	PC mag previous IP	n/a
Rate	n/a	Phrase type	n/a
Sex	n/a	Sex	n/a
Strength of break	n/a	Strength of break	n/a
VS		ZH	
Focus*	78%	Focus*	89%
Phrase type*	8%	Rate*	6%
Emotion*	7%	Duration of previous IP*	5%
Sex*	4%	Emotion	n/a
Strength of break*	4%	PC mag previous IP	n/a
Duration of previous IP	n/a	Phrase type	n/a
PC mag previous IP	n/a	Sex	n/a
Rate	n/a	Strength of break	n/a

In the ZH variety, the three predictors *focus*, *rate*, and *duration of previous IP* produce an adjusted R^2 of .15, which again means that the linear model explains 15% of the variance in PC duration.¹⁶² By far, the best predictor is again *focus* (89%), while *rate* (6%) and *duration of previous IP* (5%) occupy much weaker positions.¹⁶³

12.3 Accent component

12.3.1 AC amplitude

The effects used to create the AC amplitude models are easy to follow as well. The predictors *focus* and *phrase type*, however, should be explained further. The variable *AC focus type* (i.e. no focus, pre-focus, focus, or post-focus, excluding exceptions)³ is nested in the factor *focus* (two levels: narrow focus/broad focus).

3. Exceptions refer to focus constituents which are not spanned by ACs.

That is, the effect of whether the AC is a no focus, pre-focus, focus, or post-focus AC in a narrow focus IP is fitted in the main effect *focus*. Furthermore, the variable *phrase type* (two levels: continuing/terminating) contains the newly created nested variable *phrase type subgroup*, which contains six levels: continuing, continuing penultimate, continuing ultimate, terminating, terminating penultimate, and terminating ultimate. In other words, all IPs with first and medial ACs, as well as single ACs were pooled together to form two groups: a pool of terminating and a pool of continuing ACs. This measure was undertaken for three reasons:

1. The statistical models are likely to become more stable the less levels there are in a predictor.
2. The unfavorable modeling effect, i.e. the increase of AC amplitude in the course of an IP, can be leveled to a degree by means of pooling together certain AC positions in an IP.
3. As for the analysis of phrase type, it is particularly the penultimate and ultimate ACs that are most important. This is why these positions are assigned separate levels.

Furthermore, it should be noted that even though the explanatory variable *position of first stressed syllable* showed significant effects in the bivariate tests for all dialects during the effect screening process, the variable is not included in any of the models. This is due to the fact that it is collinear with the predictor stress (i.e. $VIF > 10$), which includes the three levels 0, 1, and 2 or more stressed syllables.⁴ The predictor word class contains the usual levels: 0, 1, and 2 or more lexical syllables. The radar charts illustrating the models and the corresponding relative importance values within the models are shown in Figure 12.4 and Table 12.3, respectively.

In the BE dialect, the five explanatory variables show an adjusted R^2 of .13 for the prediction of AC amplitude.¹⁶⁴ The model explains 13% of the variance in AC amplitude. The single best predictor is *phrase type* (34%), followed by *focus* (24%), *stress* (18%), *rate* (17%), and *emotion* (7%).¹⁶⁵

In the GR dialect, we only find three significant predictors for AC amplitude: *focus*, *articulation rate*, and *phrase type* which produce an adjusted R^2 of .12.¹⁶⁶ The model explains 12% of the variation in AC amplitude. *Focus* is the best predictor (41%), followed by *articulation rate* (32%) and *phrase type* (27%).¹⁶⁷

4. The collinearity between the predictors *number of stressed syllables in the AC* and the variable *position of first stressed syllable in the AC* occurs regardless of the variable in the response. Hence, in this model as well as in all subsequent models, if applicable, only the predictor *number of stressed syllables in the AC* is considered.

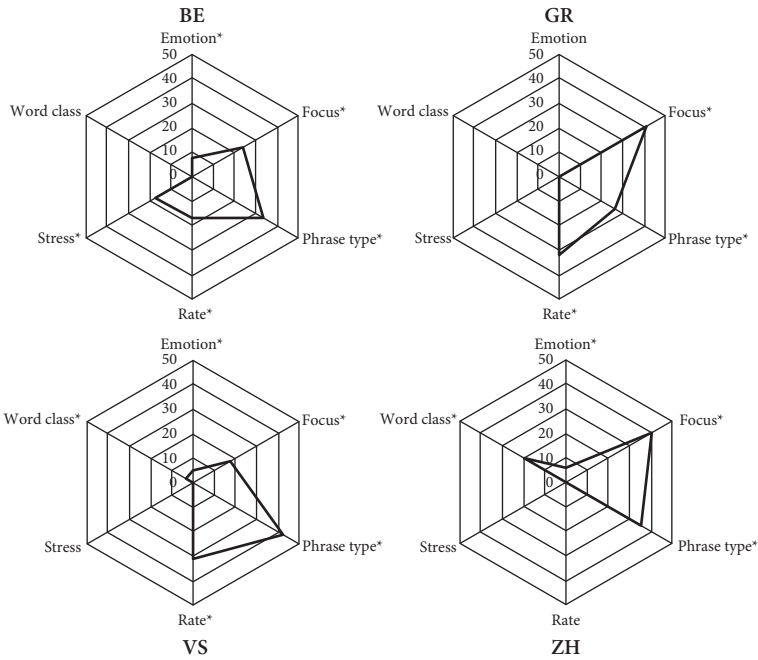


Figure 12.4. Radar charts illustrating the MLRs of AC amplitude for all four dialects

Table 12.3. Relative importance of effects within the AC amplitude models

BE		GR	
Phrase type*	34%	Focus*	41%
Focus*	24%	Rate*	32%
Stress*	18%	Phrase type*	27%
Rate*	17%	Emotion	n/a
Emotion*	7%	Stress	n/a
Word class	n/a	Word class	n/a
VS		ZH	
Phrase type*	43%	Focus*	40%
Rate*	31%	Phrase type*	35%
Focus*	18%	Word class*	20%
Emotion*	5%	Emotion*	6%
Word class*	3%	Rate	n/a
Stress	n/a	Stress	n/a

In the VS dialect, the predictors *phrase type*, *rate*, *focus*, *emotion*, and *word class* produce an adjusted R^2 of .09.¹⁶⁸ Thus, these predictors explain 9% of the variance in AC amplitude. The best predictor of AC magnitude is *phrase type* (43%), followed by *rate* (31%), *focus* (18%), *emotion* (5%), and *word class* (3%).¹⁶⁹

In the ZH variety, the four predictors *focus*, *phrase type*, *word class*, and *emotion* produce an adjusted R^2 of .09, which again means that the linear model explains 9% of the variance in AC amplitude.¹⁷⁰ The best predictor is *focus* (40%), followed by *phrase type* (35%), *word class* (20%), and *emotion* (6%).¹⁷¹

12.3.1.1 AC duration

The effects used to create models of AC duration models are the same as those for AC amplitude models, except that for AC duration, the variables *stress* and *word class* are not included since ACs are inherently longer the more stressed and/or lexical syllables they contain. The radar charts illustrating the models and the corresponding relative importance values within the models are shown in Figure 12.5 and Table 12.4, respectively.

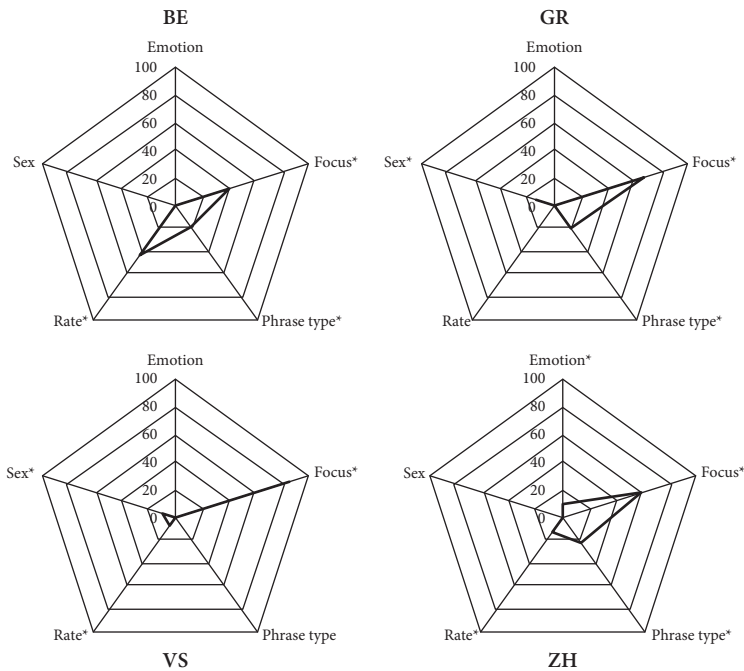


Figure 12.5. Radar charts illustrating the MLRs of AC duration for all four dialects

Table 12.4. Relative importance of effects within the AC duration models

BE		GR	
Rate*	43%	Focus*	67%
Focus*	39%	Phrase type*	19%
Phrase type*	18%	Sex*	14%
Emotion	n/a	Emotion	n/a
Sex	n/a	Rate	n/a
VS		ZH	
Focus*	84%	Focus*	56%
Sex*	9%	Phrase type*	21%
Rate*	7%	Rate*	13%
Emotion	n/a	Emotion*	10%
Phrase type	n/a	Sex	n/a

In the BE dialect, the three explanatory variables show an adjusted R^2 of .06 for the prediction of AC duration.¹⁷² The model explains 6% of the variance in AC duration. The best predictor is *articulation rate* (43%), followed by *focus* (39%) and *phrase type* (18%).¹⁷³

In the GR dialect, the three significant predictors for AC duration, namely *focus*, *phrase type*, and *sex*, show an adjusted R^2 of .06.¹⁷⁴ The model explains 6% of the variation in AC duration. *Focus* is the best predictor (67%), followed by *phrase type* (19%), and *sex* (14%).¹⁷⁵

In the VS dialect the predictors *focus*, *sex*, and *articulation rate* produce an adjusted R^2 of .05.¹⁷⁶ The predictors therefore explain 5% of the variance in AC duration. The best predictor is by far *focus* (84%), while the predictive power of *sex* (9%) and *articulation rate* (7%) are comparatively low.¹⁷⁷

In the ZH variety, the four predictors *focus*, *phrase type*, *rate*, and *emotion* produce an adjusted R^2 of .06, which again means that the linear model explains 6% of the variance in AC duration.¹⁷⁸ The best predictor is *focus* (56%), followed by *phrase type* (21%), *rate* (13%), and *emotion* (10%).¹⁷⁹

12.3.1.2 AC timing

The effects used to create the *T1dist_rise* models are the same as those applied for the AC duration models. The reason why the variables *stress* and *word class* were not considered is that, again, they correlate strongly with AC duration. AC duration was shown to have an effect on *T1dist_rise*: the longer the AC, the later the

rise relative to segment onset. Since AC duration is a model-internal parameter, the variables *stress* and *word class* are not included in the models. Figure 12.6 displays the radar charts illustrating the models for *T1dist_rise*, while Table 12.5 gives the corresponding relative importance values within the models.

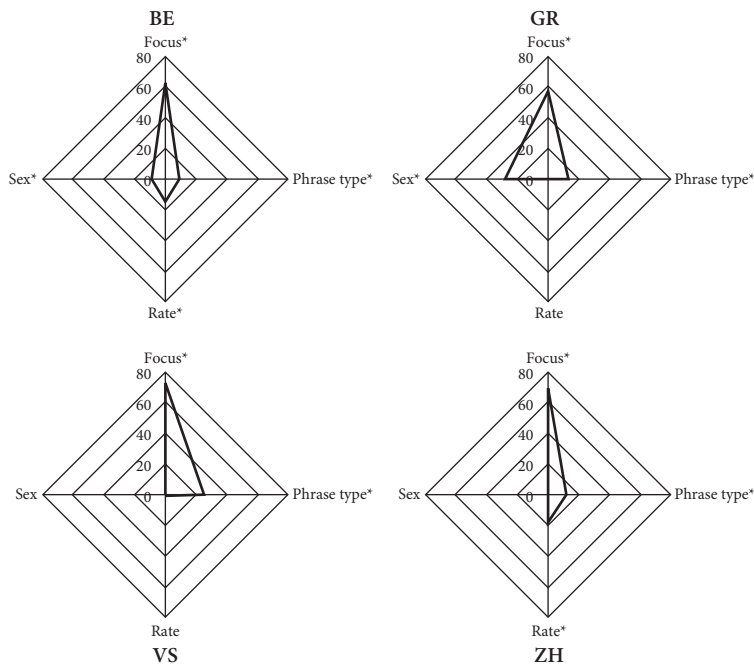


Figure 12.6. Radar charts illustrating the MLRs of *T1dist_rise* for all four dialects

Table 12.5. Relative importance of effects within the *T1dist_rise* models

BE		GR	
Focus*	67%	Focus*	59%
Rate*	15%	Sex*	28%
Phrase type*	9%	Phrase type*	13%
Sex	9%	Rate	n/a
VS		ZH	
Focus*	75%	Focus*	70%
Phrase type*	25%	Rate*	18%
Rate	n/a	Phrase type*	12%
Sex	n/a	Sex	n/a

In the BE dialect, the four explanatory variables show an adjusted R^2 of .04 for the prediction of *T1dist_rise*.¹⁸⁰ The model explains 4% of the variance in AC duration. The single best predictor is *focus* (67%), *articulation rate* (15%), *phrase type* (9%), and *sex* (9%).¹⁸¹

In the GR dialect, the three significant predictors for *T1dist_rise*, which are *phrase type*, *focus*, and *sex*, produce an adjusted R^2 of .05.¹⁸² The model explains 5% of the variation in *T1dist_rise*. *Focus* is the single best predictor (59%), followed by *sex* (28%), and *phrase type* (13%).¹⁸³

In the VS dialect, there are only two significant predictors, *phrase type* and *focus*, which produced an adjusted R^2 of .03.¹⁸⁴ The predictors explain 3% of the variance in *T1dist_rise*. The best predictor is by far *focus* (75%), while *phrase type* is a much weaker predictor (25%).¹⁸⁵

In the ZH variety, the three predictors *articulation rate*, *phrase type*, and *focus* produce an adjusted R^2 of .03, which again means that the linear model explains 3% of the variance in *T1dist_rise*.¹⁸⁶ The best predictor is again by far *focus* (70%), followed by *rate* (18%), and *phrase type* (12%).¹⁸⁷

Dialect profiles

The aim of this section is to collate the most important findings into what I call *dialect profiles*. Essentially, these profiles include the most important *exceptional features* of the dialects, findings on the *dialect-internal structures*, as well as a last section called *signature features*, which gives a multi-page summary of all the results and interpretations contrived in this study. It should be pointed out again that the following specifications and interpretations are by all means hypothetical and require further support. After all, this study aims to provide a first description of *f0* contours of four Swiss German dialects in the framework of the Command-Response model. In the light of the inexistent material for comparison and the amount of findings contrived in this study, many linguistic interpretations of the results, even if highly intriguing, are not sufficiently informed. This is not the main concern of this study.

The dialects will be presented in alphabetical order and discussed individually. First, in the *exceptional features* section, a profile for the *f0* contours of the respective dialect is created. The profile summarizes all these instances per dialect presented in the previous statistics chapters (excluding the linear models and logistic regressions) in which the dialect behaved extraordinarily from a cross-dialectal perspective and/or from the perspective of *f0* behavior in the context of a given variable.

The next section on *dialect-internal structure* shifts to a discussion of the models generated for each dialect. This involves a discussion of the ranking of the predictors and the power of the *coefficients of determination* of the variables for the individual dialect but also across dialects. It is crucial to note that certain characteristics discussed in the exceptional features section may not occur as predictors in the models, while others will. This is because the weight of the variables in the models of the different parameters does not directly translate to the findings on the same variables given in the statistics section earlier (although they are likely to approximate). It needs to be borne in mind that the ranking of the variables is based solely on *calculations within one parameter value and within one dialect*. In other words, if the VS dialect exhibits a special behavior with regard to a certain variable and, thus, stands out particularly in the statistics section because of distinct cross-dialectal differences, this does not mean that this variable is automatically a predictor in the VS models. This is because the variables in

the models are not weighed against absolute values of the *same variables in other dialects*, but are weighed against *other variables within the same dialect*. If a dialect, for example, is exceptional in its very strong distinction in PC magnitude between C and T phrases, it is likely that *phrase type* occurs as a predictor in the model of that dialect, probably a predictor which is ranked higher than in the model of other dialects. However, this need *not* be the case, since each model lists the *relative weight* of each predictor for a specific dialect. This dialect in question may exhibit other (and possibly more important) predictors than *phrase type* when it comes to explaining PC magnitude variance.

In the third subsection, *signature features* are distilled for each dialect. The idea is to provide a short, linguistically informed summary of those dialectal features that stand out in a cross-dialectal comparison. This section incorporates both findings on the dialect-specific, exceptional features as well as cross-dialectal variation of model predictors and ranking. The question to be answered is: in which respect does dialect W differ with regard to dialects X, Y, and Z?

Towards the end of this section, the focus will shift to a geolinguistic structuring of the dialects. Here, the dialects' *exceptional features* as well as *dialect-internal structures* are placed into a geographic context in order to assess the validity of an Alpine-Midland and an East-West divide (see Hotzenköcherle 1961; Lötscher 1983).

13.1 Bern

13.1.1 Exceptional features

Phrase component

Magnitude:

Phrase type: compared to the other dialects, the BE make a strong distinction between C and T phrases.

Duration:

Phrase type: When compared to the other dialects, the BE make a strong distinction between C and T phrases.

Emotion: within the variable, bored IPs with 3 or more ACs are distinctly longer than other emotion phrase types.

Accent component

Amplitude:

Stress: within the variable, the BE make a strong distinction between low amplitude ACs with 0 stressed syllables and high amplitude ACs with 1 stressed syllable. Strong

distinction between high-amplitude ACs with stress in first syllable position and very low-amplitude ACs with 0 position stress.

Word class: within the variable, the differences in amplitude are distinct. ACs with 0 lexical syllables are significantly lower than ACs with 1 or 2 or more lexical syllables.

Duration:

Stress: compared to the other dialects they show overall long durations for all AC types.

Focus: within the variable, increase-decrease-increase pattern from one AC type to the next.

Timing:

Stress: compared to the other dialects, the BE show very late rises overall in all AC types.

Phrase type: compared to the other dialects, the BE generally exhibit late rises in T phrases. In ultimate ACs of T phrases which begin with stressed syllables, the BE exhibit by far the latest rises relative to segment onset.

13.1.2 Dialect-internal structure

Phrase component

Magnitude: Given the relative weight of the predictors, the variable *PC magnitude of the previous IP* explains the greatest amount of variance in *PC magnitude* (55%), followed by *phrase type* (20%), *focus* (13%), and *sex* (12%). The BE dialect is the only group for which *sex* is a significant predictor and *strength of break* and *rate* are not significant predictors. *Focus* and *sex* are ranked high and are thus more important in explaining variation in the BE model than in models of other dialects. As to the actual effect sizes, a comparison with the other dialect models reveals that *phrase type* as well as *focus* have the strongest effects in the BE model. With 14%, this model explains the least variation in PC magnitude when compared to the models of the other dialects.

Duration: Given the relative weight of the predictors, *focus* explains by far the most variance in *PC duration* (94%), followed by *PC magnitude of previous IP* (6%). BE is the only dialect in which *PC magnitude of the previous IP* is a significant predictor. With 7%, this model explains the least variation in PC duration when compared to the models of the other dialects.

Accent component

Amplitude: Given the relative weight of the predictors, *phrase type* shows the greatest effects with *AC amplitude* (34%), followed by *focus* (24%), *stress* (18%),

articulation rate (17%), and *emotion* (7%). As to the actual effect sizes, a comparison with the other dialect models reveals that *stress* bears a critical role only in the BE model. No other dialect model lists *stress* as a significant predictor. With 13%, this model explains most variation in AC amplitude when compared to the models of the other dialects.

Duration: Given the relative weight of the predictors, *articulation rate* explains the most variance in AC *duration* (43%), followed by *focus* (39%) and *phrase type* (18%). *Articulation rate* is ranked highest and is thus more important in explaining variation in the BE model than in models of other dialects. As to the actual effect sizes, a comparison with the other dialect models reveals that in no other dialect model does the variable *rate* generate such strong effects. Also, compared to those dialect models which list *phrase type* as a significant predictor, this variable shows least effects in the BE model. With 6%, this model, along with the GR and ZH models, explains the most variation in AC duration when compared to the model of the VS dialect.

Timing: Given the relative weight of the predictors, *focus* explains the most variance in *T1dist_rise* (67%), followed by *articulation rate* (15%), *phrase type* (9%), and *sex* (9%). As to the actual effect sizes, a comparison with the other dialect models reveals that *phrase type* produces the least effects in this model.

13.2 Grisons

13.2.1 Exceptional features

Phrase component

Magnitude:

Phrase type: very high magnitudes in both C and T phrases if compared to the magnitudes of the other dialects.

Prosodic paragraphing: contrary to the trend in other dialects, strength of break preceding the IP critically determines PC magnitude.

Duration:

Phrase type: compared to the other dialects' durations, the GR show the longest durations in T phrases.

Accent component

Amplitude:

Stress: within the variable, they show little distinction between AC types with 0, 1, or 2 or more stressed syllables. Little distinction between ACs with stress on the first syllable and ACs with 0 position stress.

Word class: within the variable, the differences in amplitude are less distinct between the AC types. ACs with 0 lexical syllables are distinctly high when compared to those of the other dialects.

Focus: within the variable, the GR show the highest amplitudes in post-focal ACs and not in focus ACs. Increase-increase-decrease pattern from one AC type to the next.

Phrase type: compared to the other dialects, the GR is the only group in which C phrase ACs in first AC position are lower than their T phrase counterparts. Ultimate ACs in T phrases are high when compared to ultimate ACs in T phrases of the other dialects.

Timing:

Stress: when compared to the other dialects' timing, the Grisons speakers show comparatively early rises in all AC types.

13.2.2 Dialect-internal structure

Phrase component

Magnitude: Given the relative weight of the predictors, *PC magnitude of the previous IP* explains the greatest amount of variance in *PC magnitude* (56%), followed by *strength of break* (35%), and *articulation rate* (9%). The GR is the only dialect in which *phrase type* and *focus* are not significant predictors. As to the actual effect sizes, a comparison with the other dialect models reveals parallels to the ZH model in that *strength of break* shows highly significant effects in both models. The GR is the variety with the least number of predictors.

Duration: Given the relative weight of the predictors, *focus* explains by far the most variance in *PC duration* (91%), followed by *articulation rate*.

Accent component

Amplitude: Given the relative weight of the predictors, *focus* explains the greatest amount of variance in *AC amplitude* (41%), followed by *articulation rate* (32%) as well as *phrase type*. No other dialect shows such few predictors. GR is the only dialect in which *emotion* is not a significant predictor. *Phrase type* is ranked low and is thus less important in explaining variation in the GR model than in models of other dialects. As to the actual effect sizes, a comparison with the other dialect models reveals that *focus* as well as *articulation rate* effects are greatest in this model. *Phrase type*, on the other hand, explains comparatively little variation when compared to the other three dialect models.

Duration: Given the relative weight of the predictors, *focus* explains the most variance in *AC duration* (67%), followed by *phrase type* (19%) and *sex* (14%). The GR variety is the only dialect in which *rate* is not a significant predictor. The only

other dialect in which *sex* is also listed as a significant predictor is the VS dialect. On average, the GR male produces longer ACs than the female. With 6%, this model, along with the BE and ZH models, explains the most variation in *AC duration* when compared to the VS model.

Timing: Given the relative weight of the predictors, *focus* explains the most variance in *T1dist_rise* (59%), followed by *sex* (28%) and *phrase type* (13%). *Sex* is ranked high and is thus more important in explaining variation in the GR model than in models of other dialects. As to the actual effect sizes, a comparison with the other dialect models reveals that *focus* shows least effects in this model. *Sex*, on the other hand, explains much variance if compared to the BE model, the only other group in which *sex* is also a significant factor. With 5%, this model explains the most variation in *T1dist_rise* when compared to the models of the other dialects.

13.3 Valais

13.3.1 Exceptional features

Phrase component

Magnitude:

Phrase type: strong distinction between C and T phrases when compared to the other dialects.

Duration:

Phrase type: strong distinction between C and T phrases when compared to the other dialects.

Accent component

Amplitude:

Stress: within the variable, they make little distinction between AC types with 0, 1, and 2 or more stressed syllables. Little distinction between ACs with stress on the first syllable and ACs with 0 position stress.

Word class: within the variable, the differences in amplitudes are less distinct between the AC types. Cross-dialectally, the amplitudes for ACs with 0 lexical syllables are distinctly high.

Focus: within the variable, they produce distinctly high amplitudes in pre-focal ACs, higher than the group's no focus ACs.

Phrase type: shows a rise-rise-fall pattern in T phrases (compared to the rise-fall-rise pattern in the other dialects). Within the variable, ultimate ACs in T phrases are much lower in amplitude than their C phrase counterparts.

Duration:

Focus: within the variable and cross-dialectally, the VS show distinctly short pre-focal accents, yet distinctly long focal accents. Decrease-increase-decrease pattern from one AC type to the next.

Timing:

Phrase type: within the variable they do not make distinct differences in T1dist_rise between C and T phrase ACs.

13.3.2 Dialect-internal structure***Phrase component***

Magnitude: Given the relative weight of the predictors, *PC magnitude of the previous IP* explains the greatest amount of variance in *PC magnitude* (58%), followed by *phrase type* (14%), *strength of break* (9%), *focus* (8%), *duration of the previous IP* (6%), and *articulation rate* (4%). The VS is the only dialect in which *duration of the previous IP* is a significant predictor. In no other dialect is variance explained by such an array of variables. *Articulation rate* is ranked low and is thus less important in explaining variation in the VS model than in models of other dialects that also list *rate* as a significant predictor. With 16%, this model explains the second least variation in *PC magnitude* when compared to the models of the other dialects.

Duration: Given the relative weight of the predictors, *focus* explains the most variance (78%), followed by *phrase type* (8%), *emotion* (7%), *sex* (4%), and *strength of break* (4%). *Phrase type* is ranked high and is thus more important in explaining variation in the VS model than in models of other dialects. No other dialect exhibits such a great number of significant predictors. Accordingly, *focus* shows a less distinct effect than in the other dialect models. The VS is the only dialect in which *strength of break*, *sex*, and *emotion* are significant predictors. With 16%, this model explains the most variation in *PC duration* when compared to the models of the other dialects.

Accent component

Amplitude: Given the relative weight of the predictors, *phrase type* explains the most variance (43%), followed by *articulation rate* (31%), *focus* (18%), *emotion* (5%), and *word class* (3%). In this parameter, the VS group shows the greatest number of predictors equal to the BE group. *Focus* is ranked low and is thus less important in explaining variation in the VS model than in models of other dialects. As to the actual effect sizes, a comparison with the other dialect models reveals that *phrase type* generated the highest effects in this model, *focus*, on the other hand, the lowest effects. With 9%, this model, along with the ZH model,

explains the least variation in *AC amplitude* when compared to the models of the other dialects.

Duration: Given the relative weight of the predictors, *focus* explains by far the most variance in *AC duration* (84%). *Sex* (9%) is the second best predictor: female speakers, on average, exhibit longer ACs. *Articulation rate* explains an additional 9% in variance. The VS is the only dialect in which *phrase type* is not a significant predictor. *Sex* is ranked high and is thus more important in explaining variation in the BE model than in models of other dialects. As to the actual effect sizes, a comparison with the other dialect models reveals that there is no other model in which *focus* explains such a great deal of variance in *AC duration*. With 5%, this model explains the least variation in *AC duration* when compared to the models of the other dialects.

Timing: Given the relative weight of the predictors, *focus* explains by far the most variance in *T1dist_rise* (75%), followed by *phrase type* (25%). *Phrase type* is ranked high and is thus more important in explaining variation in the VS model than in models of other dialects. As to the actual effect sizes, a comparison with the other dialect models reveals that *focus* and *phrase type* show the greatest effects in this model. With 3%, this model, along with the ZH model, explains least variation in *T1dist_rise* when compared to the models of the other dialects.

13.4 Zurich

13.4.1 Exceptional features

Phrase component

Magnitude:

Phrase type: within the variable, they make almost no distinction between C and T phrases. From a cross-dialectal point of view, the ZH exhibit the lowest magnitudes in both C and T phrases.

Duration:

Phrase type: within the variable, they make almost no distinction between C and T phrases. Compared to the other dialects, the ZH show the longest durations in C phrases.

Articulation rate: fast speakers show the longest IPs.

Emotion: within the variable, bored IPs with 3 or more ACs are distinctly longer than other emotion phrase types.

Accent component**Amplitude:**

Stress: within the variable, they make a strong distinction between low-amplitude ACs with 0 stressed syllables and high-amplitude ACs with 1 stressed syllable. Strong distinction between high-amplitude ACs with stress on the first syllable and very low-amplitude ACs with 0 position stress.

Word class: within the variable, the differences in amplitude are distinct. ACs with 0 lexical syllables are significantly lower than ACs with 1 or 2 or more lexical syllables.

Duration:

Focus: increase-increase-decrease pattern from 1 AC type to the next.

13.4.2 Dialect-internal structure**Phrase component**

Magnitude: Given the relative weight of the predictors, *strength of break* explains the greatest amount of variance in *PC magnitude* (37%), followed by *PC magnitude of the previous IP* (31%), *articulation rate* (17%), *phrase type* (9%), and *focus* (6%). *Strength of break* is ranked highest and is thus more important in explaining variation in the ZH model. *PC magnitude of the previous IP*, on the other hand, is ranked low and is thus less important in explaining variation in the ZH model. As to the actual effect sizes, a comparison with the other dialect models reveals that *PC magnitude of the previous IP* explains relatively little variance in the ZH dialect, while *strength of break* shows the greatest effects in this variety. With 18%, this model explains the most variation in *PC magnitude* when compared to the models of the other dialects.

Duration: Given the relative weight of the predictors, *focus* explains the greatest amount of variance in *PC duration* (89%), followed by *rate* (6%) and *duration of previous IP* (5%). The ZH is the only dialect in which *duration of previous IP* is a significant predictor.

Accent component

Amplitude: Given the relative weight of the predictors, *focus* explains the greatest amount of variance in *AC amplitude* (40%), followed by *phrase type* (35%), *word class* (20%), and *emotion* (6%). ZH is the only dialect in which *rate* is not a significant predictor. *Word class* is ranked high and is thus more important in explaining variation in the ZH model than in the VS model, the only other model in which *word class* is included as a significant predictor. As to the actual effect sizes, a comparison of this and the GR model with the other dialect models reveals that *focus*

explains the greatest amount of variance in *AC duration* in the ZH and GR models. With 9%, this model, along with the VS model, explains the least variation in *AC amplitude* when compared to the models of the other dialects.

Duration: Given the relative weight of the predictors, *focus* explains most variance in *AC duration* (56%), followed by *phrase type* (21%), *articulation rate* (13%), and *emotion* (10%). The ZH is the only dialect in which *emotion* is a significant predictor. As to the actual effect sizes, a comparison with the other dialect models reveals that *phrase type* in the ZH dialect shows the greatest effects. With 6%, this model, along with the GR and BE models, explains the most variation in *AC duration* when compared to the model of the VS dialect.

Timing: Given the relative weight of the predictors, *focus* explains by far the most variance in *T1dist_rise* (70%) followed by *rate* (18%), and *phrase type* (12%). As to the actual effect size of *articulation rate*, a comparison with the BE model, the only dialect in which *rate* is also a significant predictor, reveals that the effect explains 18% of the variance in *T1dist_rise* in the ZH model, compared to the explained 15% in the BE variety. With 3%, this model, along with the VS model, explains least variation in *T1dist_rise* when compared to the models of the other dialects.

13.5 Discussion

An overall assessment of the *coefficients of determination* and the relative weight of the predictors observed for the different dialect models yields two interesting discoveries.

Firstly, the coefficients of determination, or the fractions of the variances explained in the dependent variables, are generally low. In other words, the predictive power of all the models (R^2 s) is relatively low. On the one hand, this has to do with the small number of potential predictors chosen to create the models. After all, several model-internal effects were not included in the models for reasons of circularity and suboptimal modeling, even though the adding of these variables to the multiple regressions would have increased the R^2 values. However, the predictors for the models were chosen in order to allow for linguistically sensible interpretations. The aim was to create models in which the contribution of each investigated linguistic, paralinguistic, and non-linguistic variable towards explaining variance in the f_0 contour can be assessed individually. A further possible reason for the low R^2 s involves the spontaneous nature of the data analyzed in this study. Spontaneous speech is highly multi-layered and variable, which is why Shriberg (2005:1781) aptly refers to it as “speech in the wild”. It is characterized by a constant interplay and overlap of properties

that can be categorized into a *linguistic* layer (whose representatives in this study are syllable nuclei, syllable structure, stress, and word class), a *paralinguistic* layer (focus, phrase type, and prosodic paragraphing), and a *non-linguistic* layer (articulation rate, emotion, and sex) (see Fujisaki 2004). Further characteristics of natural speech – to name just a few – include a high number of disfluencies, repetitions, repairs, false starts (Shriberg 2005: 1781), de-accentuation of syllables (see Mixdorff & Pfitzinger 2005), slurry speech, less distinct declination (Vaissière 1983: 57), and arbitrary focusing of constituents (see Xu 1999: 72). It is these characteristics and its multi-level structure that make it exceedingly difficult to predict and/or explain spontaneous speech *f0* patterns.

On a side note, it is very interesting to observe how the explained variability decreases from a comparatively high share of explained variability in the global components (*PC magnitude* overall 17%, *PC duration* overall 13%), to a relatively small amount of explained variability in the local component (*AC amplitude* overall 11%, *AC duration* overall 6%, and *T1dist_rise* overall 4%). This trend suggests that global contours are easier to predict with the factors at hand (keeping in mind the low R^2 s of the PC models to begin with). In contrast, the variability of local contours, particularly the timing of the local accents, is much more difficult to explain by means of the given predictors. It seems that the deeper a parameter is nested in the Command-Response model, the more difficult it is to explain its variance with the predictors applied. Out of curiosity, a tentative MLR for the BE dialect was designed for the prediction of *T2dist_rise*, one of the smallest units and most abstract parameters in the Command-Response model. The idea was to incorporate a Command-Response model-internal parameter as a predictor in order to check for increased R^2 values. This model incorporated the same predictors as previous models, yet the model-internal parameter *AC duration* was included. As a result, the coefficient of determination quadrupled, now explaining 8% of the variance in *T2dist_rise* instead of the previous 2%. The underlying hypothesis, a very hypothetical one that is, is that the more abstract the parameter to be explained, the less the predictive power of model-external predictors (linguistic, paralinguistic, and non-linguistic). In terms of wild speculations, one could argue that this result shows the more abstract the parameters the less the relationship to actual model-external, linguistic, paralinguistic, or non-linguistic factors. Instead, these parameters are system-dependent and conditioned but do not show consistent associations with system-external manifestations of intonation. After all, such parameters as T_1 and T_2 were developed to mirror the basic, underlying physiological mechanisms of phonation, and thus are a lot less likely to be closely associated with variables such as *emotion*, *phrase type*, or *focus*. Another wild speculation would be to take this finding as evidence that certain dimensions of the Command-Response model simply do not reflect linguistic reality.

If the R^2 s of the different dialect models are decomposed into the relative contributions of the predictors, a second interesting, cross-dialectal observation can be made: in a number of models, the ranking of the variables is relatively systematic across dialects. In other words, we find general trends as to which predictors play (or do not play) a crucial role in explaining variability in f_0 .

As for the *PC magnitude* models, *PC magnitude of the previous IP* as well as *strength of break* explain by far most variance, which underlines the significance of prosodic paragraphing, a paralinguistic function of intonation. The interaction of adjacent PC magnitudes suggests that the IPs interact and adjust to their particular surroundings: they are part of a “whole”, a bigger unit called prosodic paragraph. In the same vein, effects between declination resets and pausing, too, underline the paragraphing function of intonation: the longer the pause, the greater the declination reset. The greater the pitch reset, the more pragmatic the weight of the boundary in the course of the utterance/prosodic paragraph (for which the reverse effect is also true). Additionally, *focus* (another paralinguistic predictor) is particularly important to explain the variability in *PC duration*. PC length is a function of the informational load in the IP, where the greater the informational load, such as a *focus* constituent, the longer the IP. The reverse effect is also true.

In the *AC amplitude* models, the dialects vary greatly with respect to the relative weight in the models. *Phrase type* and *focus* are the only two variables that rank among the top three variables in every dialect. The fact that *phrase type* is a consistently high-ranking predictor underlines the significance of the conversational function of intonation. Adjusting local f_0 contours is used as a means of expressing communicative intent, namely signaling whether or not the speaker intends to hold the floor. By varying amplitude for focus or non-focus marking, a speaker puts more or less effort into speech production, based on whether the message to be conveyed is considered important. The importance of *focus* as a predictor in the dialect models underlines the informational function of intonation (see Gussenhoven 2002: 50). Both, *phrase type* and *focus* are again paralinguistic functions of intonation. Variability in *AC duration* models is explained particularly well by *focus*. In fact, the effects of *focus* on local accent duration are extensive and have been shown for a number of languages (see Mixdorff 1998a; Heldner 1996; Bauer et al. 2004). In the present data, focal as well as post-focal accent durations are greatly boosted, while pre-focal accent durations are suppressed. These effects are of communicative nature: the duration serves as communicative tool to highlight important information to be conveyed by the speaker.

As for the *AC timing* models, *focus* yet again fills the role of a critical predictor. Focus effects are distinct: focus ACs rise particularly late relative to segment onset, whilst post-focal accents rise distinctly early. It was noted that the contours with

very late T_I s in focus ACs are likely to peak late in the segment or possibly not until the following segment. Gussenhoven (2002: 53) reports that late peaks are perceived as more prominent than early peaks and invoke more “unusual occurrence” interpretations. Hence, late T_I s contribute to the saliency of focus ACs. Again, this emphasizes the informational function of intonation: the distinctly prominent, late-rising accents serve to highlight important information which the speaker wishes to successfully communicate.

These findings underline the importance of *paralinguistic predictors* – *focus* in particular – for explaining variability in f_0 parameters. The primary factor behind f_0 variability seems to be the urge to highlight important information and to successfully communicate one’s message by means of modulating the following parameters in particular: *PC duration*, *AC amplitude*, and *AC timing*. In stark contrast, *linguistic* factors never show the greatest effects in any of the 28 models. In fact, the results show that linguistic predictors are often ranked *even lower* than non-linguistic predictors. This finding undermines the importance of linguistic predictors such as *stress* (except for BE AC amplitudes) and *word class* (except for ZH and VS AC amplitudes) when it comes to explaining variance in f_0 contours. In the majority of the MLRs, *stress* and *word class* did not make it past the stage of the effect screening, and even if they are included in the models, they seldom show much effect.

At this point, I would like to briefly touch upon the predictive power of *stress*. Overall, *stress* was shown to be of low predictive power. This result ties in with findings of previous studies, which have shown that, for *spontaneous speech*, the linguistic predictor *stress* is comparatively weak in explaining variability in f_0 (see Silipo & Greenberg 1999; Kochanski et al. 2005). The question is whether the low predictive power is a result of the nature of the data. However, it cannot be answered whether or not a different ranking of variables could be achieved in experiments that control for specific linguistic, paralinguistic, and non-linguistic variables. It is indeed conceivable that, in such experiments, speakers would produce more canonical forms in which *stress* and distinct f_0 movements correlate to a higher degree, as it is the case in Standard German. In this case, the predictive power of *stress* within a given model would probably increase and the low values established in this study may be attributed to *spontaneous speech*. More importantly, however, I propose that the low effects of *stress* on local f_0 movements are a characteristic of the Alpine Swiss German dialects. Such an incongruence between *stress* and distinct f_0 movements has already been reported earlier (see Baumgartner 1922: 23 and Wipf 1910: 22 for the VS dialect; see Brun 1918: 28 and Meinherz 1920: 38 for the GR). The current data show that local f_0 movements are especially robust to *stress* in the Highest Alemannic variety spoken in the Valais. Possible explanations as to why linguistic predictors such as *stress* and *word class* are less

important in the Alpine dialects, yet more important in the Midland dialects, are discussed in the following section.

13.6 Signature features

Based on the above discussion of the exceptional features and model characteristics, we can now distill the most important signature features for each dialect. Figure 13.1 gives a visual summary of these features, given in keywords.

13.6.1 Bern

The signature features of the Bern dialect involve *long AC duration* as well as *late T1dist_rise*. Late rises, in particular, seem to be *the* prototypical feature of the Bern variety (see Fitzpatrick 1999). Additionally, the local accents of the Bern dialect accents are fairly sensitive to *stress*. As to the weight of variables in the models, it seems that *focus* and *sex* are particularly important in the *PC magnitude* model. *Articulation rate*, on the other hand, is shown to be particularly important for the *AC duration* model. The Bern variety is the only dialect in which *sex* is a significant predictor for *PC magnitude*. Conversely, it is the only dialect in which the variables *strength of break* and *rate* are not significant predictors. Since all of the annotated emotion phrases were shown to exert little effect on *f0* variance, and since the characteristics of bored phrases, namely low AC amplitudes and long PC durations, were only in part corroborated in the Bern *f0* behavior, we may conclude that the skewed relative proportions of emotional phrases do not result in a bias of data. In other words, the Bern *f0* contours do not seem to be affected by intonational correlates of bored phrases.

As mentioned elsewhere, the comparatively long AC durations can almost certainly be attributed to the generally slower articulation rate of the speakers of the Bern dialect. The comparatively slower rate is in turn contingent upon the longer mean durations of vowels as well as the more distinct final lengthening found in the Bern dialect (see Leemann & Siebenhaar 2007). The comparatively late T_1 s relative to segment onset may be a correlate of long segment durations: even if T_1 occurs after segment onset, a pitch peak realization within a long target segment is still possible. However, if the segments are shorter, the larynx needs to anticipate this duration and adjust accordingly (earlier T_1) so as to guarantee a peak realization on the target segment. Remember that late rising accents are believed to sound more prominent than early peaks and to invoke more “unusual occurrence” interpretations on the part of the listeners (see Gussenhoven 2002:51). Hence, these late T_1 s contribute considerably to the saliency of the Bern local

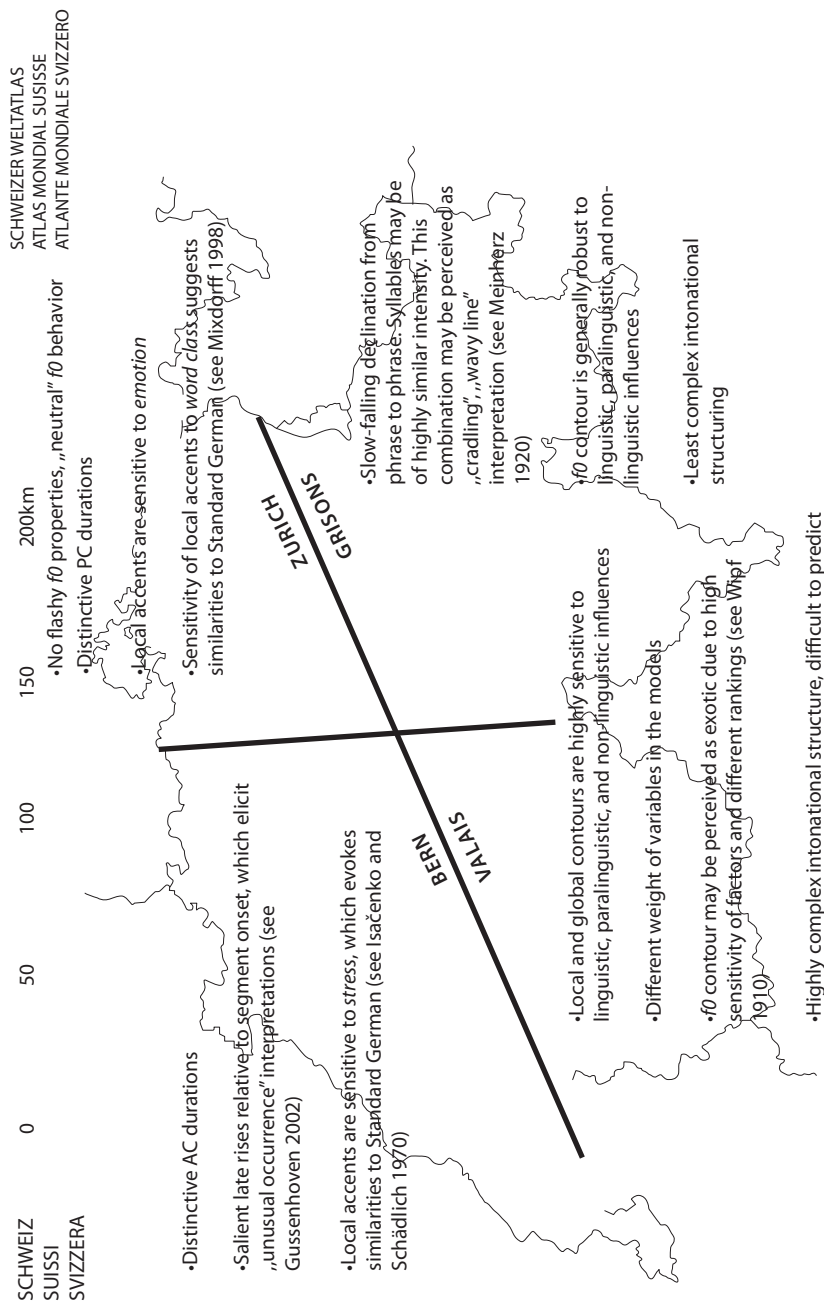


Figure 13.1. Summary of dialect-specific fundamental frequency behavior

accents. More importantly, however, f_0 contours in ACs with late T_1 onset are very likely to exhibit rising contours. Such rising contours again seem to be a characteristic of High Alemannic (which includes the Bern dialect) and Northern Alemannic varieties. Southern German varieties have been reported to exhibit rising contours in nuclear position, which are accompanied by f_0 peaks that occur late in the stressed syllable. Barker (2002) observes nuclear rises in Tyrolean German, which he labels as $L^*+H L^-$ (rising nuclear accent followed by a low phrase tone for default declarative sentences). K  gler (2004), too, provides evidence for $L^*+H L\%$ default accents in Swabian German, a Northern Alemannic variety. Gilles (2005) reports L^*+H accents in nuclear position for the Northern Alemannic dialect spoken in Freiburg, and for the Bern dialect (High Alemannic), Fitzpatrick showed low-rising L^*+H default accents in which the f_0 peak frequently aligns with the unstressed, neighboring segment. Results of the present study confirmed the late rises in the Bern dialect both for ACs on the whole, as well as for ACs with stress on the first syllable, see Figure 13.2.

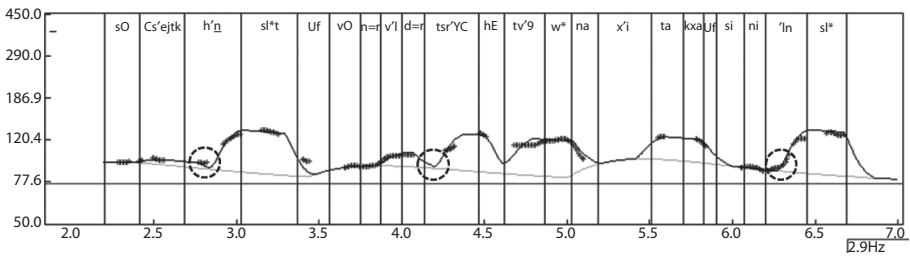


Figure 13.2. Bern-specific f_0 -behavior: Late rises on the local level

The phrase in Figure 13.2 reads *sO Cs'ejt kh'  nslet Uf VOn =r v'Id=r tsr'YC hEt v'9w* nax 'itakxa Uf sini 'Insl** (German: *so gesagt geh  nselt [auf] als er wieder zur  ck wollte nach Ithaca auf seine Insel*; Engl: *teased so to speak [to] as he tried to return to Ithaca to his Island*). The late rises in this example are particularly salient in the three circled rises of the ACs: all begin with a stressed syllable and occur late relative to syllable onset, with f_0 reaching its maximum in adjacent syllables. Therefore, these results tie in nicely with findings of previous studies on AC timing and reveal an important intonational similarity common in the High and Northern Alemannic varieties.

As for the interpretation of the sensitivity towards lexical *stress* in the context of the AC *amplitude* model, I suggest the following two explanations. Firstly, since the Bern are comparatively slow speakers, they are likely to articulate more accurately and, therefore, use less reduced forms in speech. In theory,

more accurate pronunciation evokes more canonical forms (see Werner 2004). Therefore, it could be argued that lexical stress is in fact expected to be more prevalent and thus expected to rank higher in the Bern model than in other models. Secondly, it could also be argued that the Midland dialects generally intonate more closely to the German neighbors than the Alpine varieties. For the Bern dialect, this would explain the relatively critical weight of *stress* in the AC amplitude model since in Standard German, stressed syllables frequently feature distinct *f0* movements either on the stressed syllable or in close vicinity (see for example Isačenko & Schädlich 1970).

If we look at the amount of *f0* variation that is explained by means of the Bern model, a comparison with the corresponding models of the three other dialects shows that, out of the total of seven models, three models explain the most variation and two models explain the least variation. This is an average ratio and suggests that the variability in *f0* behavior can be explained relatively well with the predictors at hand, particularly in the AC *amplitude* model. The proposed predictors seem sufficient to adequately explain *f0* variance. One possible interpretation of this ratio could be that, in comparison to the other dialects, the intonational structure of the Bern dialect is comparatively less complex since all dialects were modeled with the same potential predictors.

13.6.2 Grisons

The Grisons dialect exhibits a great number of exceptional features in *PC magnitude*, which mostly tend to induce higher *PC magnitudes*. In addition, a predominant role is ascribed to the great number of *falling ACs* that demonstrate a peculiar response to a number of variables in this variety. The variables *stress* and *word class* show comparatively little effects on local *f0* contours. In terms of the number of predictors in the models, it is striking how the GR in general only exhibit a few significant predictors per model, particularly in the models on *PC magnitude*, *PC duration*, and *AC amplitude*. In the *AC amplitude* models, the Grison dialect is the only dialect in which *emotion* is not a significant predictor. As to the weight of variables in the models, *phrase type* is not very important in the *PC magnitude* model. The Grisons group is in fact the only dialect in which *phrase type* is not a significant predictor for *phrase type*. In the *AC timing* model, *sex* is particularly important. Due to the lack of significant predictors, the ranking of effect sizes in each model differs now and again from the ranking in other models.

The finding that distinct *PC magnitudes* play a special role in this dialect makes sense for the following reasons. Earlier, it was noted that Meinherz (1920: 37) perceives the dialect's melody as slightly cradling, similar to a wavy line, and adds

that it is clearly not as melodic as the Walser variety spoken in the Canton. From the point of view of intonation, an impressionist description of cradling, wavy speech could be a somewhat soothing f_0 contour with some form of recurring rhythm or pattern. Such a contour can in fact be confirmed by the present results. It was pointed out repeatedly that the Grisons speakers stand out with their frequent, high-flat f_0 contours, which are very difficult to model. Figure 13.3 shows these high frequency, flat f_0 contours (same Figure as 7.21).

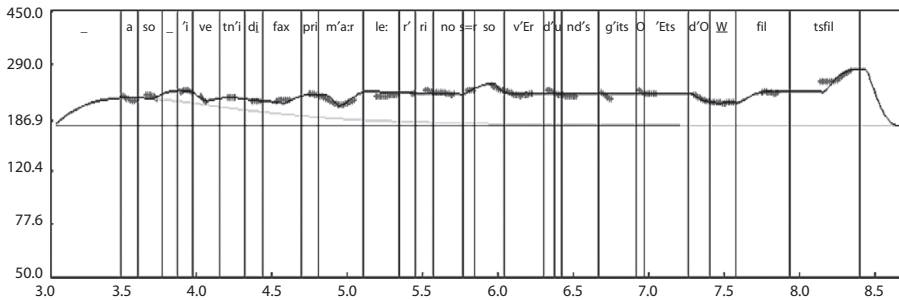


Figure 13.3. Grisons-specific f_0 -behavior: high-flat f_0 contours

The phrase in Figure 13.3 reads *aso 'I vet n'id &jfax prim'a:rl:r*rin od=r so v'Er*d* und *s g'its jO 'Ets d'O &w f'il tsf'il* (German: *also ich will nicht einfach Primarlehrerin oder so werden und es gibt ja jetzt davon auch viel zu viele*; Engl.: *well I don't want to simply become an elementary school teacher and there are way too many of them now*). The f_0 remains more or less constant between the syllables *le:* (5.3 seconds into the signal) and the *d'O* (6.9 seconds).

Proof for these high-flat contours was given by showing the correlation of the specific AC position in the IP with AC amplitude. Results showed that in no other dialect is the correlation as strong as in the Grisons variety, i.e. the later the AC in the IP, the distinctly higher its amplitude. This relationship is manifested in a very slow-falling overall declination in the phrase. Within these phrases, ACs are lined-up in sequence with increasing amplitudes towards phrase-final position. In terms of f_0 contour, those IPs start with high f_0 , which is more or less sustained throughout the phrase, then may feature a number of equal-intensity syllables, and end with a still relatively high f_0 . If these IPs occur in sequence, the gradual and recurring changes indeed can be perceived to have this wavy, cradling effect Meinherz (1920) alludes to. The small amount of predictors may further support the thesis of recurring rhythms or patterns. The model suggests that the f_0 contour is fairly robust to linguistic, paralinguistic, and non-linguistic influences, which may contribute to a more monotonous, cradling melody. Lastly, as for the

exceptionally high PC magnitudes, Meinherz (1920: 39) reports that phrase-initial words frequently are highly prominent in the Grisons dialect, even though the word in question is not of high semantic value. This observation ties in well with the distinctly high PC magnitudes found in this analysis.

As for the Grison speakers' low sensitivity towards lexical *stress* in the context of the AC amplitude models, a straightforward interpretation is difficult at this point. Sociolinguistic as well as language contact-related reasons for this peculiar behavior are worth researching in future studies. For example, a plausible explanation includes the Grisons speakers' contact with Romansh and Italian, two Romance languages also both spoken in the Canton of Grisons.¹ Italian shows penultimate and antepenultimate stress and exhibits right-headed rhythmic groups, with low-high *f0* movements on the rhythmic group (see Di Cristo 1998: 24; Rossi 1998: 220). Romansh, too, shows lexical accents in word-final or penultimate syllable position (see Cavigelli 1969). It is indeed possible that the constant contact with Italian and Romansh has affected the Grisons stress system. Since in most Germanic languages, feet are left-headed (stressed syllable plus following unstressed syllables), while Italian and Romansh are right-headed (stressed syllable plus any preceding unstressed syllables), one could argue that the Grisons dialect can be seen as a mix-version of these two stress systems. This could be what Meinherz (1920) observed when he reported that dynamic accents (i.e. stressed syllables) in the Grisons dialect are not very different from non-dynamic accents (i.e. unstressed syllables) when compared to other Swiss German dialects. In addition, note that Grisons varieties frequently feature the archaic feature of non-reduced, word-final syllables, which may too, contribute to distinct *f0* modulations in unstressed syllables. The conclusion that can be drawn from this is that if the Grisons, over centuries, alternatively incorporated both rhythmic group patterns, it could be hypothesized that the stress will eventually lose its importance, since stress is no longer perceived as discrete. Now as for the link between stress and *f0* contours, it is likely that this mix of stress systems found in the Grisons dialect does not only manifest itself in the lexical stress pattern, but also affects *f0* behavior. As we have seen, although *f0* movement does not necessarily have to be associated with stress, it oftentimes is (see Kochanski et al. 2005). Therefore, I hypothesize that the generally devalued variable stress in the Grisons dialect is likely to have an effect on the variance of *f0* contour. In other words, the only slight differences exhibited for stressed and unstressed syllables in the Grisons dialect is probably manifested

1. This is merely a possible explanation, which has to be examined further in experiments geared solely at the study of *stress* and *word class* behavior in the GR dialect.

also in *f0* behavior, since stress, in many cases was found to be associated with *f0*. Since the exact relationship between stress and *f0* is not yet clear, and the *f0* behavior for stress in Romansh and Italian was not the main concern of this study, further studies have to be conducted.

These rather small differences in stressed and unstressed ACs are likely to further contribute to the overall perception of the Grisons *f0* contour as cradling and wavy, since these ACs do not feature abrupt, distinctive changes in amplitude, but progress rather gradually with only minor changes in amplitude.

If we look at the amount of *f0* variation explained in the Grisons model, a comparison with the corresponding models of the three other dialects shows that, out of the total of seven models, three models explain the most variation and one model explains the least variation. Cross-dialectally, this is the highest ratio found and suggests that the Grisons variability in *f0* behavior can be explained very well with the predictors at hand, particularly in the PC magnitude model. Now it should be emphasized that, if compared to the other models, the Grisons model reaches the *highest* ratio of explanatory power with only the *fewest* predictors. One possible interpretation of this phenomenon is that the intonational structure of the Grisons dialect is comparatively least complex and more systematical, which again corroborates the arguments postulated above, namely that the Grisons intonation is perceived as soothing, wavy, and cradling.

13.6.3 Valais

The Valais dialect stands out with a great number of exceptional features regarding *AC amplitude*. None of the other dialects exhibit this many cross-dialectally and variable-internally exceptional results in this parameter. This means that the Valais *AC amplitude* shows distinctly different response in many factors if compared to the responses in other dialects. These include distinctly high pre-focal accents or distinctly low ultimate ACs in T phrases. As for the number of variables as well as their weight in the models, we encounter a similar picture: in the majority of models, the Valais show the greatest array of variables for explaining variation, particularly for the models of *PC magnitude*, *PC duration*, and *AC amplitude* (the latter is a tie with the BE dialect). In addition, the relative weight of the predictors often differs from the corresponding weight in other dialects: in the *PC magnitude* model, for instance, *articulation rate* bears little importance, while in the *AC amplitude* model, *focus* is less important. In the duration model, *sex* turns out to be very important, and in the *AC timing* model, *phrase type* bears more weight. The Valais dialect is the only variety in which *phrase type* is not a significant predictor in the *AC duration* model.

The finding that the Valais stands out with their peculiar AC amplitude behavior ties in nicely with the existing literature. A number of authors have reported the Valais speakers' exotic placement of accents (Wipf 1910; Weber 1987; Werlen & Matter 2004; Engeli 1971). Remember that Wipf (1910: 19) even goes so far as to say "one has [...] the almost annoying sensation as if the people place accents as strongly as possible on the most irrelevant of syllables." While in a cross-dialectal comparison the Grisons models feature the *least* predictors in a majority of the created models, the Valais models generally feature a *high* number of predictors. This is true especially for the model of *PC magnitude*, *AC amplitude*, and *PC duration*. In other words, this means that, in the Valais models, the variance in a parameter value is explained not just by a few but comparatively many predictors. Furthermore, this means that the Valais dialect's local as well as global contours are highly sensitive to linguistic, paralinguistic, and non-linguistic factors – factors, which in the other dialects were often excluded from the models during effect screening to begin with. What makes the Valais dialect even more extraordinary is the vastly different ranking of the variables when compared to the other models of the dialects. Now, in spontaneous, uncontrolled speech, each of these factors can of course occur all over the place, depending on the communicative intent and strategies, and can thus occur simultaneously on different layers of speech. If one were to judge Valais intonation based on this characteristic, one would assume the Valais speech to sound rather exotic and somewhat impalpable and maybe even chaotic, in a positive sense of course. Figure 13.4 demonstrates the typical visual representation of highly varying VS local accent contours (particularly if compared to the Grisons high-flat *f0* contours).

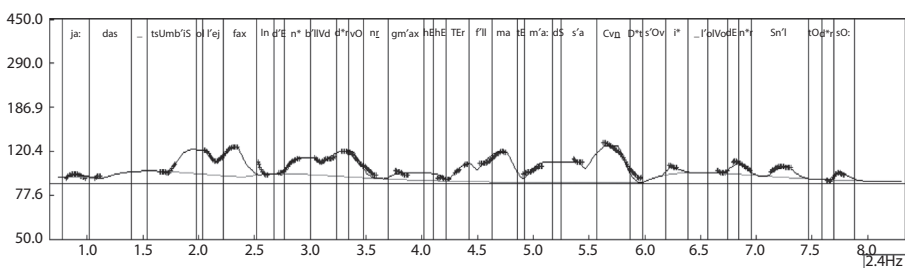


Figure 13.4. Valais-specific *f0*-behavior: varying local accent *f0* contours

The phrase reads *ja: das tsUm b'iSpIl 'ejfax In d'En* b'Id*r vOn =r gm'axt hEt hEt Er f'Il matEmatISs 'aCv'e'nD*t s'Ovi* g'oldEn*r Sn'It Od*r sO:* (German: *ja das zum Beispiel einfach in diesen Bildern die er gemacht hat hat er viel Mathematisches angewendet so wie goldener Schnitt oder so*; Engl: *yes this for example in the drawings he made he has applied lots of mathematics such as the golden rule or similar*).

As for the Valais speakers' low sensitivity towards *lexical stress* and *word class* in the context of the AC amplitude models, which was already discussed for the Grisons dialect, we could again argue from the viewpoint of language contact. From a geographical point of view, the Valais dialect is also in rather close contact with Italian to the South and French to the West, both of which are Romance languages. As mentioned earlier, Italian exhibits word-final, penultimate, and antepenultimate stress and shows right-headed feet, with low-high f_0 movements on the rhythmic group (see Di Cristo 1998:24). French is a language in which the three prominence markers *loudness*, *duration*, and *fundamental frequency* show reduced correlation. These prominence marking parameters are set according to the first and last syllable of the word: the first syllable typically shows a rise in f_0 , while the word-final syllable may exhibit a variety of prominence contrasts, frequently however a rise in f_0 (see Welby 2006). Prosodic prominence is thus bound to the first and the last syllable of a word (Vaissière 1983:63). The exposition of the Valais dialect to the prominence systems of these two Romance languages may over decades have led to an interesting mix in prominence marking for the Valais dialect. It could be argued that the Valais dialect was influenced both by the different stress assignment rules in Italian as well as by the different use of prosodic parameters for prominence marking in French. This language contact may have contributed to complex and somewhat unpredictable f_0 variability that Wipf (1910) alludes to. In addition, Valais varieties, too, commonly feature the archaic feature of non-reduced word-final syllables. These may also contribute to distinct f_0 modulations in unstressed syllables. Again, this is only a tentative hypothesis that should be examined in further studies geared solely at the explanation for this peculiar behavior.

If we look at the amount of f_0 variation explained in the Valais model, a comparison with the corresponding models of the three other dialects shows that, out of the total of seven models, two models explain the most variation and four models explain the least variation. Cross-dialectally, this is the poorest ratio and suggests that the Valais' variability in f_0 behavior can be explained relatively poorly with the predictors at hand. Apparently, further variables should be taken into consideration in order to shed more light on underlying patterns in the f_0 variability and, thus, boost the predictive power of the model. Now, given that all dialects were modeled with the same potential predictors, one could argue that the Valais' intonational structure is comparatively complex, less straightforward, and literally less *predictable*. Once more, the impressionistic description of the somewhat erratic and highly unsystematic behavior of the Valais f_0 contours can be corroborated by the statistical analyses and the analyses of the model properties conducted in this study.

13.6.4 Zurich

The Zurich dialect stands out with a number of exceptional features in *PC duration*, most of which hint at comparatively longer phrase durations. What is also worth noting is that the Zurich group is the only dialect in which *AC amplitudes* are very sensitive to *word class*. Furthermore, it is also the only dialect in which *emotion* is a significant predictor in the *AC duration* model, and in which *rate* is not a significant predictor in the *AC amplitude* model. As for the weight of the variables in the Zurich models, *strength of break* is very important in the *PC magnitude* model, while *PC magnitude of the previous IP* is less so.

Compared to the other dialect's idiosyncratic *f0* behavior, the Zurich dialect does not exhibit any truly flashy properties. Of course, exceptionally long durations of PCs, the importance of *word class*, and an idiosyncratic ranking of variables in some of the models are worth mentioning. Figure 13.5 indicates an example of a particularly long IP from Zurich data.

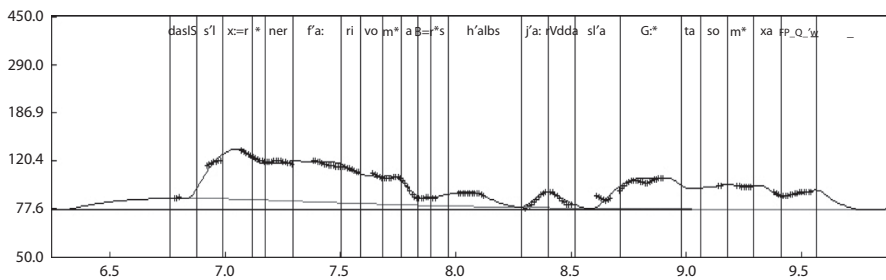


Figure 13.5. Zurich-specific *f0*-behavior: comparatively long IPs

The phrase reads *das IS s'ix=r *n erf'a:ri vo m* aB=r *s h'albs j'a:r das l'aG:*t aso m* xa &w* (German: *das ist sicher eine Erfahrung wo man aber ein halbes Jahr das reicht also man kann auch*; Engl: *that is certainly an experience where one but a half a year is enough well one can also*). The phrase spans nearly three seconds, compared to an overall mean duration (entire corpus) of 1.47 seconds per IP.

Overall, however, the dialect does not seem to be particularly marked from an intonational point of view. This is perhaps exactly one of the reasons why Zurich German is perceived by Swiss Germans as rather *neutral* (see Ris 1992). Weber (1987) further adds reports that fundamental frequency movements in the Zurich dialect are generally “calm and gradual” and do not feature as many quick intonation movements as other Swiss German dialects, except in case of emotionally-charged language. This observation hits the nail on the head: in no other dialect is *emotion* such an important variable in the models as in the Zurich

variety. The distinctive *PC durations* found for this variety could be a result of the participants' highly cooperative and self-confident attitude during the interviews. It could be argued that cooperative speakers generally show greater communicative effort, and thus also exhibit increased vocal effort and activity, which results in longer mean lengths of utterances (see Van Kleeck & Street 1982). In simple terms, maybe they actually had *more* to say in the course of one IP than speakers of other dialects. Furthermore, Zurich speakers are generally fast speakers, and higher articulation was shown to result in a reduction of phrase boundaries (see Mixdorff 1998: 176; Ladd et al. 1998). Such reductions are very likely to lead to less PC placing, especially in spontaneous speech, and may thus cause overall longer IPs.

Let me turn to the Zurich speakers' heightened sensitivity towards *word class* in the context of the *AC amplitude* models. As mentioned earlier, the Midland dialects may generally intonate more closely to the German neighbors than the Alpine varieties, which, in the Bern variety is reflected in the importance of *stress* in the *AC amplitude* model. Possibly, the Zurich speakers do so by clearly distinguishing between rather low-amplitude grammatical words and high-amplitude lexical words, in the same way the German speakers do (see Mixdorff 1998; Möbius 1993a). Additionally, it should not be forgotten that the recordings were made in Winterthur, a North-Eastern Zurich variety close to the border of Germany.

If we look at the amount of *f0* variation explained in the Zurich model, a comparison with the corresponding models of the three other dialects shows that, out of the total of seven models, three models explain the most variation and two models explain the least variation. When compared to the other dialects, we observe that the ratio for the Bern and the Zurich dialect are identical. This suggests that the variability in *f0* behavior can be explained relatively well with the predictors at hand, particularly in the *PC magnitude* model. The proposed predictors seem sufficient to adequately explain *f0* variance. One possible interpretation of this ratio could be that, in comparison to VS dialect for example, the intonational structure of the Zurich dialect is comparatively less complex since all dialects were modeled with the same potential predictors.

13.7 Alpine-Midland divide

13.7.1 *f0* behavior in variables

Phrase component

Duration:

Emotion: bored IPs with 3 or more ACs are distinctly longer than other emotion phrases in the Midland varieties.

Accent component**Amplitude:**

Stress: the Midland varieties strongly distinguish between low-amplitude ACs with 0 stressed syllables and high-amplitude ACs with 1 stressed syllable. The Alpine varieties show less of a distinction in amplitude between the three AC types. Midland varieties exhibit a particular distinction between high-amplitude ACs with stress on the first syllable and very low-amplitude ACs with 0 stress, while the Alpine varieties do less so.

Word class: the differences in amplitude are more distinct for the Midland varieties. ACs with 0 lexical syllables are significantly lower than ACs with 1 or 2 or more lexical syllables. Alpine varieties show much higher amplitudes in ACs with 0 lexical syllables than the Midland varieties.

Timing:

Word class: in ACs with 1 lexical syllable, the Alpine varieties show relatively early rises, while the Midland varieties rise later.

Focus: in pre-focal accents, the Alpine varieties rise early, the Midland varieties comparatively late.

Phrase type: in penultimate and ultimate ACs of C phrases, the Midland varieties rise late, the Alpine varieties early.

13.7.2 Features in the models**Accent component**

Amplitude: articulation rate is more important in the Alpine varieties than in the Midland dialects. Linguistic predictors, such as stress and word class, are more important in the Midland varieties.

Duration: sex is a very important variable in explaining variation in the Alpine dialects.

Timing: articulation rate is important in the Midland dialects.

13.8 East-West divide**13.8.1 f_0 behavior in variables****Phrase component****Magnitude:**

Phrase type: Western varieties make a strong distinction between C and T phrases.

Duration:

Phrase type: Western varieties differentiate strongly between the durations of C and T phrases.

Accent component

Duration:

Focus: Western varieties show a decrease-increase-decrease pattern from one AC type to the next, while the Eastern varieties clearly show an increase-increase-decrease pattern in duration.

13.8.2 Features in the models

Phrase component

Magnitude: phrase type is a much more important factor in the Western varieties, while strength of break and rate, on the other hand, are much more important in the Eastern varieties.

Duration: articulation rate is much more important in the Eastern varieties than in the Western dialects.

Accent component

Amplitude: focus is more important in the Eastern varieties, whilst phrase type is more important in the Western varieties.

Duration: phrase type is more crucial in the Eastern varieties.

Figures 13.6 and 13.7 provide a summary of the established differences between the Alpine/Midland as well as East/West varieties.

13.9 Discussion

In terms of a geolinguistic structuring of Swiss German f_0 behavior, we can establish two rather intriguing, underlying patterns. In the context of how f_0 behaves with respect to a given variable and with regard to how variables are weighed in the statistical models, the Alpine/Midland geolinguistic groups seem to differ almost exclusively in *AC parameters*, i.e. on the local contour, whereas East/West groups exhibit differences in both *AC parameters* as well as – more strikingly – in *PC parameters*, i.e. on the local and global contour. To be more precise, Alpine/Midland differences concern in particular the *AC timing* and *AC amplitudes*, while the East/West divide is characterized by differences in *PC magnitude* and *PC duration*.

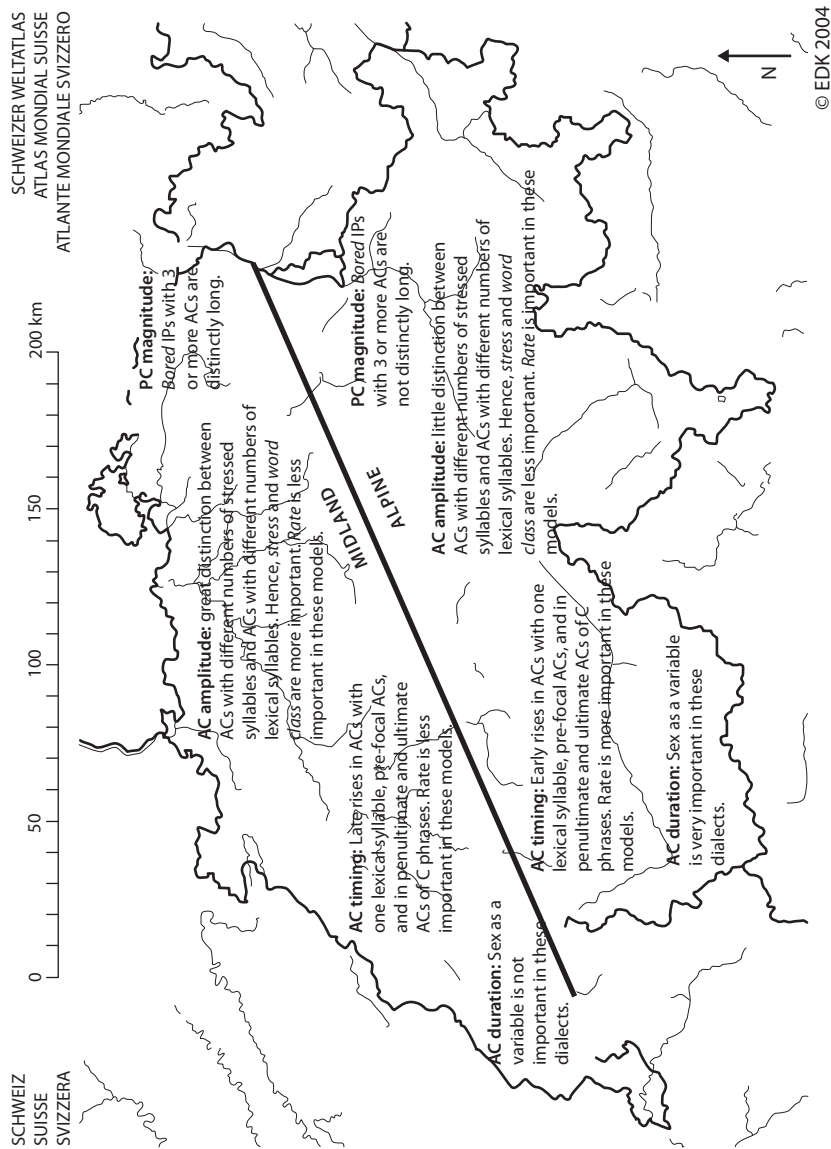


Figure 13.6. Alpine/Midland divide of fundamental frequency behavior in Swiss German

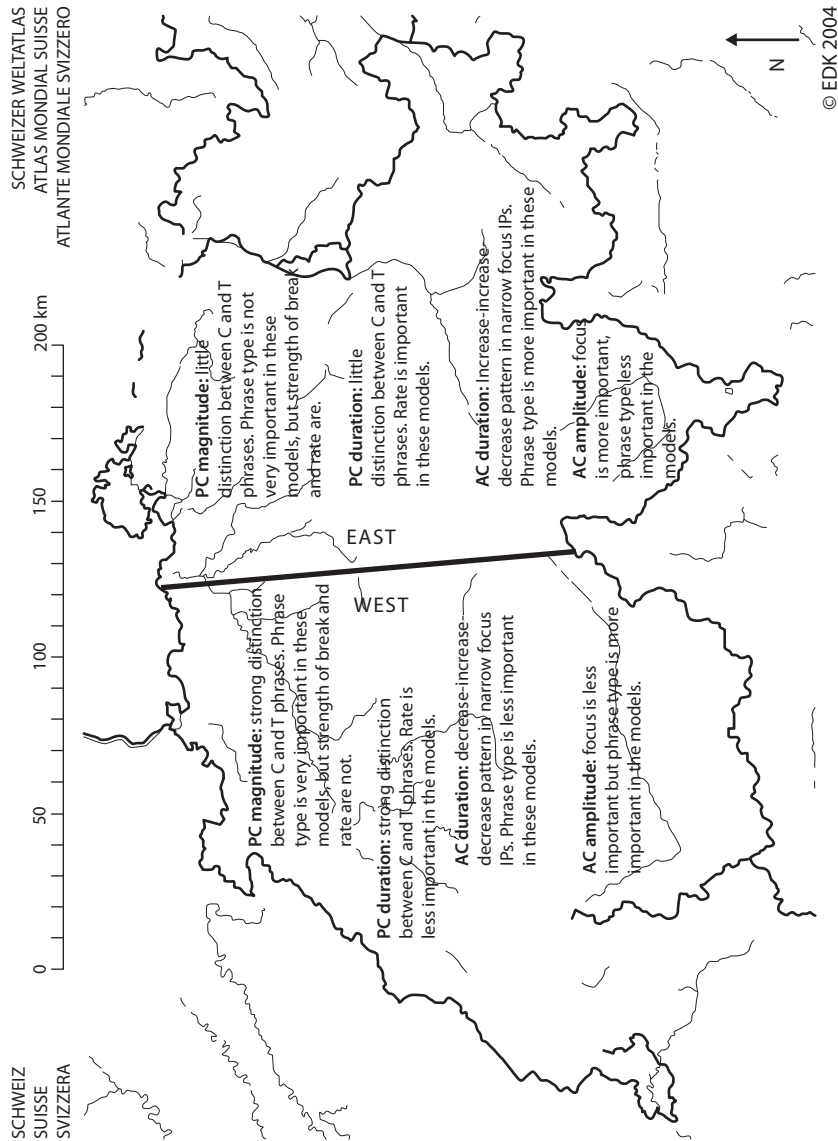


Figure 13.7 East/West divide of fundamental frequency behavior in Swiss German

The most striking difference between the Alpine and Midland groups is the relative weight of the *linguistic* predictors in the *AC amplitude* models, including the predictors *stress* and *word class*. This phenomenon and the possible causes thereof were discussed in depth in the previous subchapters. There are of course, alternate interpretations regarding the distinct difference between Alpine and Midland dialect behavior. Exploring language and migration history may provide one way of tapping into these differences. Given the mountainous terrain, Alpine varieties may have served as linguistic refuges over the past centuries and – in that sense – may represent what Johanna Nichols (1992) refers to as residual zones. Here, the highest Alemannic varieties were preserved, retaining what are now described as archaic features. On a segmental level, these differences can be reconstructed in part (Wiesinger 1983:829; Lötscher 1983:14). However, a historical reconstruction of prosodic – particularly intonational – features has, to the best of my knowledge, not been endeavored – and is of course an impossible endeavor given the obvious lack of audio data from past centuries.

The second most critical difference between Northern and Southern groups seem to be located in the domain of *local accent timing*. Here, the signature feature are the late rises of the Midland (particularly the BE) dialect in ACs with only 1 lexical syllable, in pre-focal ACs, and also in penultimate and ultimate ACs of C phrases. In these instances, the Alpine dialects behave conversely. It is possible that this trend is associated with sociolinguistic as well as language contact-related factors. This finding corroborates the call for future research on a comparison of *f0* timing behavior between the Midland and Southern German dialects, on the one hand, and further investigation of possible convergence effects between the Alpine and the Italian, French, and Romansh varieties, on the other hand.

As for the East/West divide, an analysis of the *f0* behavior for given variables and the weight of the different variables in the statistical models shows that the implementation of the PC parameters for *phrase type* demarcation is strikingly more rigorous in the Western dialects than in the Eastern dialects. Western dialects clearly distinguish between high-magnitude C phrases and low-magnitude T phrases, as well as between long T phrases and short C phrases. The Eastern dialects exhibit similar, yet much less rigorous distinctions. One may speculate as to the reasons for this phenomenon. This could possibly be attributed to the PC parameter behavior regarding *phrase type* demarcation in French, due to the close proximity of the French language to Bern and Valais Swiss German. However, at this point, this is only a hypothesis and is to be corroborated with intonation literature on French.

In sum, we can establish the following trends:

While in the segmental realm, the differences between Alpine and Midland dialects lie primarily in the use of archaic/modern forms (see Wiesinger 1983:829),

on a suprasegmental level, this divide is mirrored in the groups' different intonation behavior with respect to local timing and AC amplitude parameters.

While in the segmental realm, the differences between Eastern and Western dialects lie primarily in the morphological domain (see Lötscher 1983), this divide is mirrored in the groups' different intonation behavior with respect to global magnitude and PC duration parameters.

13.10 Overall assessment of applying the command-response model on natural dialectal speech

The current study represents the first attempt of applying the Command-Response model on natural dialectal speech. In this last section, I therefore would like to present a critical overview over the favorable/unfavorable aspects of the deployment of the Command-Response Model on this kind of data and discuss the general implications thereof.

Working with natural speech bears two obvious benefits: one, natural speech allows for idiosyncratic features to permeate and two, it permits a holistic investigation of intonational variation on linguistic, paralinguistic, and extralinguistic levels. For reasons discussed at 4.6, the somewhat unorthodox Command-Response model was selected as an ideal tool to model intonation. However, both working with spontaneous speech as well as the application of the Command-Response model bore serious difficulties.

One of the major concerns of working with spontaneous speech is the demarcation of phrase boundaries, which in the study at hand equates with the placement of phrase commands, as well as unfavorable voice quality (see 7.3.2.1 and 8.2). Phrase commands are placed according to a bundle of different features, most importantly including pausing/breathing, phrase-final lengthening, laryngealization, performance errors, filled pauses, and/or syntactic completion. Performing a systematic labeling based on these characteristics turned out to be highly intricate. A further obstacle of systematic labeling presented itself with creaky or whispery voice quality on behalf of the subjects. Roughly 8% of the recorded data, which amounts to approximately 80 syllables per speaker on average, was discarded due to poor *f0* measurements. Often, it is not entire phrases which are affected by poor voice quality but rather parts of phrases (particularly in phrase-final position). The question then is: do we discard the entire phrase or do we nevertheless attempt a modeling with the, albeit possibly faulty, extracted pitch contour?

In terms of the Command-Response model, there are a number of model-specific drawbacks which have been mentioned at 7.2.2 as well as 4.5.3. Model parameter setting is critical before starting analyses. At 7.2.2 it was

mentioned how an α of different values can critically obstruct cross-comparisons between studies applying the Command-Response model. It is highly desirable to employ an identical α value which enables cross-comparisons between different studies and different languages. Further drawbacks with regard to the modeling procedure concern the incapability of the model to capture high flat f_0 contours as well as slow global rises in a linguistically plausible way. This may be alleviated with the proposed implementation of a slow-rise component as introduced at 7.4.2.1. On the local level, this incapability to adequately model slowly falling or slowly rising local accents is somewhat discouraging. One of the most essential drawbacks of applying such an unorthodox model, however, is the difficulty in comparing the current study's findings with results from other studies on intonational variation, particularly Grabe et al.'s (1998a; 1998b) studies on intonational variation on British dialects (IViE), or the modified ToBI approaches of Auer et al. (2000), Gilles (2005), and Bergmann (2008) on German dialectal variation in intonation. Cross-comparisons to the above studies are rendered difficult due to the vastly different meta-theoretical concepts, theoretical approaches, as well as types of categories and parameters used for the codification of intonational parameters.

From the author's point of view, the advantages of working with the Command-Response model make up for its shortcomings, however. In providing a precise mathematical description of f_0 contours we can shed light on f_0 behavior, which is not feasible with abstract symbolic analyses. The detected differences between the dialects – even if minor in nature – may in the end be perceptually relevant for cross-dialectal comparisons (cf. Haas 1978). The mathematical approach allows for a straightforward quantification of intonational events, which is practical if analyses of intonational variation on a linguistic, paralinguistic, and extralinguistic level are desired. The current study represents a novel, holistic account of the multilayered characteristics of natural speech intonation, incorporating statistical modeling of the intonational variation on the aforementioned levels. The variation is thoroughly accounted for and systematic patterns in the f_0 contours of the investigated dialects are detected; a procedure which, until today, has not been conducted in this manner. In describing the intonational variation in these variables, the present study sheds light on the weight of the different factors that shape the f_0 contours of natural speech.

Conclusion

Previous to this study, no systematic, large-scale account of Swiss German dialectal intonation existed. This strongly empirical, close-to-the signal analysis of f_0 contours has brought to light detailed underlying patterns of Swiss German dialectal f_0 behavior which would not have been conceivable several centuries, or even decades ago. Most of the existing studies on the fundamental frequency and prosodic characteristics of Swiss German dialects were impressionistic and featured descriptions such as the “singing of the shepherds in the high mountains of [the] Valais” (Stalder 1819:7–8), the “slightly cradling” melody of the Grisons dialect, which is “comparable to a wavy line” (Meinherz 1920:37), the “homely”, “snug”, and “slow” idiom of the Bernese (Schwarzenbach 1969), and the “fast”, “neutral”, and “adaptive” dialect of Zurich (Ris 1992). Hotzenköcherle (1962:240) therefore rightly noted in the introductory volume of the *Linguistic Atlas of German-speaking Switzerland* (1962–2003) that the study of suprasegmental features of Swiss German dialects is “an important and alluring task for future monographic research”. This is what this study set out to do: to tackle this important and alluring task.

In the framework of a research project at the Linguistics Department of the University of Bern (2005–2008),¹ 40 subjects from four different regions of German-speaking Switzerland were interviewed. The decision to work with spontaneous speech data, as opposed to laboratory speech, is two-fold: firstly, spontaneous speech data allows for the idiosyncratic dialectal features to permeate uninhibitedly (see Gilles 2005) and, secondly, it allows for an investigation of f_0 variability in the context of linguistic, as well as paralinguistic and non-linguistic functions of intonation, since paralinguistic and non-linguistic components of intonation – unless intentionally induced or portrayed – only surface in spontaneous speech. The purpose of this study was to provide a broad account of the dialectal features of fundamental frequency contours in Swiss German dialects, which is why a great number of variables were incorporated. This study is meant to lay the

1. SNF-Project 100011-116271/1: “Quantitative Ansätze zu einer Sprachgeographie der schweizerdeutschen Prosodie”. Institute of Linguistics, University of Berne, 2005–2008.

foundation for new, stimulating research geared solely towards the investigation of specific variables and parameters.

The recorded data was processed as follows. First, the speech was transcribed, segmented, and annotated with variables of interest. *f0* contours were then explored via analysis-by-synthesis (Bell et al. 1961) procedure using the Fujisaki intonation model (Fujisaki & Hirose 1982). *f0* behavior in each of the linguistic, paralinguistic, and non-linguistic variables was analyzed using parametric and non-parametric statistical tests against the background of detecting dialect-specific patterns as well as cross-dialectal differences. *f0* variation in the predictors was tested via an analysis of two global intonation parameters, *PC magnitude* and *PC duration*, as well as three local parameters, *AC amplitude*, *AC duration*, and *AC timing*. Dialect-specific multiple linear regression models and logistic regressions for all parameters were then developed. These models allowed for a distillation of the relative contribution of independent variables towards explaining *f0* variability in a given parameter value. At the core of the study are the dialect profiles presented in Chapter 13, in which the major findings are revisited, discussed and placed into a broader, dialectological and sociolinguistic context. A short summary and discussion of each dialect is given in the subchapter termed *signature features*, which sums up all the major results and interpretations contrived in this study.

The descriptions of the prosodic properties of Swiss German dialects reiterated at the beginning of this section undoubtedly sound very unscientific and impressionistic. Hence, it is all the more astonishing that these descriptions can essentially be corroborated in this study via fundamental frequency analysis based on Command-Response modeling (Fujisaki & Hirose 1982), bivariate statistical analyses, and multiple linear regression models.

The signature properties of the BERN dialect involve *long AC durations* as well as *late T1dist_rise values*. Late rises, in particular, seem to be the prototypical feature of the Bern variety. Comparatively long AC durations can almost certainly be attributed to the generally slower articulation rate of Bern speakers (a trend that was empirically confirmed in this study as well as by Leemann & Siebenhaar 2007). This corroborates the stereotype of the Bern speakers as “slow” speakers (Schwarzenbach 1969). It is indeed plausible that the combination of late rises, long AC durations, and slow articulation rate would evoke the notion of a tranquility and ease, of “snugginess” or “homeliness”, as alluded to in previous accounts – fertile grounds for future research.

The GRISONS dialect exhibits a great number of exceptional features in *PC magnitude*, i.e. the global declination parameter. Prototypical IPs start with a high *f0*, which is more or less sustained throughout the phrase, then may feature a number of equal intensity syllables, and end with a still relatively high *f0*. In sequence,

such IPs can be conceived to have this “wavy”, “cradling” melody Meinherz (1920) alludes to. In addition, a number of models in the Grisons variety exhibit only a few significant predictors, suggesting that *f0* contour is fairly robust to linguistic, paralinguistic, and non-linguistic influences. This may well be argued to contribute to a more monotonous, cradling melody. If we look at the amount of *f0* variation explained in the Grisons model, we find that the variability in *f0* behavior can be explained very well with the predictors at hand. Therefore, it was argued that the dialect is comparatively least complex, more systematic, and thus easiest to predict.

The VALAIS dialect stands out with a great number of exceptional features regarding *AC amplitude*. In bivariate tests, their *AC amplitudes* reflect distinctly different responses in many factors. More importantly, in the linear regression models, they show the greatest array of variables for explaining variation. In other words, Valais *f0* behavior is highly sensitive to external variables, particularly if paralinguistic and non-linguistic in nature. Moreover, the relative weight of the predictors often differs from the corresponding weight in other dialects. In spontaneous speech, this peculiar, dialect-internal structuring of the VS intonation results in what may be commonly perceived as an exotic and rather impalpable melody. Possibly, this is what characterizes the Valais variety as a ‘singing’ variety (Stalder 1819: 7–8). If we look at the amount of *f0* variation explained in the Valais model, it becomes evident that the Valais’ intonational structure is comparatively complex and literally less *predictable*. As for stress placement, Wipf (1910: 19) reported that “when first listening [to Valais Swiss German speakers], [...] one is overcome with an almost annoying sensation, as if the people place accents as strongly as possible on the most irrelevant of syllables” (translation AL). This early, impressionistic description on the somewhat erratic and highly unsystematic behavior of the Valais *f0* contours, too, can be corroborated by the statistical analyses in this study.

The ZURICH dialect stands out with a number of exceptional features in *PC durations*, most of which hint at comparatively longer PC durations. Yet, if compared to the fairly idiosyncratic *f0* behavior of other dialects, Zurich Swiss German does not feature any truly flashy properties. This is perhaps one of the reasons as to why Zurich German is known as a rather *neutral* dialect (see Ris 1992). The distinctive *PC durations* found for this variety may be a result of the participants’ cooperative and self-confident attitude during the interviews. Cooperative speakers generally show greater communicative effort, and thus also exhibit increased vocal effort and activity, which results in longer mean lengths of utterances (see Van Kleeck & Street 1982). Furthermore, Zurich speakers are generally fast speakers – another stereotype (see Ris 1992) that was empirically confirmed in this study – and faster articulation rate was shown to result in a

reduction of phrase boundaries (see Mixdorff 1998:176; Ladd et al. 1998), thus contributing to more distinct overall phrase length.

In the course of the analyses, distinct geolinguistic features of the Alpine/Midland and Eastern/Western dialects emerged. The most striking difference between the Alpine and Midland groups is found in the relative weight of the *linguistic* predictors in the *AC amplitude* models, including the predictor *stress*. It was argued that the comparatively high importance of these linguistic predictors in the Midland dialects is a result of the dialects' proximity to Germany since, in German, stressed syllables are clearly differentiated from unstressed syllables by means of *f0* movements, intensity, and duration (see Isačenko & Schädlich 1970; Mixdorff 1998). Possible reasons for the low importance of these linguistic predictors in the Alpine varieties' models could be found in the dialects' respective contacts with the right-headed Romance languages French (syllable-timed), Italian, and Romansh. The clash of two systems with different prosodic means for prominence marking, and a clash of the rhythmical patterns of Germanic and Romance languages may have "neutralized" the importance of *stress* and its link to *f0* movements, resulting in insignificant effects of *stress* in the Alpine dialects' *AC amplitude* models. The differences between Eastern and Western dialects are reflected in the groups' dissimilar intonation behavior with respect to global magnitude and duration parameters. The implementation of *PC* parameters for *phrase type* demarcation is strikingly more rigorous in the Western dialects than in the Eastern dialects. Possible sociolinguistic as well as language contact-related factors were again postulated.

As for the findings on the statistical models, a dialect-overarching assessment of the coefficients of determination of the created models reflected that the fractions of the variances explained in the dependent variables are generally low. It was argued that the principle reason behind the low coefficients is the spontaneous nature of the data analyzed in this study. Spontaneous speech is highly multi-layered and variable. It is characterized by a constant interplay and overlap of properties that were categorized into a *linguistic* layer (whose representatives in this study are syllable nuclei, syllable structure, stress, and word class), a *paralinguistic* layer (focus, phrase type, and prosodic paragraphing), and a *non-linguistic* layer (articulation rate, emotion, and sex) (see Fujisaki 2004). It is these characteristics and its multi-level structure that make it exceedingly difficult to predict and/or explain spontaneous speech *f0* patterns.

An overall assessment of the relative weight of the predictors underlined the importance of *paralinguistic predictors* – *focus* in particular – for explaining variability in *f0* parameters. The primary factor behind *f0* variability seems to be caused by a speaker's urge to highlight important information and to successfully communicate one's message by means of modulating the following parameters in

particular: *PC duration*, *AC amplitude*, and *timing*. In stark contrast, *linguistic* predictors seem to generally cause little effects in spontaneous speech *f0* contours.

The results of this study serve as a first starting point for more controlled experiments in which *f0* behavior of Swiss German dialectal speech can be tested for specific linguistic, paralinguistic, or non-linguistic predictors. This particularly concerns the non-linguistic predictors *articulation rate* as well as *emotion*, since, in the current study, both factors resulted in ambivalent results on a cross-dialectal scale.

Moreover, it would certainly be worth conducting controlled experiments in order to investigate in more detail the acoustic correlates of metrical *stress* in the context of the given dialects, particularly the roles of *intensity* and *f0*. Until today, the acoustic correlates of *stress* are only poorly understood, let alone studied in the context of Swiss German dialects. Ulbrich and Hirschfeld (2002) point to highly peculiar intensity patterns in Swiss German, intensity being a primary marker for *stress*. In several instances during this study, the dialects', particularly the Alpine varieties', incongruence between metrical *stress* and *f0 movements* was highlighted, and informal intensity measurements have been performed which confirmed the hypothesis that intensity differences from nucleus to nucleus are minor in the Grisons dialect, for example, regardless of whether the syllable is stressed or not. Vaissière (1983:66) observed that the specific interrelations between the three suprasegmental features (*f0*, duration, and intensity) for prominence marking may be the most salient characteristic by which to differentiate the dialects. Now that the technical means are available, a study on the acoustic correlates of stress would certainly be worthwhile as this would open doors to a more comprehensive and a more accurate picture of our idioms' prosodic behavior – a highly fascinating task for future research.

Further, it would be highly appealing to conduct a number of perception experiments. Leemann and Siebenhaar (2008) have already conducted such perception experiments based on the current data. Subjects were presented with filtered dialectal speech material that contained samples of all four dialects (frequencies between 250 and 7000 Hz were filtered). The speech was devoid of segmental cues and the subjects were asked to locate the speaker's dialectal origin based on what they heard. It turned out that an identification based on merely prosodic cues alone was possible, yet with relatively poor rates, for dialects except for the Grisons variety. Perception experiments could further aid to test the results contrived in this study and to assess whether or not the fundamental frequency modeling as well as the statistical modeling is valid not only from a speech production point of view, but also from the angle of speech perception. From a more sociolinguistic, dialectological perspective, it would be highly intriguing to examine whether the dialects' specific *f0* patterns, if tested individually, would

indeed evoke the notions described in the impressionistic accounts postulated by the mentioned scholars.

A final, admittedly ambitious, task for future research would be to develop a comprehensive regional atlas of Swiss German dialectal intonation, based on either controlled or natural data, similar to Auer et al.'s (2000) project on regional German intonation. Certain geolinguistic trends in f_0 behavior that surfaced throughout this study could be examined, corroborated, or refuted in greater detail if more dialects from all corners of German-speaking Switzerland were included. The established methods for an f_0 modeling of spontaneous speech based on the Command-Response model (Fujisaki & Hirose 1982) as well as the quantitative statistical methods applied in this study seem to be well suited for such an investigation.

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Appendix

¹ BE $r(2600) = .3, p < .0001$; GR $r(2866) = .43, p < .0001$; VS $r(2894) = .26, p < .0001$; ZH $r(2788) = .3, p < .0001$.

² $R^2 = .06$, $F(1, 2392) = 153, p < .0001$.

³ $F(3, 4826) = 47.5, p < .0001$.

⁴ GR with BE $t(4826) = 3.5, p = .0004$; VS with BE $t(4826) = -4.3, p < .0001$; VS with GR $t(4826) = -7.8, p < .0001$; ZH with BE $t(4826) = -7.6, p < .0001$; ZH with GR $t(4826) = -11.1, p = .0000$; ZH with VS $t(4826) = -3.4, p = .0007$.

⁵ $F(3, 4831) = 5.7, p = .0006$.

⁶ GR with BE $t(4831) = -2.5, p = .014$, ZH with GR $t(4831) = 3.9, p < .0001$, VS with ZH $t(4831) = 2.7, p = .0066$.

⁷ $F(3, 4830) = 20.9, p < .0001$.

⁸ VS with BE $t(4830) = 4.5, p < .0001$, VS with GR $t(4830) = 6.1, p < .0001$, ZH with BE $t(4830) = 6.1, p < .0001$, and ZH with GR $t(4830) = 6.5, p < .0001$.

⁹ $F(3, 4830) = 13, p < .0001$.

¹⁰ VS with BE $t(4830) = 4.4, p < .0001$, VS with GR $t(4830) = 2.7, p = .007$, ZH with BE $t(4830) = 5.7, p < .0001$, and ZH with GR $t(4830) = 4.0, p < .0001$.

¹¹ $R^2 = .013$, $F(1, 11065) = 155, p < .0001$.

¹² $R^2 = .02$, $F(1, 11065) = 234, p < .0001$.

¹³ $R^2 = .02$, $F(1, 11090) = 285, p < .0001$.

¹⁴ $R^2 = .02$, $F(1, 11070) = 197, p < .0001$.

¹⁵ $F(3, 11222) = 18.9, p < .0001$.

¹⁶ GR with BE $t(11222) = 6.9, p < .0001$, VS with BE $t(11222) = 3.9, p < .0001$, VS with GR $t(11222) = -3.05, p = .0023$, ZH with GR $t(11222) = -5.7, p < .0001$, and ZH with VS $t(11222) = -2.7, p = .0083$.

¹⁷ $t(11222) = 1.3, p = .19$.

¹⁸ $F(3, 11240) = 8.9, p < .0001$.

¹⁹ GR with BE: $t(11240) = -27, p = .0053$; VS with BE: $t(11240) = -4.8, p < .0001$; ZH with BE: $t(11240) = -4.1, p < .0001$.

²⁰ $F(3, 11240) = 11.5, p < .0001$.

²¹ VS with BE: $t(11240) = 4.6, p < .0001$; VS with GR: $t(11240) = 3.7, p = .0003$; ZH with BE: $t(11240) = 4.6, p < .0001$; ZH with GR: $t(11240) = 3.7, p = .0001$.

²² $F(3, 9619) = 25, p < .0001$.

²³ GR with BE $t(9619) = 7.7, p < .0001$; VS with BE $t(9619) = -7.3, p < .0001$; ZH with BE $t(9619) = -5.7, p < .0001$; ZH with GR $t(9619) = 2, p = .04$.

²⁴ $R^2 = .19$, $F(1,11144) = 1726$, $p < .0001$.

²⁵ *T1dist_rise* and *AC duration*: Pearson's correlation $r(9655) = .21$, $p < .0001$, *T2dist_rise* and *AC duration*: Pearson's correlation $r(9655) = -.26$, $p < .0001$.

²⁶ Pearson's correlation $r(1519) = -.7$, $p < .0001$.

²⁷ $X^2(6, 11245) = 45$, $p < .0001$.

²⁸ Pearson's correlation $r(11135) = .06$, $p < .0001$.

²⁹ BE: 0 ($M = .39$, $SD = .11$), 1 ($M = .44$, $SD = .12$), 2 or more ($M = .43$, $SD = .11$), ANOVA $F(2, 2617) = 48$, $p < .0001$. GR: 0 ($M = .43$, $SD = .13$), 1 ($M = .45$, $SD = .14$), 2 or more ($M = .45$, $SD = .13$), ANOVA $F(2, 2887) = 3$, $p = .04$. VS: 0 ($M = .43$, $SD = .13$), 1 ($M = .44$, $SD = .12$), 2 or more ($M = .44$, $SD = .11$), ANOVA $F(2, 2906) = 3$, $p = .04$. ZH: 0 ($M = .40$, $SD = .11$), 1 ($M = .44$, $SD = .13$), 2 or more ($M = .43$, $SD = .11$), ANOVA $F(2, 2802) = 24$, $p < .0001$.

³⁰ T-tests: 1 with 0 $t(2617) = 9.3$, $p < .0001$ and 2 or more with 0 $t(2617) = 6.4$, $p < .0001$.

³¹ T-test: 1 with 0 $t(2887) = 2.5$, $p = .02$.

³² T-test: 1 with 0 $t(2906) = 2.5$, $p = .01$.

³² T-test: 1 with 0 $t(2802) = 7$, $p < .0001$, 2 or more 0 $t(2802) = 3.1$, $p = .002$, and 2 or more with 1 $t(2802) = -2.3$, $p = .03$.

³⁴ 0 stressed syllables: ANOVA $F(3, 3741) = 28$, $p < .0001$, 2 or more stressed syllables: ANOVA $F(3, 1851) = 3$, $p = .03$.

³⁵ T-tests: GR with BE $t(3741) = 8$, $p < .0001$, VS with BE $t(3741) = 7$, $p < .0001$, ZH with BE $t(3741) = 2.3$, $p = .02$, ZH with GR $t(3741) = -6$, $p < .0001$, and ZH with VS $t(3741) = -4.2$, $p < .0001$.

³⁶ T-test: $t(1851) = -3$, $p = .003$.

³⁷ BE: Pearson's correlation $r(2609) = .54$, $p < .0001$, GR: Pearson's correlation $r(2867) = .59$, $p < .0001$, VS: Pearson's correlation $r(2900) = .61$, $p < .0001$, ZH: Pearson's correlation $r(2780) = .61$, $p < .0001$.

³⁸ 0 stressed syllables: BE ($M = .36$, $SD = .2$), GR ($M = .32$, $SD = .17$), VS ($M = .31$, $SD = .18$), ANOVA $F(3, 3757) = 8$, $p < .0001$, 1 stressed syllable: BE ($M = .49$, $SD = .23$), GR ($M = .46$, $SD = .21$), VS ($M = .45$, $SD = .21$), ZH ($M = .45$, $SD = .19$), ANOVA $F(3, 5625) = 12$, $p < .0001$, 2 or more stressed syllables: BE ($M = .83$, $SD = .38$), GR ($M = .81$, $SD = .42$), VS ($M = .73$, $SD = .34$), ZH ($M = .75$, $SD = .32$), ANOVA $F(3, 1851) = 8$, $p < .0001$.

³⁹ T-tests: GR with BE $t(3757) = -4$, $p = .0002$, VS with BE $t(3757) = -4.5$, $p < .0001$, and ZH with BE $t(3757) = -3.5$, $p = .0004$.

⁴⁰ T-tests: GR with BE $t(5625) = -3.8$, $p = .0002$, VS with BE $t(5625) = -5.1$, $p < .0001$, and ZH with BE $t(5625) = -5.2$, $p < .0001$.

⁴¹ T-tests: VS with BE $t(1851) = -4$, $p < .0001$, VS with GR $t(1851) = -3.5$, $p = .0005$, ZH with BE $t(1851) = -3.3$, $p = .0008$, and ZH with GR $t(1851) = -2.6$, $p = .009$.

⁴² BE: Pearson's correlation $r(2252) = .13$, $p < .0001$, GR: Pearson's correlation $r(2408) = .11$, $p < .0001$, VS: Pearson's correlation $r(2510) = .18$, $p < .0001$, ZH: Pearson's correlation $r(2441) = .14$, $p < .0001$.

⁴³ 0 stressed syllables in AC $F(3, 2784) = 7$, $p < .0001$, 1 stressed syllable in AC $F(3, 5012) = 19$, $p < .0001$, 2 or more stressed syllables in AC $F(3, 1821) = 5$, $p = .00013$.

⁴⁴ T-tests: 0 stressed syllables in AC: GR with BE $t(2784) = -3.4$, $p = .0008$, VS with BE $t(2784) = -4$, $p < .0001$, and ZH with BE $t(2784) = -4$, $p = .0001$. ACs with 1 stressed syllable: GR with BE

$t(5012) = -6.7, p < .0001$, VS with BE $t(5012) = -6.6, p < .0001$, ZH with BE $t(5012) = -4.5, p < .0001$, ZH with GR $t(5012) = 2.3, p = .03$, and ZH with VS $t(5012) = 2.3, p = .03$.

⁴⁵ $X^2(6, 11182) = 41, p < .0001$.

⁴⁶ BE: 0 ($M = .39, SD = .11$), first ($M = .45, SD = .13$), second or later ($M = .41, SD = .11$), ANOVA $F(2, 2600) = 70, p < .0001$. GR: 0 ($M = .43, SD = .13$), first ($M = .46, SD = .14$), second or later ($M = .42, SD = .13$), ANOVA $F(2, 2867) = 17, p = .04$. VS: 0 ($M = .43, SD = .13$), first ($M = .45, SD = .12$), second or later ($M = .42, SD = .11$), ANOVA $F(2, 2894) = 15, p < .0001$. ZH: 0 ($M = .40, SD = .11$), first ($M = .45, SD = .13$), second or later ($M = .41, SD = .11$), ANOVA $F(2, 2788) = 44, p < .0001$.

⁴⁷ T-tests: first with 0 $t(2600) = 12, p < .0001$, second or later with 0 $t(2600) = 2.8, p = .005$, and second or later with first $t(2600) = -6.2, p < .0001$.

⁴⁸ T-tests: first with 0 $t(2788) = 8.5, p < .0001$ and second or later with first $t(2788) = -7, p < .0001$.

⁴⁹ T-tests: GR first with 0 $t(2867) = 4.3, p < .0001$ and second or later with first $t(2867) = -5, p < .0001$; VS first with 0 $t(2894) = 4.3, p < .0001$ and second or later with first $t(2894) = -5, p < .0001$.

⁵⁰ 0: ANOVA $F(3, 3812) = 28, p < .0001$, second or later: ANOVA $F(3, 2269) = 3, p = .04$.

⁵² T-tests: GR with BE $t(3812) = 8, p < .0001$, VS with BE $t(3812) = 6.6, p < .0001$, ZH with BE $t(3812) = 2.4, p = .02$, ZH with GR $t(3812) = -5.6, p < .0001$, and ZH with VS $t(3812) = -4, p < .0001$.

⁵² T-tests: GR with BE $t(2269) = 2, p = .03$ and ZH with GR $t(2269) = -2.5, p = .02$.

⁵³ $X^2(6, 11169) = 40, p < .0001$.

⁵⁴ BE: 0 ($M = .38, SD = .09$), 1 ($M = .42, SD = .12$), 2 or more ($M = .44, SD = .12$), ANOVA $F(2, 2600) = 41, p < .0001$. GR: 0 ($M = .42, SD = .12$), 1 ($M = .45, SD = .14$), 2 or more ($M = .46, SD = .14$), ANOVA $F(2, 2867) = 15, p < .0001$. VS: 0 ($M = .41, SD = .12$), 1 ($M = .43, SD = .13$), 2 or more ($M = .44, SD = .12$), ANOVA $F(2, 2894) = 15, p < .0001$. ZH: 0 ($M = .38, SD = .10$), 1 ($M = .44, SD = .13$), 2 or more ($M = .43, SD = .12$), ANOVA $F(2, 2788) = 44, p < .0001$.

⁵⁵ T-tests: BE: 1 with 0 $t(2600) = 6, p < .0001$, 2 or more with 0 $t(2600) = 9, p < .0001$, and 2 or more with 1 $t(2600) = 3, p < .0001$; GR: 1 with 0 $t(2867) = 4, p < .0001$ and 2 or more with 0 $t(2867) = 5.4, p < .0001$; VS: 1 with 0 $t(2894) = 3.6, p < .0001$, 2 or more with 0 $t(2894) = 5.5, p < .0001$, and 2 or more with 1 $t(2894) = 2, p = .049$; ZH: 1 with 0 $t(2788) = 8.5, p < .0001$ and 2 or more with 0 $t(2788) = 8.5, p < .0001$.

⁵⁶ 0: ANOVA $F(3, 2204) = 15, p < .0001$, 1: ANOVA $F(3, 3605) = 6, p = .0006$, 2 or more ANOVA $F(3, 5340) = 7, p = .0002$.

⁵⁷ T-tests: GR with BE $t(2204) = 5.5, p < .0001$, VS with BE $t(2204) = 3.8, p < .0001$, ZH with GR $t(2204) = -5.3, p < .0001$, and ZH with VS $t(2204) = -3.7, p = .0003$.

⁵⁸ T-tests: GR with BE $t(3605) = 4, p < .0001$, VS with GR $t(3605) = -2.3, p = .03$, and ZH with BE $t(3605) = 2.6, p = .009$.

⁵⁹ T-tests: GR with BE $t(5340) = 3.2, p = .001$, VS with GR $t(5340) = -2.4, p = .02$, ZH with GR $t(5340) = -4.3, p < .0001$, and ZH with VS $t(5340) = -2, p = .044$.

⁶⁰ BE: no focus ACs ($M = .38, SD = .11$), pre-focus ACs ($M = .38, SD = .11$), focus ACs ($M = .46, SD = .13$), post-focus ACs ($M = .44, SD = .11$), ANOVA $F(3, 1535) = 36, p < .0001$. GR: no focus

ACs ($M = .40$, $SD = .11$), pre-focus ACs ($M = .40$, $SD = .12$), focus ACs ($M = .48$, $SD = .14$), post-focus ACs ($M = .49$, $SD = .12$), ANOVA $F(3, 1823) = 48$, $p < .0001$. VS: no focus ACs ($M = .41$, $SD = .10$), pre-focus ACs ($M = .42$, $SD = .11$), focus ACs ($M = .46$, $SD = .13$), post-focus ACs ($M = .45$, $SD = .11$), ANOVA $F(3, 1793) = 18$, $p < .0001$. ZH: no focus ACs ($M = .40$, $SD = .11$), pre-focus ACs ($M = .39$, $SD = .10$), focus ACs ($M = .47$, $SD = .13$), post-focus ACs ($M = .45$, $SD = .11$), ANOVA $F(3, 1535) = 39$, $p < .0001$.

⁶¹ T-tests show the following significant differences: focus with no focus ACs $t(1535) = 6.9$, $p < .0001$, focus with pre-focus ACs $t(1535) = 9$, $p < .0001$, post-focus with no focus ACs $t(1535) = 4.9$, $p < .0001$, post-focus with pre-focus, $t(1535) = 6.4$, $p < .0001$.

⁶² T-tests show the following significant differences: focus with no focus ACs: $t(1819) = 7.8$, $p < .0001$, focus with pre-focus ACs $t(1819) = 9.2$, $p < .0001$, post-focus with no focus ACs $t(1819) = 7.6$, $p < .0001$, and post-focus with pre-focus ACs $t(1819) = 8.7$, $p < .0001$.

⁶³ T-tests show the following significant differences: focus with no focus ACs $t(1789) = 6$, $p < .0001$, focus with pre-focus ACs $t(1789) = 5.4$, $p < .0001$, post-focus with no focus ACs $t(1789) = 3.8$, $p < .0001$, post-focus with pre-focus ACs $t(1789) = 3.1$, $p = .0002$, and post-focus with focus ACs $t(1789) = -2.3$, $p = .02$.

⁶⁴ T-tests show the following significant differences: focus with no focus ACs $t(1765) = 7.7$, $p < .0001$, focus with pre-focus ACs $t(1765) = 9.3$, $p < .0001$, post-focus with no focus ACs $t(1765) = 4.9$, $p < .0001$, post-focus with pre-focus ACs $t(1765) = 6$, $p < .0001$, and post-focus with focus ACs $t(1765) = -2.8$, $p = .005$.

⁶⁵ Pre-focus ACs: $F(3, 1146) = 6$, $p = .0005$, focus ACs: $F(3, 3268) = 7$, $p < .0001$, $F(3, 1633) = 14$, $p < .0001$.

⁶⁶ T-tests: GR with BE $t(1146) = 2.6$, $p = .01$, VS with BE $t(1146) = 4.2$, $p < .0001$, ZH with BE $t(1146) = 2$, $p = .046$, and ZH with VS $t(1146) = -2.3$, $p = .02$.

⁶⁷ T-tests: GR with BE $t(3268) = 4.5$, $p < .0001$, VS and GR $t(3268) = -3$, $p < .003$, and ZH and BE $t(3268) = 2.4$, $p = .02$.

⁶⁸ T-tests: GR with BE $t(1633) = 5.6$, $p < .0001$, VS and GR $t(1633) = -5.3$, $p < .0001$, and ZH and GR $t(1633) = -4.7$, $p < .0001$.

⁶⁹ BE: no focus ACs ($M = .41$, $SD = .24$), pre-focus ACs ($M = .40$, $SD = .23$), focus ACs ($M = .56$, $SD = .31$), post-focus ACs ($M = .48$, $SD = .28$), ANOVA $F(3, 1540) = 27$, $p < .0001$. GR: no focus ACs ($M = .38$, $SD = .19$), pre-focus ACs ($M = .40$, $SD = .22$), focus ACs ($M = .56$, $SD = .31$), post-focus ACs ($M = .47$, $SD = .28$), ANOVA $F(3, 1818) = 37$, $p < .0001$. VS: no focus ACs ($M = .41$, $SD = .23$), pre-focus ACs ($M = .37$, $SD = .21$), focus ACs ($M = .54$, $SD = .29$), post-focus ACs ($M = .45$, $SD = .24$), ANOVA $F(3, 1793) = 37$, $p < .0001$. ZH: no focus ACs ($M = .37$, $SD = .19$), pre-focus ACs ($M = .42$, $SD = .24$), focus ACs ($M = .54$, $SD = .28$), post-focus ACs ($M = .45$, $SD = .25$), ANOVA $F(3, 1768) = 35$, $p < .0001$.

⁷⁰ T-tests show the following significant differences: focus with no focus ACs $t(1540) = 6.2$, $p < .0001$, focus with pre-focus ACs $t(1540) = 7.8$, $p < .0001$, post-focus with no focus ACs $t(1540) = 2.5$, $p = .01$, post-focus with pre-focus ACs $t(1540) = 3.3$, $p = .0004$, and post-focus with focus ACs $t(1540) = -4$, $p < .0001$.

⁷¹ T-tests show the following significant differences: focus with no focus ACs $t(1818) = 8$, $p < .0001$, focus with pre-focus ACs $t(1818) = < .0001$, post-focus with no focus ACs $t(1818) = 3.8$, $p = .0001$, post-focus with pre-focus ACs $t(1818) = 3$, $p = .003$, and post-focus with focus ACs $t(1818) = -5.3$, $p < .0001$.

⁷² T-tests show the following significant differences: focus with no focus ACs $t(1793) = 6.6$, $p < .0001$, focus with pre-focus ACs $t(1793) = 9.2$, $p < .0001$, post-focus with pre-focus ACs $t(1793) = 4$, $p < .0001$, and post-focus with focus ACs $t(1793) = -5.7$, $p < .0001$.

⁷³ T-tests show the following significant differences: pre-focus with no focus ACs $t(1768) = 2.3$, $p = .02$, focus with no focus ACs $t(1768) = 9$, $p < .0001$, focus with pre-focus ACs $t(1768) = 6.7$, $p < .0001$, post-focus with no focus ACs $t(1768) = 3.9$, $p < .0001$, and post-focus with focus ACs $t(1768) = -5.6$, $p < .0001$.

⁷⁴ Pre-focus ACs: $F(3, 1150) = 2.9$, $p = .04$.

⁷⁵ T-tests: $t(1150) = 2.9$, $p = .004$.

⁷⁶ BE: C phrases ($M = .47$, $SD = .12$) and T phrases ($M = .42$, $SD = .13$), $t(1231) = -6.5$, $p < .0001$; VS: C phrases ($M = .45$, $SD = .13$) and T phrases ($M = .40$, $SD = .11$), $t(1185) = -6.7$, $p < .0001$; ZH: C phrases ($M = .43$, $SD = .13$) and T phrases ($M = .39$, $SD = .14$), $t(1098) = -3.7$, $p = .0002$.

⁷⁷ A t-test showed that the differences between the PC magnitude of C and T phrases are significant for all dialects ($p < .0002$) except GR.

⁷⁸ C phrases: ANOVA $F(3, 3208) = 26.8$, $p < .0001$; T phrases: ANOVA $F(3, 1576) = 26.5$, $p < .0001$.

⁷⁹ T-tests: GR ($M = .48$, $SD = .13$) and ZH ($M = .43$, $SD = .13$): $t(3208) = -8.1$, $p < .0001$; BE ($M = .48$, $SD = .12$) and ZH: $t(3208) = 7.1$, $p < .0001$; GR and VS ($M = .46$, $SD = .13$): $t(3208) = -4.4$, $p < .0001$; VS and ZH: $t(3208) = -3.7$, $p = .0002$; BE and VS: $t(3208) = -4.4$, $p = .0007$.

⁸⁰ T-tests: GR ($M = .47$, $SD = .12$) and BE ($M = .43$, $SD = .13$): $t(1576) = 5$, $p < .0001$; VS ($M = .4$, $SD = .11$) and BE: $t(1576) = -2.2$, $p = .02$; VS and GR: $t(1576) = -7.3$, $p < .0001$; ZH ($M = .4$, $SD = .13$) and BE: $t(1576) = -2.8$, $p = .005$; ZH and GR: $t(1576) = -7.8$, $p < .0001$.

⁸¹ BE C phrases ($M = 1.45$, $SD = .9$), T phrases ($M = 1.6$, $SD = 1.16$); VS C phrases ($M = 1.37$, $SD = .85$), T phrases ($M = 1.63$, $SD = .93$); GR C phrases ($M = 1.39$, $SD = .96$), T phrases ($M = 1.41$, $SD = 1$); ZH C phrases ($M = 1.55$, $SD = .96$), T phrases ($M = 1.58$, $SD = 1.14$); BE: $t(1231) = 2.6$, $p = .005$; VS: $t(1185) = 4.7$, $p < .0001$.

⁸² A t-test shows that the PC duration differences between C and T phrases is significant only for the BE group $t(1231) = 2.6$, $p = .01$ and the VS group $t(1185) = 4.7$, $p = .0001$.

⁸³ C phrases: ANOVA $F(3, 3210) = 5.8$, $p = .0006$; T phrases: ANOVA $F(3, 1578) = 3.4$, $p = .02$.

⁸⁴ T-tests: ZH ($M = 1.55$, $SD = .96$) and BE ($M = 1.45$, $SD = .91$): $t(3210) = 2.2$, $p = .03$; ZH and GR ($M = 1.39$, $SD = .96$): $t(3210) = 3.4$, $p = .0008$; ZH and VS ($M = 1.37$, $SD = .85$): $t(3210) = 3.8$, $p < .0001$.

⁸⁵ T-tests: GR ($M = 1.4$, $SD = 1.0$) and BE ($M = 1.6$, $SD = 1.15$): $t(1578) = -2.6$, $p = .02$; GR and VS ($M = 1.62$, $SD = .92$): $t(1578) = 2.8$, $p = .005$; GR and ZH ($M = 1.58$, $SD = 1.14$): $t(1578) = 2.3$, $p = .03$.

⁸⁶ C phrases: BE ($M = .43$, $SD = .12$), GR ($M = .46$, $SD = .14$), VS ($M = .45$, $SD = .13$), and ZH ($M = .44$, $SD = .13$); T phrases: BE ($M = .41$, $SD = .11$), GR ($M = .44$, $SD = .12$), VS ($M = .42$, $SD = .11$), and ZH ($M = .42$, $SD = .11$). T-test between AC amplitude C phrases and AC amplitudes T phrases: BE $t(2605) = -3.3$, $p = .0008$, GR $t(2859) = -2.3$, $p = .02$, VS $t(2889) = -6.7$, $p < .0001$, ZH $t(2786) = -4$, $p < .0001$.

⁸⁷ A t-test showed that the differences in PC amplitudes for C and T phrases are significant for all dialects ($p < .02$).

⁸⁸ C phrases: ANOVA $F(3, 7412) = 11$, $p < .0001$; T phrases: ANOVA $F(3, 3727) = 10.4$, $p < .0001$.

⁸⁹ T-tests: GR ($M = .44$, $SD = .14$) with BE ($M = .43$, $SD = .12$) $t(4712) = 5$, $p < .0001$; VS ($M = .44$, $SD = .12$) with BE $t(7412) = 4.3$, $p < .0001$; ZH ($M = .43$, $SD = .12$) with GR $t(7412) = -3.7$, $p = .0002$, and ZH with VS $t(7412) = -2.9$, $p = .003$.

⁹⁰ T-tests: GR ($M = .44$, $SD = .12$) with BE ($M = .41$, $SD = .11$) $t(3727) = 4.8$, $p < .0001$; VS ($M = .41$, $SD = .11$) with GR $t(3727) = -4.3$, $p < .0001$; ZH ($M = .41$, $SD = .11$) with GR $t(3727) = -4.3$, $p < .0001$.

⁹¹ BE C phrases: $F(3, 1536) = 70.9$, $p < .0001$, T phrases: $F(3, 713) = 20.2$, $p < .0001$; GR C phrases: $F(3, 1646) = 99.8$, $p < .0001$, T phrases: $F(3, 811) = 16.6$, $p < .0001$; VS C phrases: $F(3, 1659) = 58.2$, $p < .0001$, T phrases: $F(3, 925) = 7.3$, $p < .0001$; ZH C phrases: $F(3, 1673) = 62.4$, $p < .0001$, T phrases: $F(3, 827) = 12.7$, $p < .0001$.

⁹² BE: C phrases: first ($M = .37$, $SD = .1$), medial ($M = .46$, $SD = .1$), penultimate ($M = .45$, $SD = .12$), ultimate ($M = .47$, $SD = .13$), T phrases: first ($M = .37$, $SD = .11$), medial ($M = .46$, $SD = .12$), penultimate ($M = .42$, $SD = .12$), ultimate ($M = .43$, $SD = .1$); GR: C phrases: first ($M = .37$, $SD = .11$), medial ($M = .5$, $SD = .13$), penultimate ($M = .5$, $SD = .14$), ultimate ($M = .49$, $SD = .14$), T phrases: first ($M = .4$, $SD = .12$), medial ($M = .47$, $SD = .11$), penultimate ($M = .46$, $SD = .12$), ultimate ($M = .46$, $SD = .13$); VS: C phrases: first ($M = .39$, $SD = .11$), medial ($M = .46$, $SD = .11$), penultimate ($M = .47$, $SD = .11$), ultimate ($M = .49$, $SD = .14$), T phrases: first ($M = .39$, $SD = .11$), medial ($M = .43$, $SD = .11$), penultimate ($M = .43$, $SD = .09$), ultimate ($M = .41$, $SD = .11$); ZH: C phrases: first ($M = .37$, $SD = .10$), medial ($M = .46$, $SD = .11$), penultimate ($M = .46$, $SD = .12$), ultimate ($M = .47$, $SD = .14$), T phrases: first ($M = .38$, $SD = .11$), medial ($M = .43$, $SD = .1$), penultimate ($M = .42$, $SD = .09$), ultimate ($M = .44$, $SD = .11$).

⁹³ A t-test shows that the differences in amplitude BE tween first and medial ACs are $p < .0001$ for all dialects.

⁹⁴ The distinction in amplitude BE tween medial and penultimate ACs in T phrases is only significant in the case of the BE dialect: $t(713) = -2.3$, $p = .02$. In all other instances, the differences are not significant.

⁹⁵ None of the differences in amplitude BE tween penultimate and ultimate ACs in T phrases are significant.

⁹⁶ A t-test reveals that the differences in amplitude BE tween first and medial ACs are $p < .0001$ for all dialects.

⁹⁷ None of these differences in amplitude BE tween medial and penultimate ACs in C phrases are significant.

⁹⁸ In the case of the BE dialect, the difference in AC amplitude BE tween penultimate and ultimate ACs in C phrases are significant: $t(794) = 2$, $p = .038$.

⁹⁹ A t-test of AC amplitude by phrase type, AC position, and dialect yielded the following significant results: BE: C and T phrase penultimate AC $t(411) = -2$, $p = .04$ and C and T phrase ultimate AC $t(746) = -4.4$, $p < .00001$; GR: C and T phrase first AC $t(743) = 3.1$, $p = .0019$, C and T phrase medial AC $t(539) = -21$, $p = .035$, C and T phrase penultimate AC $t(433) = -3$, $p = .0032$, and C and T phrase ultimate AC $t(742) = -2.9$, $p = .0044$; VS: C and T phrase medial AC $t(562) = -3.4$, $p = .0008$, C and T phrase penultimate AC $t(464) = -3.9$, $p < .0001$, and C and T phrase ultimate AC $t(780) = -7$, $p < .0001$; ZH: C and T phrase medial AC $t(602) = -2.3$, $p = .02$, C and T phrase penultimate AC $t(457) = -3.6$, $p = .0004$, and C and T phrase ultimate AC $t(721) = -3.6$, $p = .0003$.

¹⁰⁰ 4 AC positions (*first*, *medial*, *penultimate*, and *last*) were calculated under two conditions: T and C.C phrases: first AC position $F(3, 1973) = 4.7, p = .003$; medial position $F(3, 1380) = 11.3, p < .0001$; penultimate position: $F(3, 1182) = 7.8, p < .0001$. T phrases: first AC position $F(3, 1008) = 3.3, p = .019$; medial position $F(3, 675) = 6, p = .0005$; penultimate position $F(3, 583) = 3.1, p = .03$; ultimate position $F(3, 1010) = 5.8, p = .00006$.

¹⁰¹ C phrases: first AC position VS with BE $t(1973) = 3.3, p = .0009$, VS with GR $t(1973) = 3.1, p = .0020$, and ZH with VS $t(1973) = 3.3, p = .01$. Medial AC position GR with BE $t(1380) = 4.5, p < .0001$, VS with GR $t(1380) = -4, p < .0001$, and ZH with GR $t(1380) = -5, p < .0001$. Penultimate AC position GR with BE $t(1182) = 4.4, p < .0001$, VS with BE $t(1182) = 2.2, p = .03$, VS with GR $t(1182) = -2.2, p = .03$, and ZH with GR $t(1182) = -3.8, p = .0000$. Ultimate AC position GR with BE $t(1979) = 2.1, p = .04$. T phrases: first AC position GR with BE $t(1008) = 2.9, p = .003$ and VS with BE $t(1008) = 2.3, p = .02$. Medial AC position VS with BE $t(675) = -2, p = .04$, VS with GR $t(675) = -3.7, p = .0002$, and ZH with GR $t(675) = -3.5, p = .0005$. Penultimate AC position GR with BE $t(583) = 2.3, p = .02$ and ZH with GR $t(583) = -2.9, p = .0037$. Ultimate AC position GR with BE $t(1010) = 3, p = .003$, VS with GR $t(1010) = -4, p < .0001$, and ZH with GR $t(1010) = 1.6, p = .047$.

¹⁰² For the BE and ZH groups, the differences in *T1dist_rise* values BE tween the two phrase types is significant. BE $t(2247) = -2.6, p = .01$, ZH $t(2443) = -3.7, p = .0002$.

¹⁰³ C phrases: ANOVA $F(3, 6485) = 21, p < .0001$; T phrases: ANOVA $F(3, 3094) = 5, p = .002$.

¹⁰⁴ T-tests: GR ($M = -3, SD = 94$) and BE ($M = 18, SD = 92$) $t(6485) = -6.8, p < .0001$; VS ($M = -3, SD = 81$) and BE $t(6485) = -6.9, p < .0001$; ZH ($M = 5, SD = 88$) and BE $t(6485) = -4.3, p < .0001$; ZH and GR $t(6485) = 2.6, p = .005$; ZH and VS $t(6485) = 2.7, p = .008$.

¹⁰⁵ T-tests: GR ($M = -6.5, SD = 89$) with BE ($M = 7.6, SD = 93$) $t(3094) = -3, p = .003$; VS ($M = -4.7, SD = 79$) with BE $t(3094) = -2.7, p = .006$; ZH ($M = -9, SD = 88$) with BE $t(3094) = -3.7, p = .0003$.

¹⁰⁶ BE C phrases: $F(3, 1339) = 4.5, p = .0036$; BE T phrases: $F(3, 603) = .7, p = .5$; GR C phrases: $F(3, 1450) = 10.7, p < .0001$; GR T phrases: $F(3, 656) = 3.1, p = .03$; VS C phrases: $F(3, 1467) = 7, p = .0001$; VS T phrases: $F(3, 780) = 3, p = .03$; ZH C phrases: $F(3, 1514) = 5.3, p = .0012$; ZH T phrases: $F(3, 712) = 3.8, p = .02$.

¹⁰⁷ BE: C phrases: first ($M = 21, SD = 93$), medial ($M = -4, SD = 73$), penultimate ($M = 7, SD = 80$), ultimate ($M = 14, SD = 88$); T phrases: first ($M = -1, SD = 95$, medial ($M = 4, SD = 87$), penultimate ($M = 2, SD = 84$), ultimate ($M = 11, SD = 91$). GR: C phrases: first ($M = 12, SD = 100$), medial ($M = -20, SD = 5$), penultimate ($M = -22, SD = 85$), ultimate ($M = -5, SD = 84$); T phrases: first ($M = -7, SD = 94$), medial ($M = -25, SD = 80$), penultimate ($M = -20, SD = 87$), ultimate ($M = .46, SD = .13$). VS: C phrases: first ($M = 9, SD = 85$), medial ($M = -8, SD = 76$), penultimate ($M = -20, SD = 75$), ultimate ($M = -8, SD = 82$); T phrases: first ($M = 6, SD = 90$), medial ($M = -12, SD = 73$), penultimate ($M = -17, SD = 70$), ultimate ($M = -4, SD = 80$). ZH: C phrases: first ($M = -10, SD = 87$), medial ($M = 2, SD = 83$), penultimate ($M = 13, SD = 87$), ultimate ($M = 9, SD = 82$); T phrases: first ($M = -27, SD = 91$), medial ($M = -2, SD = 80$), penultimate ($M = -14, SD = 98$), ultimate ($M = .5, SD = 85$).

¹⁰⁸ A t-test shows that the differences BE tween ultimate and medial $t(656) = 2.7, p = .03$ as well as ultimate and penultimate $t(656) = 2.2, p = .03$ are significant.

¹⁰⁹ A t-test shows that the differences BE tween first and medial $t(780) = -2.2, p = .02$ as well as penultimate and first $t(780) = -2.8, p = .006$ are significant.

¹¹⁰ A t-test confirms significant differences BE tween medial and first ACs $t(712) = 2.7, p = .008$ as well as ultimate and first ACs $t(712) = 3.1, p = .002$.

¹¹¹ A t-test provides the following p values: the differences BE tween medial and first $t(1339) = -3.5, p = .0005$, penultimate and first $t(1339) = -2.2, p = .03$, as well as ultimate and medial $t(1339) = 2.3, p = .012$ are significant.

¹¹² In this group, a number of significant differences BE tween the positions are established: differences BE tween medial and first $t(1450) = -4.8, p < .0001$, penultimate and first $t(1450) = -4.8, p < .0001$, ultimate and first $t(1450) = -2.7, p = .006$, ultimate and medial $t(1450) = 2.3, p = .003$, as well as ultimate and penultimate $t(1450) = 2.5, p = .012$.

¹¹³ A t-test shows that the differences BE tween first AC positions and all other positions are significant: medial and first $t(1467) = -2.8, p = .005$, penultimate and first $t(1467) = -4.4, p < .0001$, as well as ultimate and first $t(1467) = -3, p = .003$.

¹¹⁴ T-test $t(575) = -2.7, p = .008$.

¹¹⁵ T-test $t(547) = -2, p = .043$.

¹¹⁶ T-test $t(552) = -2, p = .046$.

¹¹⁷ T-test $t(423) = -2.8, p = .005$.

¹¹⁸ C phrases: first position $F(3, 1573) = 8.4, p < .0001$, medial position $F(3, 1263) = 4.5, p = .004$, first position $F(3, 1089) = 13.2, p < .0001$, first position $F(3, 1845) = 7.3, p < .0001$. T phrases: first position $F(3, 720) = 4.2, p = .006$, medial position $F(3, 602) = 3.4, p = .02$, penultimate position $F(3, 517) = 1.1, p = .33$, ultimate position $F(3, 912) = 1.3, p = .24$.

¹¹⁸ T-tests show the following inter-level differences: ZH with BE $t(720) = -2.5, p = .01$, ZH with VS $t(720) = -3.4, p = .0007$.

¹²⁰ A t-test shows significant differences BE tween GR and BE $t(602) = -2.8, p = .005$ as well as ZH and GR $t(602) = 2.5, p = .011$.

¹²¹ T-tests indicate significant differences BE tween VS and BE $t(1573) = -2, p = .04$, ZH and BE $t(1573) = -5, p < .001$, ZH and GR $t(1573) = -3.3, p = .001$, as well as ZH and VS $t(1573) = -2.9$.

¹²² T-tests show significant differences for GR and BE $t(1263) = -2.2, p = .02$ as well as ZH and GR $t(1263) = 3.6, p = .0003$.

¹²³ Significant differences are established for GR and BE $t(1089) = -4, p < .0001$, VS and BE $t(1089) = -3.7, p = .0003$, ZH and GR $t(1089) = 5.1, p < .0001$, as well as ZH with VS $t(1089) = 4.7, p < .0001$.

¹²⁴ T-tests show significant differences BE tween GR and BE $t(1845) = -3.5, p = .0006$, VS and BE $t(1845) = -3.9, p < .0009$, ZH and GR $t(1845) = 2.5, p = .01$, as well as ZH and VS $t(1845) = 3, p = .003$.

¹²⁵ BE: $R^2 = .002, F(1, 1219) = 2.3, p = .13$, GR: $R^2 = .05, F(1, 1257) = 66, p < .0001$, VS: $R^2 = .02, F(1, 1182) = 14, p = .0002$, ZH: $R^2 = .04, F(1, 1087) = 45, p < .0001$.

¹²⁶ $R^2 = .01, F(1, 1182) = 13, p = .0003$.

¹²⁷ BE: $R^2 = .09, F(1, 1227) = 120, p < .0001$, GR: $R^2 = .10, F(1, 1266) = 141, p < .0001$, VS: $R^2 = .11, F(1, 1192) = 150, p < .0001$, ZH: $R^2 = .07, F(1, 1096) = 80, p < .0001$.

¹²⁸ $R^2 = .02, F(1, 1182) = 21, p < .0001$.

¹²⁹ $R^2 = .01, F(1, 1087) = 11, p < .0009$.

¹³⁰ $R^2 = .006$, $F(1, 1227) = 8$, $p = .005$.

¹³¹ Fast ($M = 1.9$, $SD = 1.3$), normal ($M = 1.46$, $SD = .93$) slow ($M = 1.5$, $SD = .97$), ANOVA $F(2, 1109) = 13$, $p < .0001$.

¹³² T-tests: normal with fast $t(1106) = -5$, $p < .0001$ and slow with fast $t(1106) = -4.3$, $p < .0001$.

¹³³ *Pearson's correlation* analysis of specific articulation rate and AC duration by dialect: BE: $r(2609) = -.2$, $p < .0001$, GR: $r(2867) = -.03$, $p = .08$, VS: $r(2900) = -.1$, $p < .0001$, and ZH: $r(2793) = -.14$, $p < .0001$.

¹³⁴ $\chi^2(12, 4835) = 192$, $p < .0001$.

¹³⁵ Bored ($M = 1.4$, $SD = 1.2$), disgust ($M = 1.3$, $SD = .7$), happy ($M = 1.6$, $SD = .9$), neutral ($M = 1.5$, $SD = .97$), ANOVA $F(3, 1215) = 1.6$, $p = .2$.

¹³⁶ Bored ($M = 2.8$, $SD = 1.75$), disgust ($M = 1.9$, $SD = .65$), happy ($M = 2.1$, $SD = .95$), and neutral ($M = 2.2$, $SD = .92$), ANOVA $F(3, 410) = 4.8$, $p = .003$.

¹³⁷ T-tests: disgust with bored $t(410) = -3.2$, $p = .002$, happy with bored $t(410) = -2.7$, $p = .009$, and neutral with bored $t(410) = -3.5$, $p = .0005$.

¹³⁸ T-test: females ($M = .46$, $SD = .12$), males ($M = .44$, $SD = .14$): $t(4829) = -6.5$, $p < .0001$.

¹³⁹ 0 ACs in IP ($M = .47$, $SD = .11$), 1 AC in IP ($M = .46$, $SD = .12$), 2 ACs in IP ($M = .45$, $SD = .12$), 3 ACs in IP ($M = .46$, $SD = .13$), 4 or more ACs in IP ($M = .48$, $SD = .12$), ANOVA $F(4, 2412) = 3.5$, $p = .007$.

¹⁴⁰ T-tests: 2 with 0 $t(2412) = -2$, $p = .04$, 4 or more with 1 $t(2412) = 2.8$, 4 or more with 2 $t(2412) = 3.4$, $p = .0007$, and 4 or more with 3 $t(2412) = 2$, $p = .004$.

¹⁴¹ T-tests: 0 $t(467) = -3.3$, $p = .003$, 1 $t(1333) = -2.6$, $p = .01$, 3 $t(783) = -2.3$, $p = .03$, 4 or more $t(996) = -5.5$, $p < .0001$.

¹⁴² Males ($M = .432$, $SD = .12$) and females ($M = .431$, $SD = .13$).

¹⁴³ Males ($M = .43$, $SD = .12$) and females ($M = .41$, $SD = .12$) $t(2618) = 3.8$, $p = .0002$.

¹⁴⁴ For the calculations in this subchapter, IPs with only one AC are not taken into consideration.

¹⁴⁵ Females: first ($M = .38$, $SD = .1$), medial ($M = .47$, $SD = .12$), penultimate ($M = .46$, $SD = .13$), ultimate ($M = .47$, $SD = .13$), ANOVA $F(3, 4810) = 186$, $p < .0001$. Males: first ($M = .39$, $SD = .11$), medial ($M = .46$, $SD = .11$), penultimate ($M = .46$, $SD = .11$), ultimate ($M = .46$, $SD = .13$), ANOVA $F(3, 5072) = 131$, $p < .0001$.

¹⁴⁶ Females: medial with first $t(4810) = 17$, $p < .0001$, penultimate with first $t(4810) = 16$, $p < .0001$, and ultimate with first $t(4810) = 21$, $p < .0001$. Males: medial with first $t(5072) = 15$, $p < .0001$, penultimate with first $t(5072) = 14$, $p < .0001$, and ultimate with first $t(5072) = 17$, $p < .0001$.

¹⁴⁷ T-test $t(3012) = 3$, $p = .002$.

¹⁴⁸ $F(6, 1195) = 33$, $p < .0001$. Bivariate tests confirmed that *strength of break* and *duration of previous IP* are not significant predictors, which is why the variables were not included in the model. Despite a significant association with the dependent variable in a bivariate test, *articulation rate* was removed as a predictor due to insignificant effects within the model.

¹⁴⁹ Effect tests: *sex* $F(1, .28) = 20$, $p < .0001$, *phrase type* $F(1, .47) = 32$, $p < .0001$, *focus* $F(1, .31) = 7$, $p < .0001$, and *PC magnitude of previous IP* $F(1, 1.32) = 90$, $p < .0001$.

¹⁵⁰ $F(4, 1239) = 66$, $p < .0001$. Bivariate tests confirmed that *focus*, *phrase type*, *sex*, and *duration of previous IP* are not significant predictors of the dependent variable and were therefore not included in the model.

¹⁵¹ Effect tests: *PC magnitude of previous IP* $F(1, 1.9) = 138, p < .0001$, *strength of break* $F(1, 1.2) = 87, p < .0001$, *rate* $F(2, .3) = 11, p < .0001$.

¹⁵² $F(9, 1161) = 27, p < .0001$. A bivariate test of *sex* and *emotion* confirmed no significant effects. Therefore *sex* was not included in the model.

¹⁵³ Effect tests: *PC magnitude of previous IP* $F(1, 1.5) = 114, p < .0001$, *strength of break* $F(1, .25) = 19, p < .0001$, *focus* $F(3, .2) = 5, p = .002$, *phrase type* $F(1, .38) = 28, p < .0001$, *duration of previous IP* $F(1, .17) = 13, p = .0004$, and *rate* $F(2, .12) = 4.5, p = .02$.

¹⁵⁴ $F(8, 1047) = 29, p < .0001$. A bivariate test confirmed that *duration of previous IP* is not a significant predictor of the dependent variable, which is why the variable was not included in the model. Despite a significant association with the dependent variable in a bivariate test, *sex* was removed as a predictor due to insignificant effects within the model.

¹⁵⁵ Effect tests: *rate* $F(2, .51) = 18, p < .0001$, *phrase type* $F(1, .27) = 19, p < .0001$, *focus* $F(3, .18) = 4, p = .006$, *strength of break* $F(1, 1.14) = 80, p < .0001$, *PC magnitude of previous IP* $F(1, .95) = 67, p < .0001$.

¹⁵⁶ $F(2, 1199) = 46, p < .0001$. Bivariate tests confirmed that *sex*, *emotion*, *duration of previous IP*, and *strength of break* are not significant predictors. These variables were therefore not included in the model. Despite a significant association with the dependent variable in bivariate tests, *articulation rate* and *phrase type* were removed as predictors due to insignificant effects within the model.

¹⁵⁷ Effect tests: *focus* $F(1, 79) = 83, p < .0001$ and *PC magnitude of previous IP* $F(1, 5) = 5, p < .02$.

¹⁵⁸ $F(3, 1254) = 60, p < .0001$. Bivariate tests confirmed that *sex*, *emotion*, *phrase type*, *duration of previous IP*, *strength of break*, and *PC magnitude of the previous IP* are not significant predictors. These variables were therefore not included in the model.

¹⁵⁹ Effect tests: *focus* $F(1, 133) = 154, p < .0001$, *rate* $F(2, 13) = 7, p = .0006$.

¹⁶⁰ $F(7, 1164) = 33, p < .0001$. Bivariate tests confirmed that *articulation rate*, *duration of the previous IP*, and *PC magnitude of the previous IP* are not significant predictors. These variables were therefore not included in the model.

¹⁶¹ Effect tests: *focus* $F(1, 96) = 149, p < .0001$, *sex* $F(1, 4.4) = 7, p = .009$, *strength of break* $F(1, 5) = 8, p = .006$, *phrase type* $F(1, 10) = 15, p < .0001$, *emotion* $F(1, 8) = 4, p = .005$.

¹⁶² $F(4, 1053) = 46, p < .0001$. Bivariate tests confirmed that *emotion*, *phrase type*, *strength of break*, and *PC magnitude of the previous IP* are not significant predictors. These variables were therefore not included in the model. Despite a significant association with the dependent variable in a bivariate test, *sex* was removed as a predictor due to insignificant effects within the model.

¹⁶³ Effect tests: *focus* $F(1, 134) = 153, p < .0001$, *rate* $F(2, 9) = 5, p = .005$, and *duration of previous IP* $F(1, 8) = 9, p = .003$.

¹⁶⁴ $F(14, 2537) = 29, p < .0001$. Despite a significant association with the dependent variable in bivariate tests, *word class* and *sex* were removed as predictors due to insignificant effects.

¹⁶⁵ Effect tests: *focus* $F(3, .96) = 25, p < .0001$, *phrase type* $F(4, 1.4) = 27, p < .0001$, *rate* $F(2, .7) = 27, p < .0001$, *emotion* $F(3, .3) = 8, p < .0001$, *stress* $F(2, .7) = 28, p < .0001$.

¹⁶⁶ $F(9, 2797) = 44, p < .0001$. Bivariate tests confirmed that *sex*, *emotion*, and *stress* are not significant predictors. These variables were therefore not included in the model. Despite a significant

association with the dependent variable in a bivariate test, *word class* was removed as a predictor due to insignificant effects within the model.

¹⁶⁷ Effect tests: *focus* $F(3, 2.1) = 43, p < .0001$, *phrase type* $F(4, 1.4) = 21, p < .0001$, *rate* $F(2, 1.6) = 50, p < .0001$.

¹⁶⁸ $F(14, 2827) = 21, p < .0001$. Bivariate tests confirmed that *sex* and *stress* are not significant predictors. These variables were therefore excluded.

¹⁶⁹ Effect tests: *focus* $F(3, .62) = 15, p < .0001$, *phrase type* $F(4, 1.5) = 27, p < .0001$, *articulation rate* $F(2, 1) = 39, p < .0001$, *emotion* $F(3, .17) = 4.4, p = .004$, and *word class* $F(2, .1) = 4, p = .03$.

¹⁷⁰ $F(12, 2718) = 26, p < .0001$. Bivariate tests confirmed that *articulation rate* and *sex* are not significant predictors. These variables were therefore excluded. Despite a significant association with the dependent variable in a bivariate test, *stress* was removed as a predictor due to insignificant effects.

¹⁷¹ Effect tests: *focus* $F(3, 1.1) = 28, p < .0001$, *phrase type* $F(4, 1) = 19, p < .0001$, *emotion* $F(3, .17) = 4, p = .006$, and *word class* $F(2, .6) = 21, p < .0001$.

¹⁷² $F(9, 2551) = 19, p < .0001$. Bivariate tests confirmed that *emotion* and *sex* are not significant predictors. These variables were therefore not included in the model.

¹⁷³ Effect tests: *articulation rate* $F(2, 5.7) = 35, p < .0001$, *phrase type* $F(4, 2.3) = 7, p < .0001$, and *focus* $F(3, 5.1) = 21, p < .0001$.

¹⁷⁴ $F(8, 2799) = 25, p < .0001$. A bivariate test confirmed that *emotion* is not a significant predictor. This variable was therefore not included in the model. Despite a significant association with the dependent variable in a bivariate test, *articulation rate* was removed as a predictor due to insignificant effects within the model.

¹⁷⁵ Effect tests: *phrase type* $F(4, 2.6) = 8.5, p < .0001$, *focus* $F(3, 8.5) = 39, p < .0001$, and *sex* $F(1, 1.8) = 23, p < .0001$.

¹⁷⁶ $F(18, 2841) = 28, p < .0001$. A bivariate test confirmed that *emotion* is not a significant predictor. This variable was therefore not included in the model. Despite a significant association with the dependent variable in a bivariate test, *phrase type* was removed as predictor due to insignificant effects within the model.

¹⁷⁷ Effect tests: *focus* $F(3, 8.33) = 41, p < .0001$, *sex* $F(1, .9) = 13, p = .0002$, and *articulation rate* $F(2, .64) = 4.8, p = .008$.

¹⁷⁸ $F(12, 2723) = 17, p < .0001$. A bivariate test confirmed that *sex* is not a significant predictor. This variable was therefore not included in the model.

¹⁷⁹ Effect tests: *focus* $F(3, 6.2) = 32, p < .0001$, *rate* $F(2, 1.4) = 11, p < .0001$, *emotion* $F(3, 1.1) = 5.7, p = .0007$, and *phrase type* $F(4, 2.3) = 8.9, p < .0001$.

¹⁸⁰ $F(10, 2198) = 11, p < .0001$. Despite a significant association with the dependent variable in a bivariate test, *emotion* was removed as predictor due to insignificant effects within the model.

¹⁸¹ Effect tests: *sex* $F(1, 72248) = 9.2, p = .002$, *rate* $F(2, 122615) = 7.8, p = .0004$, *phrase type* $F(4, 76044) = 2.5, p = .046$, and *focus* $F(3, 545291) = 23, p < .0001$.

¹⁸² $F(8, 2348) = 17, p < .0001$. A bivariate test confirmed that *emotion* is not a significant predictor. This variable was therefore not included in the model. Despite a significant association with the dependent variable in a bivariate test, *articulation rate* was removed as predictor due to insignificant effects within the model.

¹⁸³ Effect tests: *Phrase type* $F(4, 121445) = 4, p = .003$, *focus* $F(3, 54244) = 24, p < .0001$, and *sex* $F(1, 256896) = 34, p < .0001$.

¹⁸⁴ $F(7, 2458) = 13, p < .0001$. Bivariate tests confirmed that *emotion*, *articulation rate*, and *sex* are not significant predictors. These variables were therefore not included in the model.

¹⁸⁵ Effect tests: *phrase type* $F(4, 139566) = 5.7, p = .0001$ and *focus* $F(3, 429691) = 23, p < .0001$.

¹⁸⁶ $F(9, 2401) = 10, p < .0001$. A bivariate test confirmed that *focus* is not a significant predictor. This variable was therefore not included in the model. Despite a significant association with the dependent variable in a bivariate test, *articulation rate* was removed as predictor due to insignificant effects within the model.

¹⁸⁷ Effect tests: *articulation rate* $F(2, 100487) = 7.1, p = .0008$, *phrase type* $F(4, 67694) = 2.4, p = .048$, and *focus* $F(3, 394833) = 19, p < .0001$.

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